Active Directory Domain Services Installation and Security Reinforcement

The project's primary focus was the end-to-end process of installing and hardening Active Directory on a newly provisioned Windows Server. After establishing a private NAT Network, the server was promoted to a Domain Controller, creating the nxgluxurylimited.local. The administrative structure was then organized with OUs, Groups, and Users, followed by the successful integration of a client endpoint. The crucial hardening phase involved creating and linking restrictive Group Policy Objects (GPOs) to limit user and computer capabilities such as disabling shutdown and command prompt access ensuring a secure and centrally managed domain environment.

Visual Structure (Conceptual Diagrams)

Diagram A: Domain Hierarchy

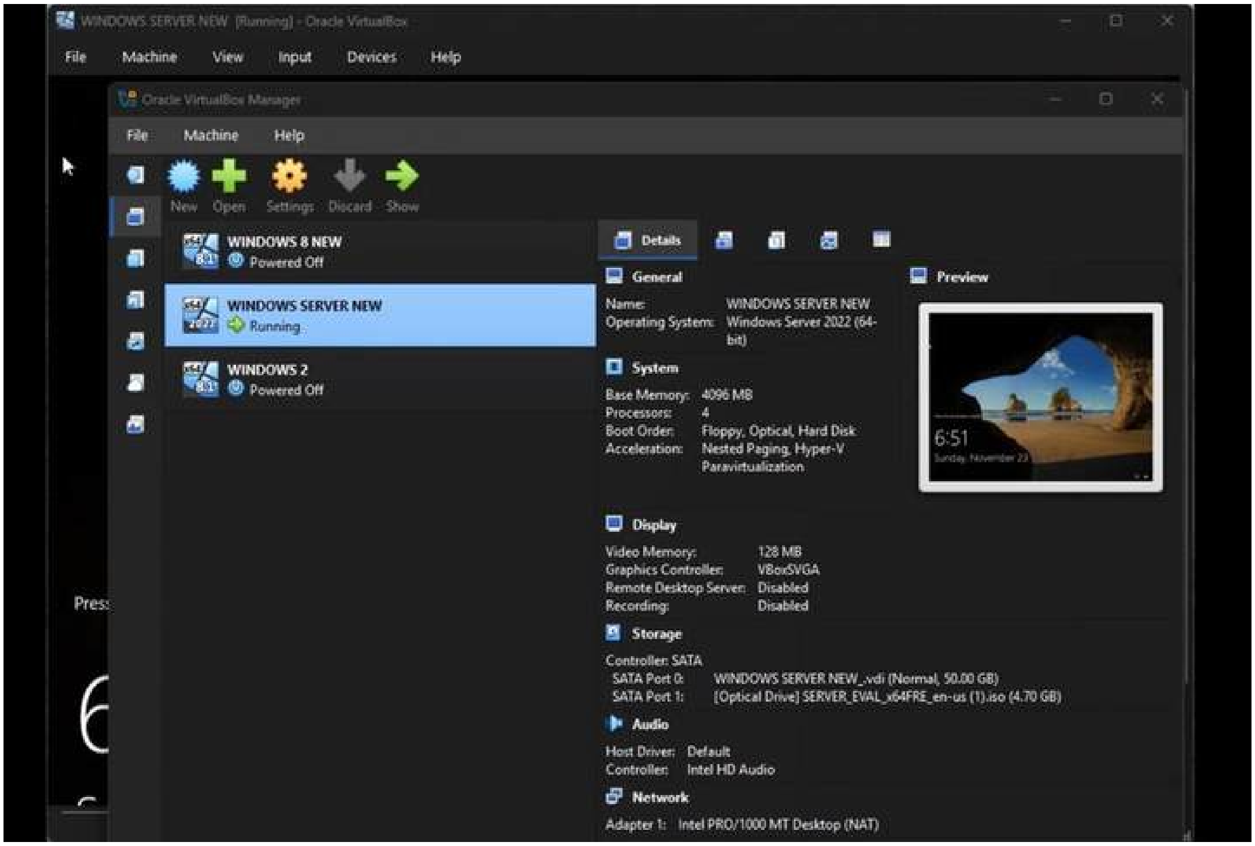
Illustrates the logical structure created within Active Directory to manage users and resources.

```
              [nxgluxurylimited.local] Domain Root
                            |
    ---------------------________________________________
      |                                    |
    [Uk] OU                            [Canada] OU
       |                                   |
    [IT] Group.                          [Marketing] Group
       |                                   |
   (debz.IT)        (Username )        (Jay.Market)
```

Let me walk through all the steps;

Stage 1: Secure VM Access

The process commenced by powering on the Windows Server VM from the VirtualBox Manager.

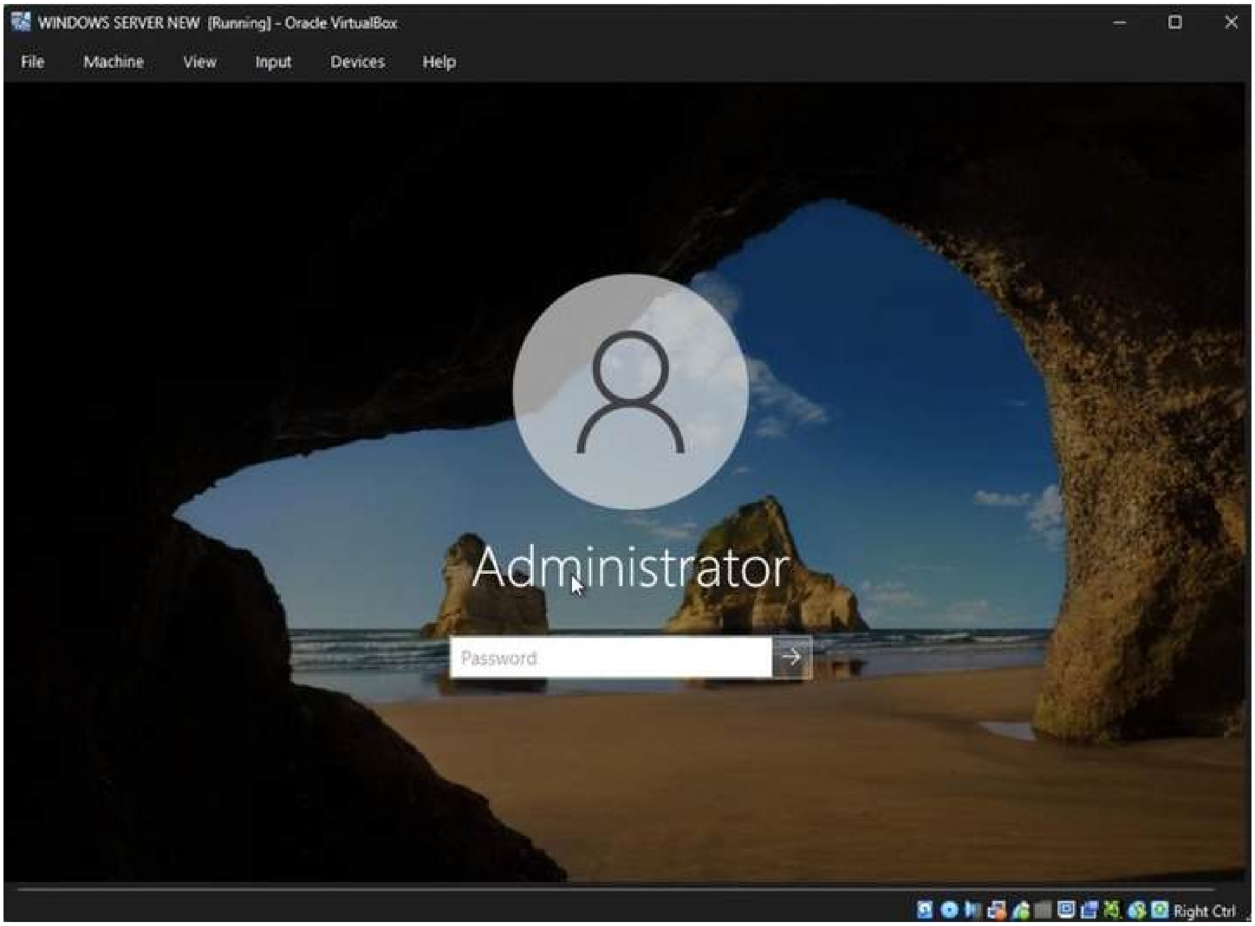


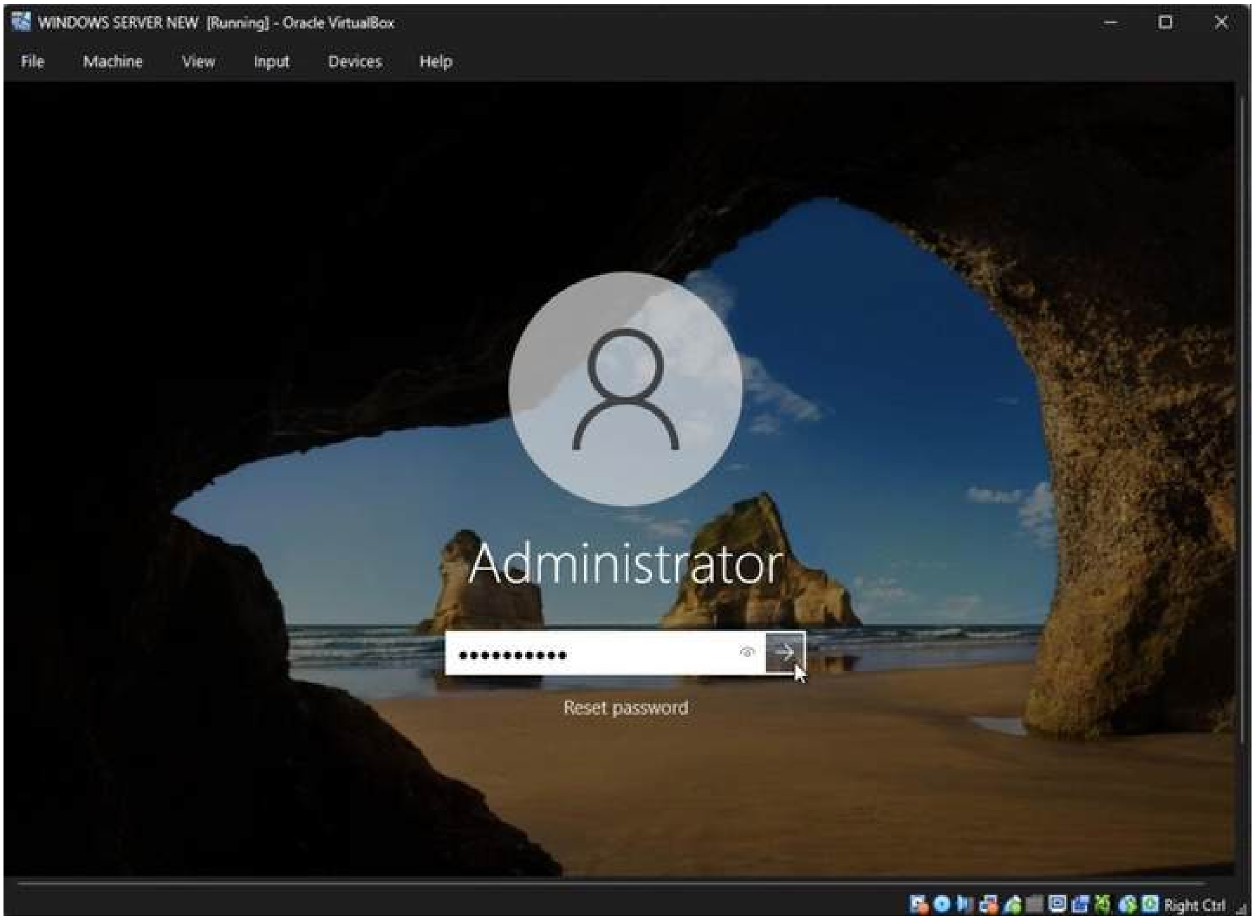
Upon reaching the lock screen prompt for the Secure Attention Sequence (Ctrl+Alt+Del),

Then I correctly navigated the VirtualBox interface (Input > Keyboard) to execute the virtual command.

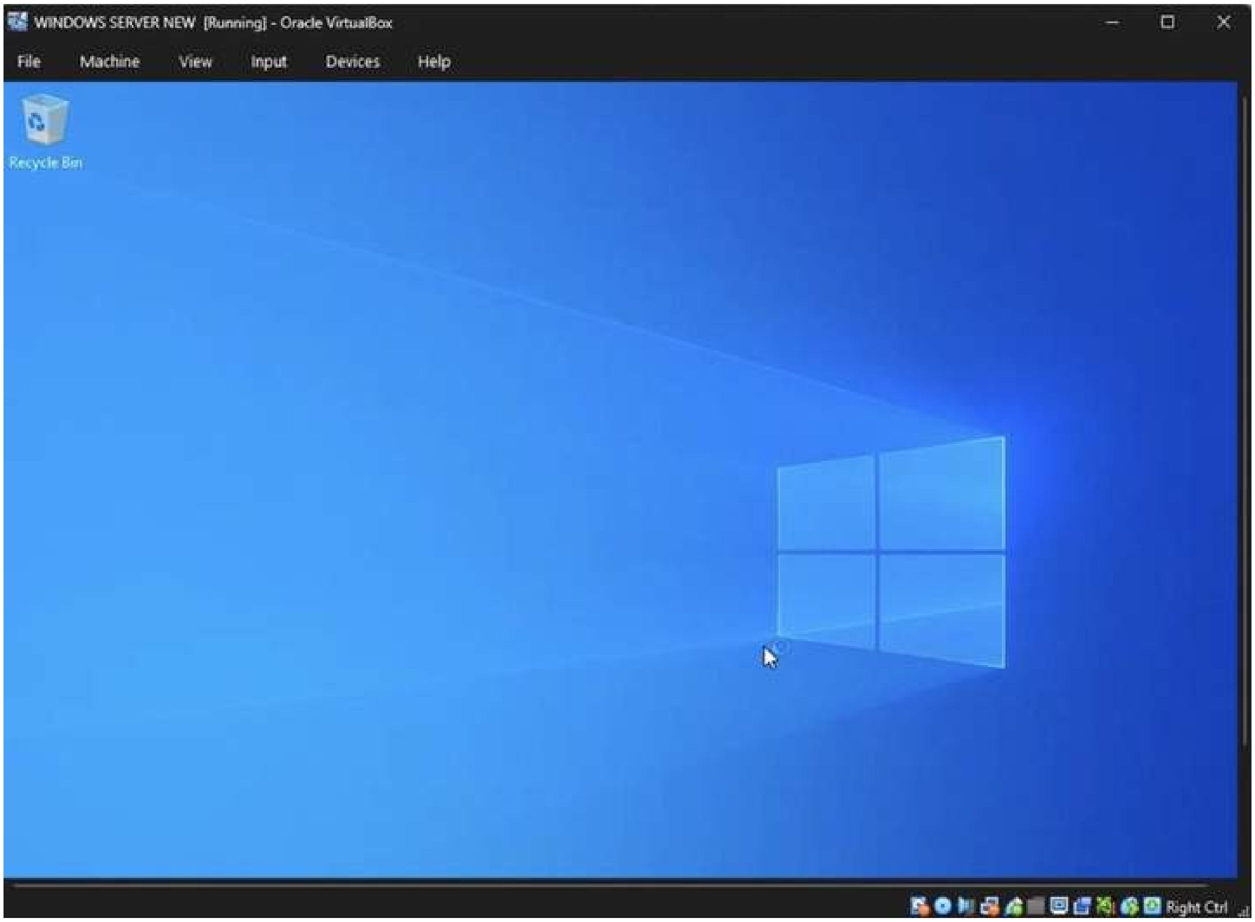
“This is a necessary step, as the physical host keyboard command is intercepted by the host operating system.”

The required administrative credentials were input to successfully log into the Windows Server desktop.


WINDOWS SERVER NEW [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Administrator
Password
Right Ctrl

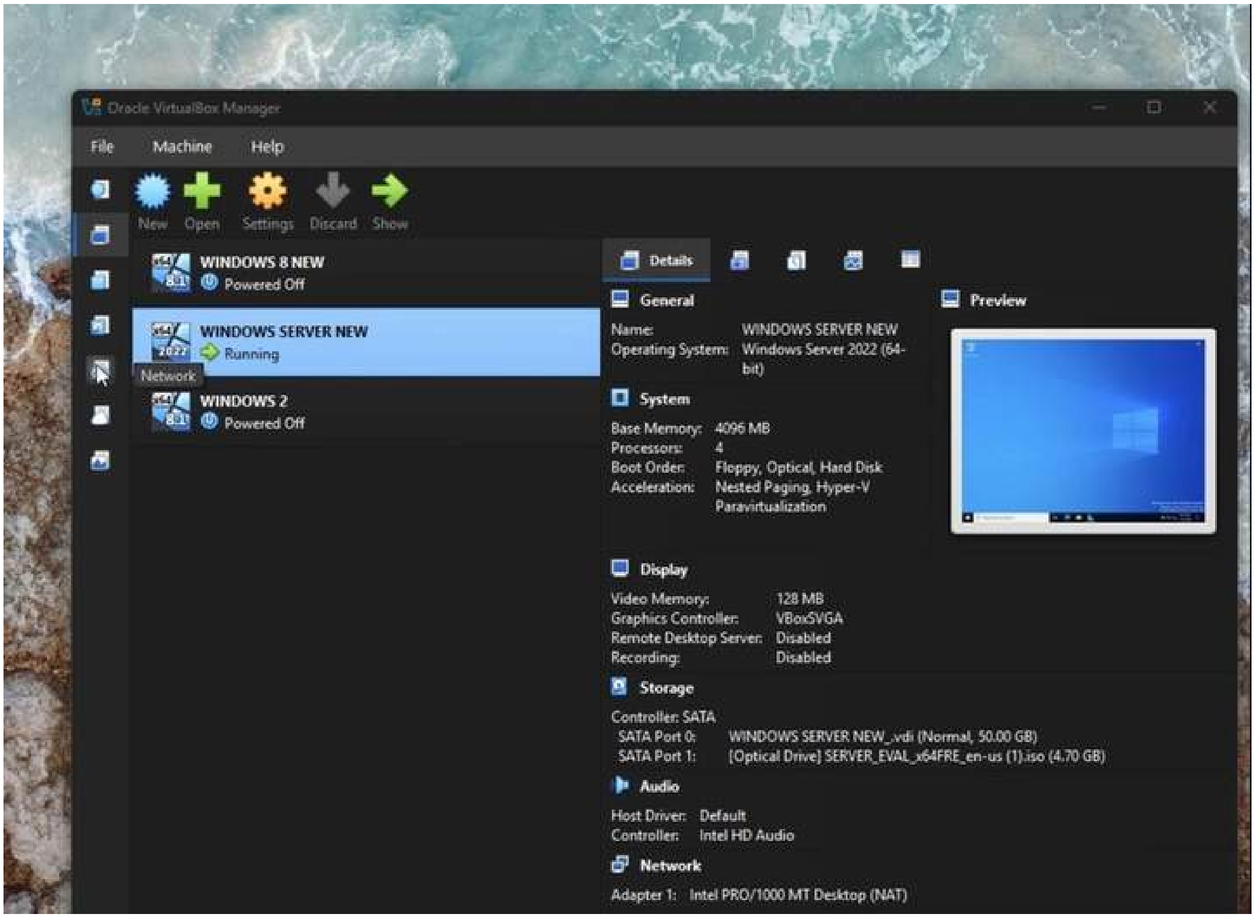


The Server Manager console loaded, confirming the operating system was fully operational and ready for configuration.

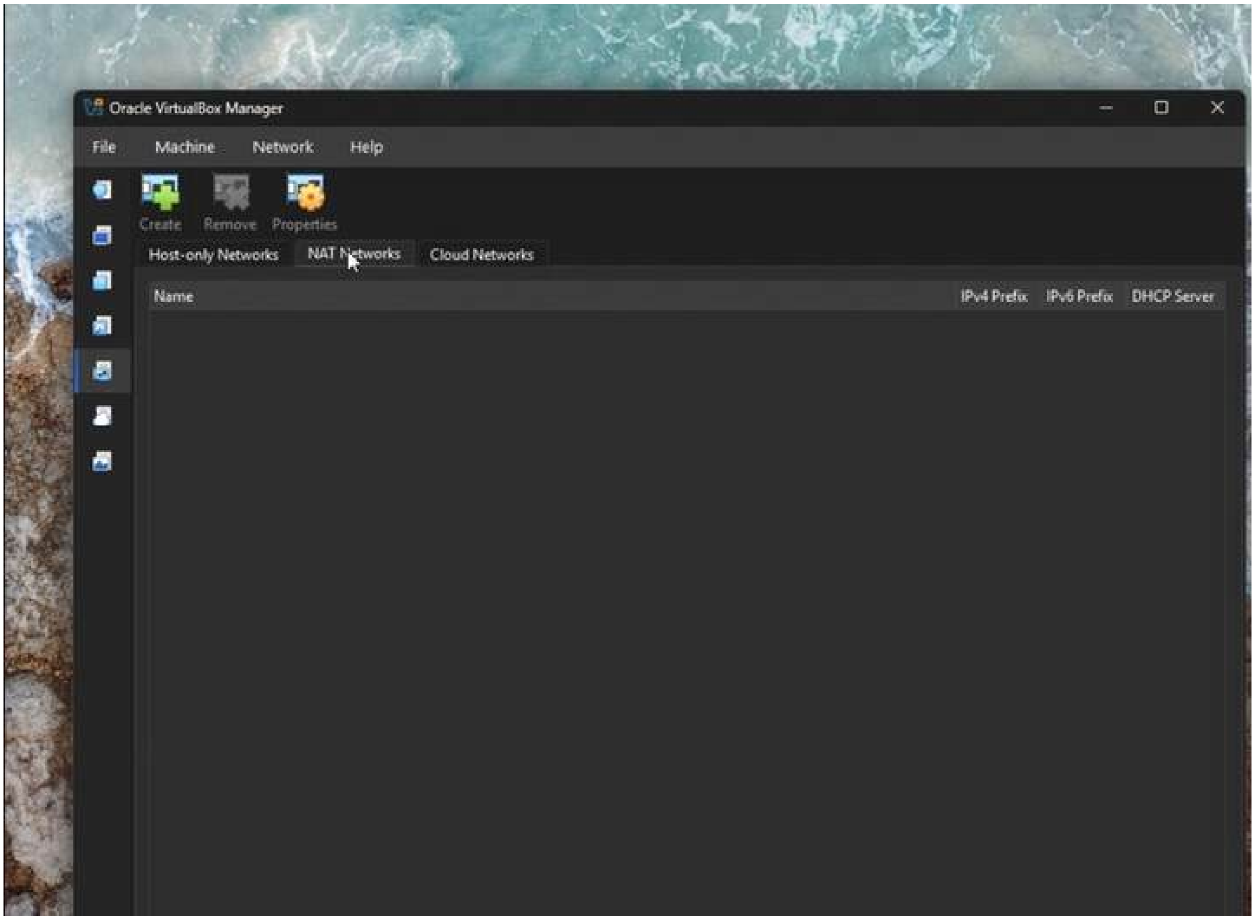


Stage 2: Creating the NAT Network for Subnet Isolation

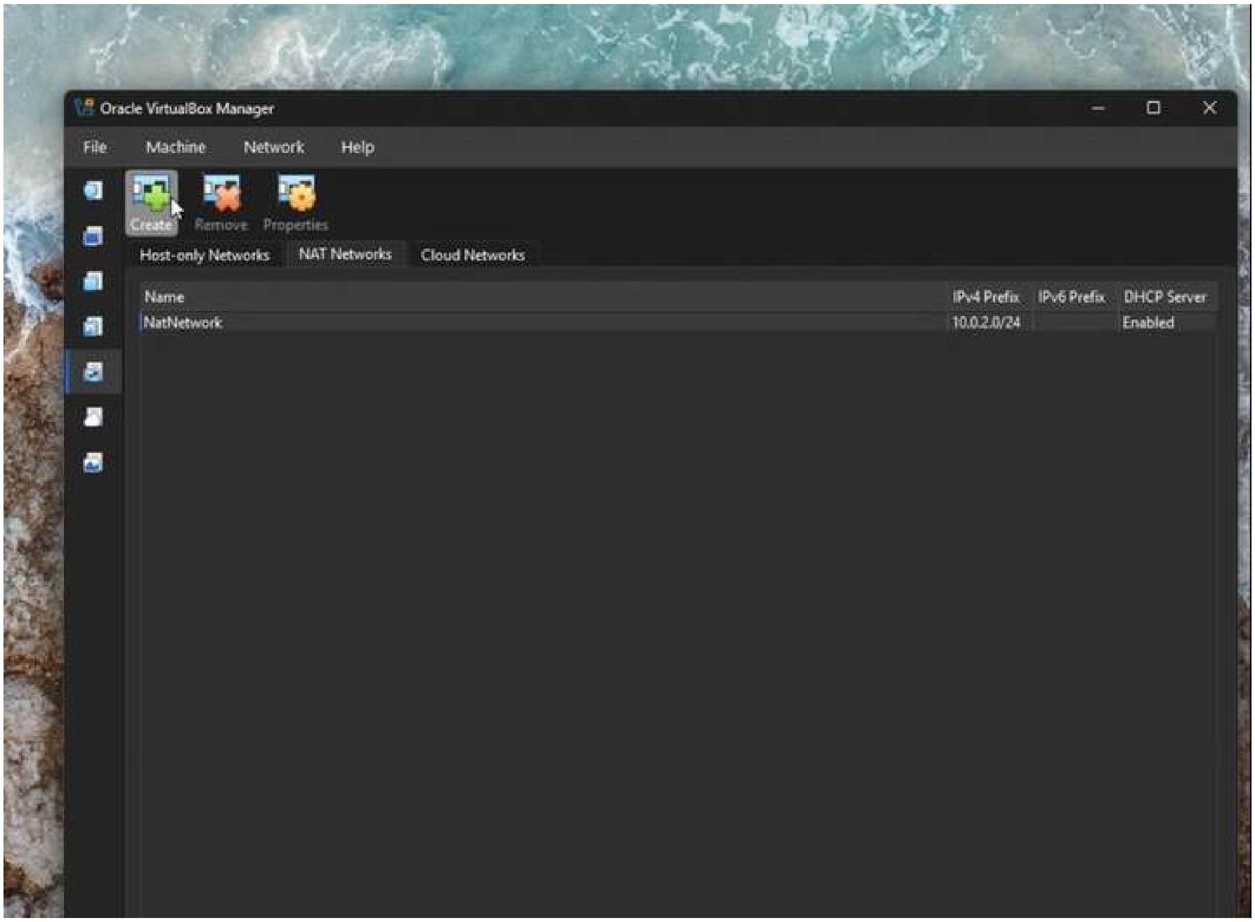
To ensure all machines operate within the same subnet and to prevent configuration errors, a dedicated NAT Network was established within the Hypervisor. I accessed the VirtualBox network settings by navigating to the main menu (Tools) and selecting Network (or the corresponding icon).


Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Show
WINDOWS 8 NEW
Powered Off
WINDOWS SERVER NEW
Running
Network
WINDOWS 2
Powered Off
Details
General
Name: WINDOWS SERVER NEW
Operating System: Windows Server 2022 (64-bit)
Preview
System
Base Memory: 4096 MB
Processors: 4
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, Hyper-V Paravirtualization
Display
Video Memory: 128 MB
Graphics Controller: VBoxSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: SATA
SATA Port 0: WINDOWS SERVER NEW_.vdi (Normal, 50.00 GB)
SATA Port 1: [Optical Drive] SERVER_EVAL_x64FRE_en-us (1).iso (4.70 GB)
Audio
Host Driver: Default
Controller: Intel HD Audio
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)

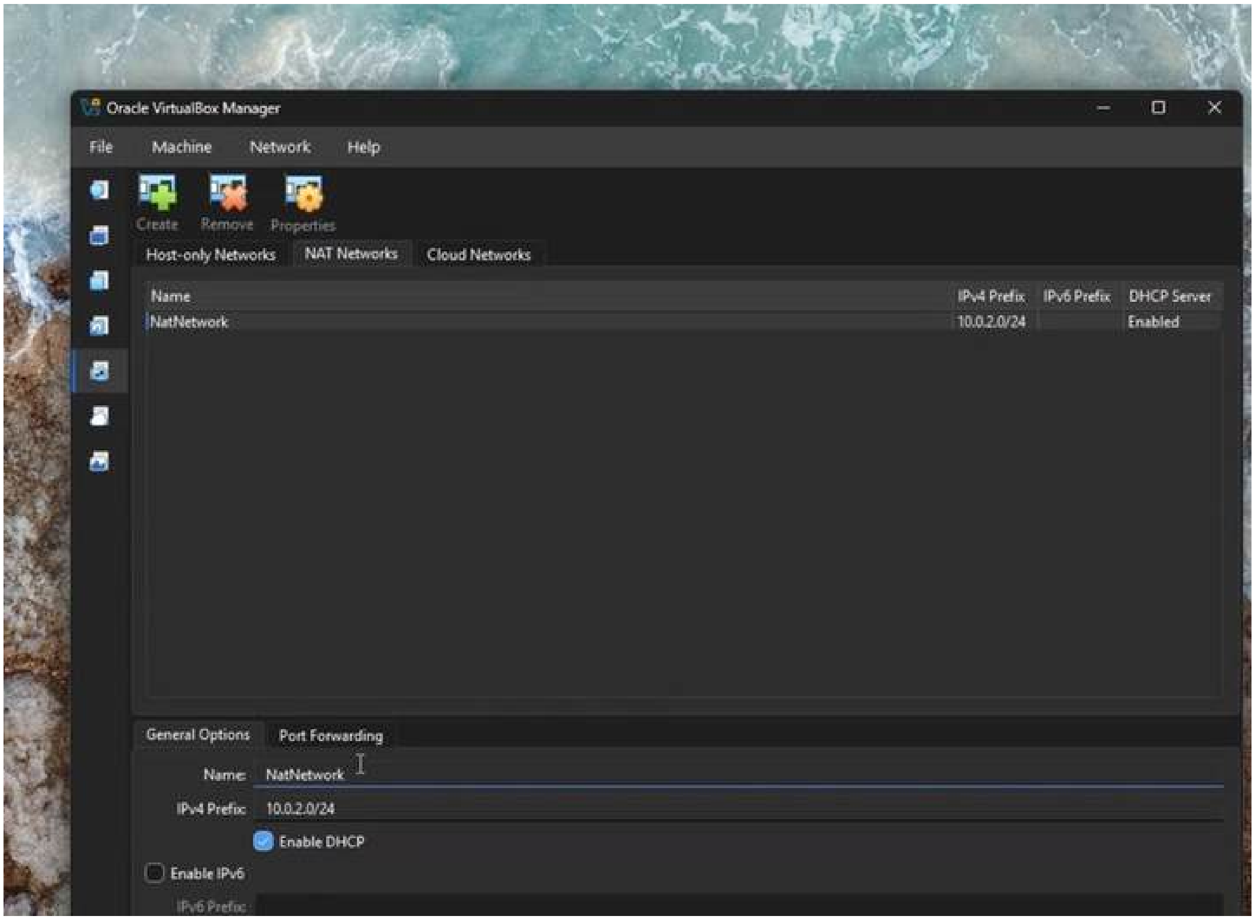
Within the network options, I choose the NAT Network option

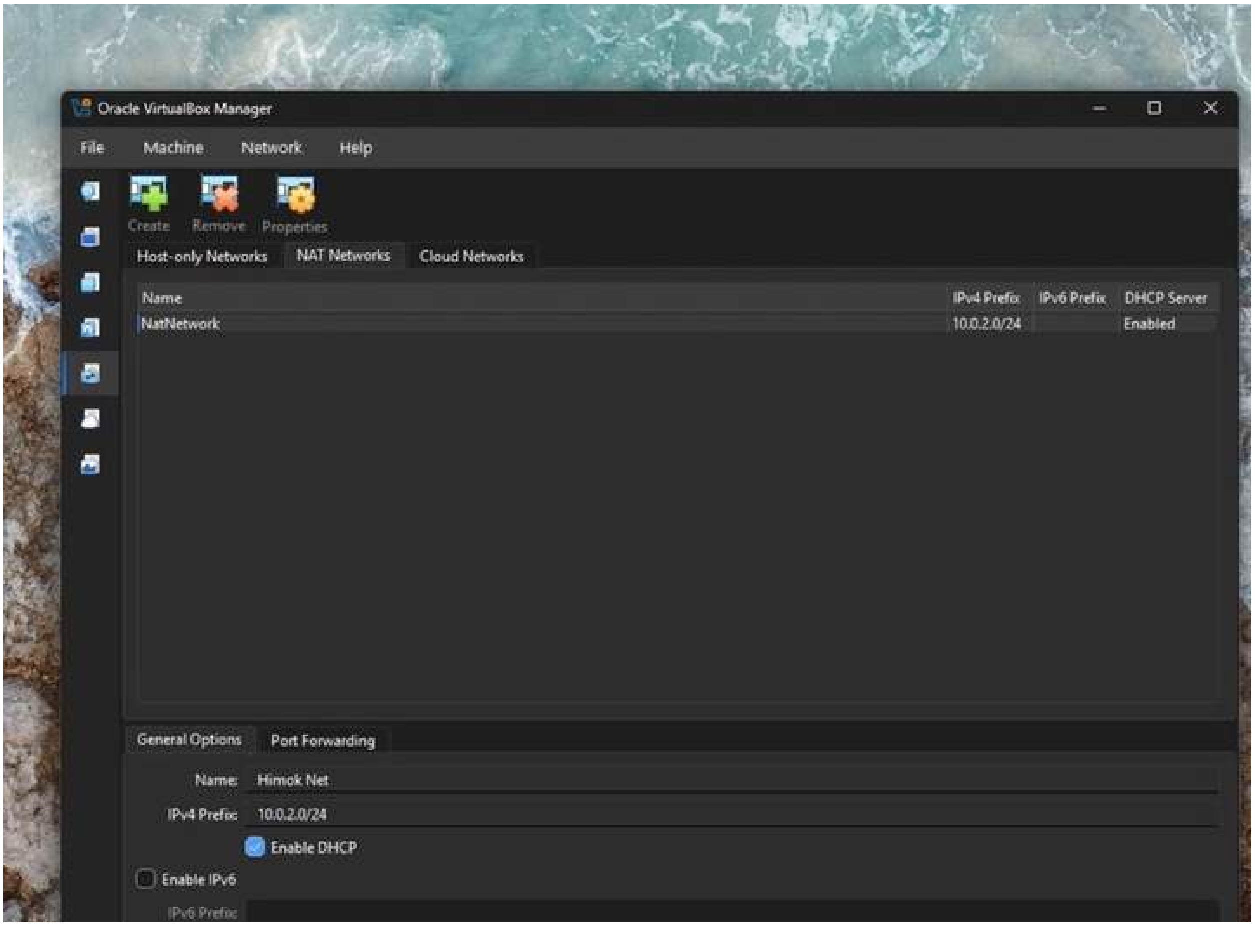


and initiated the creation of a new network.

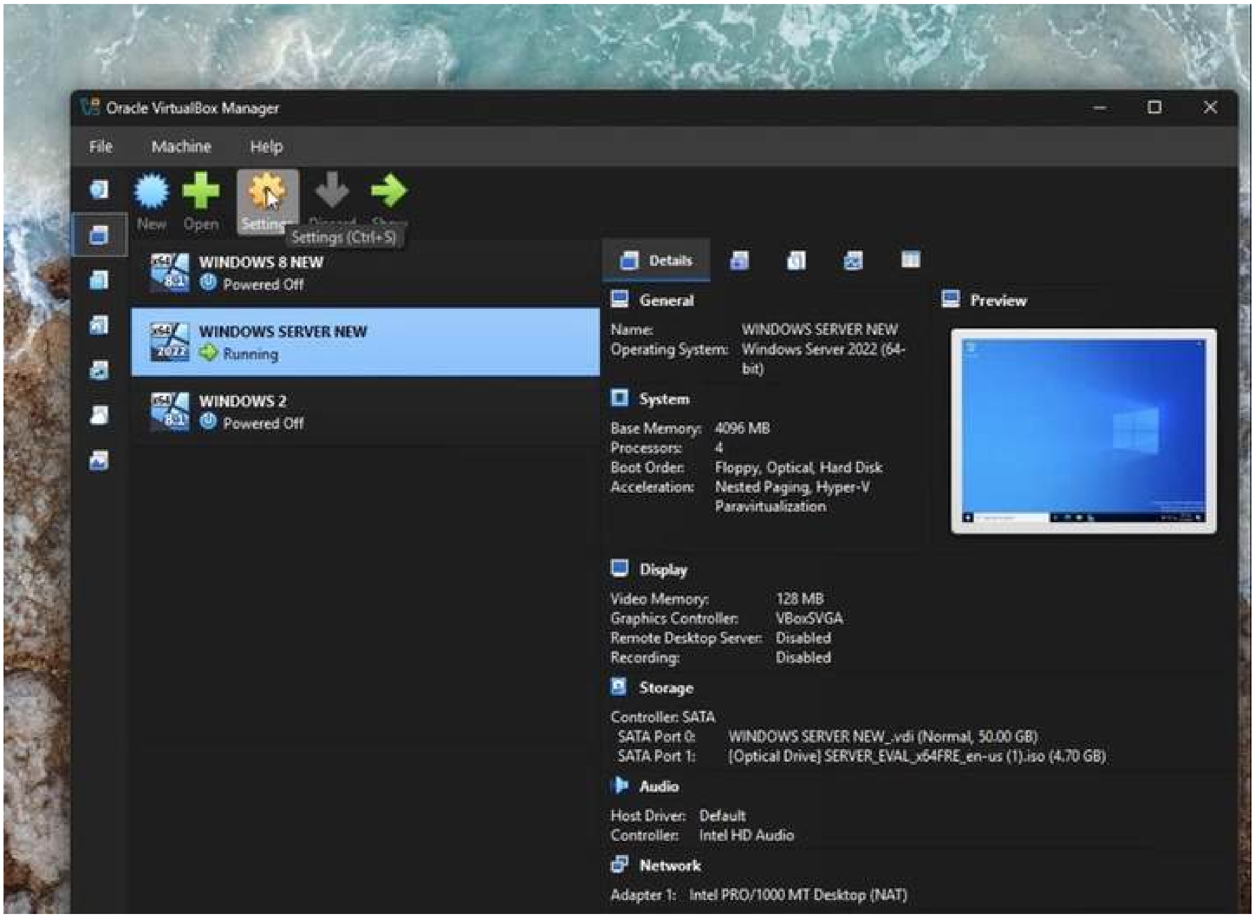


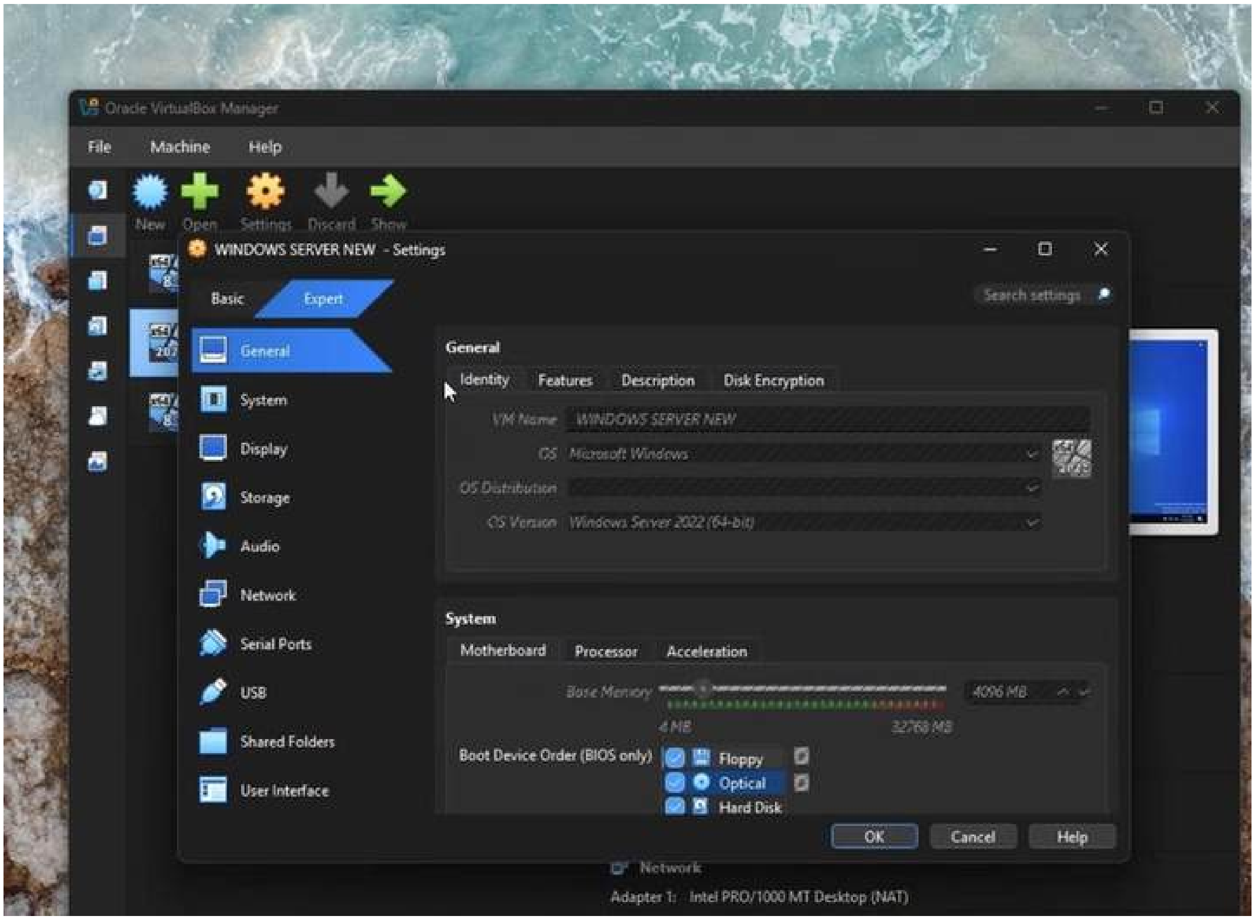
The new network was professionally named HIMOK NET. The default IPv4 range was accepted, and importantly, the Dynamic Host Configuration Protocol (DHCP) service was explicitly enabled to automatically assign IP addresses to the connecting virtual machines, simplifying the network setup.


Oracle VirtualBox Manager
File
Machine
Network
Help
Create
Remove
Properties
Host-only Networks
NAT Networks
Cloud Networks
Name
IPv4 Prefix
IPv6 Prefix
DHCP Server
NatNetwork
10.0.2.0/24
Enabled
General Options
Port Forwarding
Name:
NatNetwork
IPv4 Prefix:
10.0.2.0/24
Enable DHCP
Enable IPv6
IPv6 Prefix:

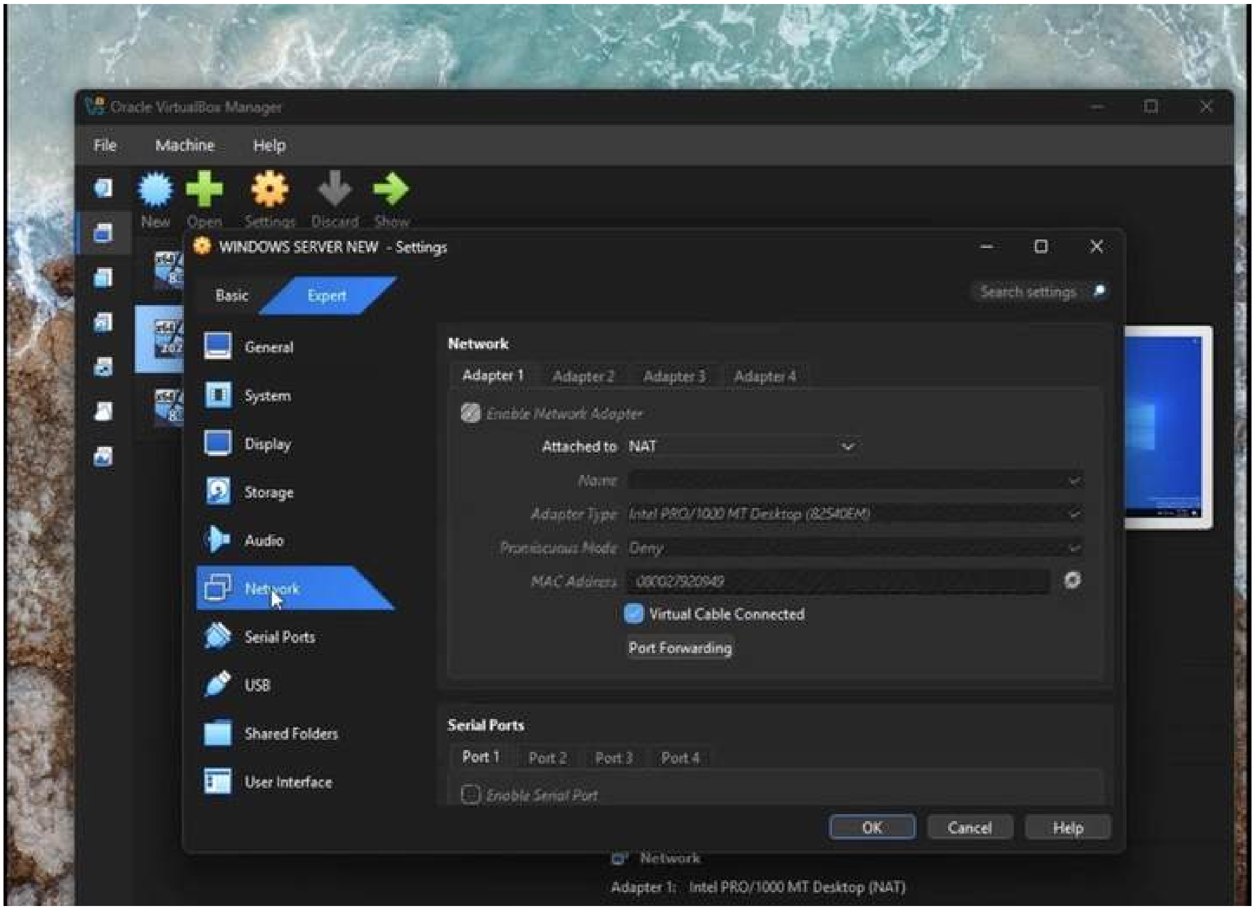


The Windows Server VM was then configured to use this new network. The VM's network adapter was changed from the default setting to NAT Network. Starting from the windows server, I selected setting on the top left.

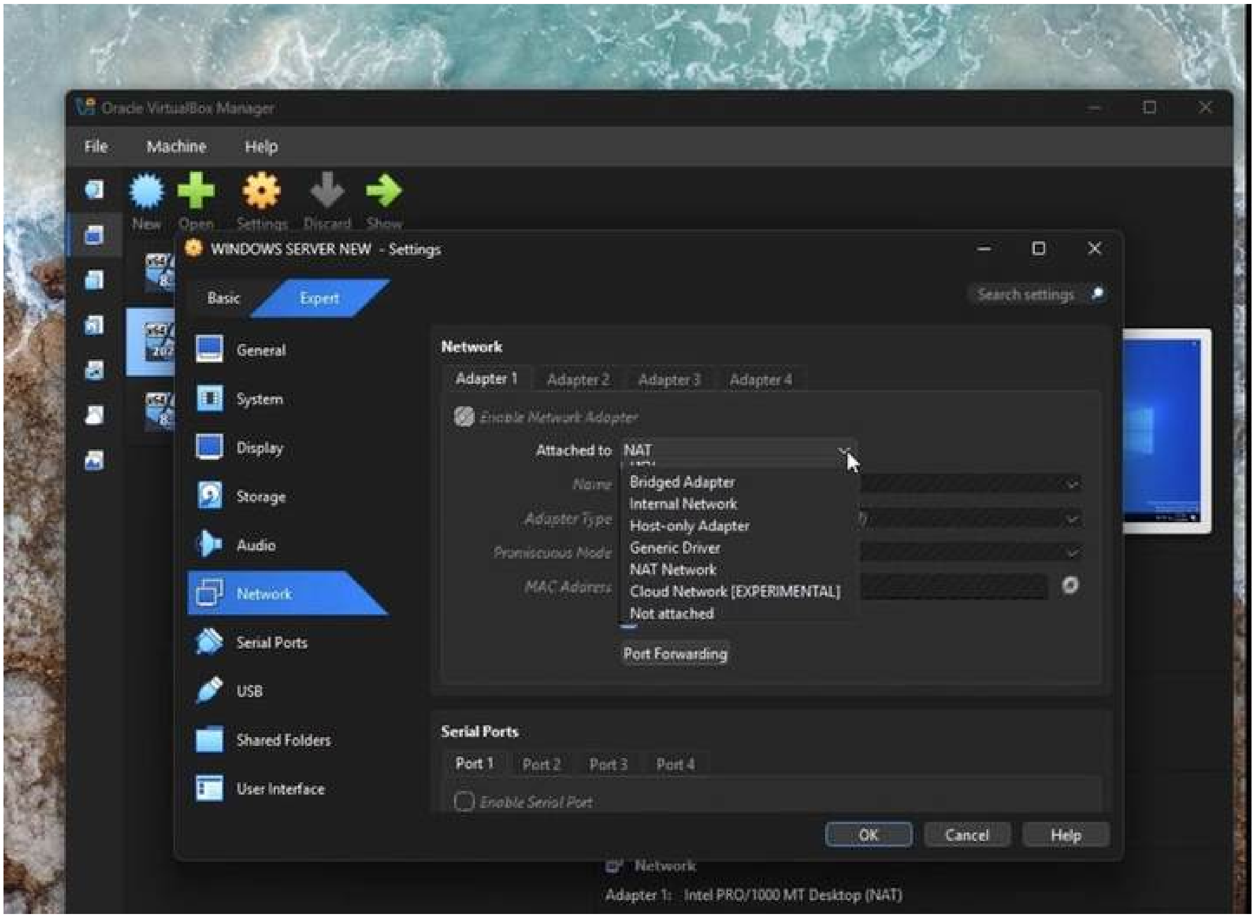

Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings (Ctrl+S)
WINDOWS 8 NEW
Powered Off
WINDOWS SERVER NEW
Running
WINDOWS 2
Powered Off
Details
General
Name: WINDOWS SERVER NEW
Operating System: Windows Server 2022 (64-bit)
System
Base Memory: 4096 MB
Processors: 4
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, Hyper-V Paravirtualization
Preview
Display
Video Memory: 128 MB
Graphics Controller: VBoxSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: SATA
SATA Port 0: WINDOWS SERVER NEW_.vdi (Normal, 50.00 GB)
SATA Port 1: [Optical Drive] SERVER_EVAL_x64FRE_en-us (1).iso (4.70 GB)
Audio
Host Driver: Default
Controller: Intel HD Audio
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)



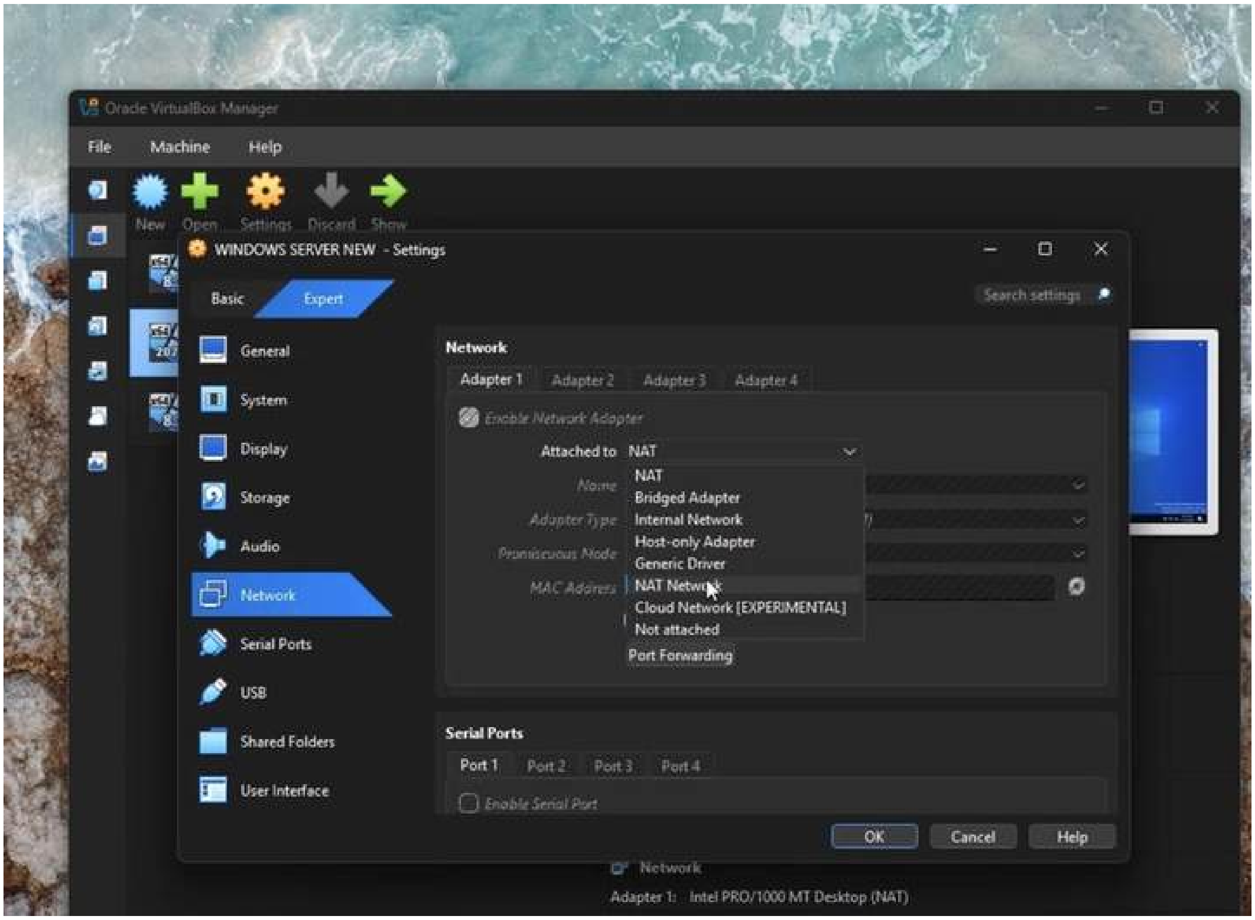
I selected Network from the options



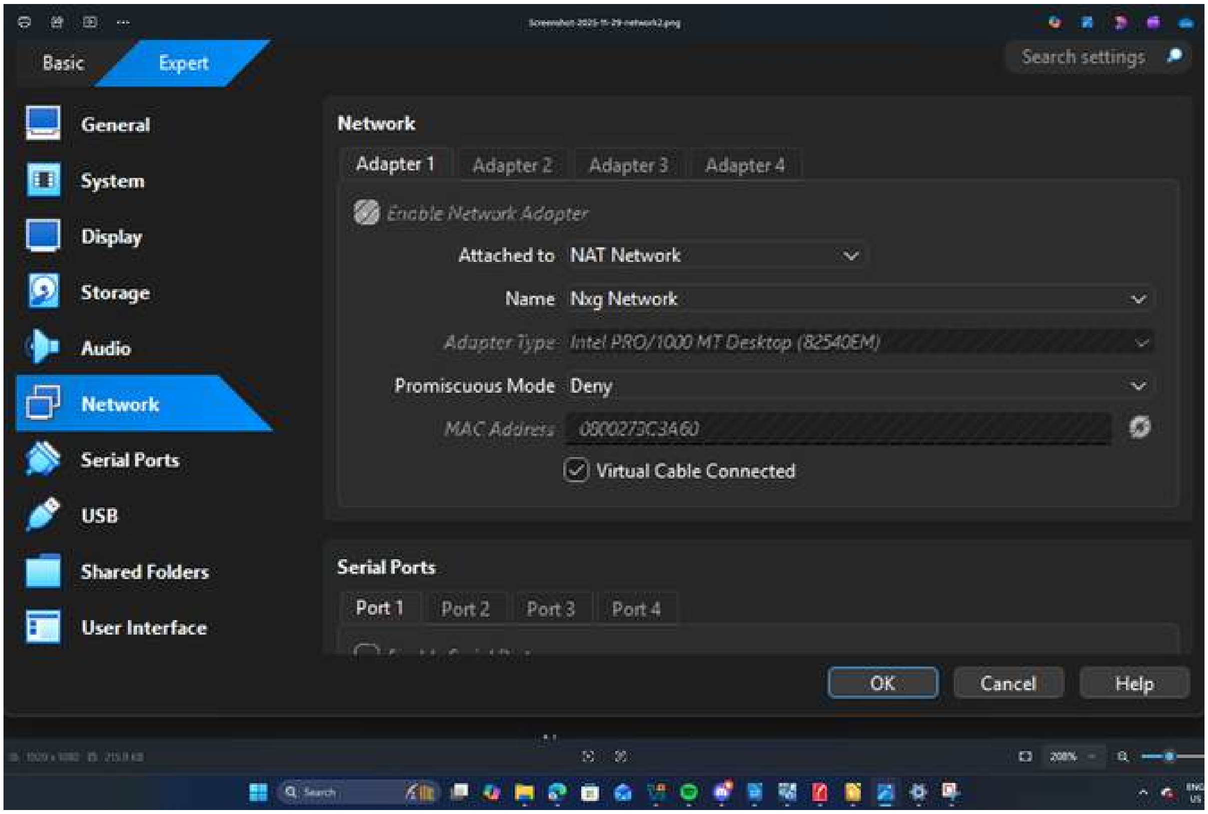
from the options available in Attached to



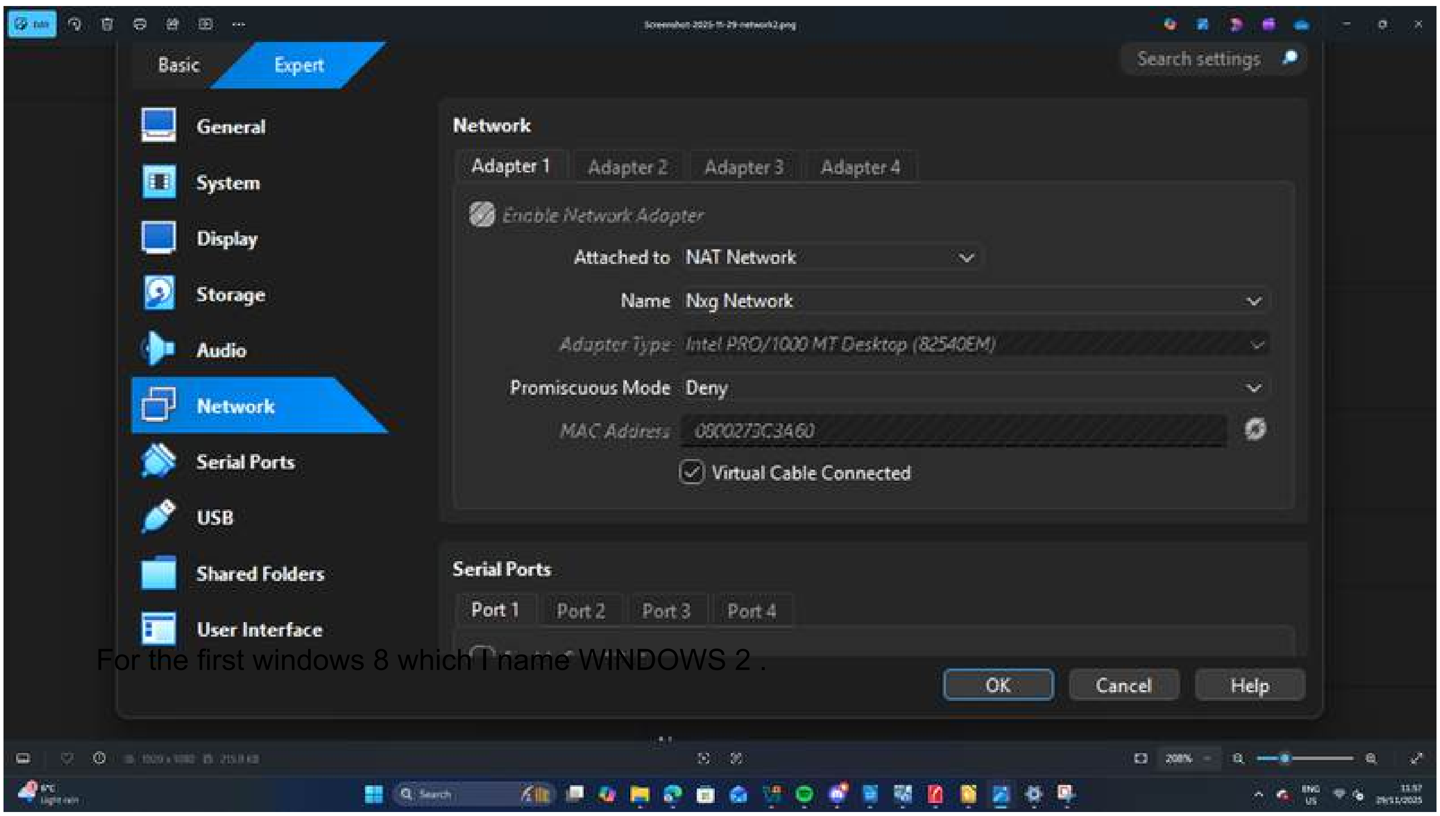
I selected Nat Network



The specific network name Nxg Network was selected, effectively connecting the Server to the new private subnet.

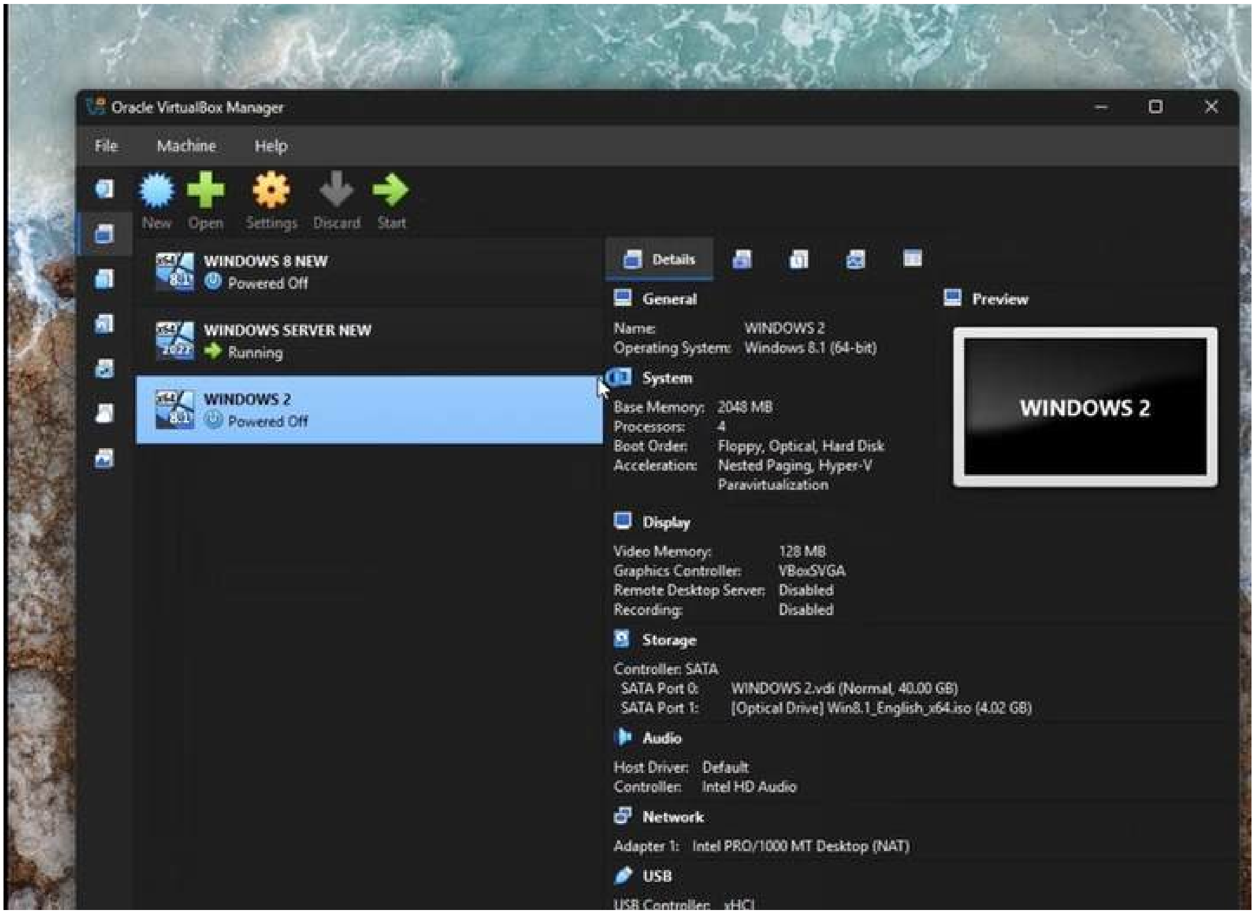

Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to NAT Network
Name Nxg Network
Adapter Type Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode Deny
MAC Address 0800273C3A60
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
OK
Cancel
Help

Then ok

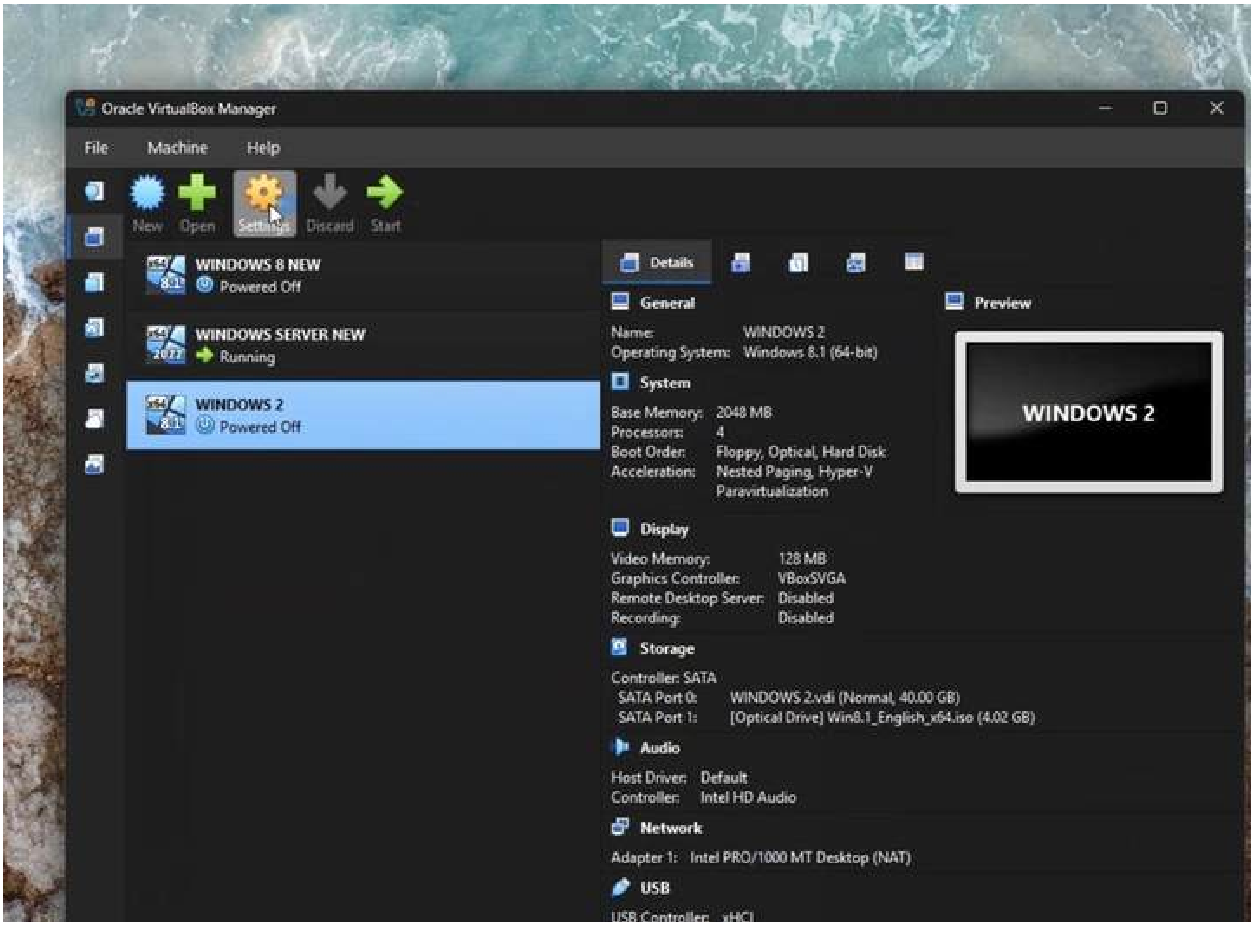


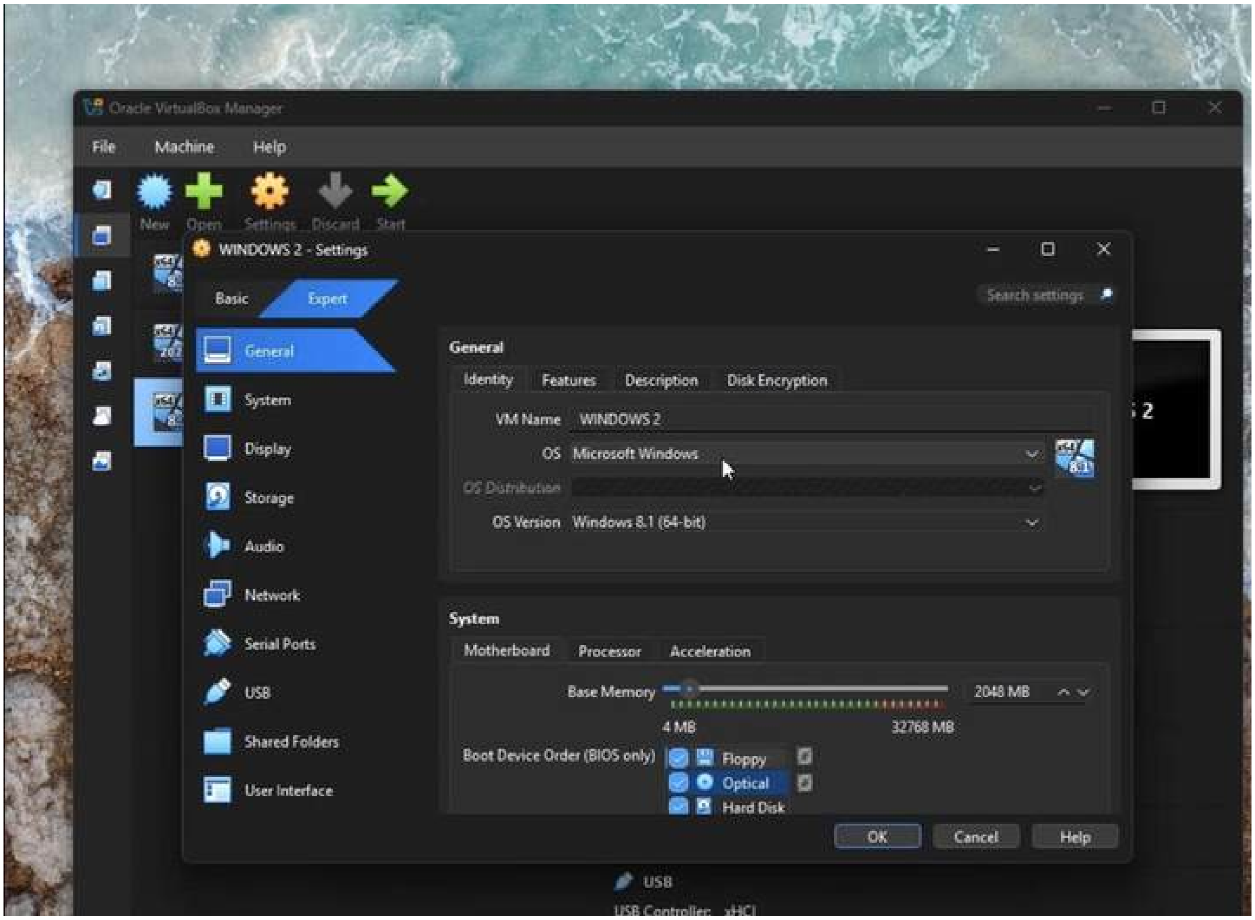
For the first windows 8 which I name WINDOWS 2 .

This critical network configuration was then replicated for both the first and second Windows 8 client VMs, ensuring all three virtual machines were logically attached to the same isolated Nxg Network.



Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
WINDOWS 8 NEW
Powered Off
WINDOWS SERVER NEW
Running
WINDOWS 2
Powered Off
Details
General
Name: WINDOWS 2
Operating System: Windows 8.1 (64-bit)
System
Base Memory: 2048 MB
Processors: 4
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, Hyper-V Paravirtualization
Preview
WINDOWS 2
Display
Video Memory: 128 MB
Graphics Controller: VBoxSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: SATA
SATA Port 0: WINDOWS 2.vdi (Normal, 40.00 GB)
SATA Port 1: [Optical Drive] Win8.1_English_x64.iso (4.02 GB)
Audio
Host Driver: Default
Controller: Intel HD Audio
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB


Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
WINDOWS 8 NEW
Powered Off
WINDOWS SERVER NEW
Running
WINDOWS 2
Powered Off
Details
General
Name: WINDOWS 2
Operating System: Windows 8.1 (64-bit)
System
Base Memory: 2048 MB
Processors: 4
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, Hyper-V Paravirtualization
Preview
WINDOWS 2
Display
Video Memory: 128 MB
Graphics Controller: VBoxSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: SATA
SATA Port 0: WINDOWS 2.vdi (Normal, 40.00 GB)
SATA Port 1: [Optical Drive] Win8.1_English_x64.iso (4.02 GB)
Audio
Host Driver: Default
Controller: Intel HD Audio
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB



Oracle VirtualBox Manager
File
Machine
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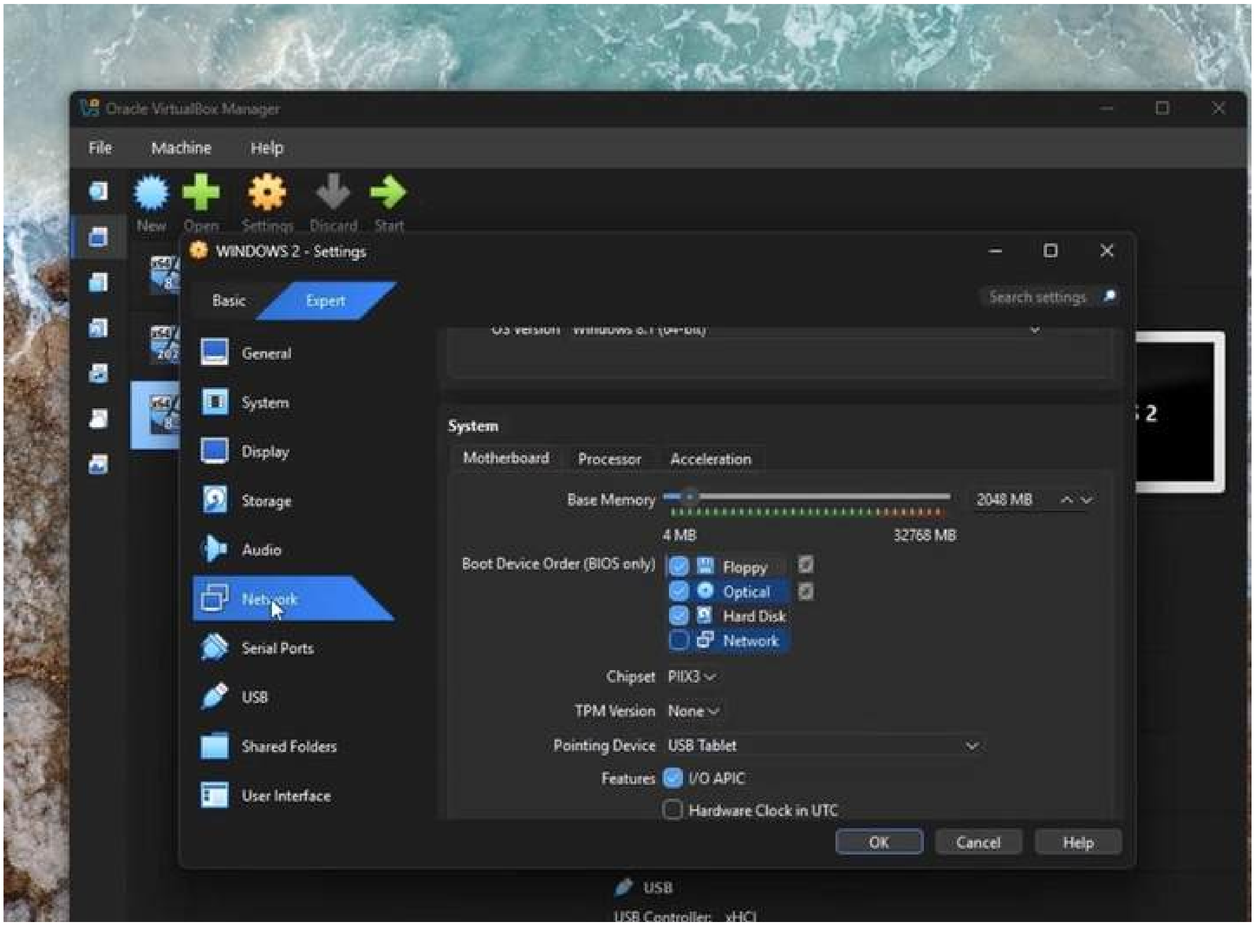
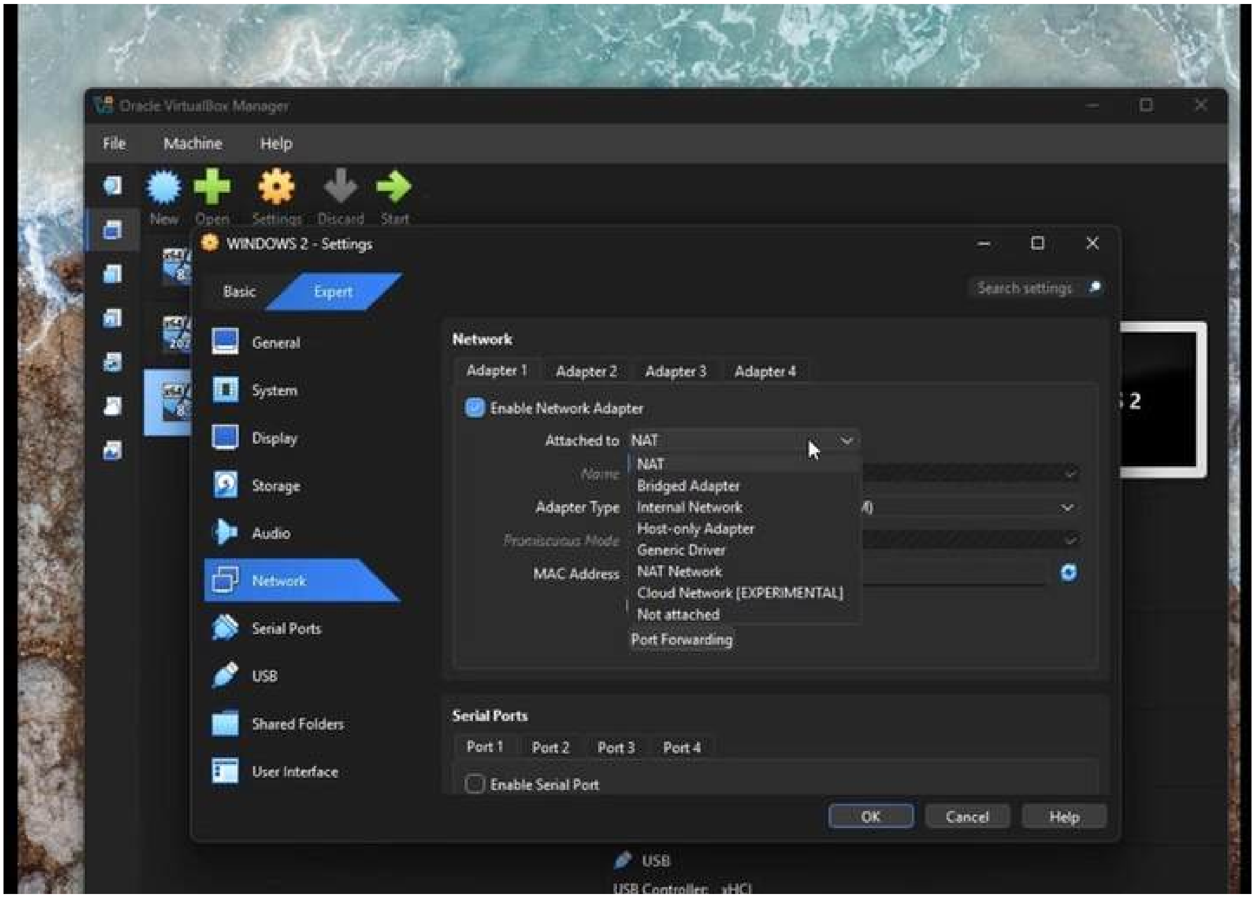
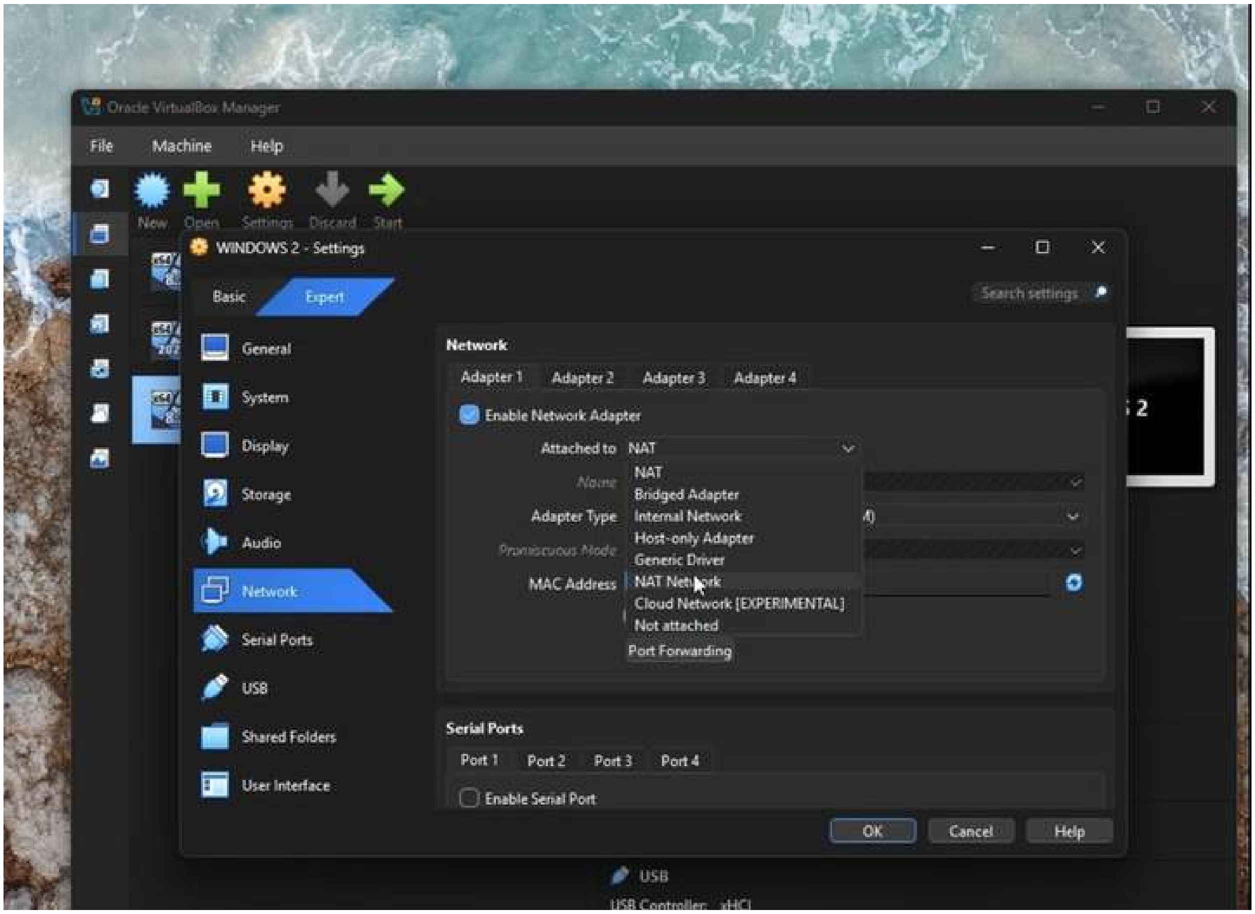
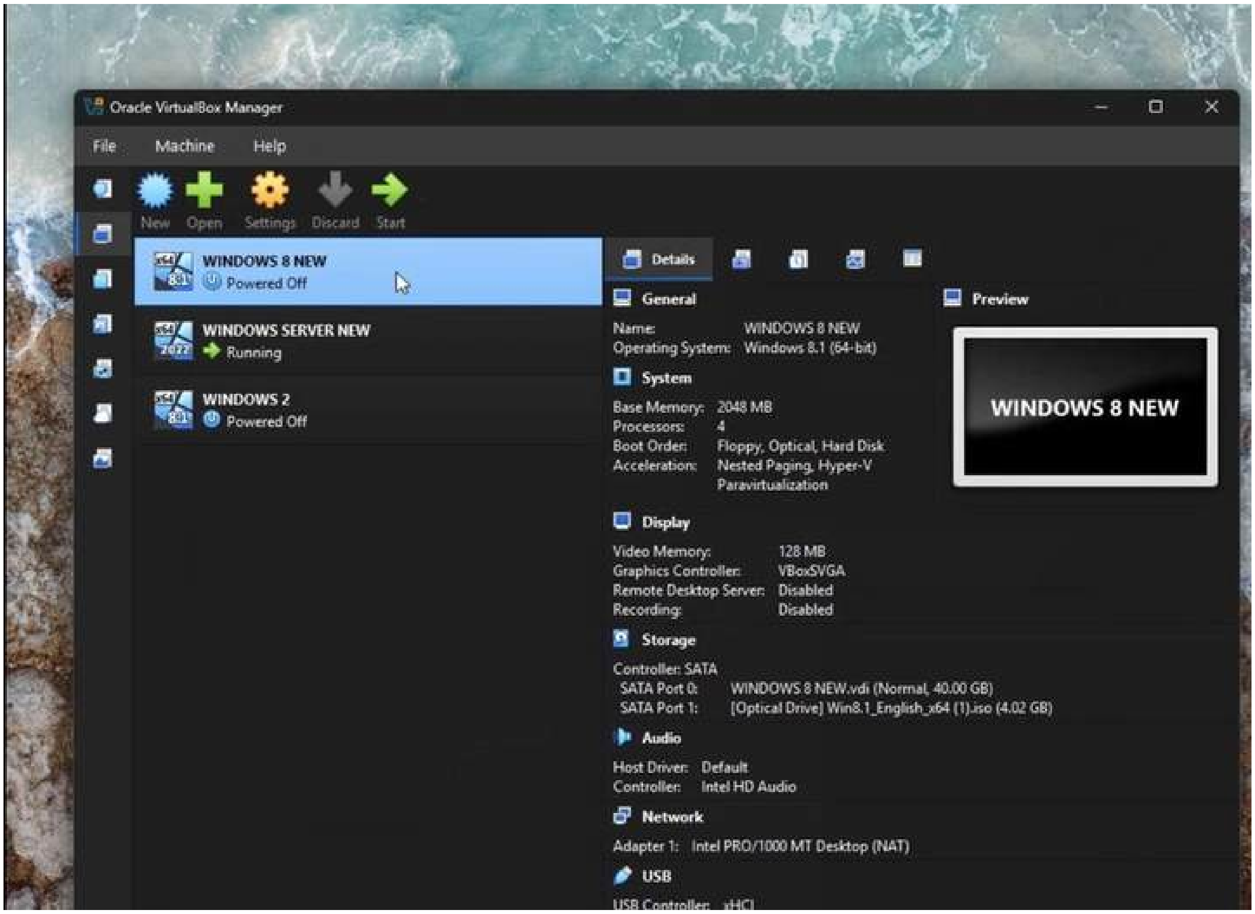
New
Open
Settings
Discard
Start
WINDOWS 2 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
General
Identity
Features
Description
Disk Encryption
VM Name
WINDOWS 2
OS
Microsoft Windows
OS Distribution
OS Version
Windows 8.1 (64-bit)
System
Motherboard
Processor
Acceleration
Base Memory
2048 MB
4 MB
32768 MB
Boot Device Order (BIOS only)
Floppy
Optical
Hard Disk
OK
Cancel
Help
USB
USB Controller: xHCI

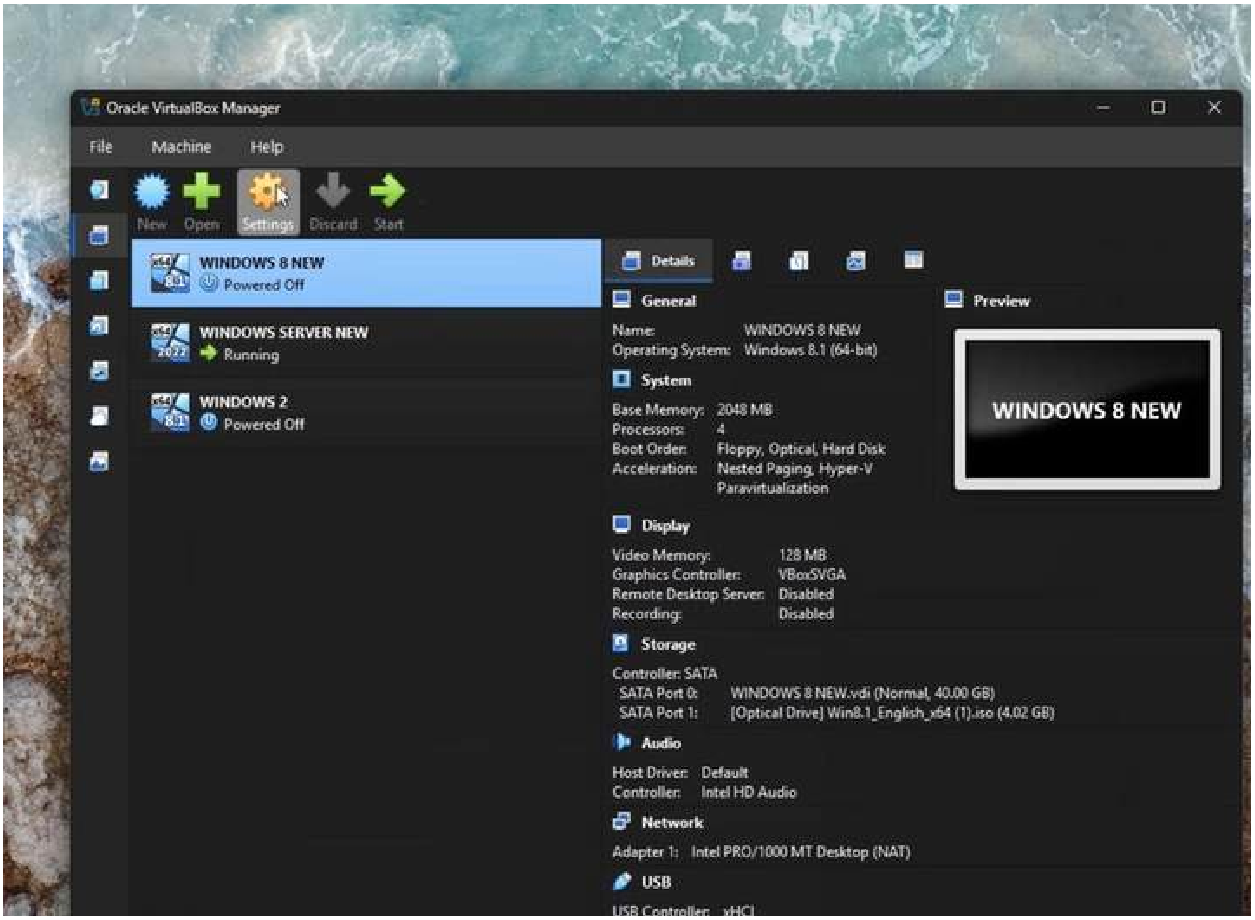

Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
WINDOWS 2 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
System
Motherboard
Processor
Acceleration
Base Memory
2048 MB
4 MB
32768 MB
Boot Device Order (BIOS only)
Floppy
Optical
Hard Disk
Network
Chipset
PIIX3
TPM Version
None
Pointing Device
USB Tablet
Features
I/O APIC
Hardware Clock in UTC
OK
Cancel
Help
USB

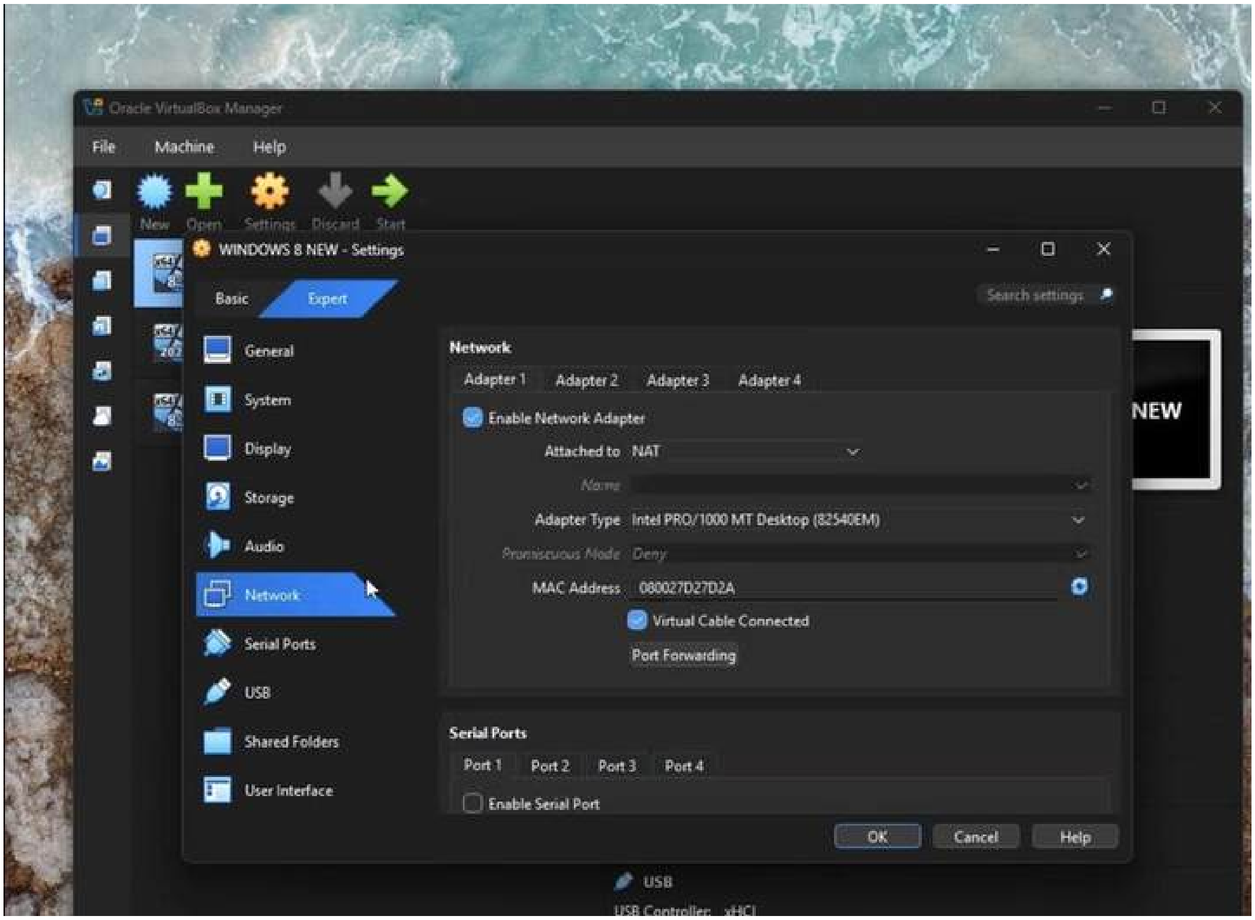

Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
WINDOWS 2 - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
NAT
Name
Adapter Type
Promiscuous Mode
MAC Address
NAT
Bridged Adapter
Internal Network
Host-only Adapter
Generic Driver
NAT Network
Cloud Network [EXPERIMENTAL]
Not attached
Port Forwarding
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
OK
Cancel
Help
USB
USB Controller: xHCI

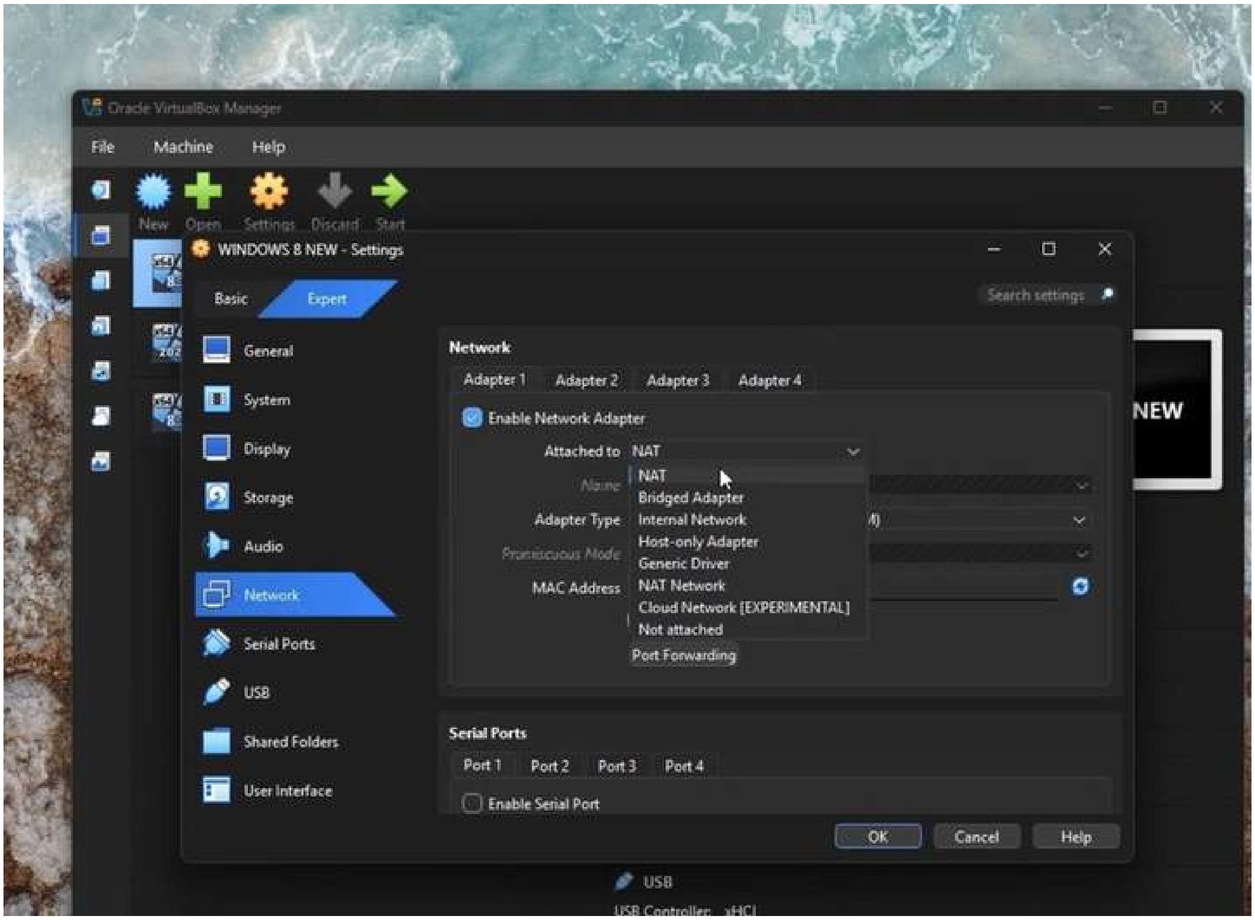


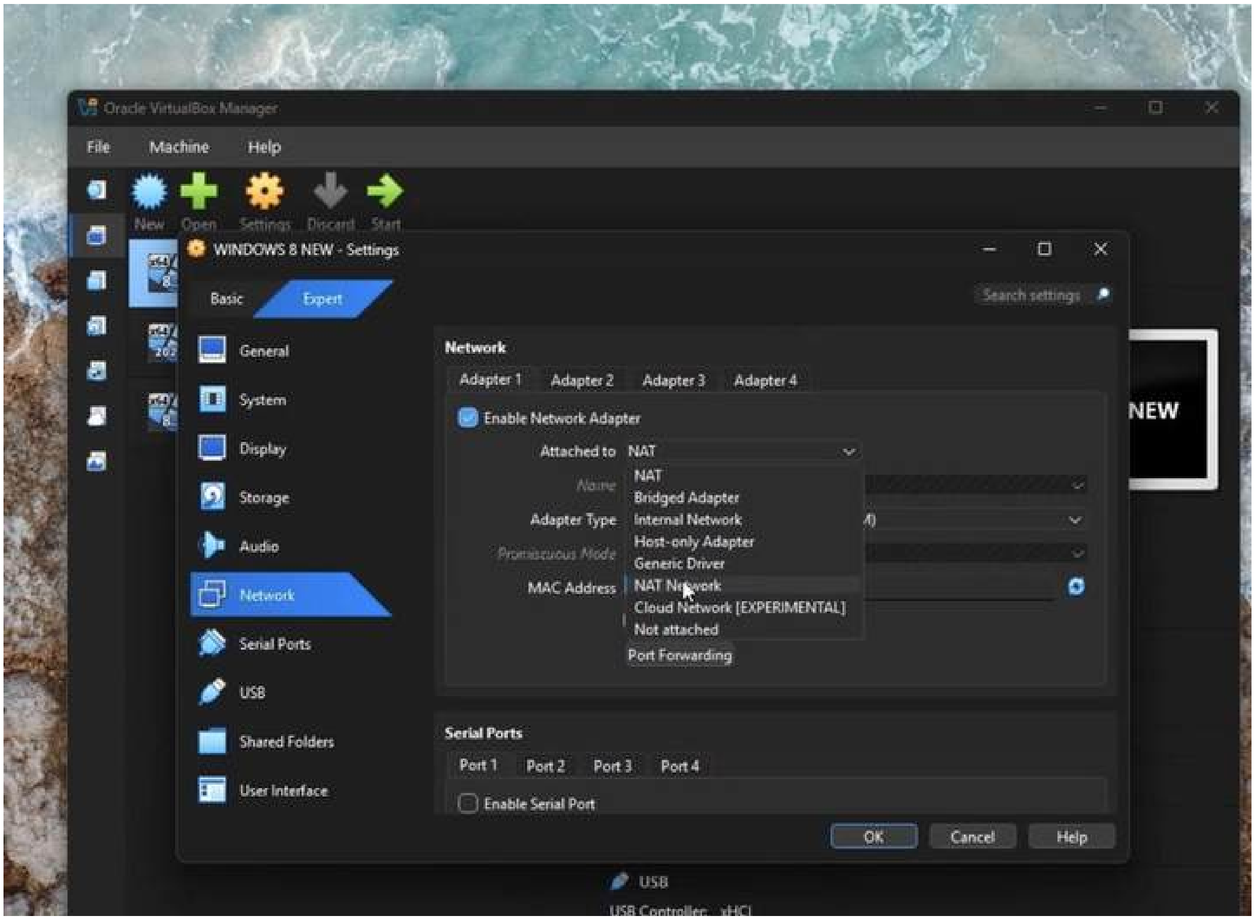
For the second windows 8 which Is name WINDOWS 8 New.


Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
WINDOWS 8 NEW
Powered Off
WINDOWS SERVER NEW
Running
WINDOWS 2
Powered Off
Details
General
Name: WINDOWS 8 NEW
Operating System: Windows 8.1 (64-bit)
System
Base Memory: 2048 MB
Processors: 4
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, Hyper-V Paravirtualization
Preview
WINDOWS 8 NEW
Display
Video Memory: 128 MB
Graphics Controller: VBoxSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: SATA
SATA Port 0: WINDOWS 8 NEW.vdi (Normal, 40.00 GB)
SATA Port 1: [Optical Drive] Win8.1_English_x64 (1).iso (4.02 GB)
Audio
Host Driver: Default
Controller: Intel HD Audio
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB

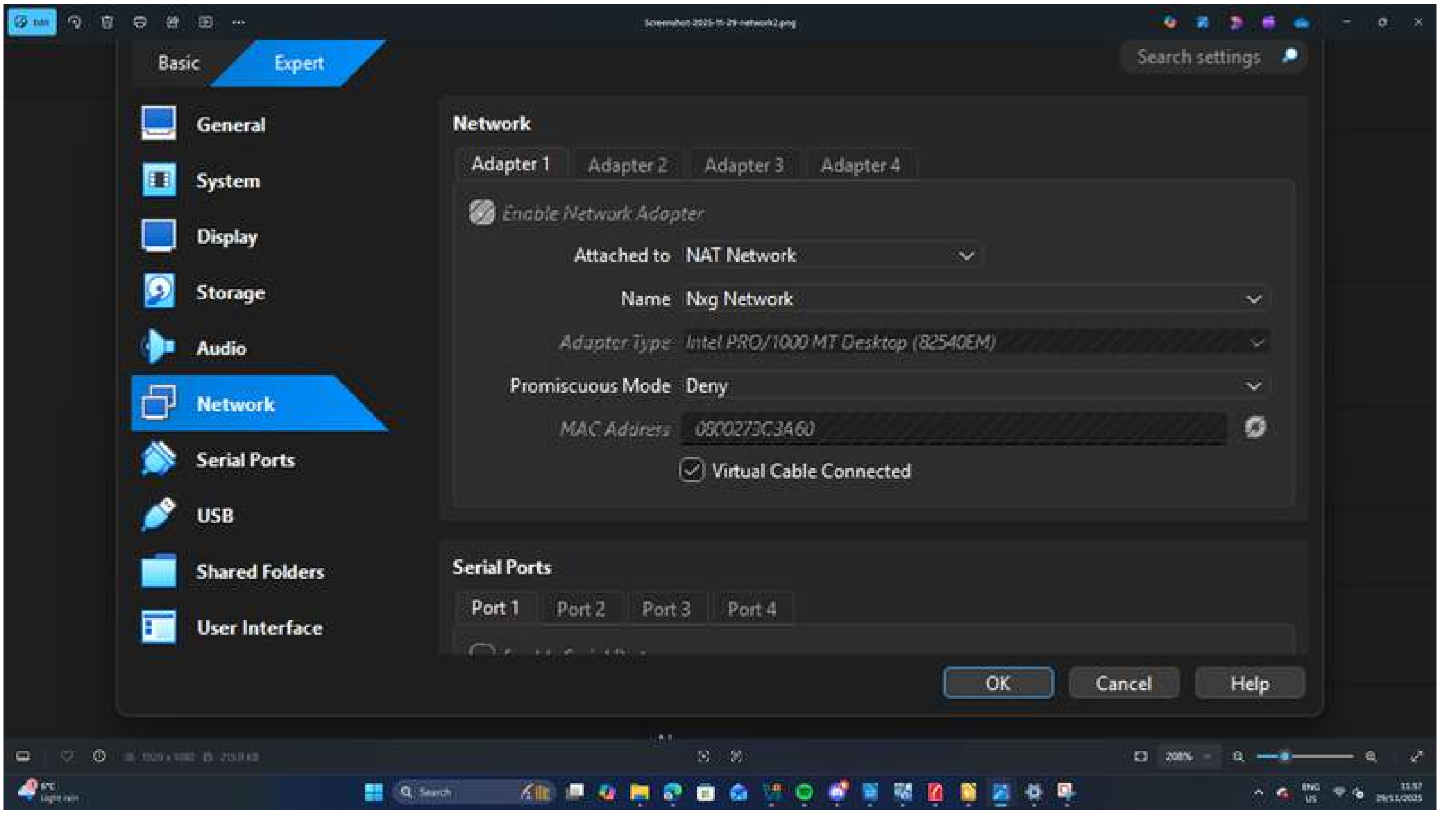

Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
WINDOWS 8 NEW
Powered Off
WINDOWS SERVER NEW
Running
WINDOWS 2
Powered Off
Details
General
Name: WINDOWS 8 NEW
Operating System: Windows 8.1 (64-bit)
System
Base Memory: 2048 MB
Processors: 4
Boot Order: Floppy, Optical, Hard Disk
Acceleration: Nested Paging, Hyper-V Paravirtualization
Display
Video Memory: 128 MB
Graphics Controller: VBoxSVGA
Remote Desktop Server: Disabled
Recording: Disabled
Storage
Controller: SATA
SATA Port 0: WINDOWS 8 NEW.vdi (Normal, 40.00 GB)
SATA Port 1: [Optical Drive] Win8.1_English_x64 (1).iso (4.02 GB)
Audio
Host Driver: Default
Controller: Intel HD Audio
Network
Adapter 1: Intel PRO/1000 MT Desktop (NAT)
USB
Preview
WINDOWS 8 NEW

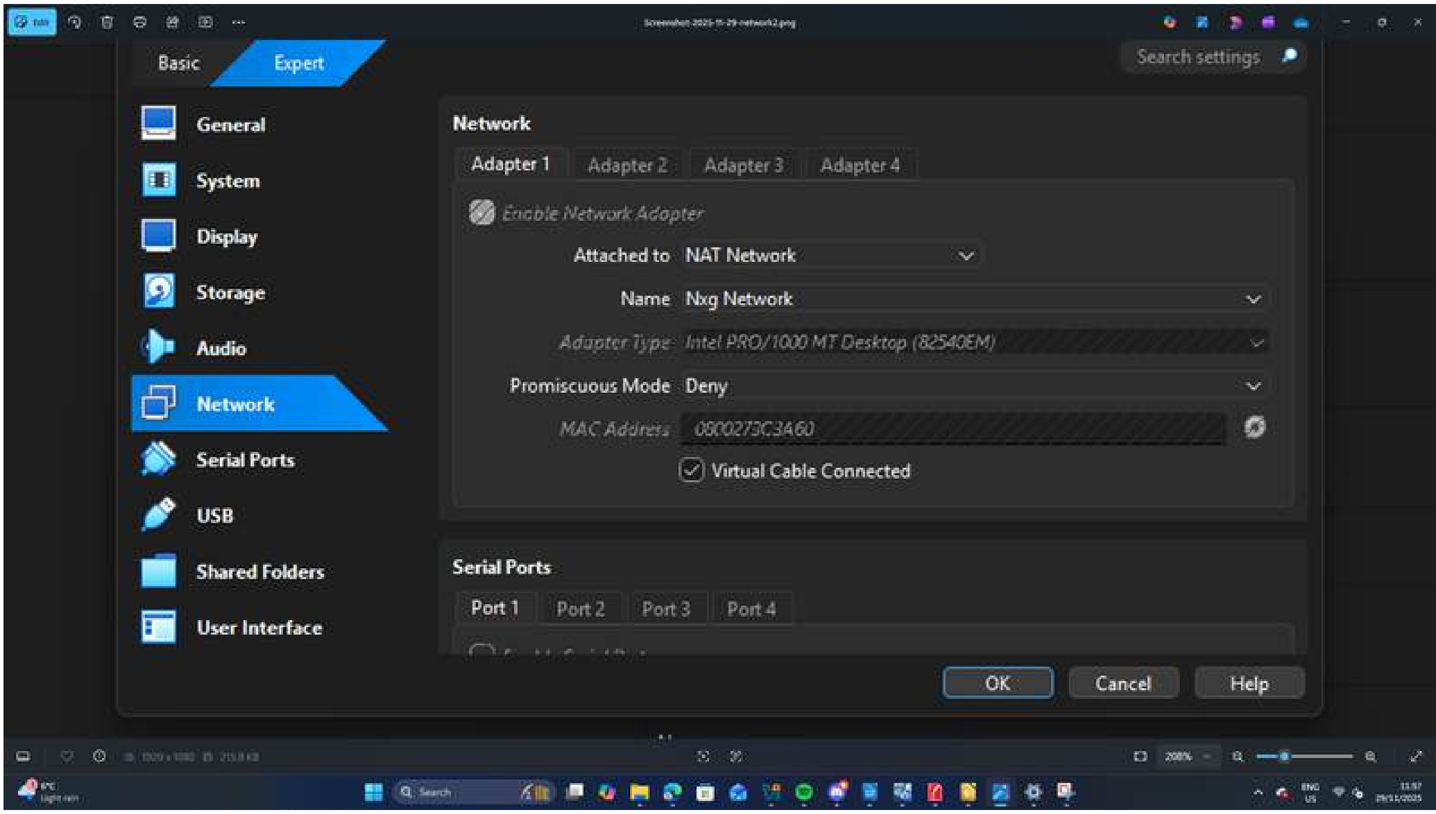

Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
WINDOWS 8 NEW - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
NAT
Name
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
080027D27D2A
Virtual Cable Connected
Port Forwarding
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
OK
Cancel
Help
NEW
USB
USB Controller:
xHCI


Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
WINDOWS 8 NEW - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
NAT
Name
Adapter Type
Promiscuous Mode
MAC Address
NAT
Bridged Adapter
Internal Network
Host-only Adapter
Generic Driver
NAT Network
Cloud Network [EXPERIMENTAL]
Not attached
Port Forwarding
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
OK
Cancel
Help
NEW
USB
USB Controller: xHCI

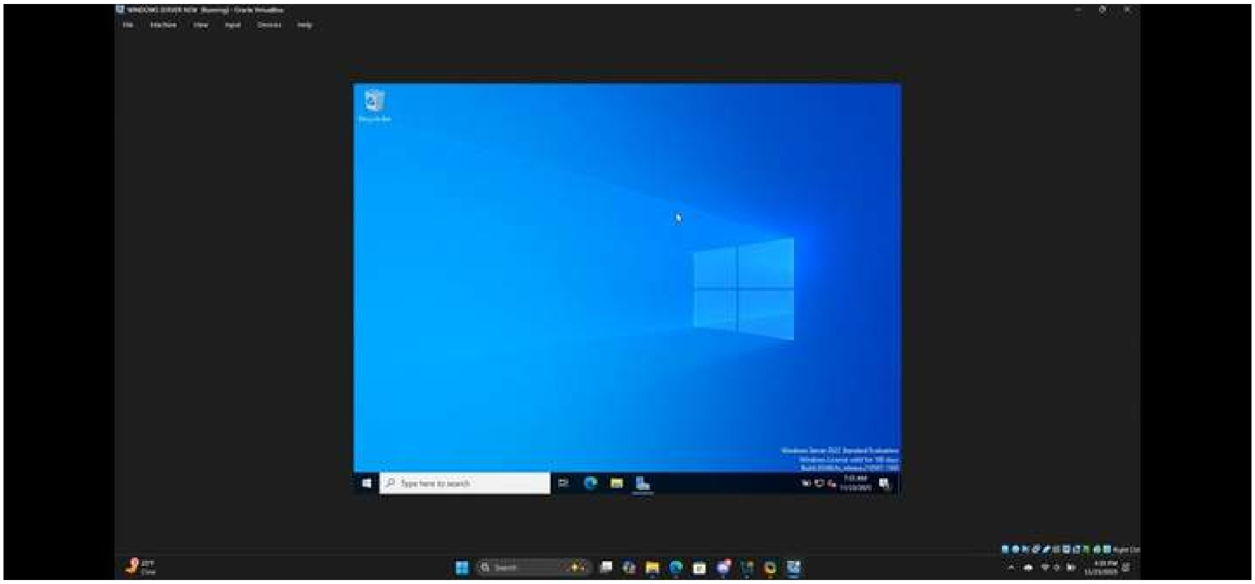


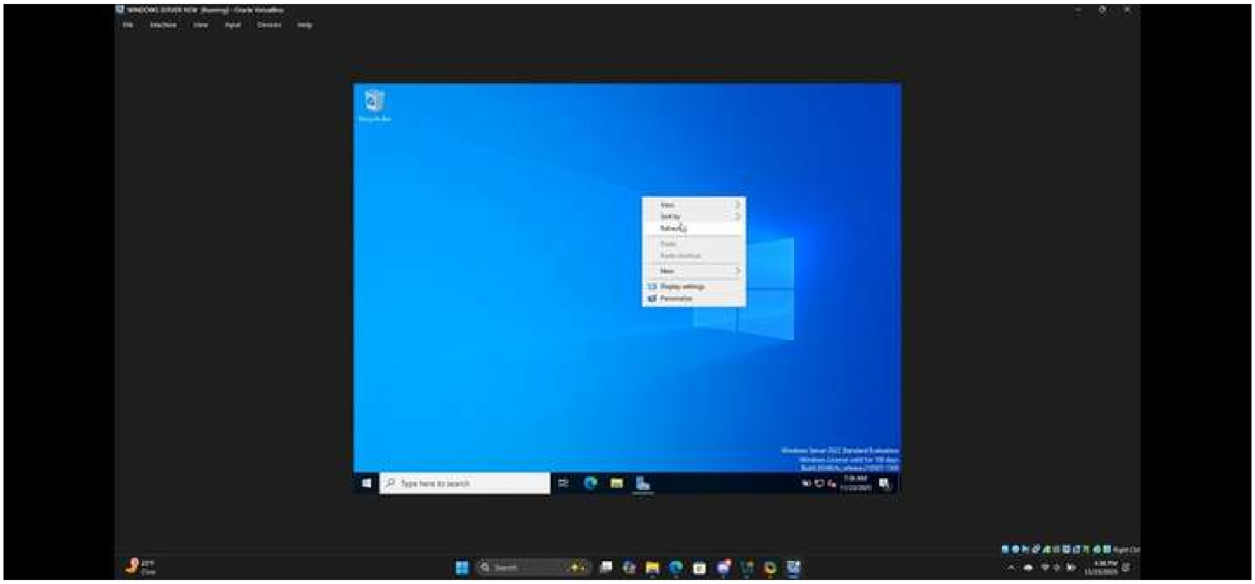
Oracle VirtualBox Manager
File
Machine
Help
New
Open
Settings
Discard
Start
WINDOWS 8 NEW - Settings
Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
NAT
Name
Adapter Type
Promiscuous Mode
MAC Address
NAT
Bridged Adapter
Internal Network
Host-only Adapter
Generic Driver
NAT Network
Cloud Network [EXPERIMENTAL]
Not attached
Port Forwarding
Serial Ports
Port 1
Port 2
Port 3
Port 4
Enable Serial Port
OK
Cancel
Help
NEW
USB
USB Controller: xHCI


Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Network
Adapter 1
Adapter 2
Adapter 3
Adapter 4
Enable Network Adapter
Attached to
NAT Network
Name
Nxg Network
Adapter Type
Intel PRO/1000 MT Desktop (82540EM)
Promiscuous Mode
Deny
MAC Address
0800273C3A60
Virtual Cable Connected
Serial Ports
Port 1
Port 2
Port 3
Port 4
OK
Cancel
Help

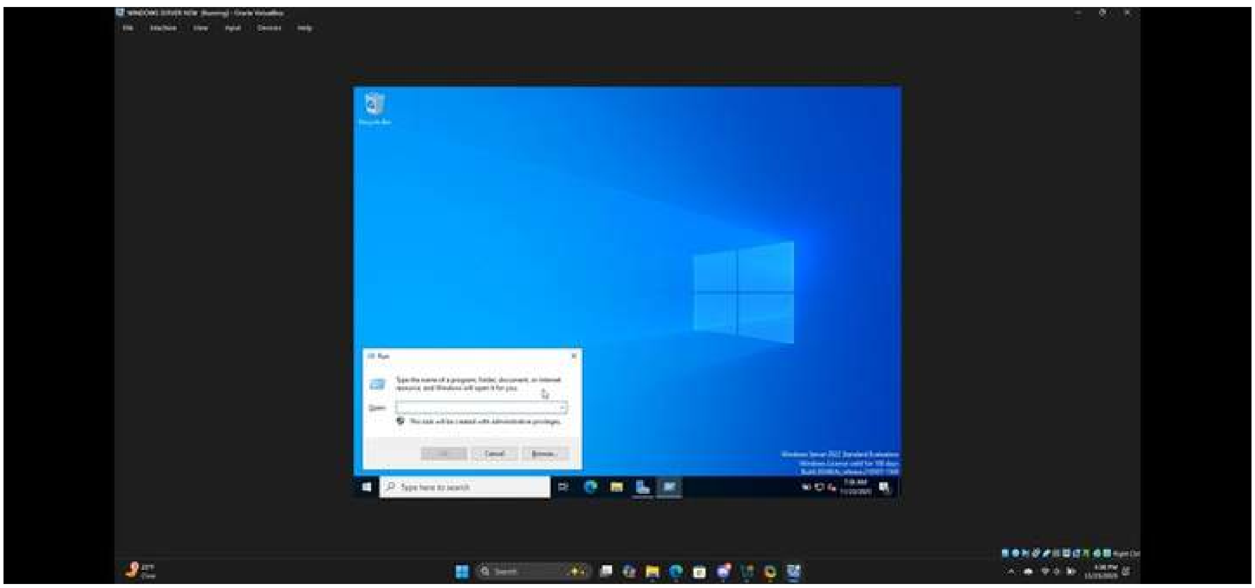


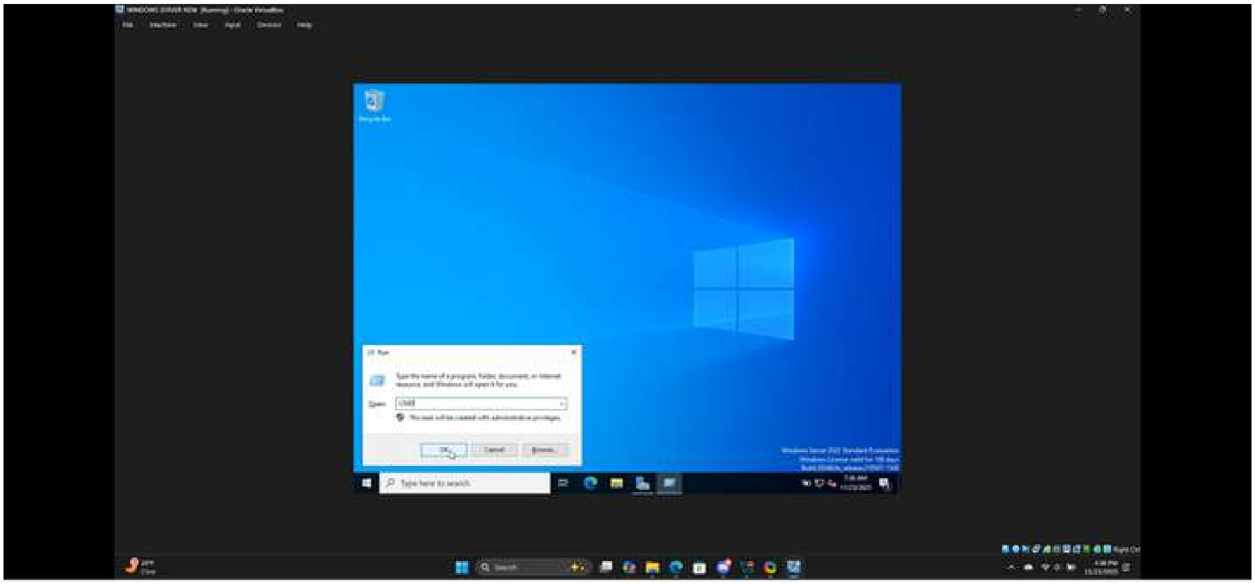
I returned to the Windows Server and refresh the page

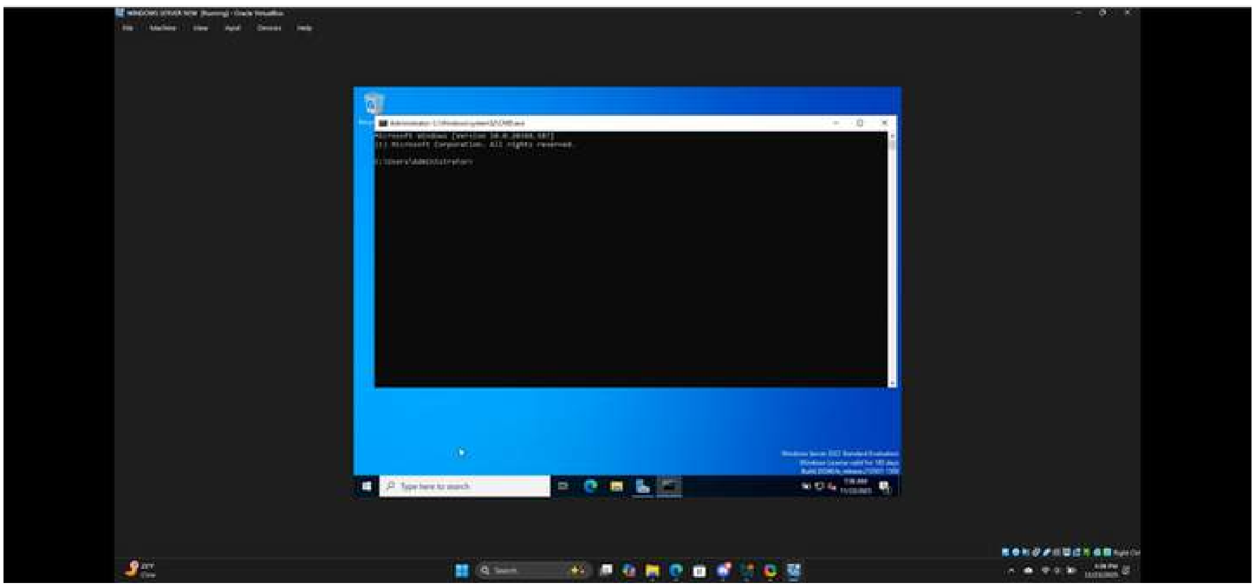




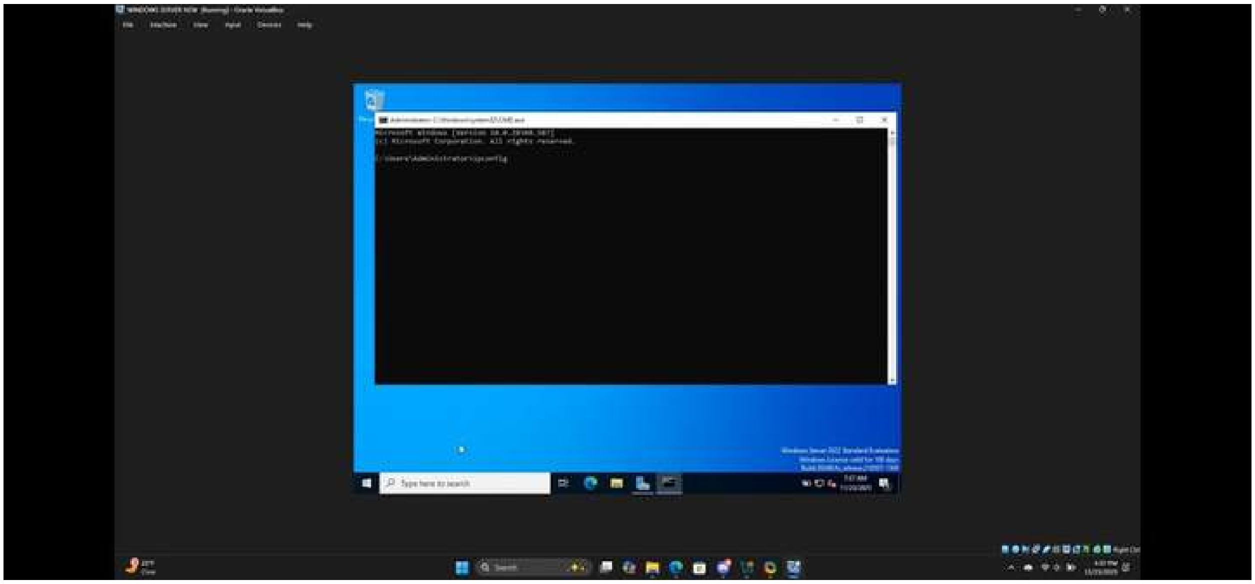
and executed ipconfig via the Command Prompt by pressing (window key +R and type CMD) to verify the dynamically assigned IP address.



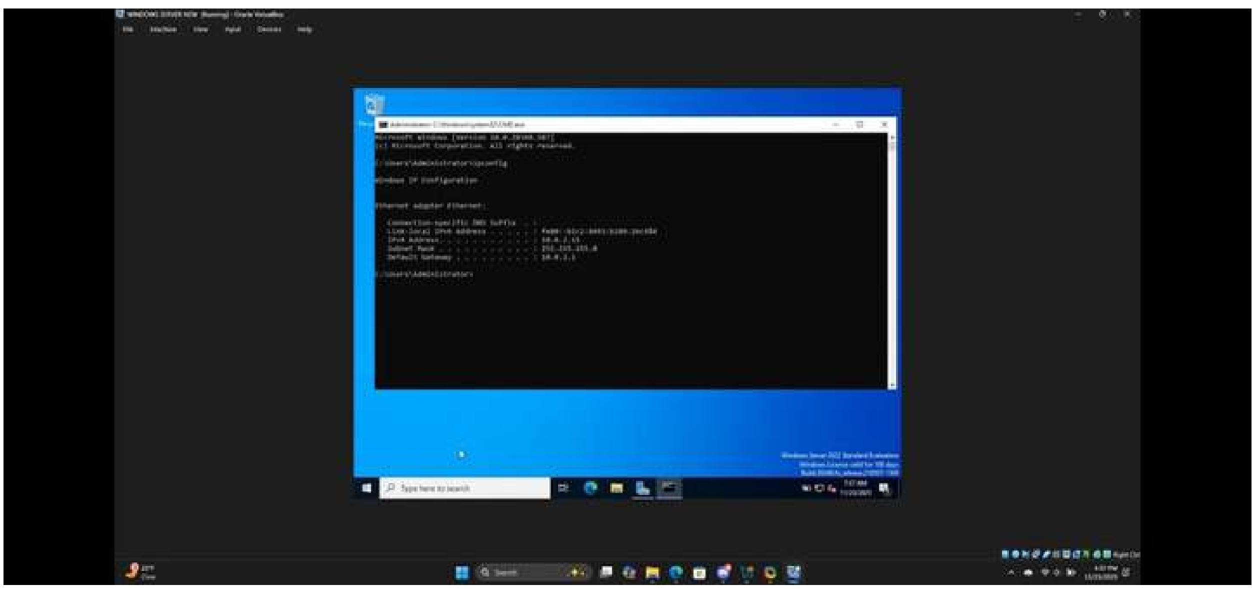




I type “ipconfig”

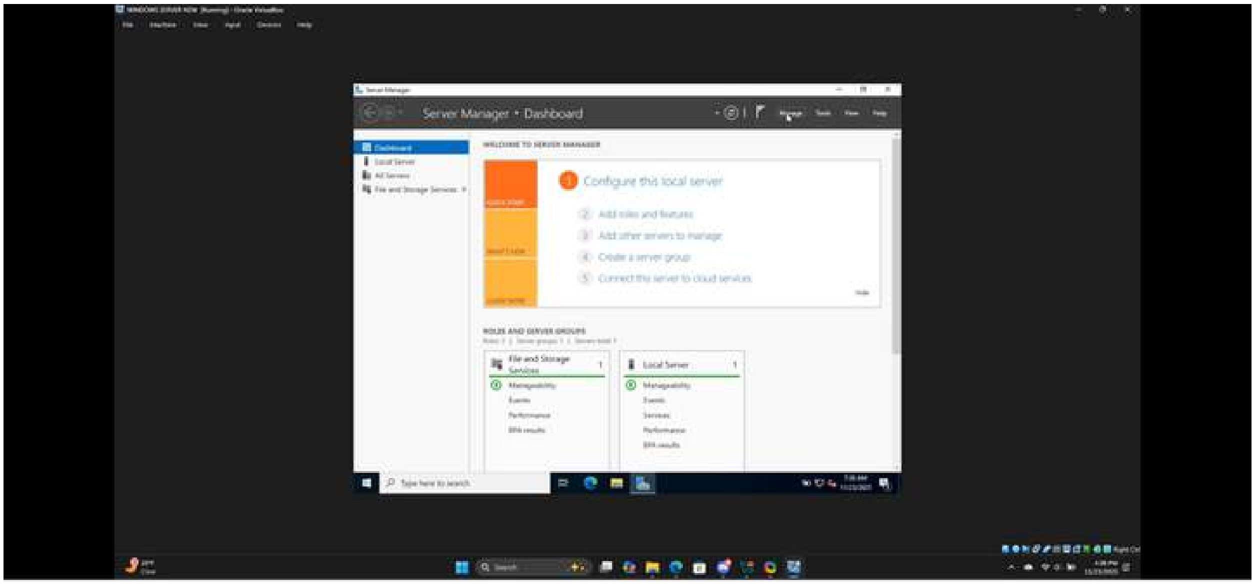


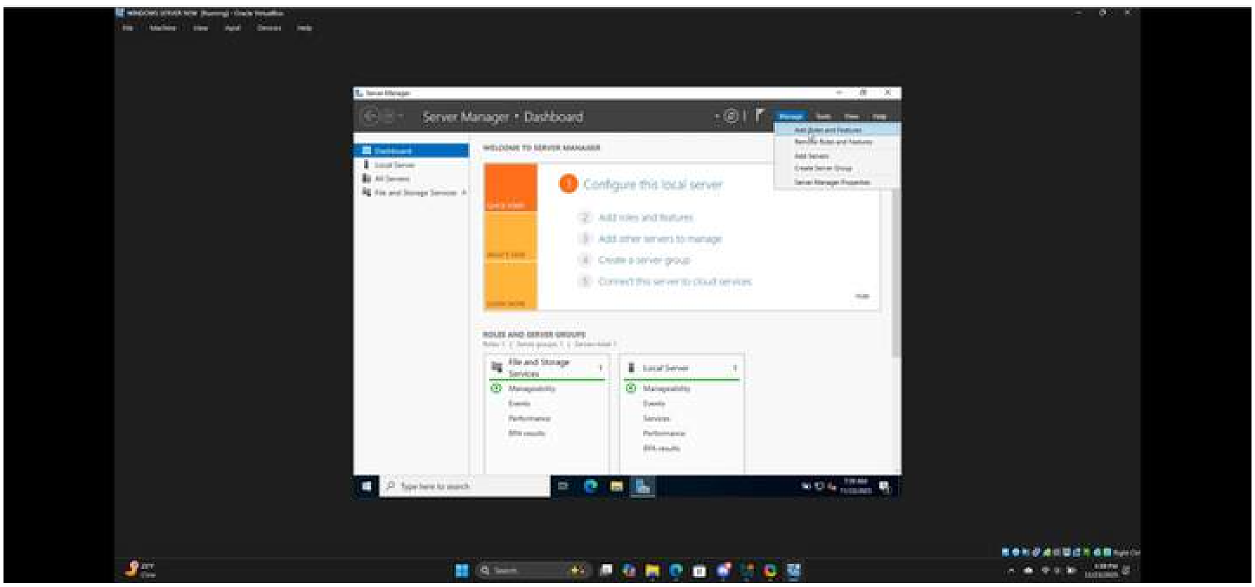
The subsequent ipconfig prompt , it confirmed successful network connectivity to the subnet gateway, concluding the foundational network setup



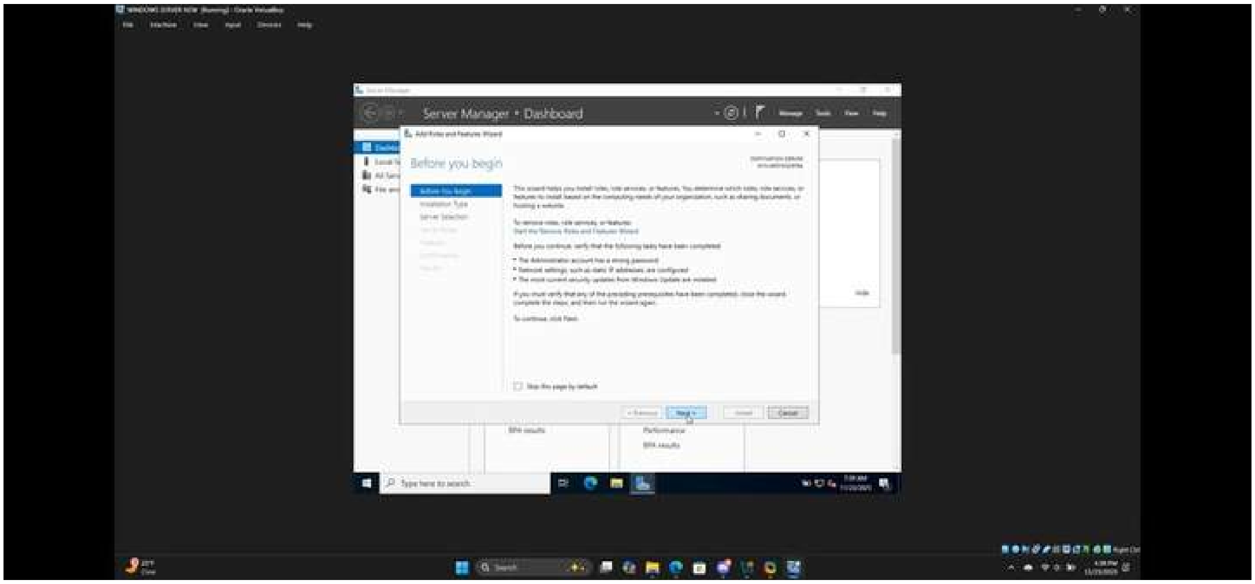
With stable network connectivity established, the project moved to installing and promoting the Windows Server to a Domain Controller, which is the central identity and security management system for the entire network.

Stage 1: Installing the Active Directory Domain Services (AD DS) Role Inside the Server Manager dashboard, i initiated the role installation process by selecting Manage and then Add Roles and Features.

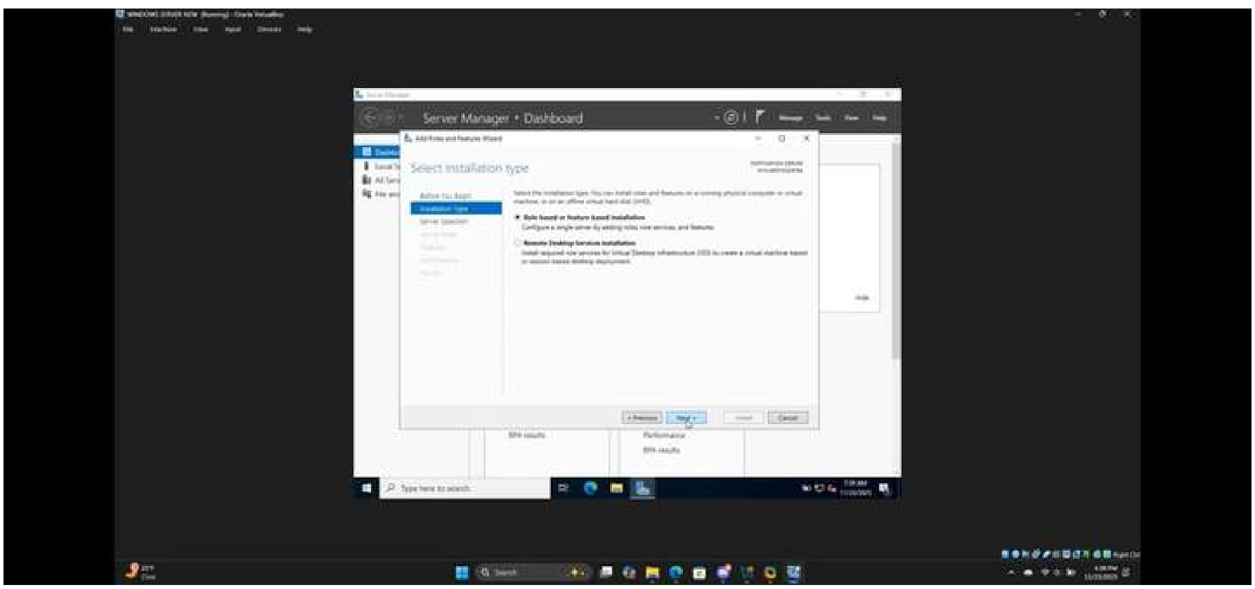




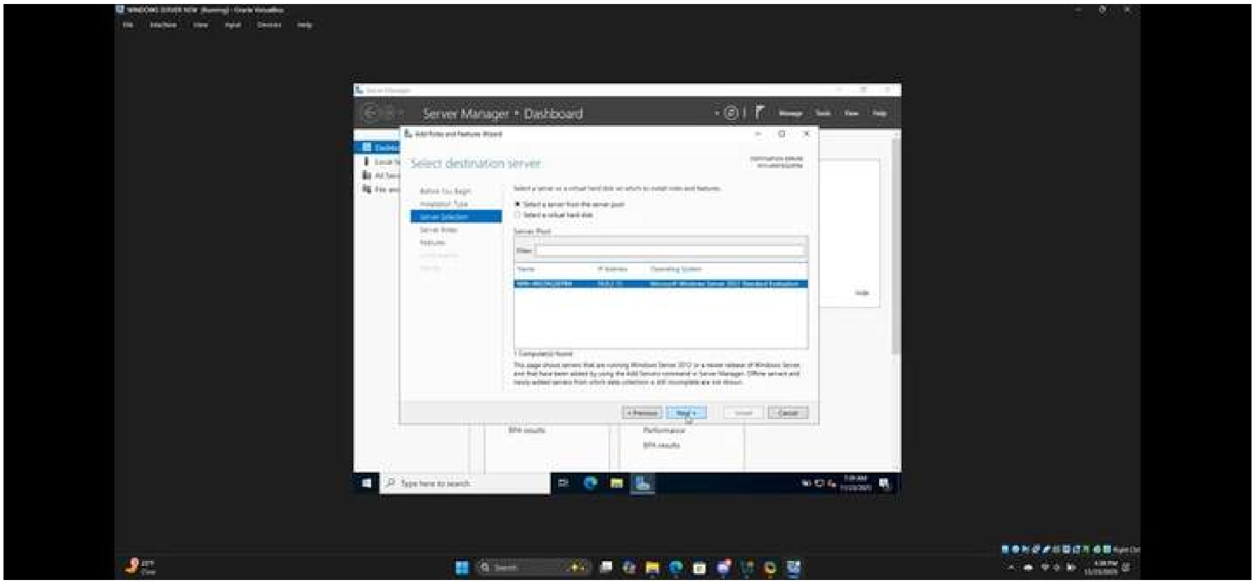
At this stage I’m navigating through the initial wizard prompts, the first page which is “before you begin” I clicked next after reading through



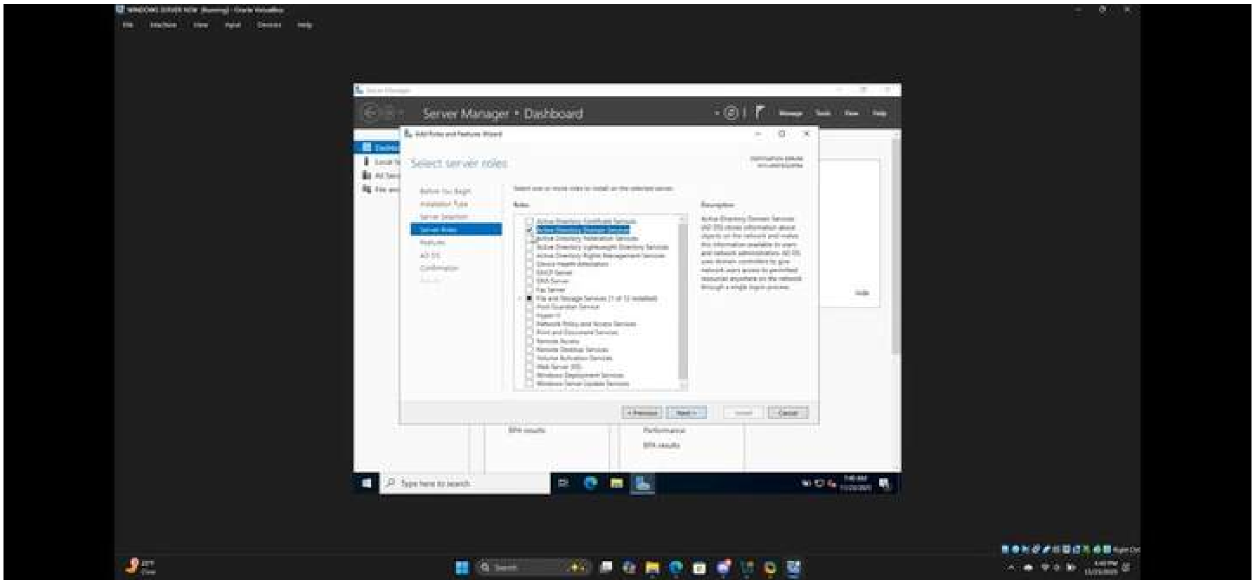
On the installation type which is the next page, I choose “Role based or future based installation” then clicked Next



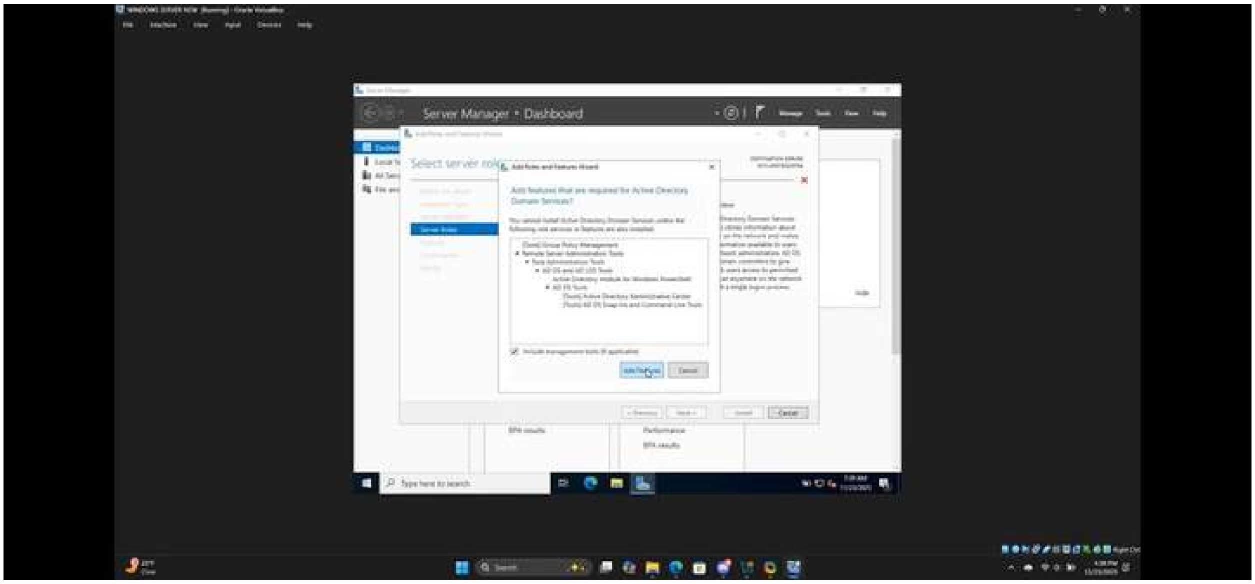
On “server selection” page , I clicked next



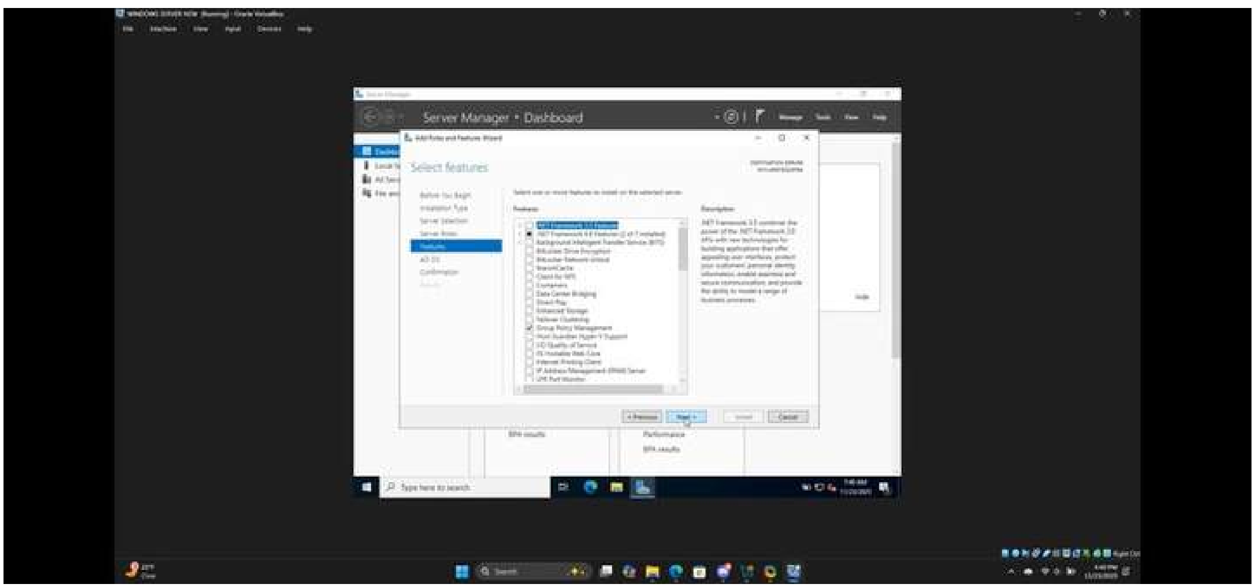
i reached the Server Roles page and selected the Active Directory Domain Services (AD DS) role.



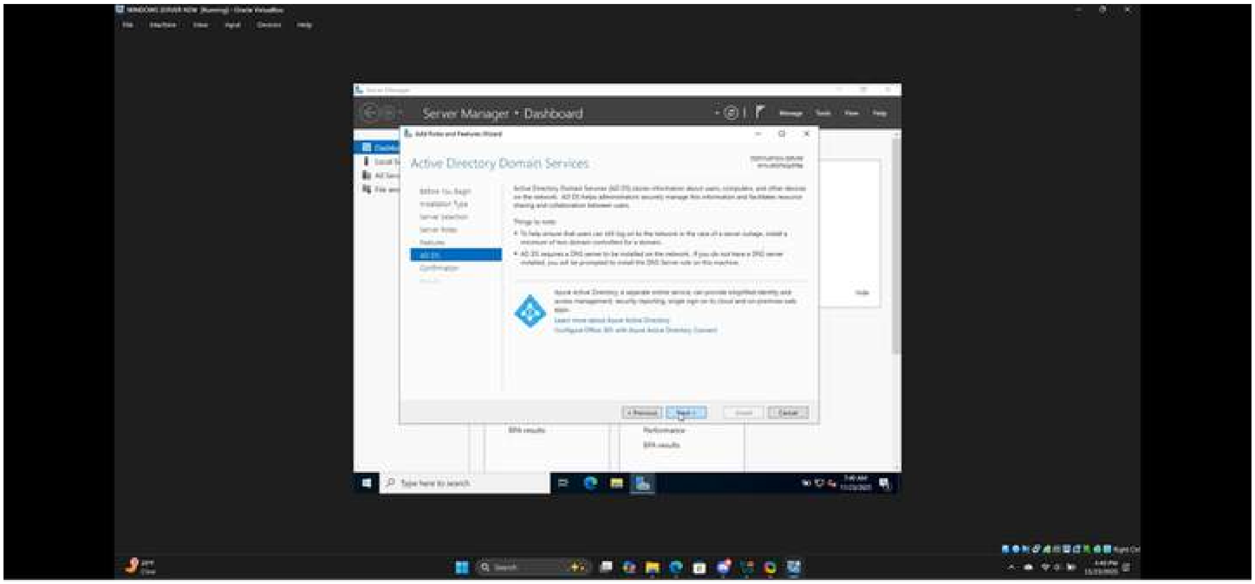
The required accompanying features were automatically added when I clicked add features

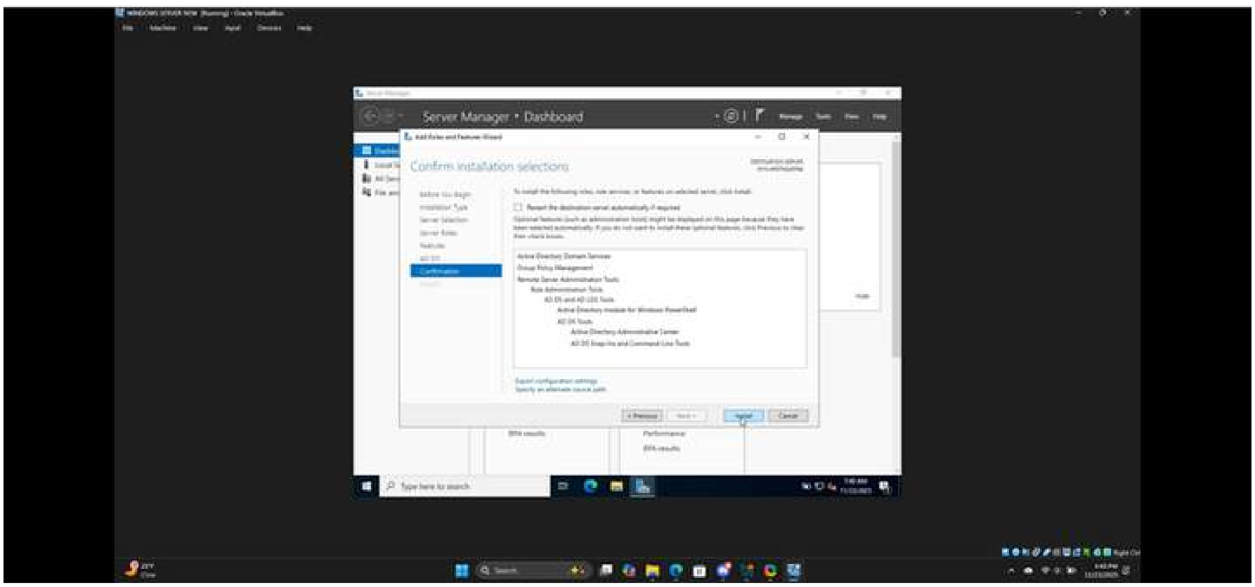


I clicked next for more added features

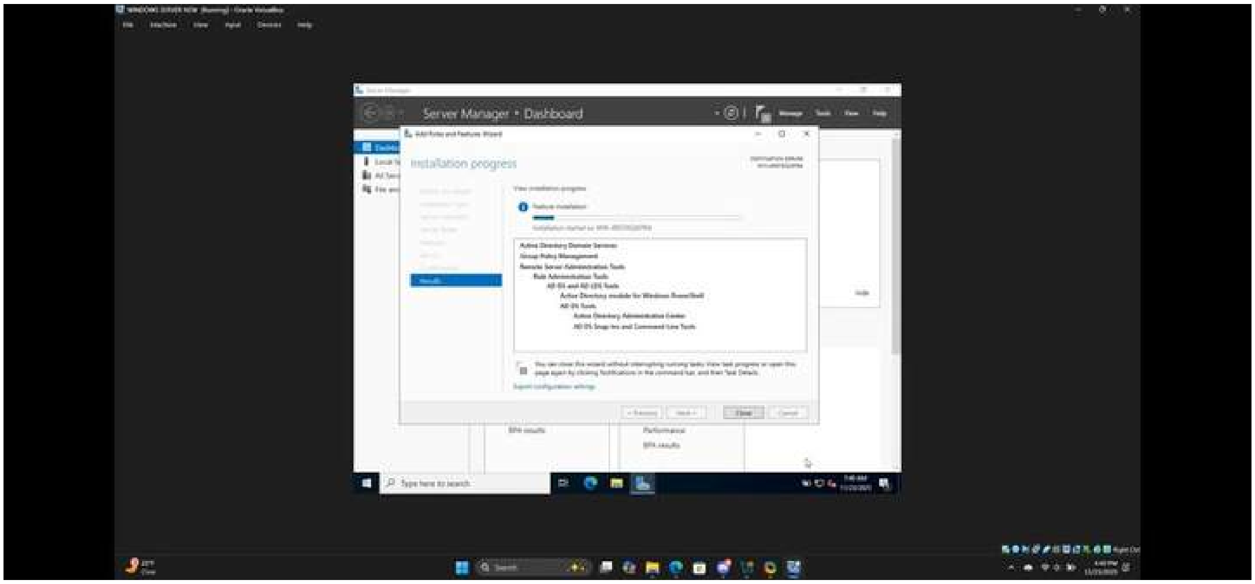


I proceeded through the final confirmation screens and initiated the installation of the AD DS.

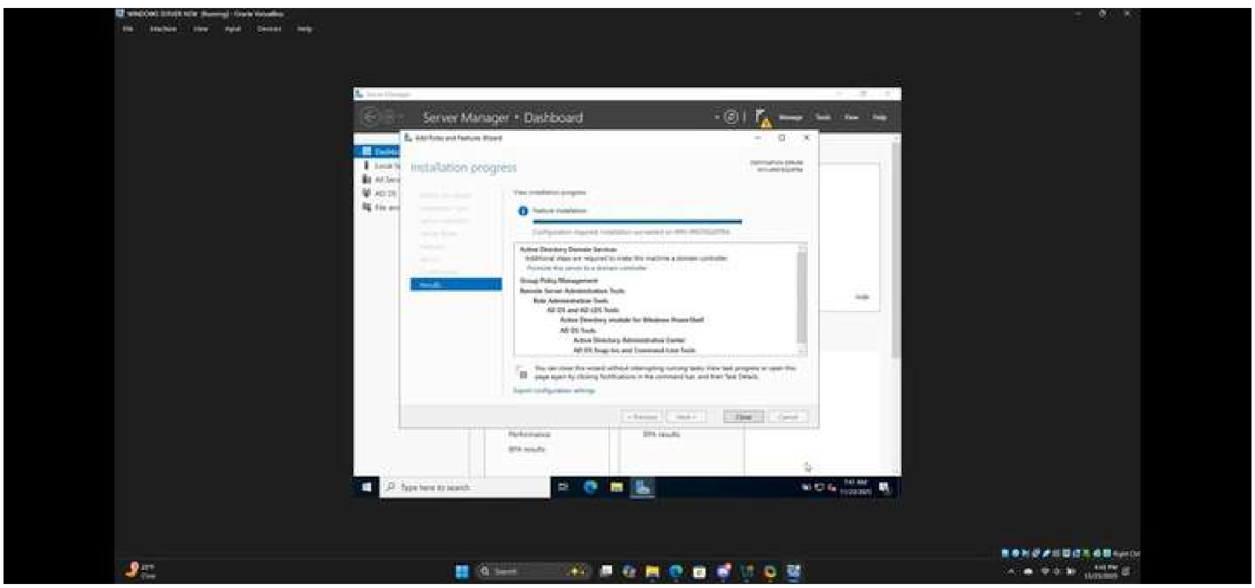




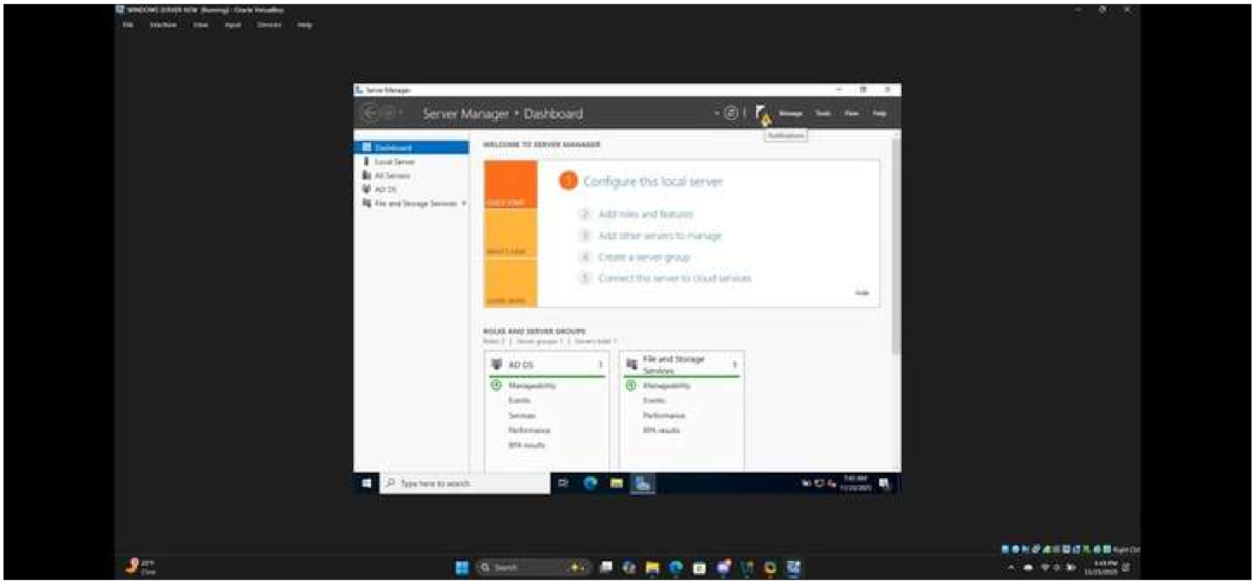
Installation begins



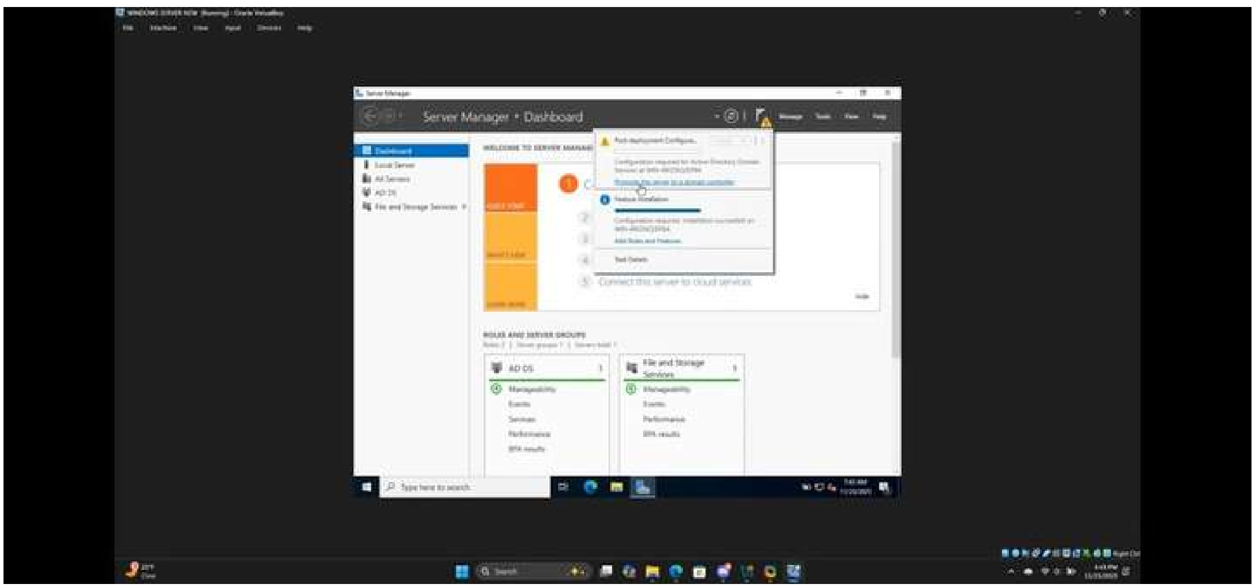
Installation Completed



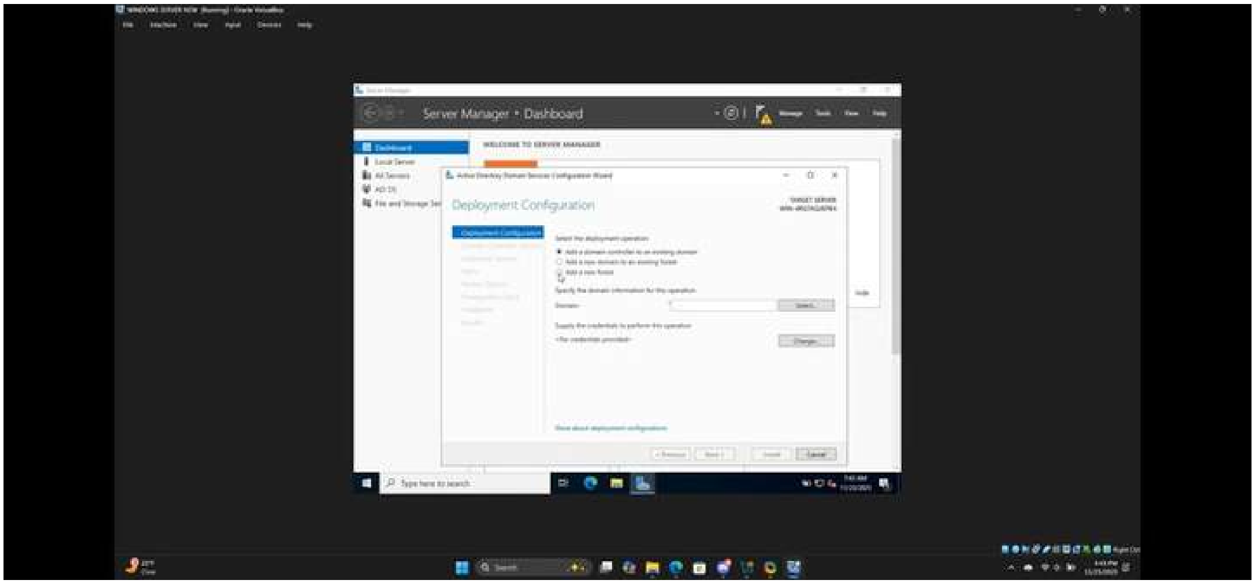
Upon completion, the role was visible on the Server Manager dashboard.



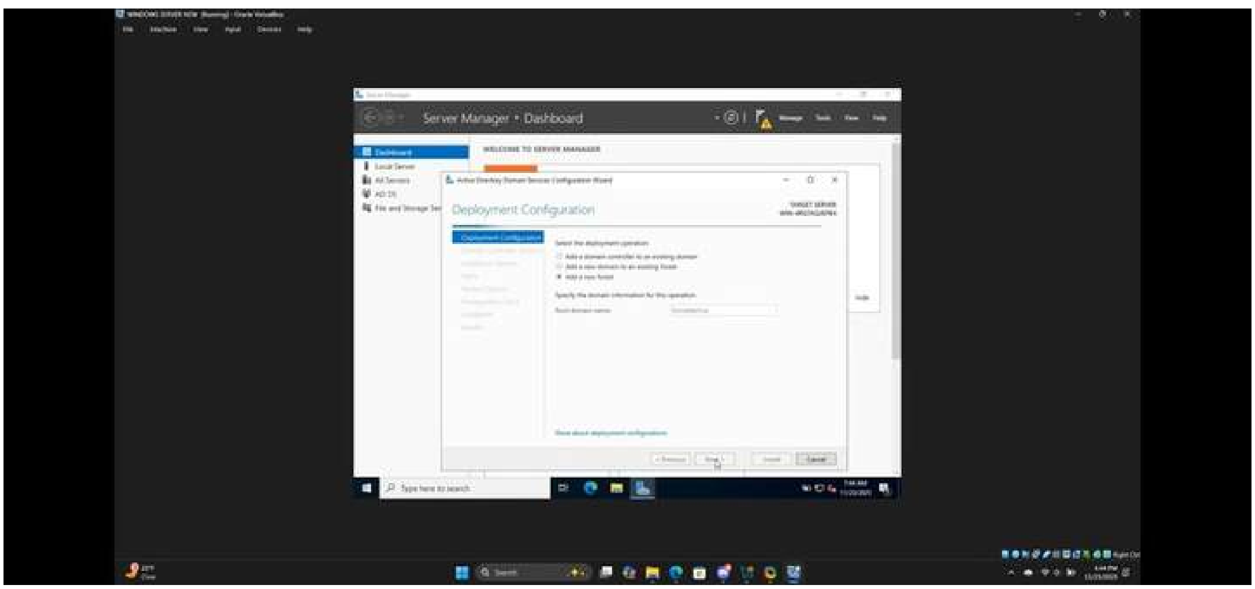
Stage 2: Promoting the Server to a Domain Controller:Installing the role is a prerequisite; the server must be promoted to actively manage the domain.The promotion wizard was launched by clicking the Notifications Flag (yellow icon) on the top right and selecting Promote this server to a domain controller.



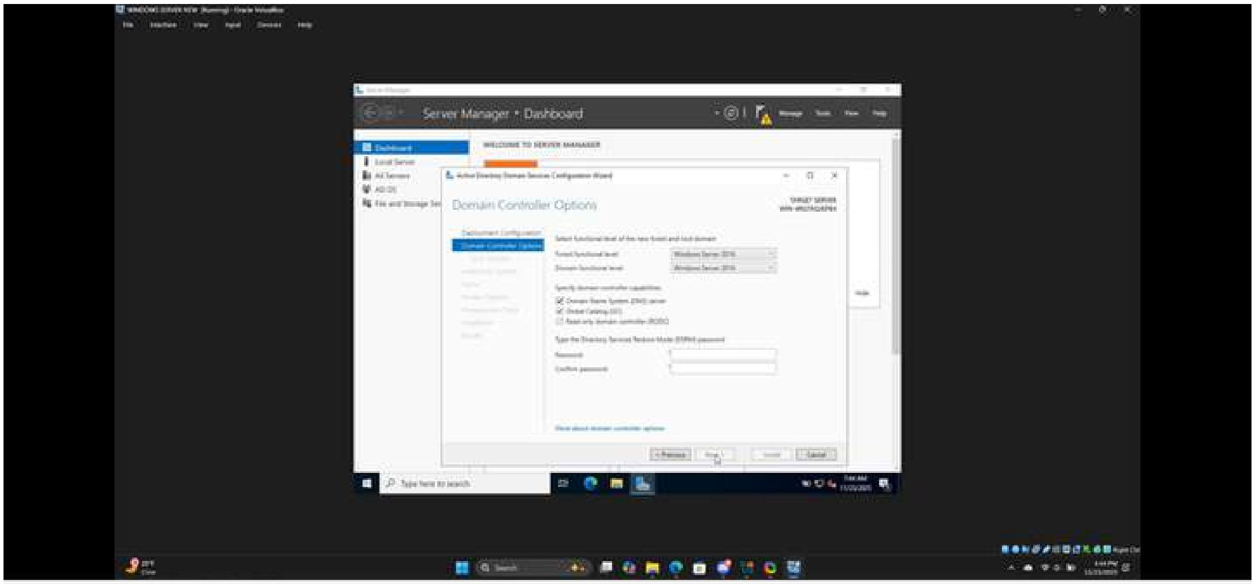
On the initial screen, the deployment operation was correctly set to Add a new forest. This establishes the server as the very first domain controller for a new organizational structure.

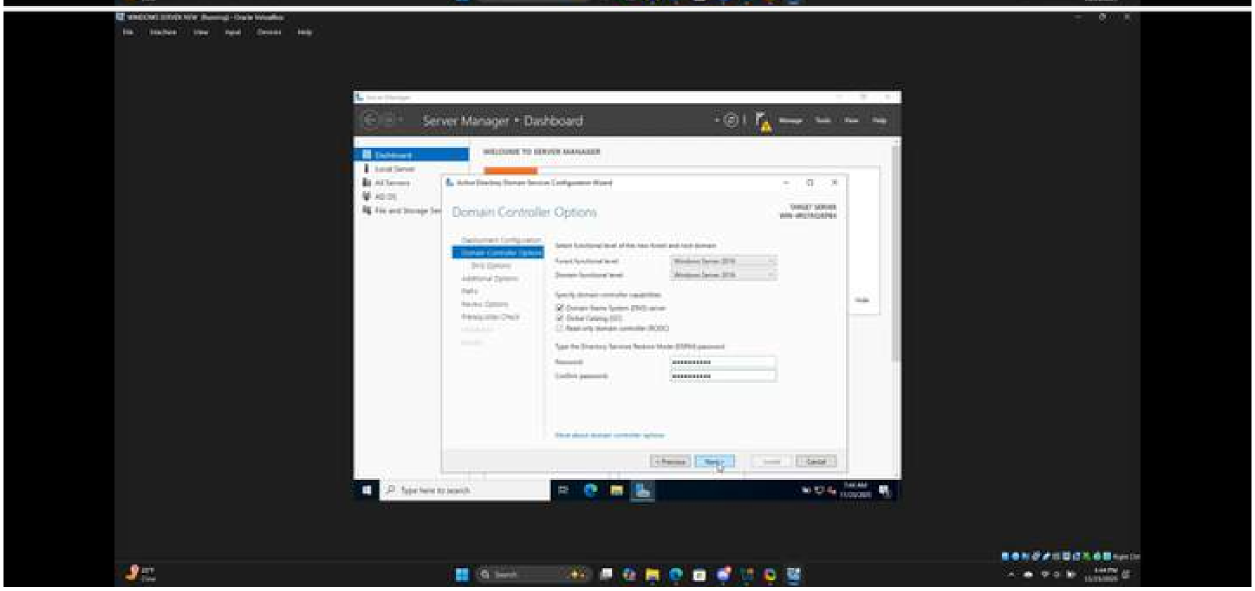


The Root Domain Name for the new forest was defined as “nxgluxurylimited.local”, providing the unique, fully qualified name for the enterprise environment and I clicked next

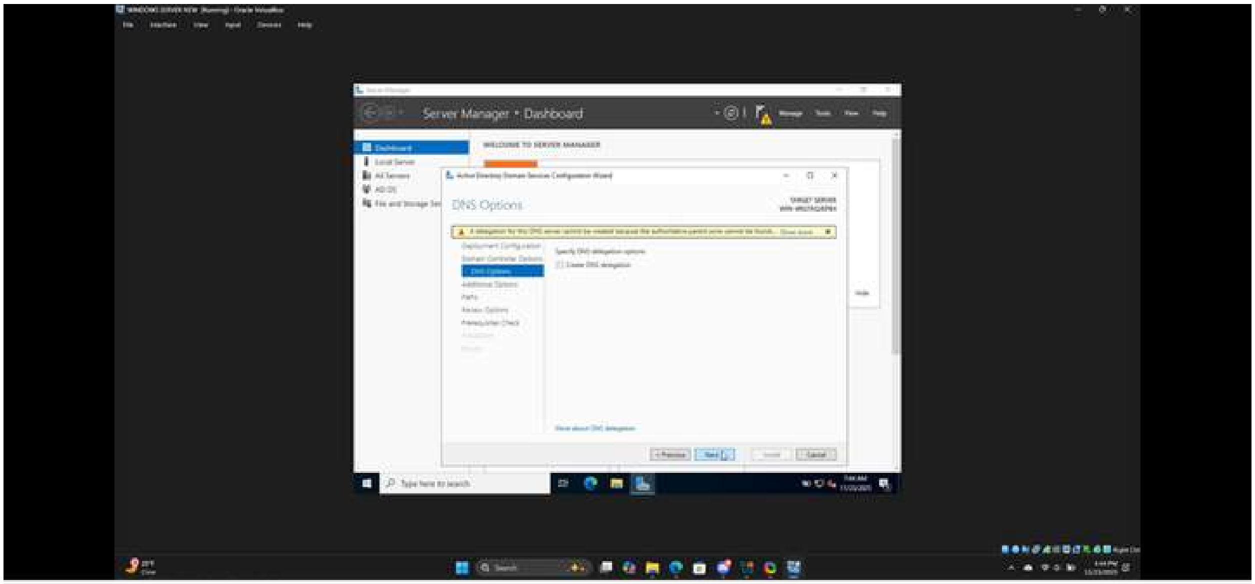


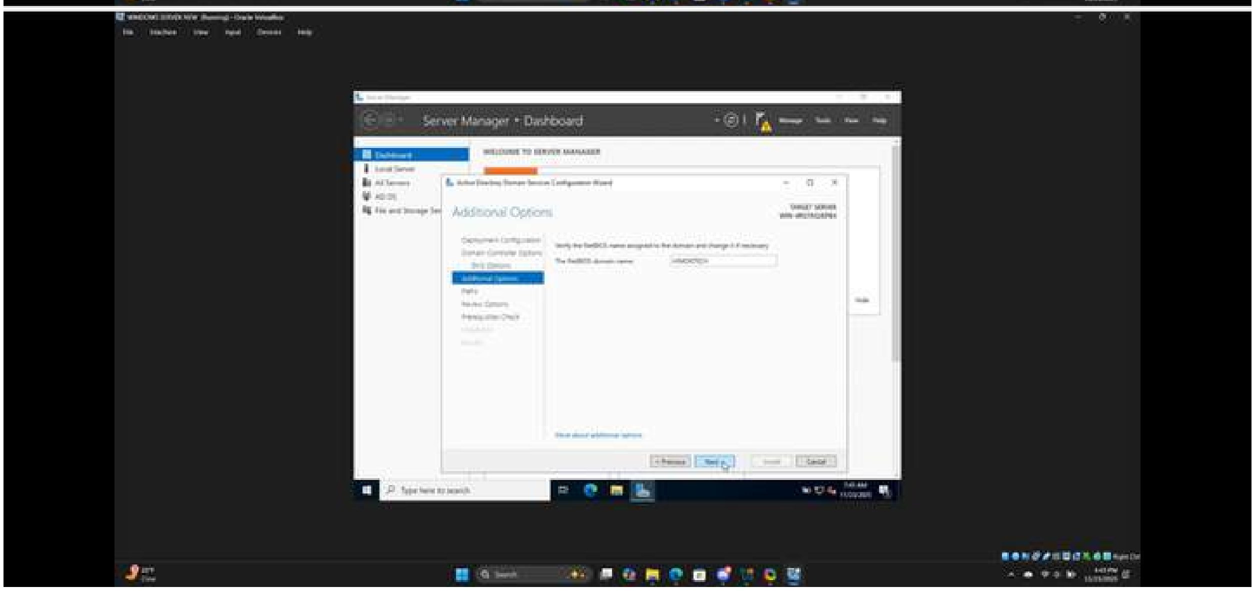
On the Domain Controller Options: I defined the Directory Services Restore Mode (DSRM) password, then I click next.



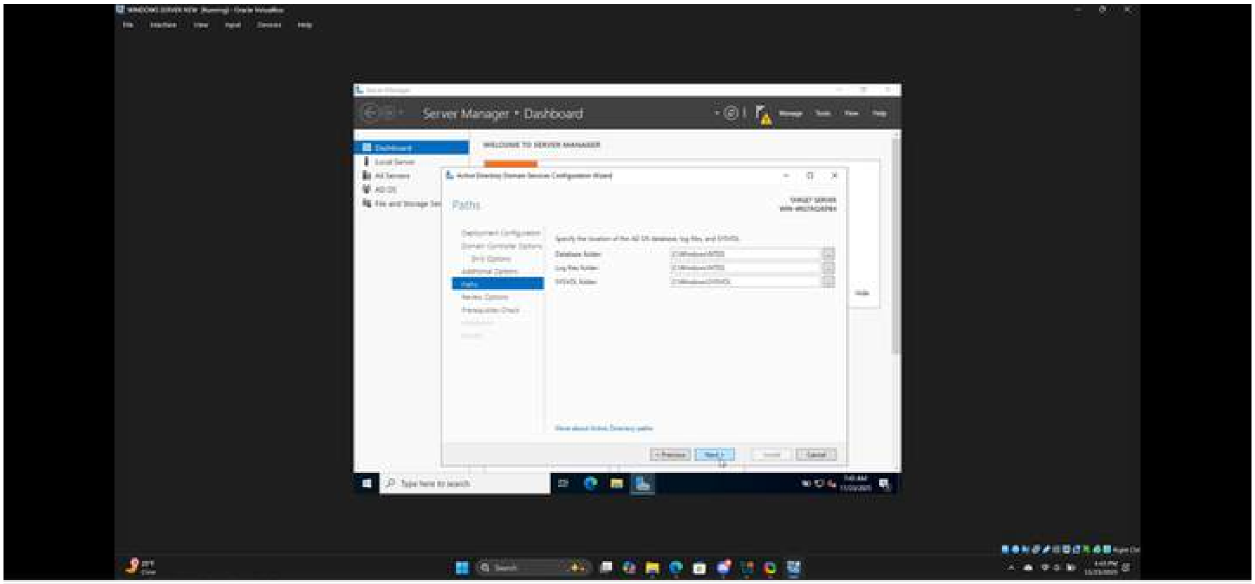


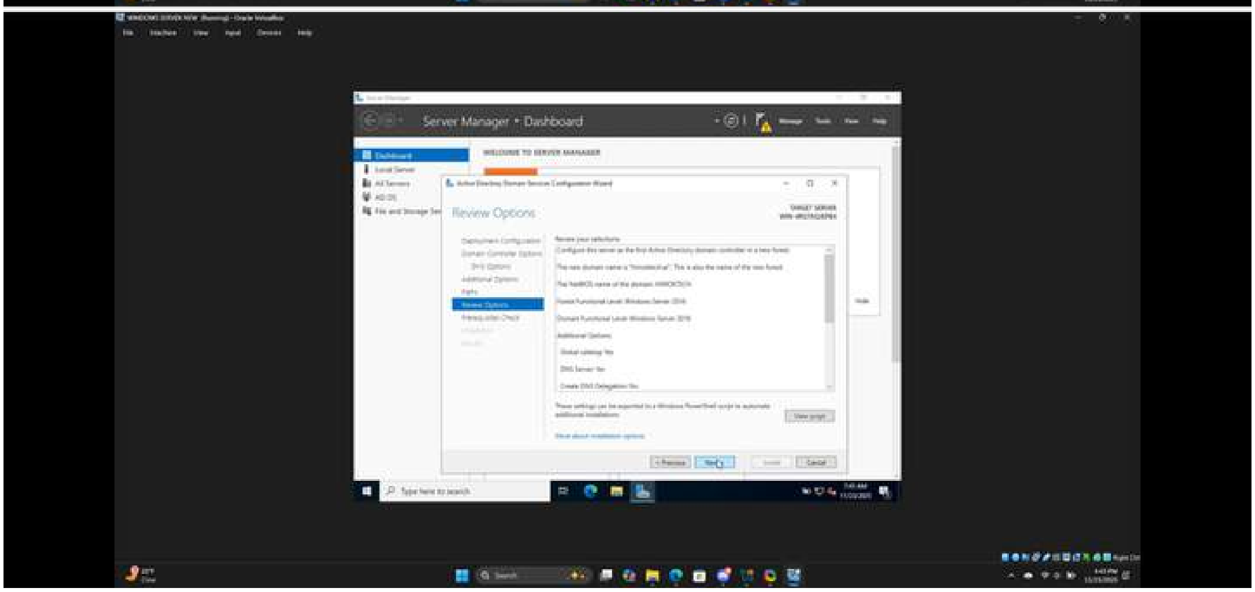
5. NetBIOS Configuration: After accepting the default DNS and additional options, you specified the NetBIOS Domain Name as nxgluxurylimited.local.



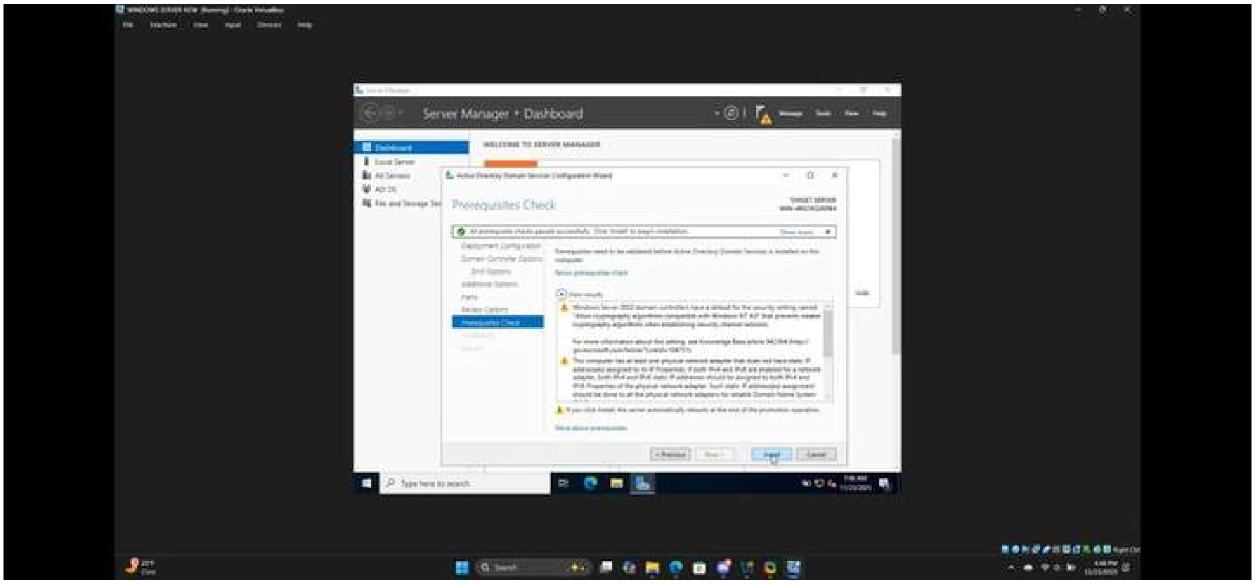


I accept the path and review options by clicking next

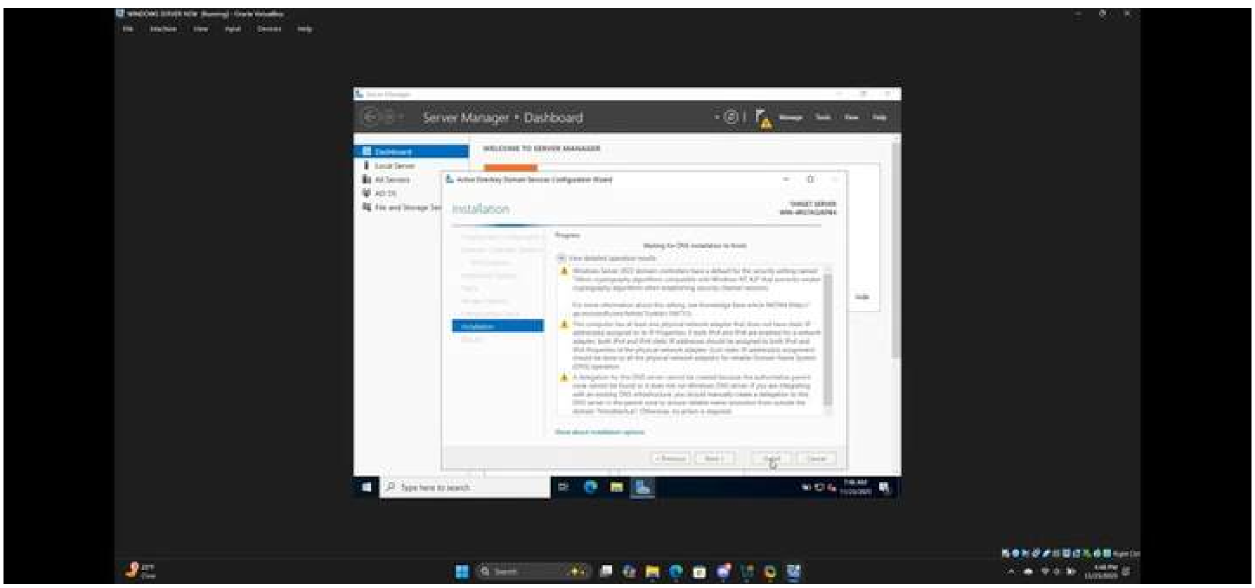




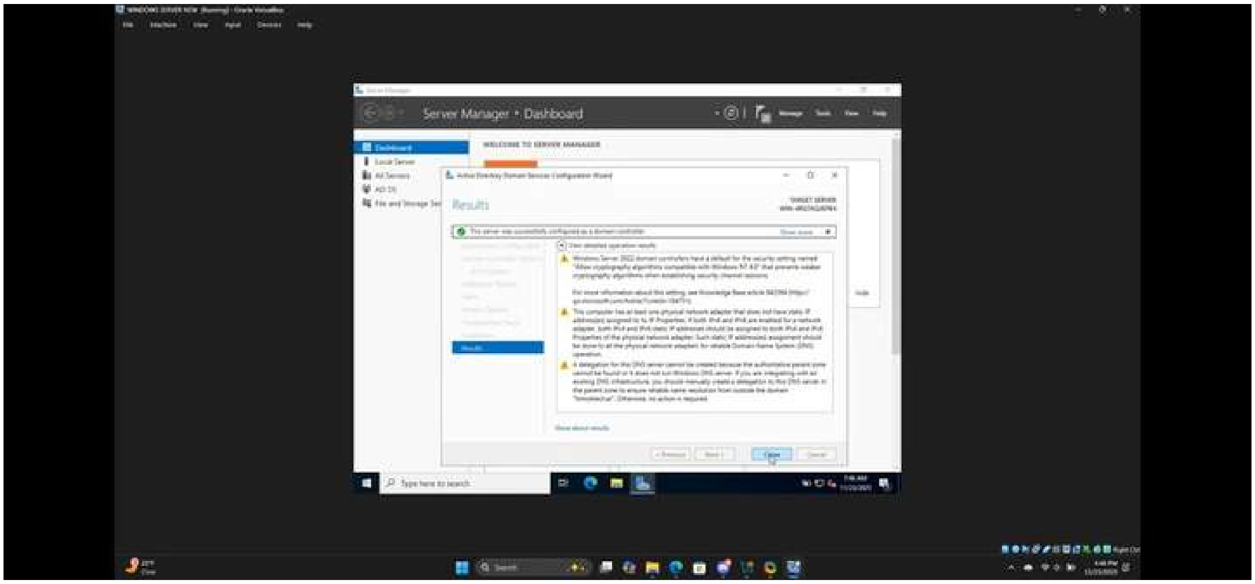
On Prerequisite Check and Final Installation: Following a successful Prerequisite Check, the final installation was initiated. This process resulted in the server automatically rebooting to finalize the new Domain Controller role and configuration.



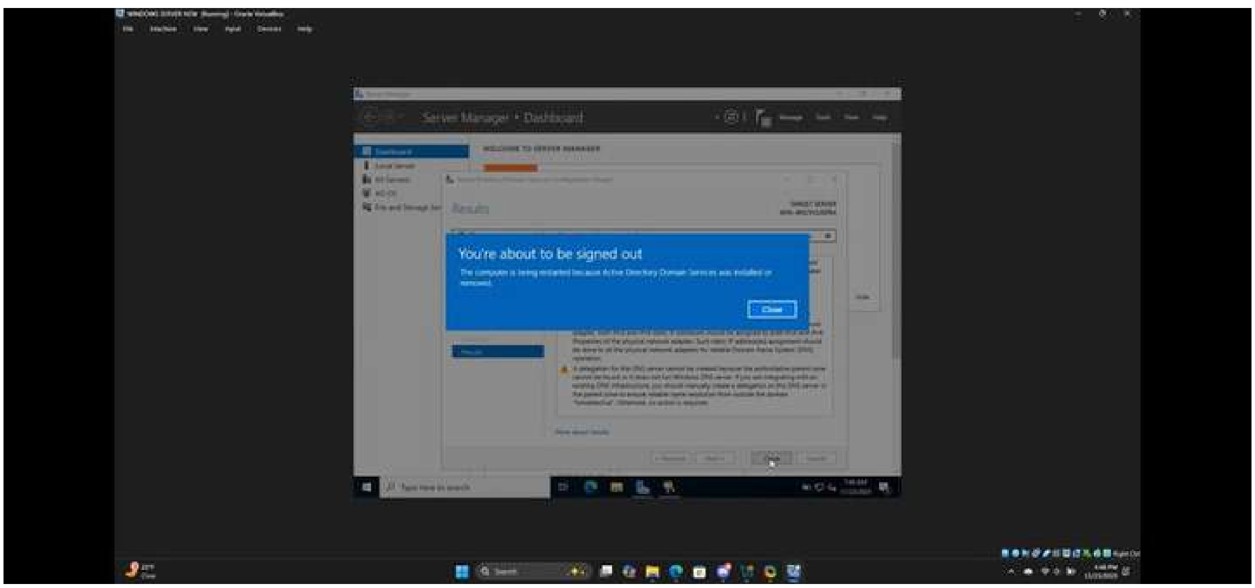
Installing….



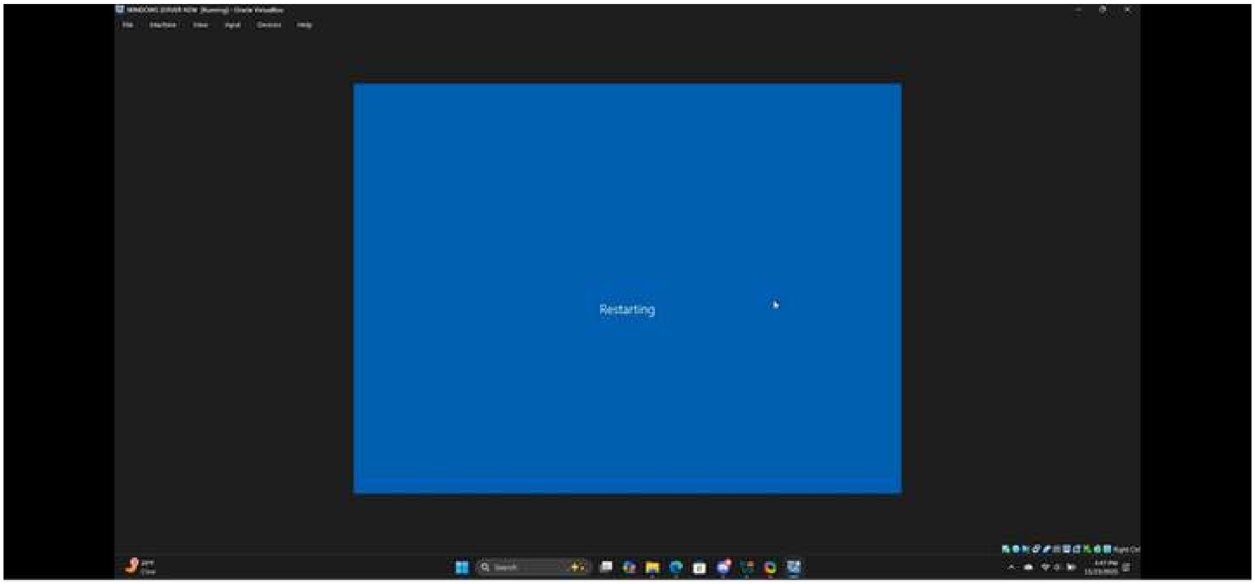
Installation completed

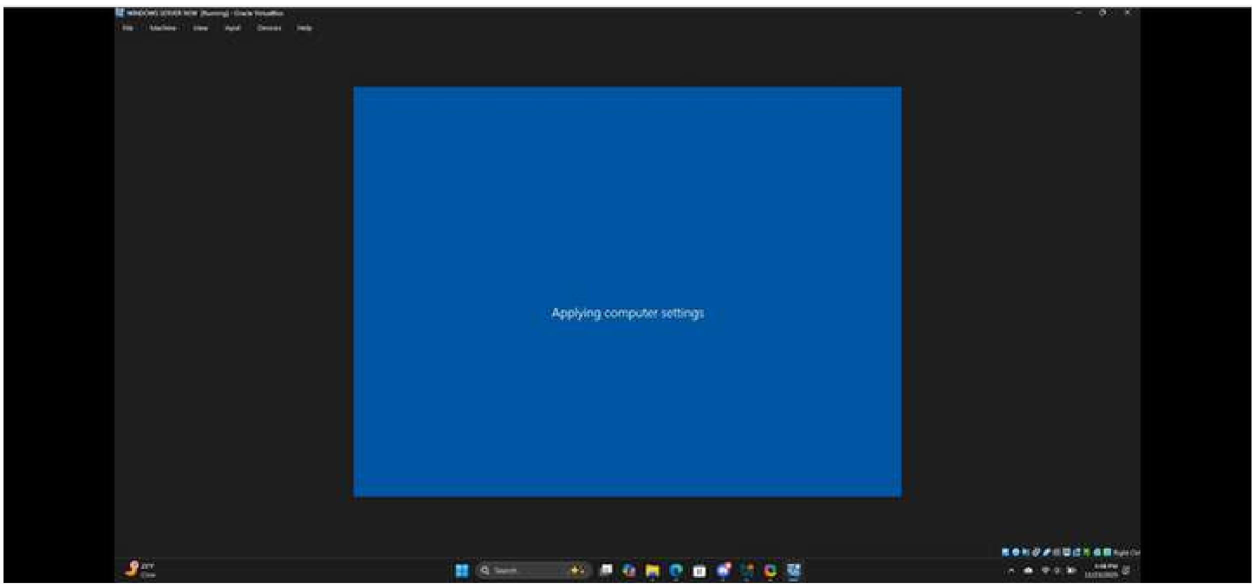


I closed it to



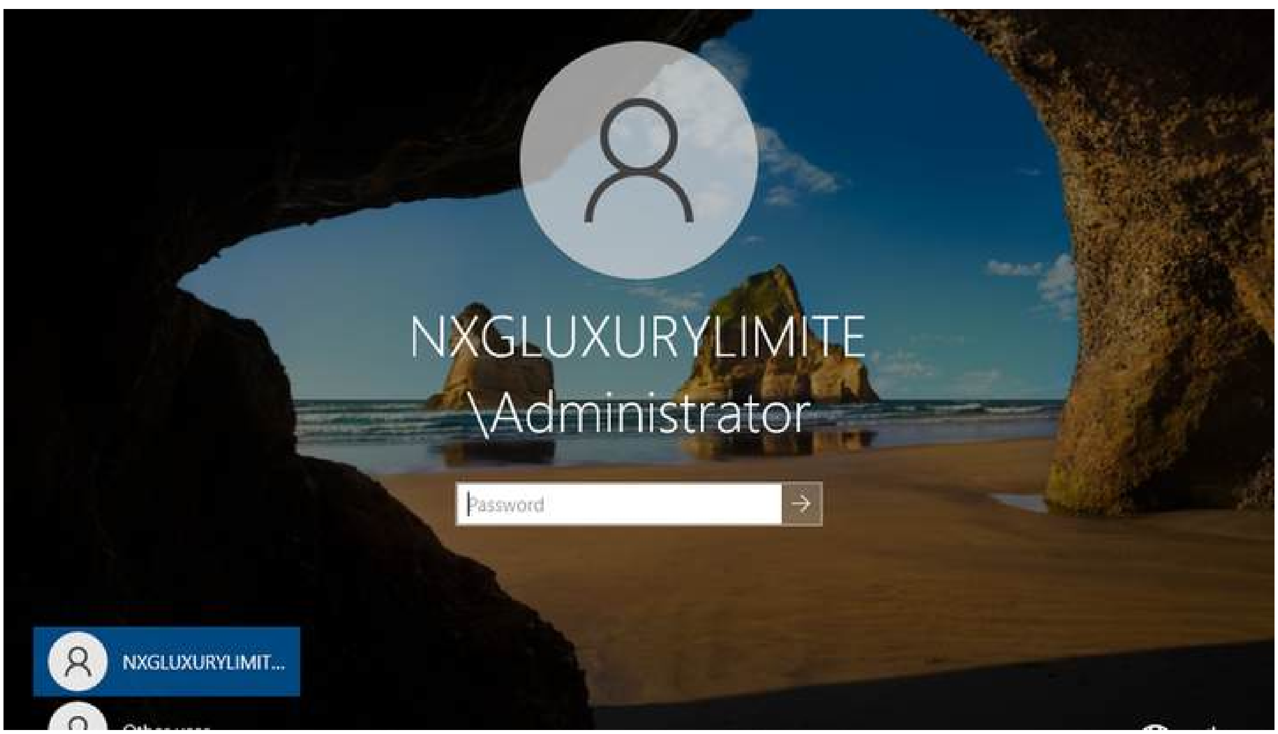
Restarting process started



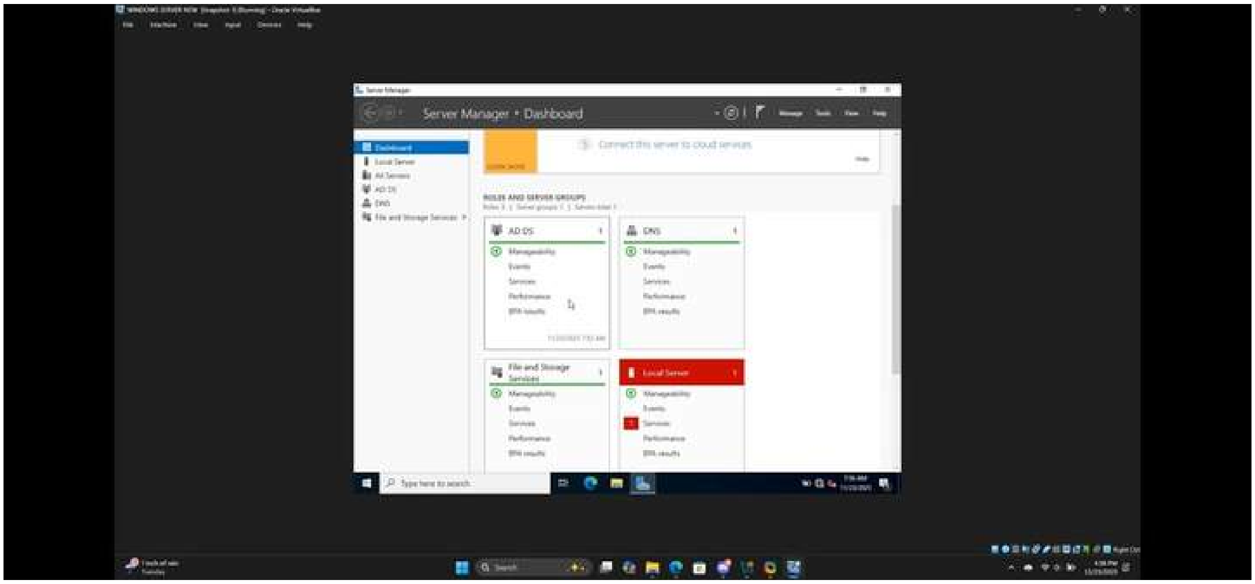


Successfully Restarted.

Upon restart, the secure login screen presented the new domain context (NXGLUXURYLIMITED\Administrator), confirming the successful promotion.

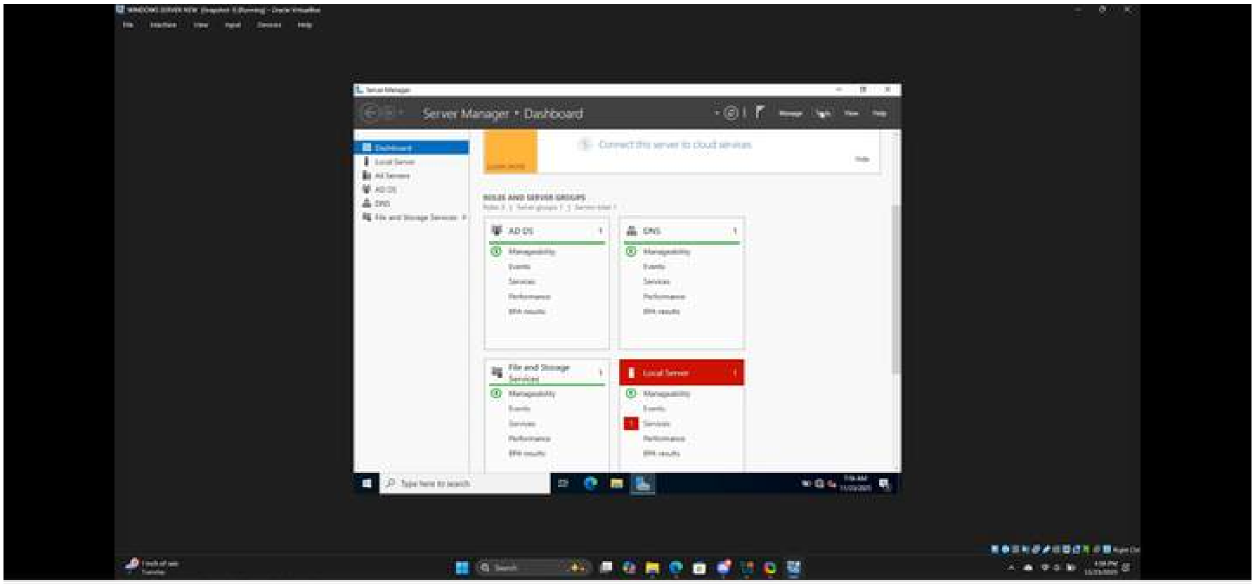


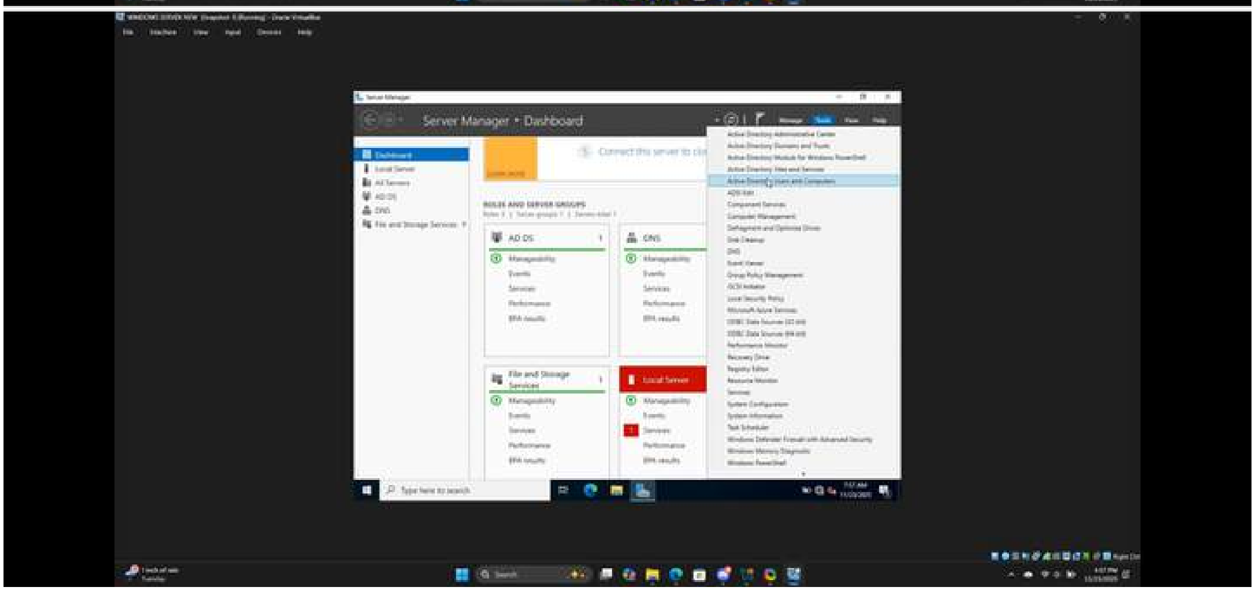
The server was then logged into using the newly configured administrative password, confirming the Active Directory service was fully operational.



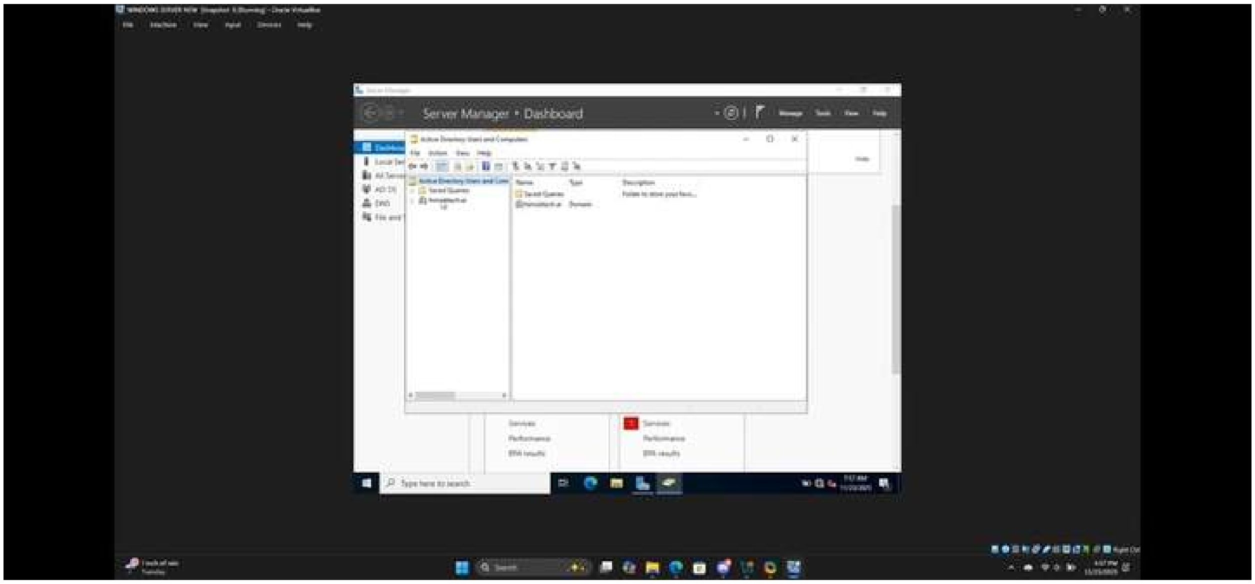
This phase I want to establishes the hierarchical administrative structure within the himoktech.ai domain and integrates the first client endpoint, allowing for centralized security and user management. Firstly I started with Creating Organizational Units, Groups, and Users. The administration of the domain begins with creating a logical framework to manage users and resources efficiently.

From the Server Manager dashboard, the Active Directory Users and Computers (ADUC) tool was accessed via the Tools menu.

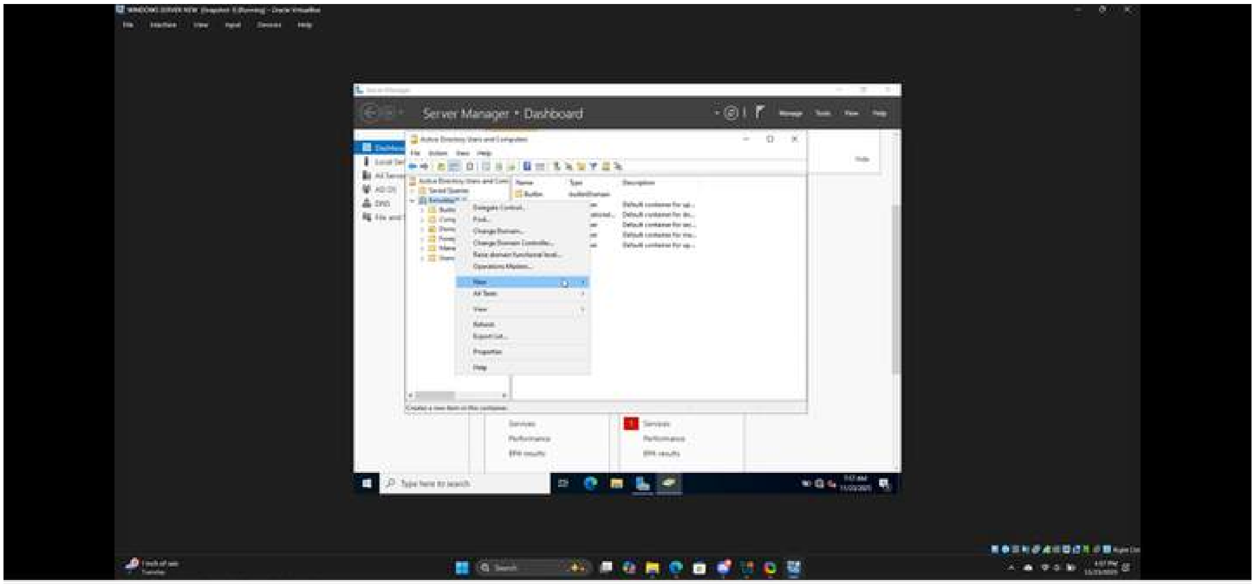


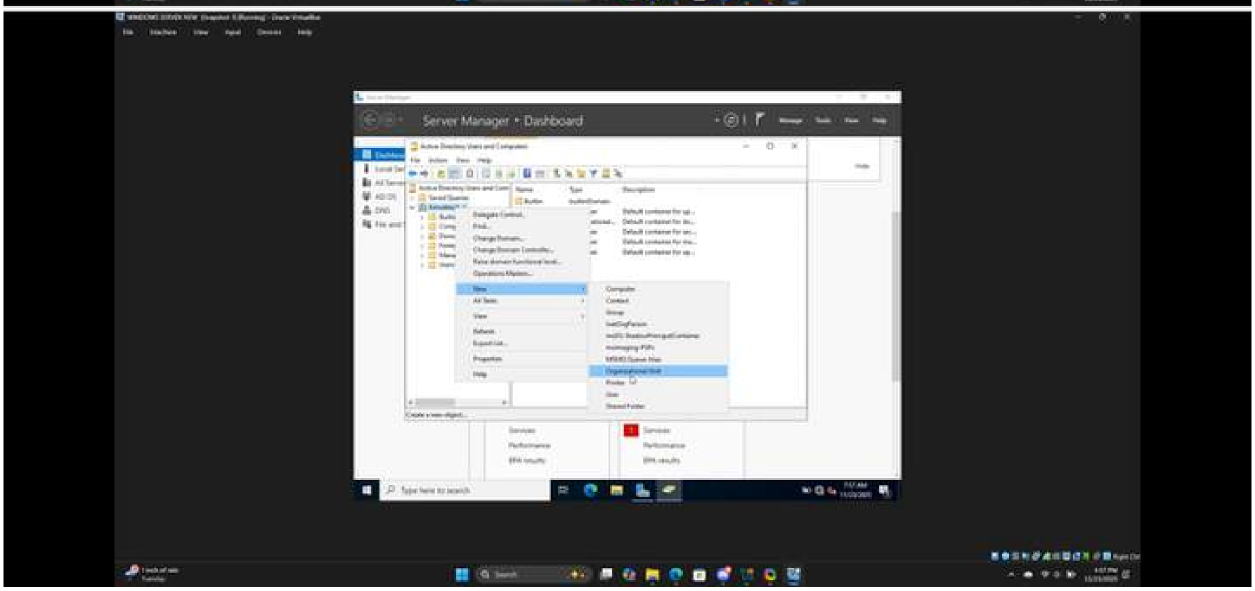


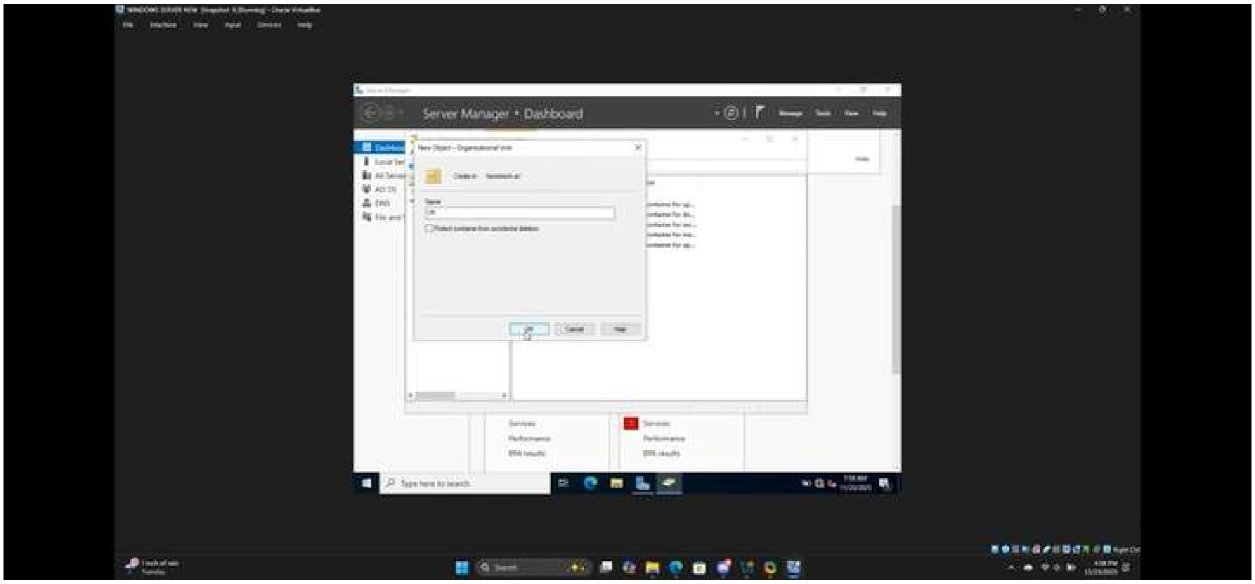
The console immediately displayed the new nxgluxurylimited.local



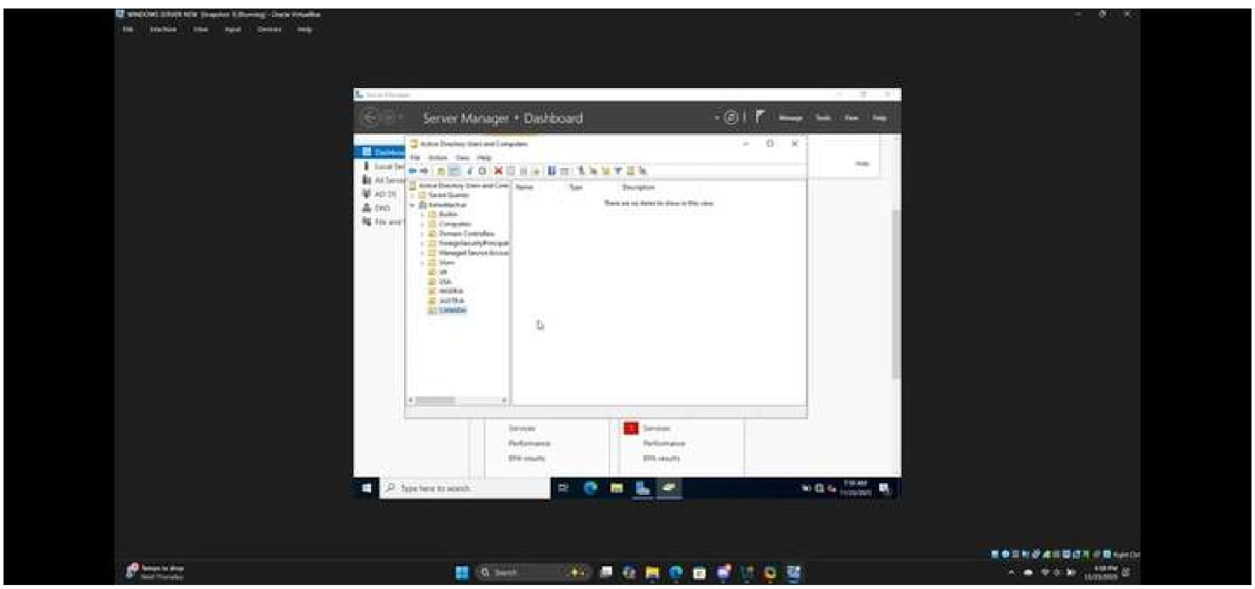
To create Organizational Unit (OU) : I navigated to the himoktech.ai root, right-clicked, and selected New > Organizational Unit.


Server Manager • Dashboard
Active Directory Users and Computers
Delegate Control...
Find...
Change Domain...
Change Domain Controller...
Raise domain functional level...
Operations Masters...
New
All Tasks
View
Refresh
Export List...
Properties
Help

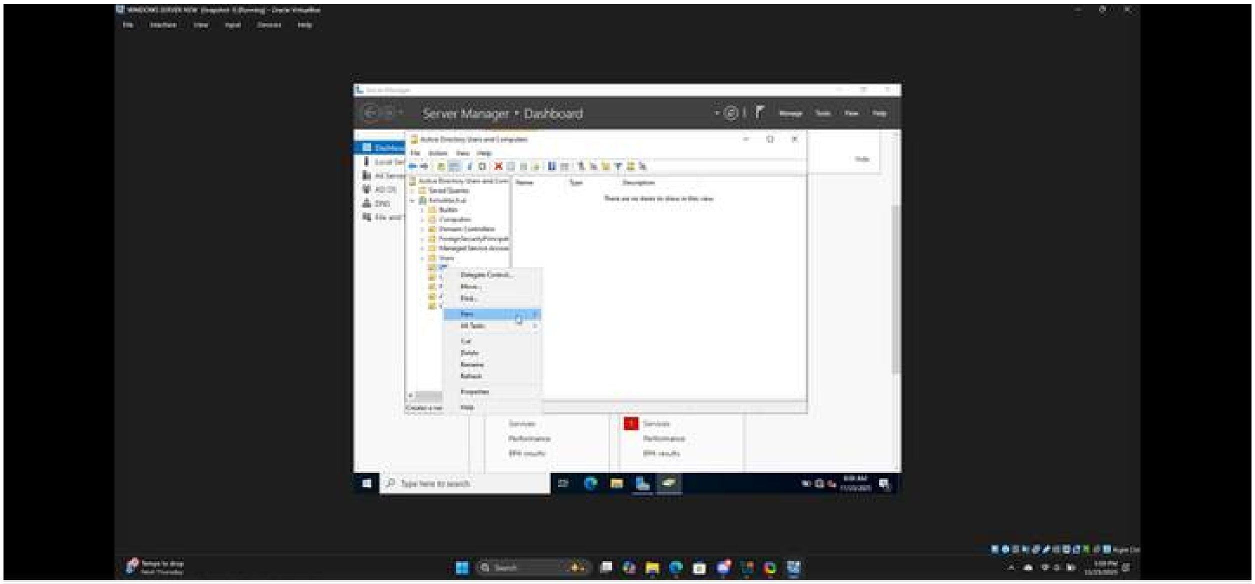

Server Manager • Dashboard
Active Directory Users and Computers
Delegate Control...
Find...
Change Domain...
Change Domain Controller...
Raise domain functional level...
Operations Masters...
New
All Tasks
View
Refresh
Export List...
Properties
Help
Computer
Contact
Group
Organizational Unit
Printer
User
Shared Folder

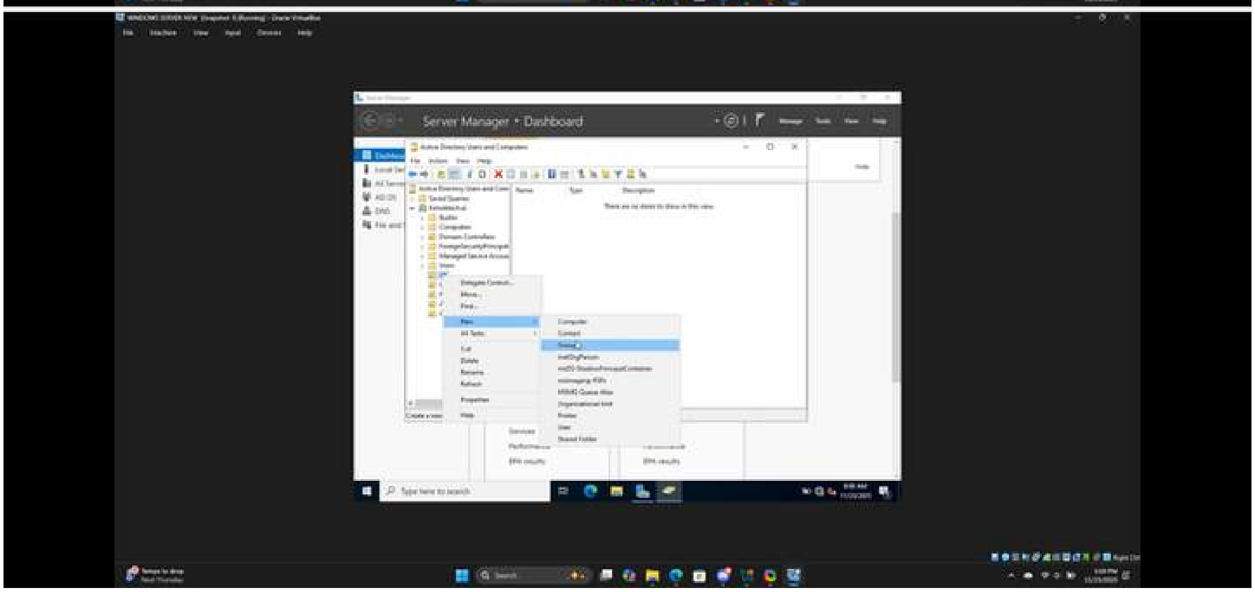


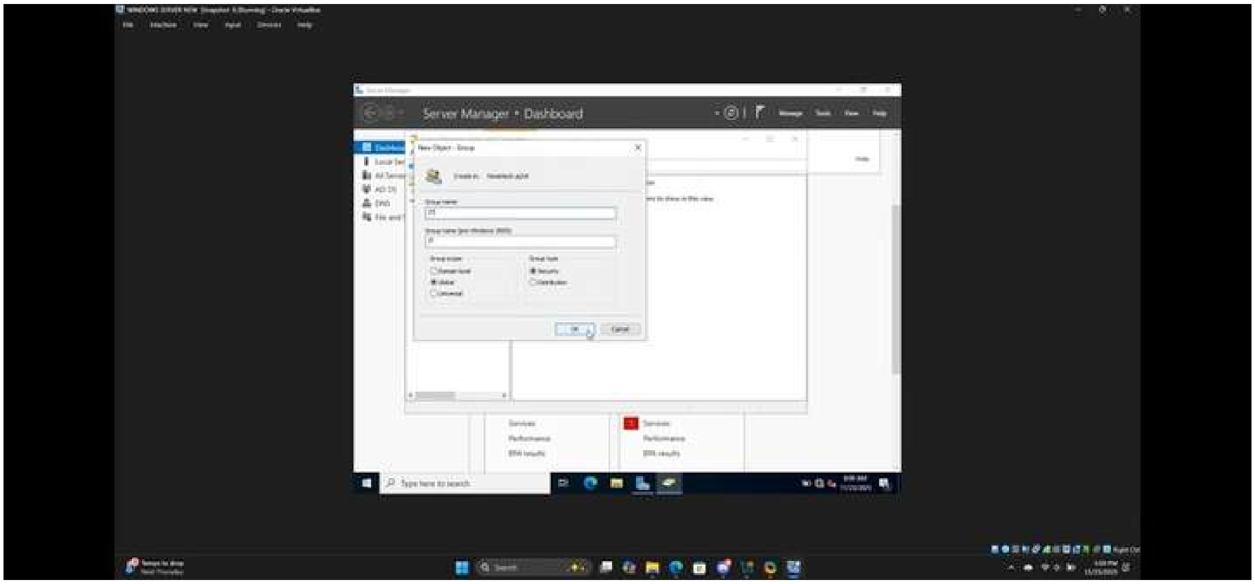
• OUs were systematically created for various geographic regions: uk, and canada.The default option to Protect container from accidental deletion was intentionally unchecked for administrative flexibility regarding future cleanup or restructuring of the OUs.



I proceed to Group Creation: Within each newly created OU (uk, and canada), a security group was created by right-clicking the OU, selecting New > Group. This step prepares the structure for role-based access control.

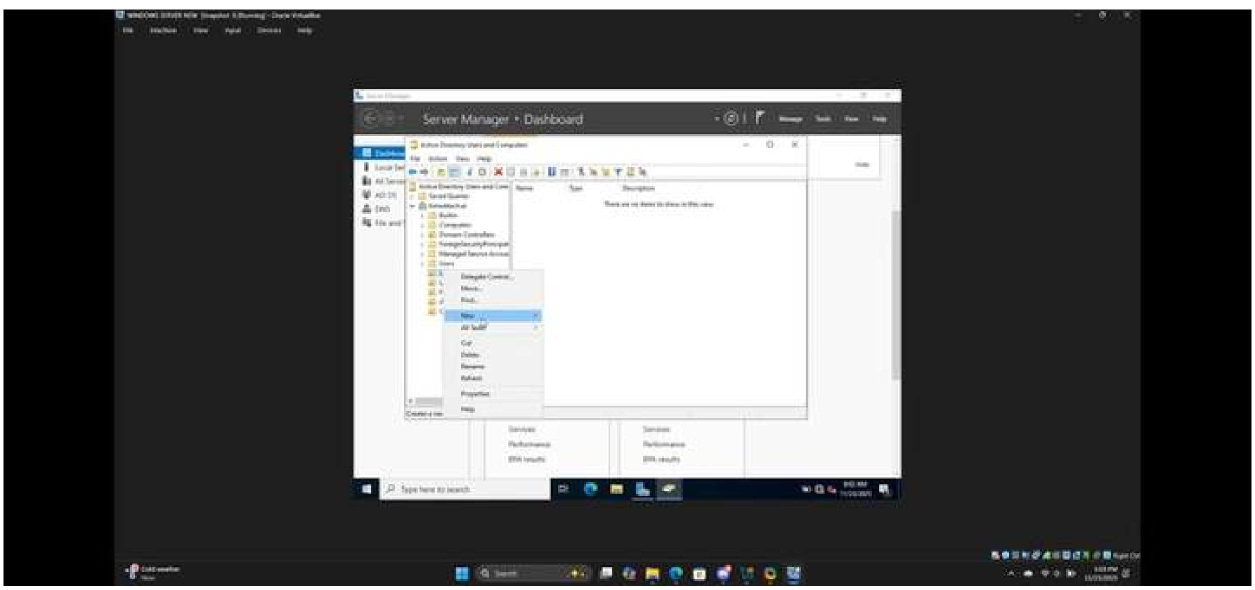

Server Manager • Dashboard


Server Manager • Dashboard

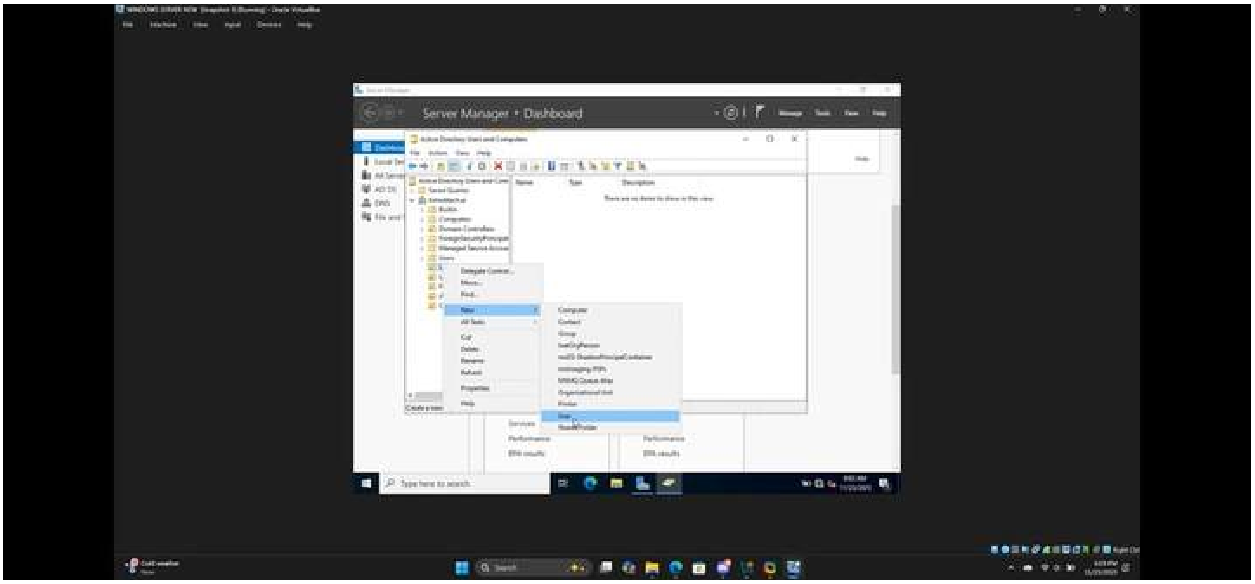


I created more groups.

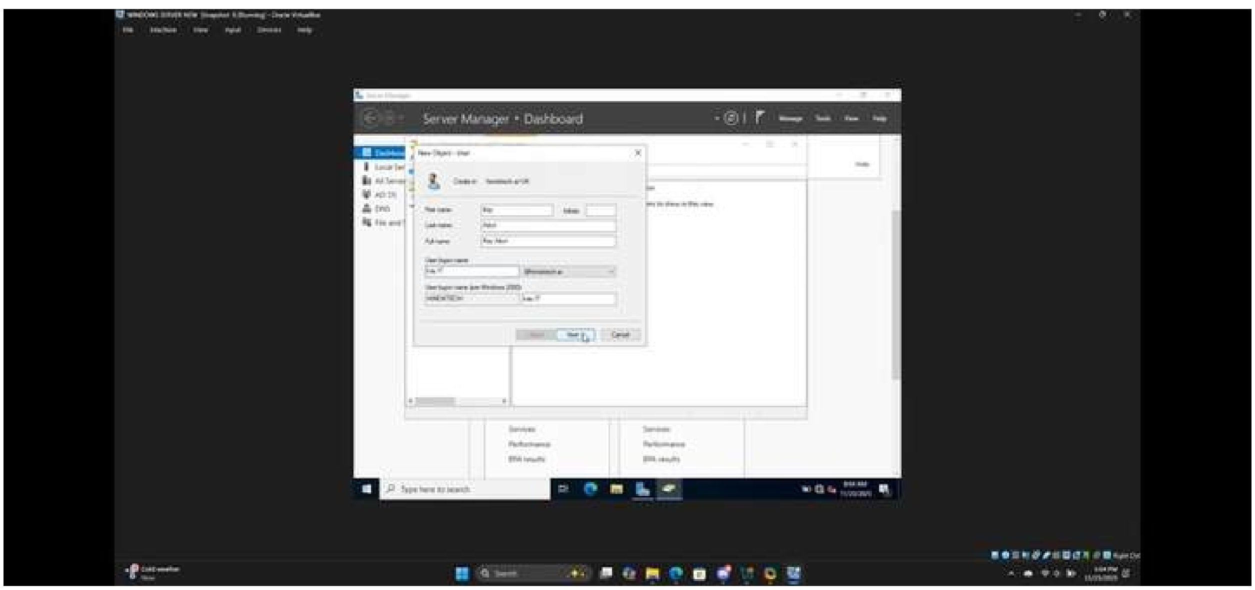
Next is creating the User Account, User accounts were created within the appropriate security groups (e.g., inside the IT group within the uk OU). I right clicked on UK organizational units and select new



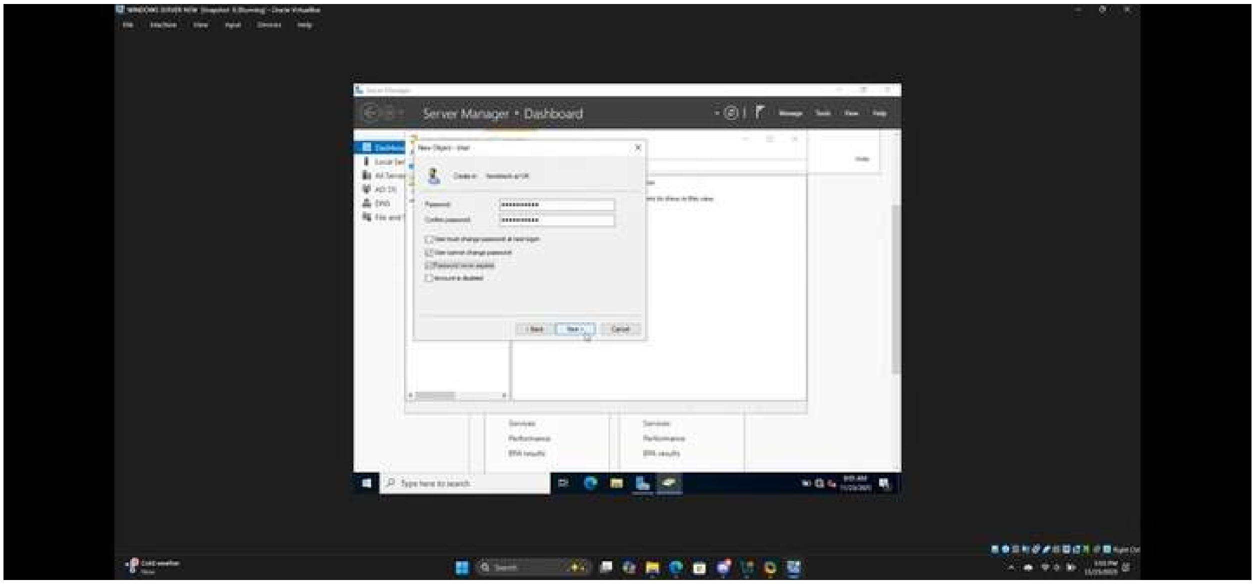
Then from New, I selected user..



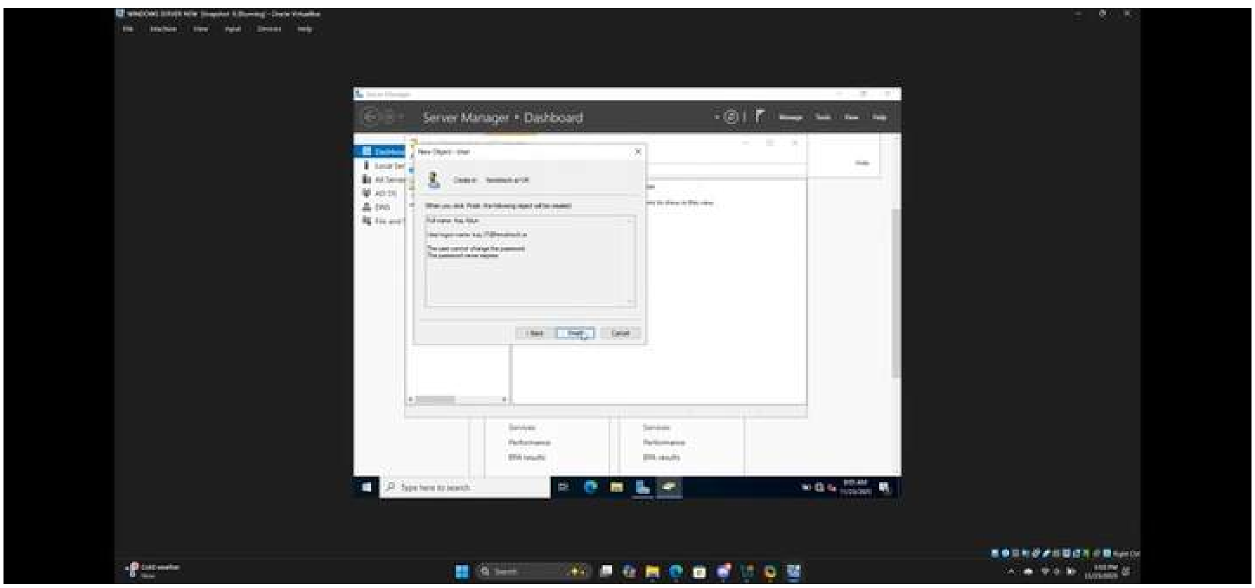
The account details were defined (e.g., First Name: deborah, Last Name: samson). A professional User Logon Name was established as debz.IT. This naming convention is a professional best practice, linking the username to the department (IT) for simplified traceability during security incident response. Then I clicked next..



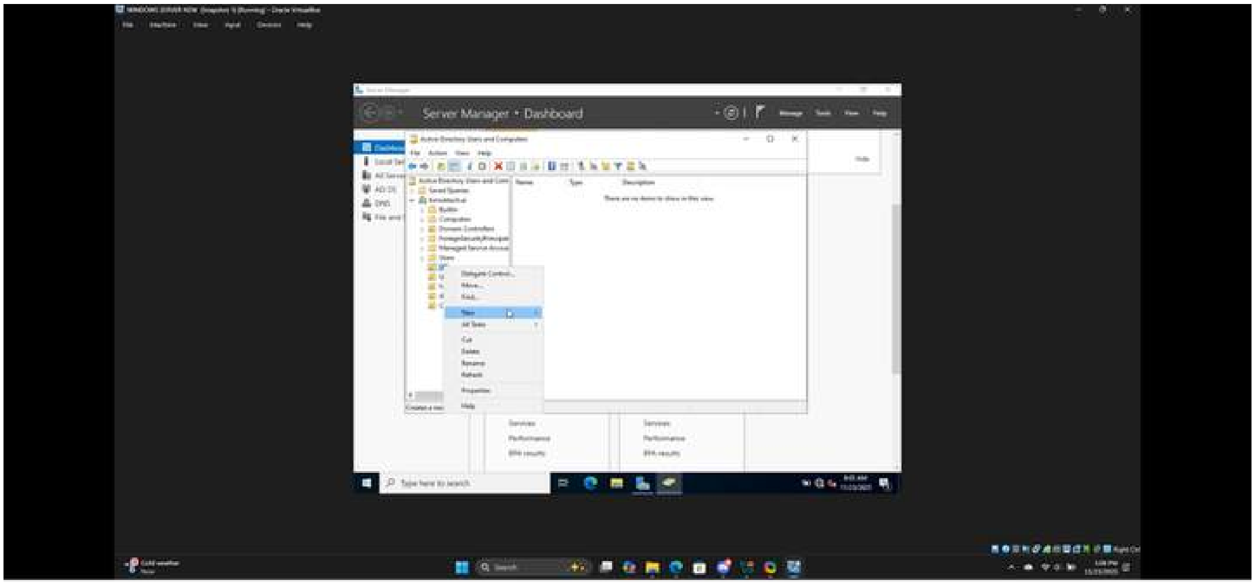
The password was set, and the security policies User cannot change password and Password never expires were selected. This setting, common in lab environments, prevents users from altering system-mandated passwords. Then I clicked next

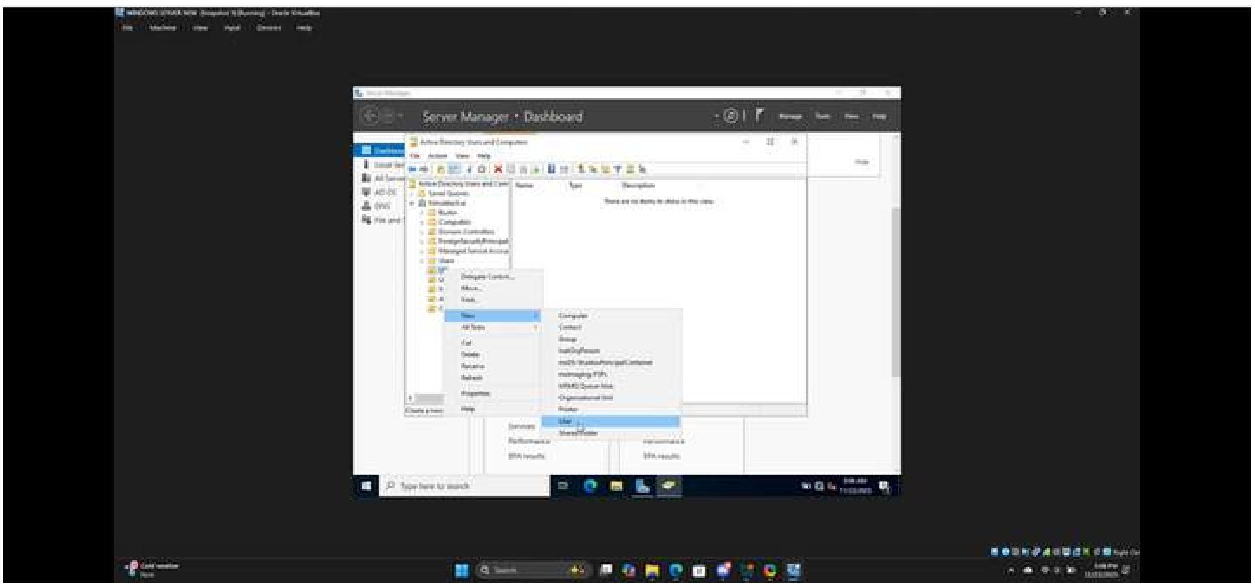


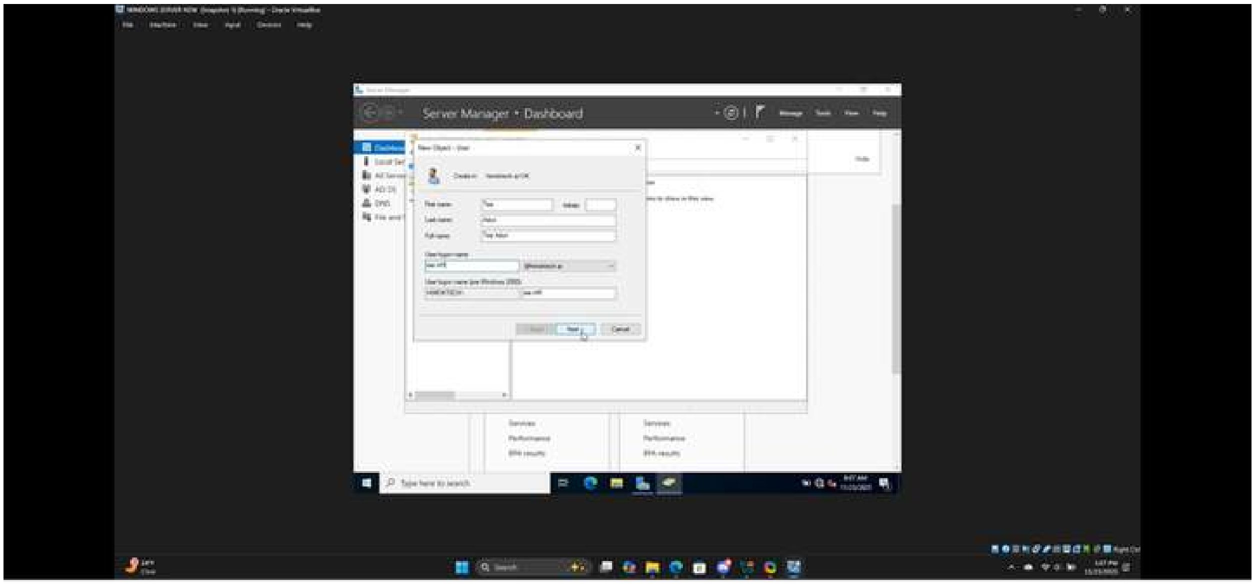
And then I clicked finish…

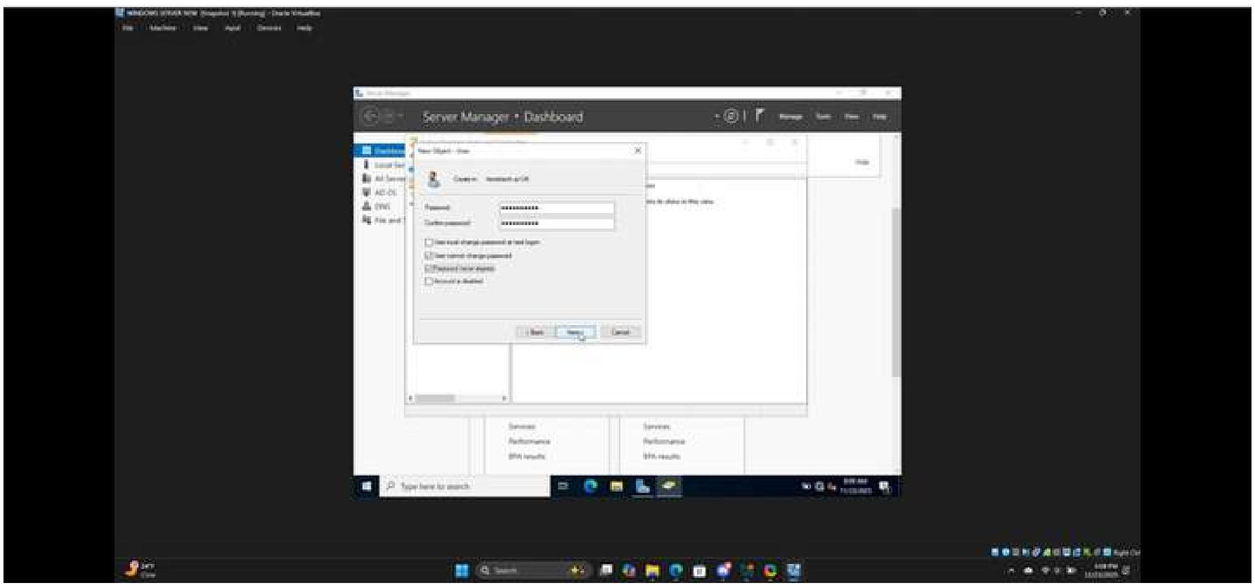


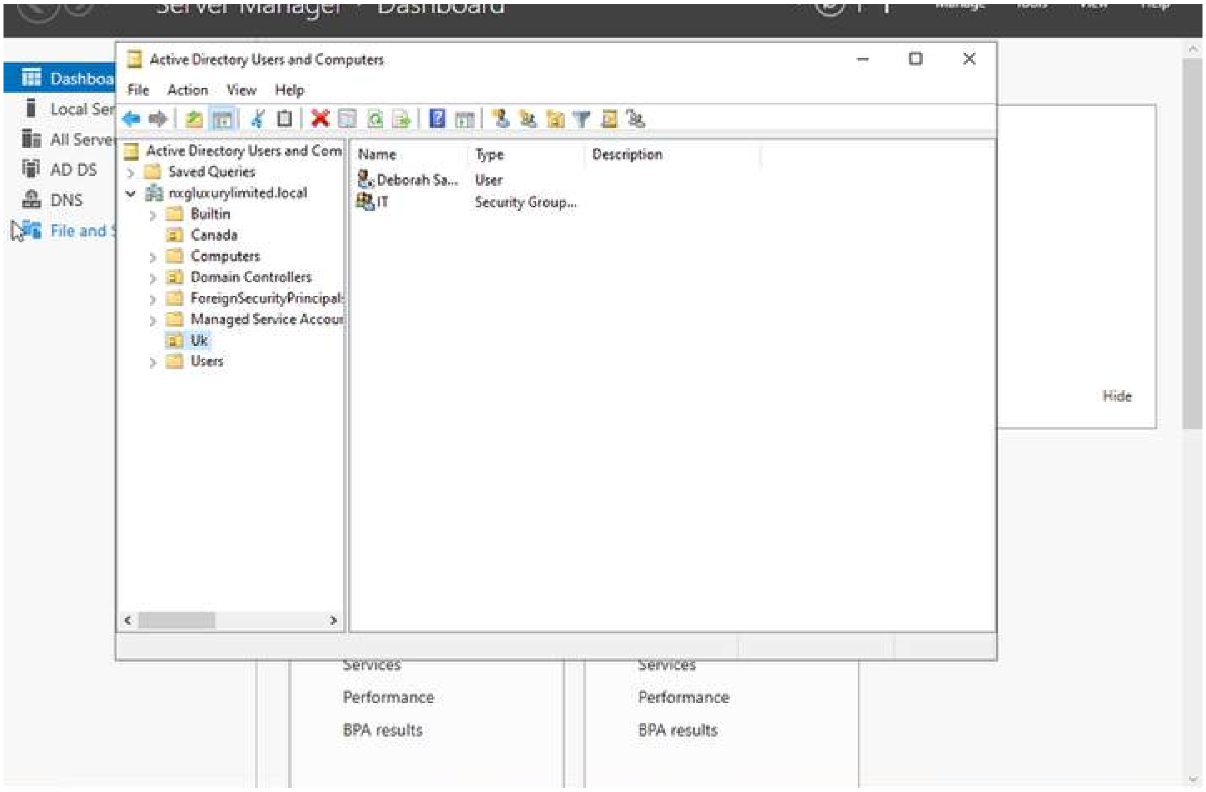
This process was meticulously repeated to provision all necessary user accounts across every defined group.


Server Manager • Dashboard
Services
Performance
BPA results
Type here to search

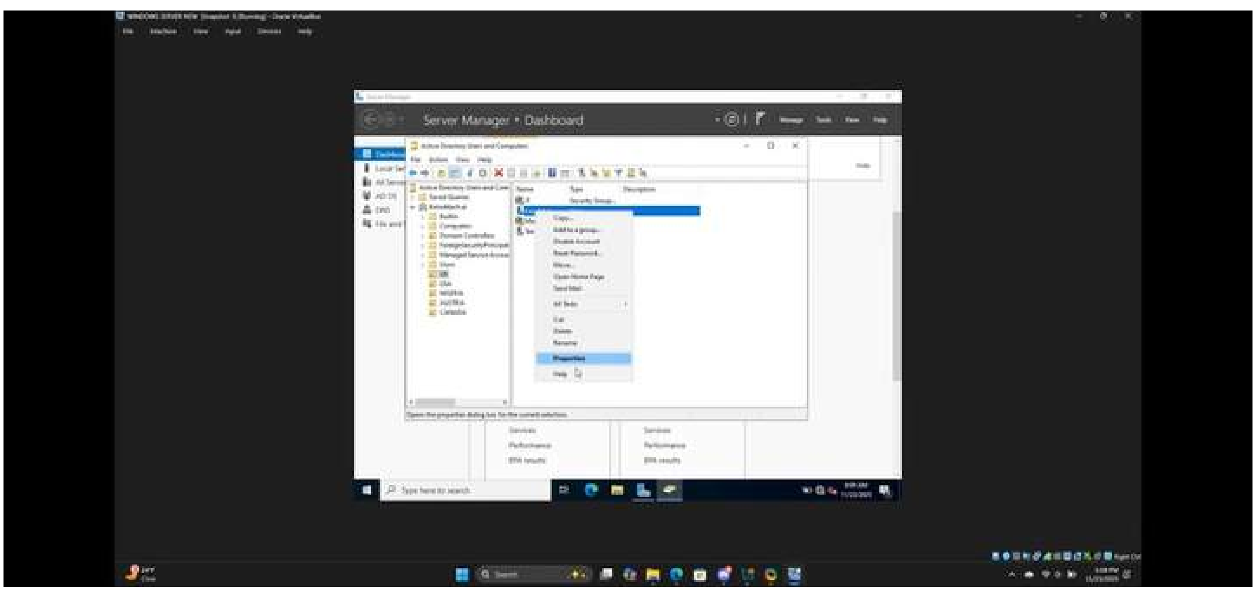

Server Manager • Dashboard
Services
Performance
BPA results
Type here to search


Server Manager • Dashboard


Server Manager • Dashboard



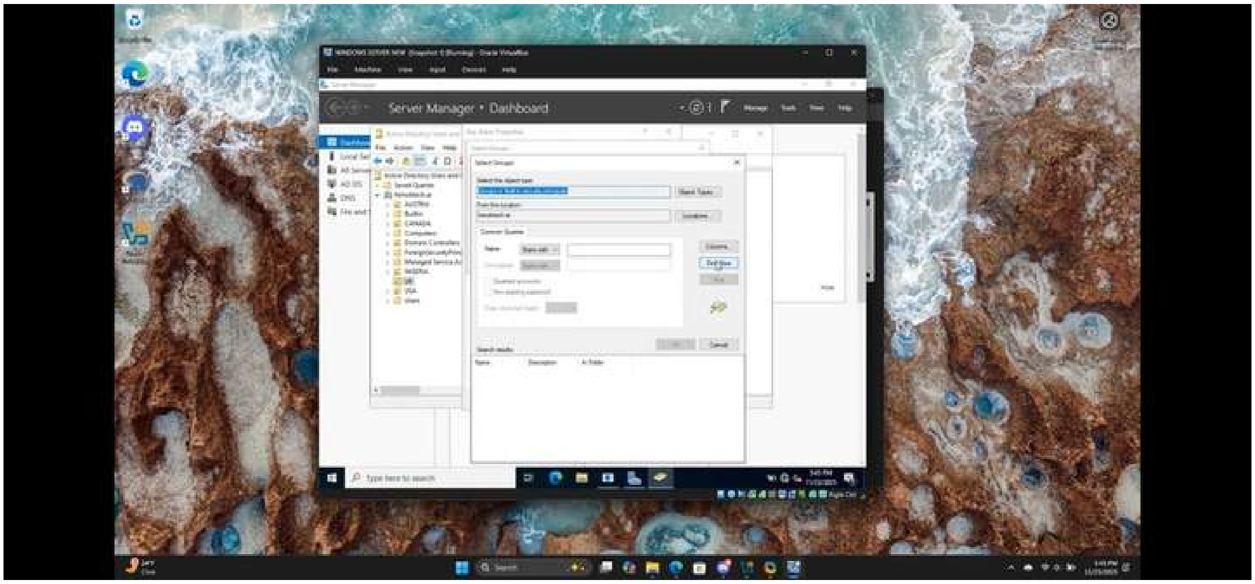
To enforce security policies, each user was assigned to their respective group. For the user Kay.IT, accessed Properties,



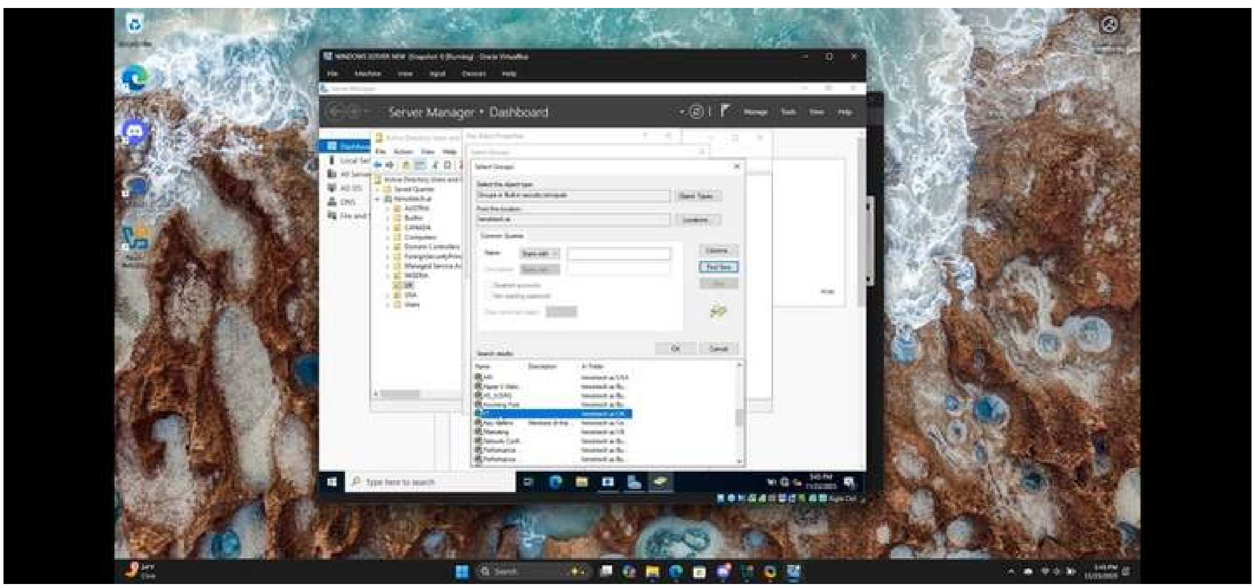
navigated to the Member Of tab and I clicked ok

and I clicked the Advanced button

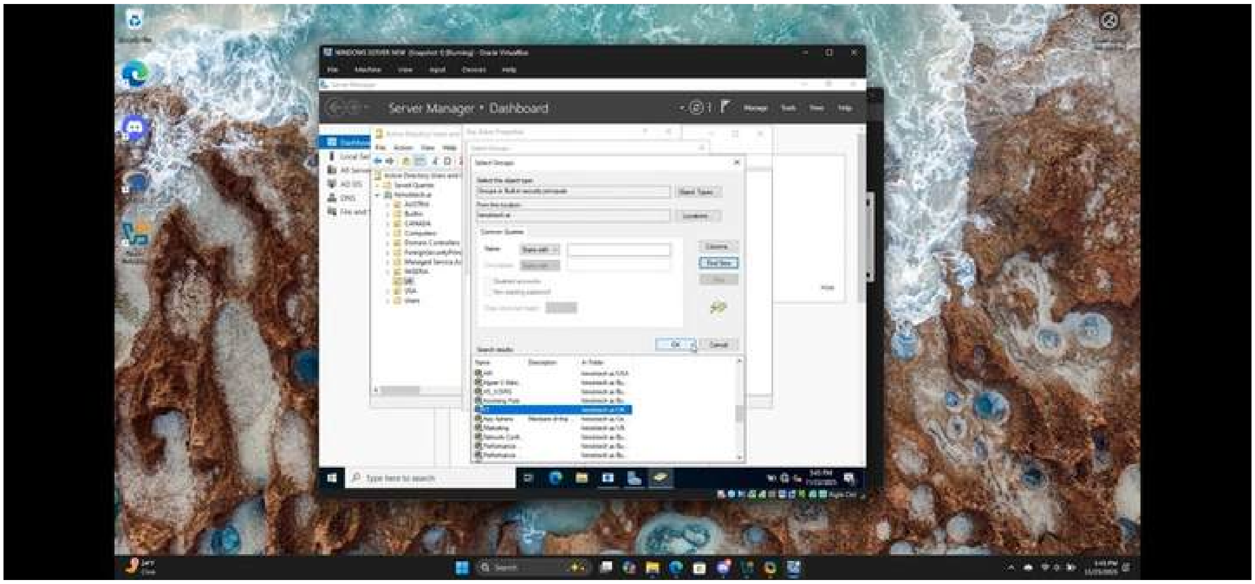
I clicked on Find Now feature to locate and assign membership to the IT group.



I assigned the IT group to debz.IT

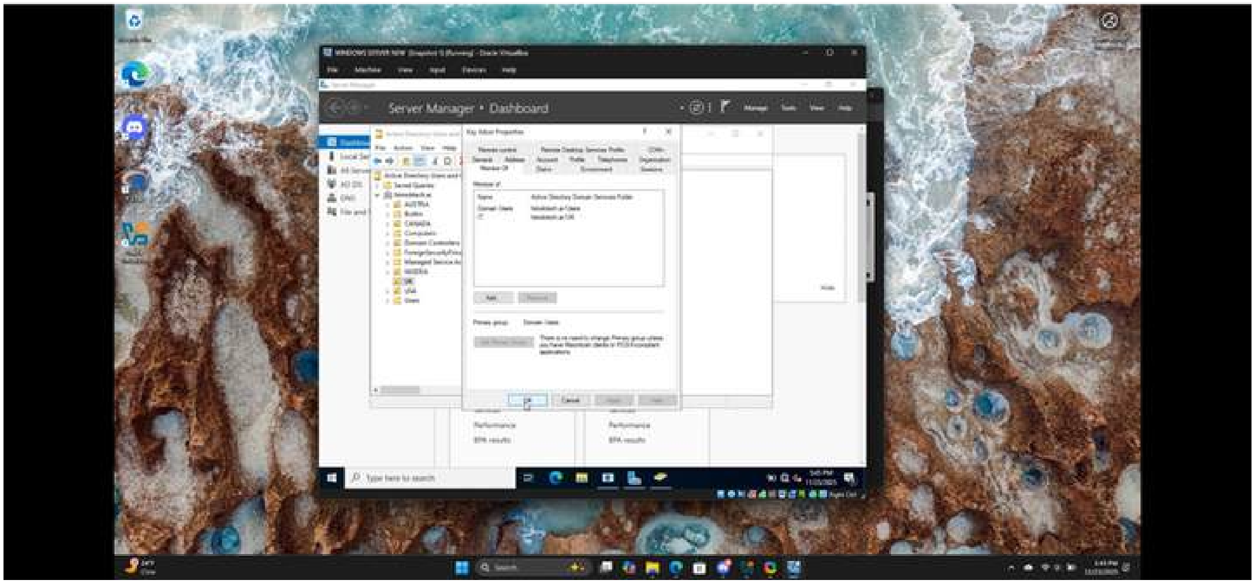


I clicked ok to confirm the change

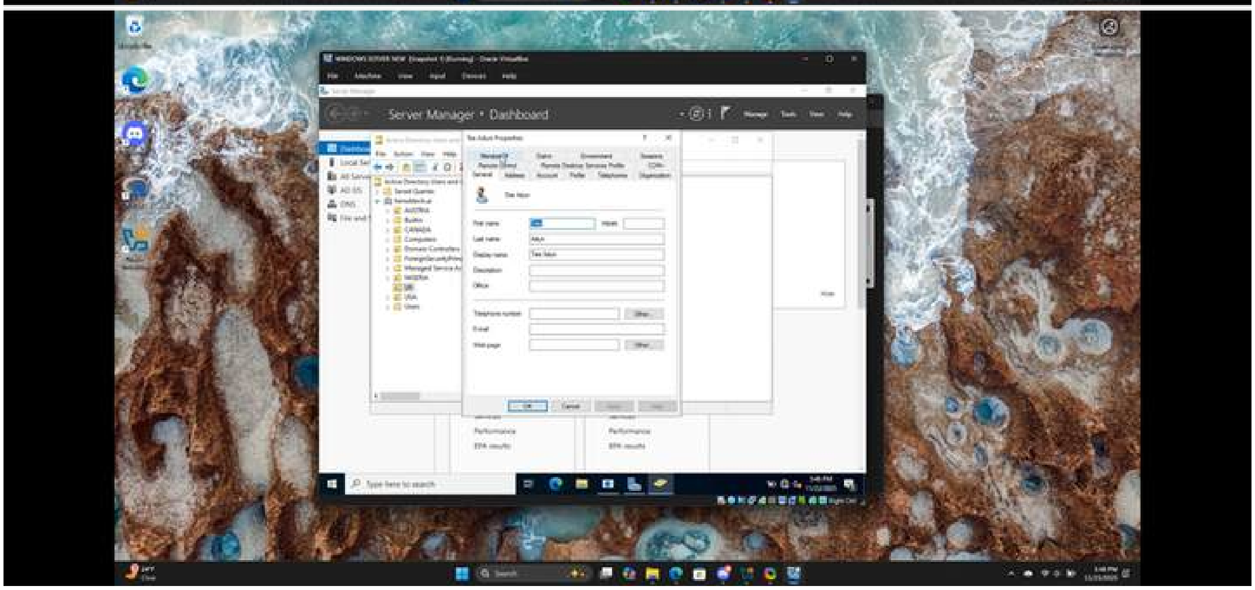


I clicked ok to affirm the changes

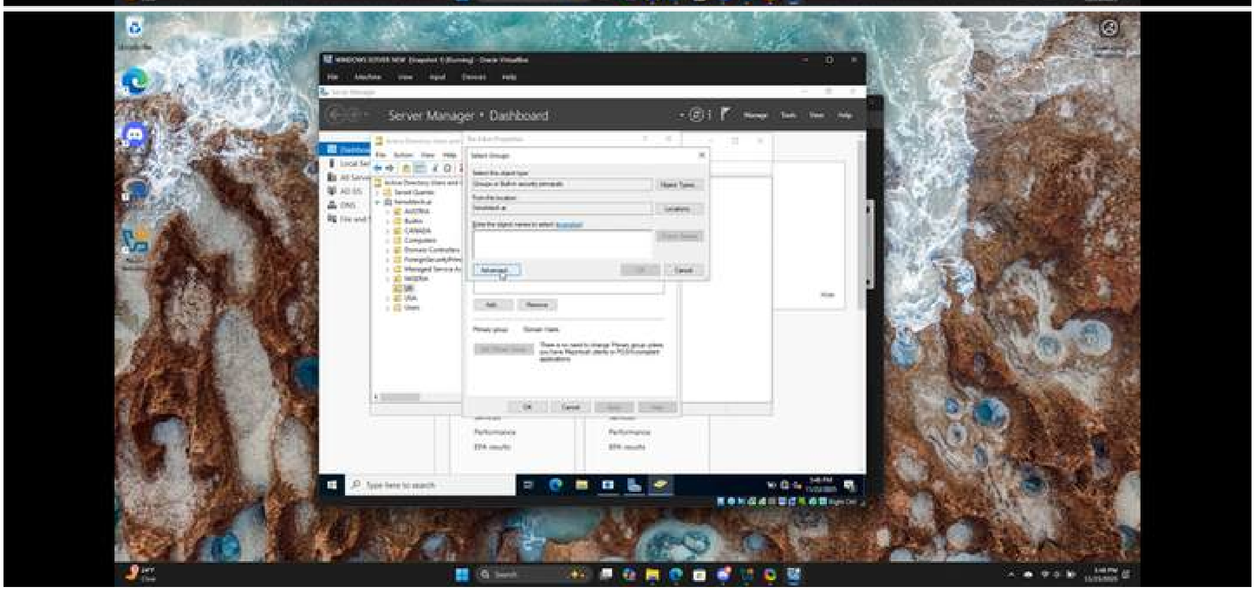
I clicked apply and ok to complete the changes. This step ensures the user inherits the group's security permissions.

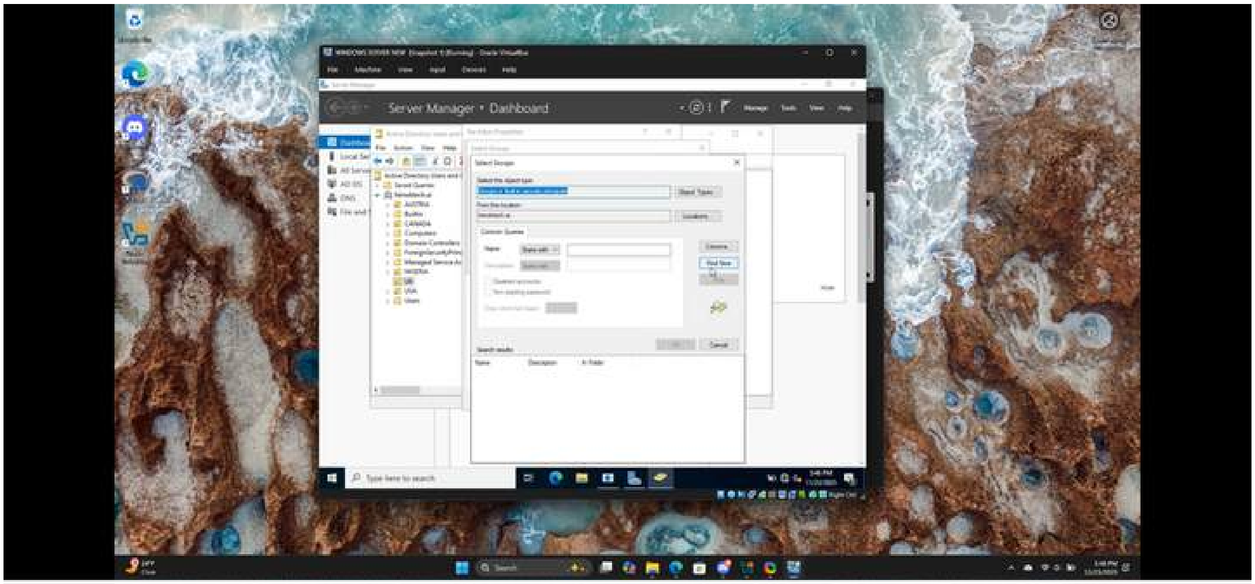


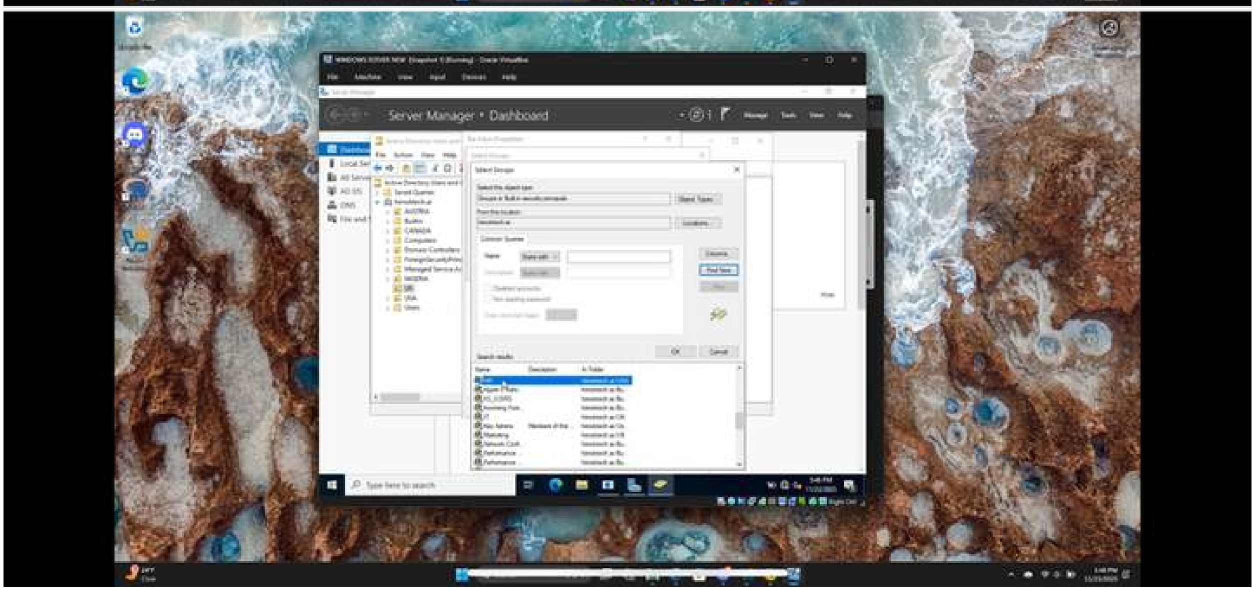
The process was repeated for Jay.Market

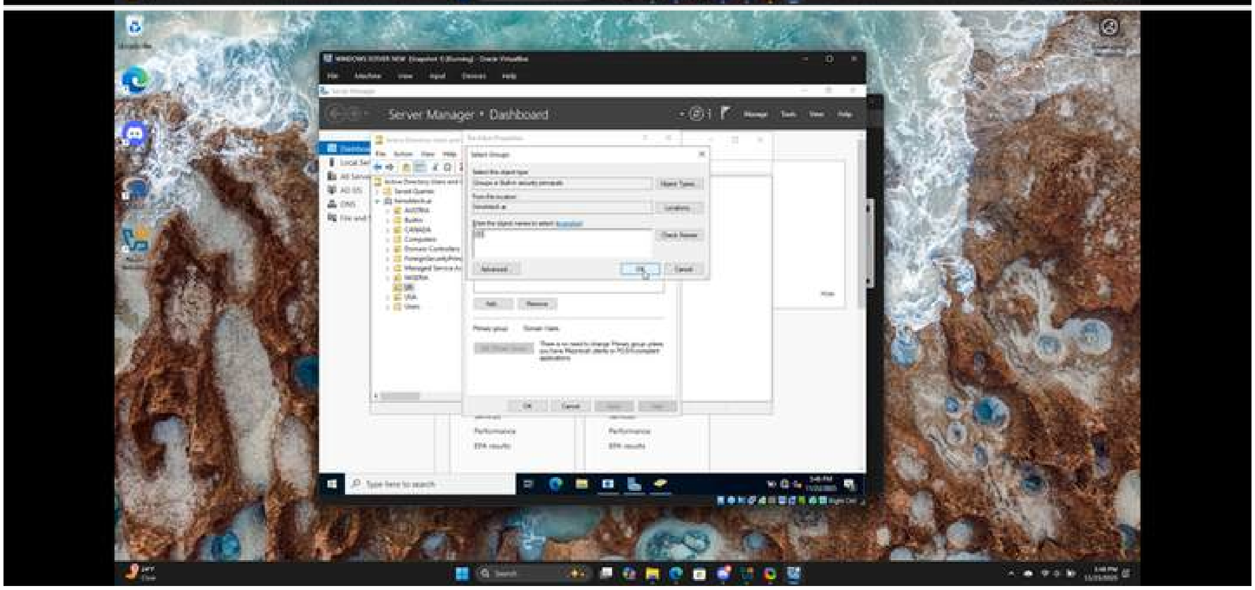



Server Manager • Dashboard
Type here to search


Server Manager • Dashboard
Type here to search

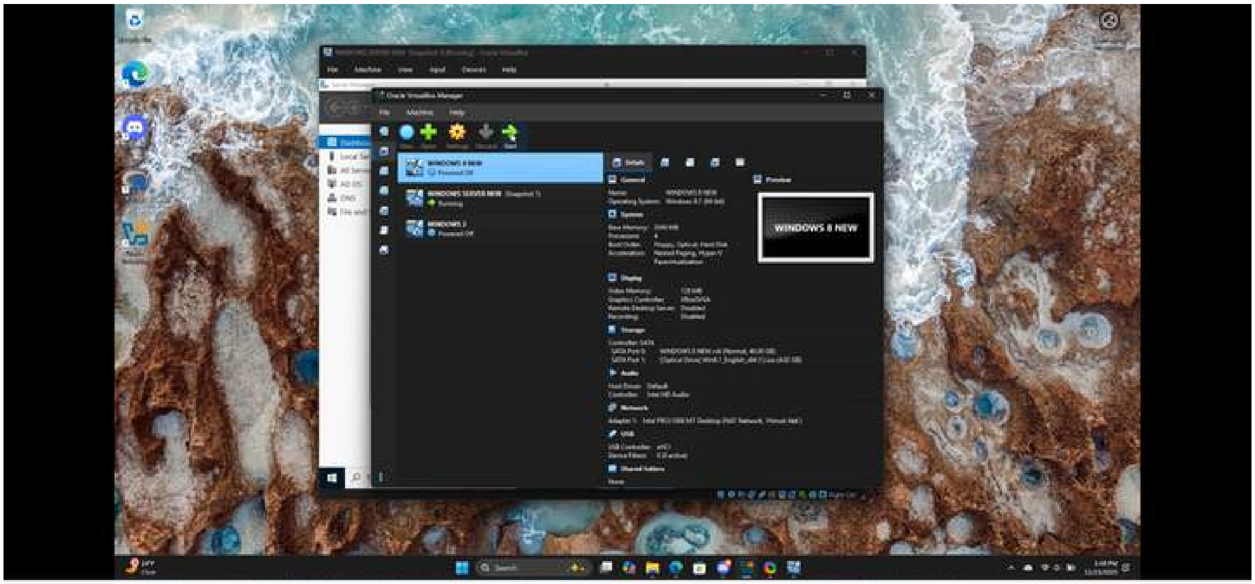

Server Manager • Dashboard

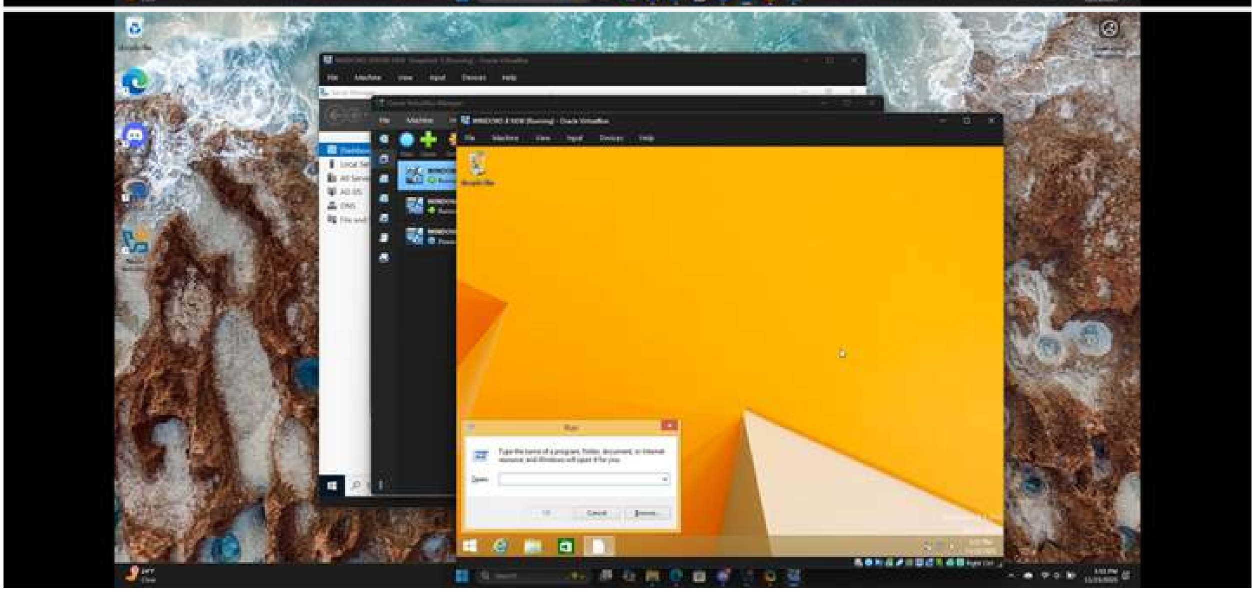

Server Manager • Dashboard

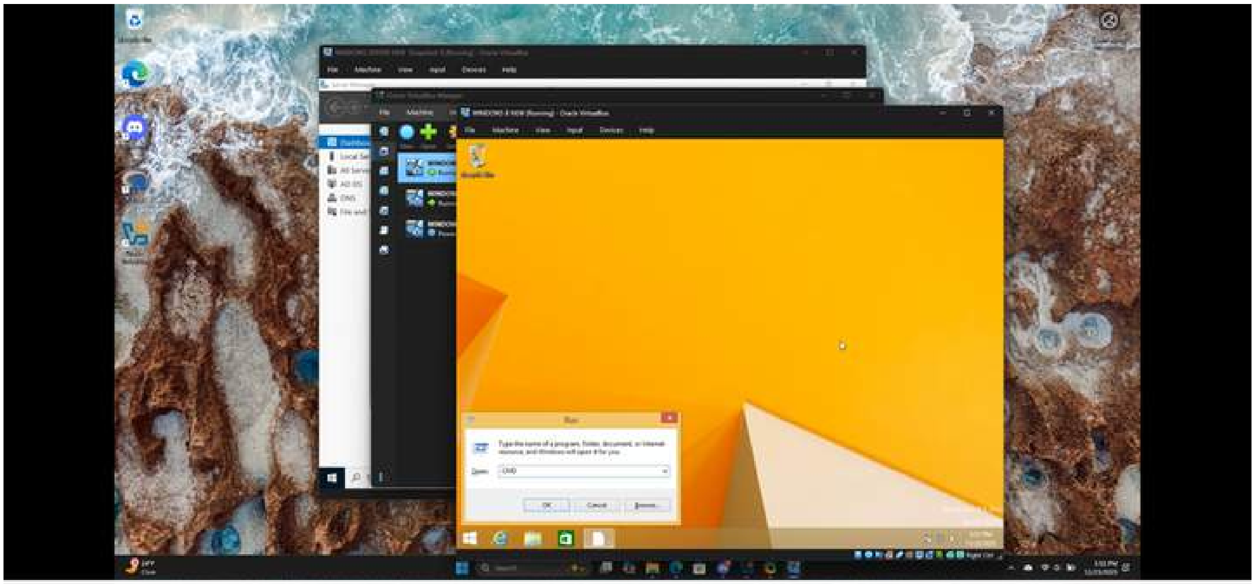

Server Manager • Dashboard

Server Manager • Dashboard

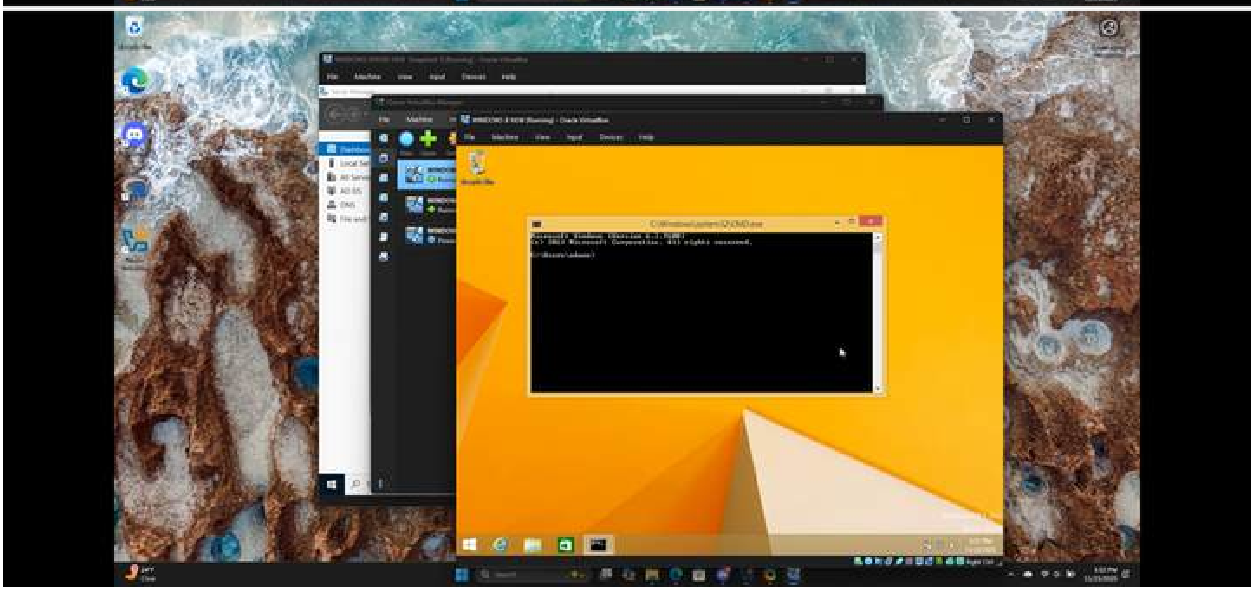
After that is concluded , The next step in this phase is the successful enrollment of a client machine (Windows 8) into the nxgluxurylimited.local domain, making it centrally manageable by the Windows Server. Let me walk through how I did that;

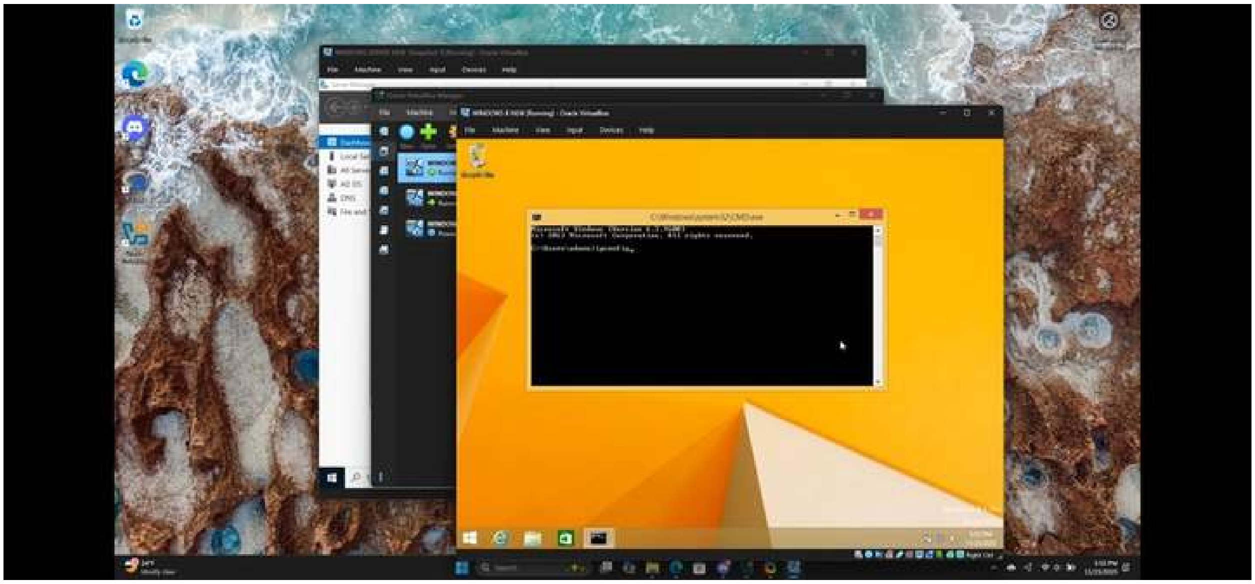
I first used the ipconfig command via the Command Prompt on the Windows 8 machine to verify its current IP address.


WINDOWS 8 NEW

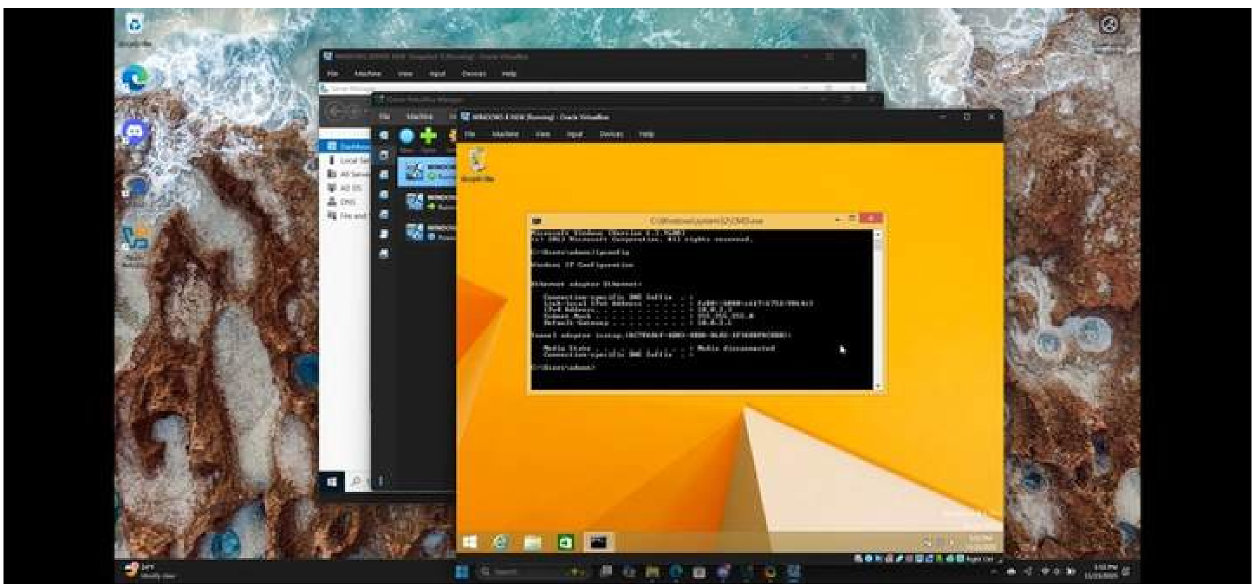

Run
Cancel



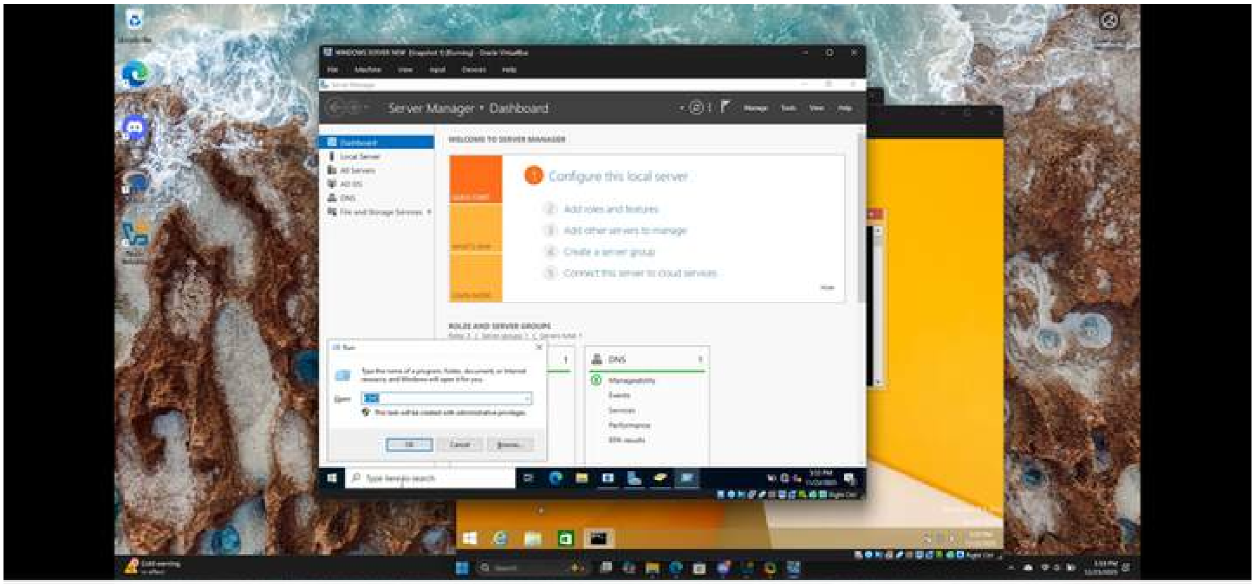


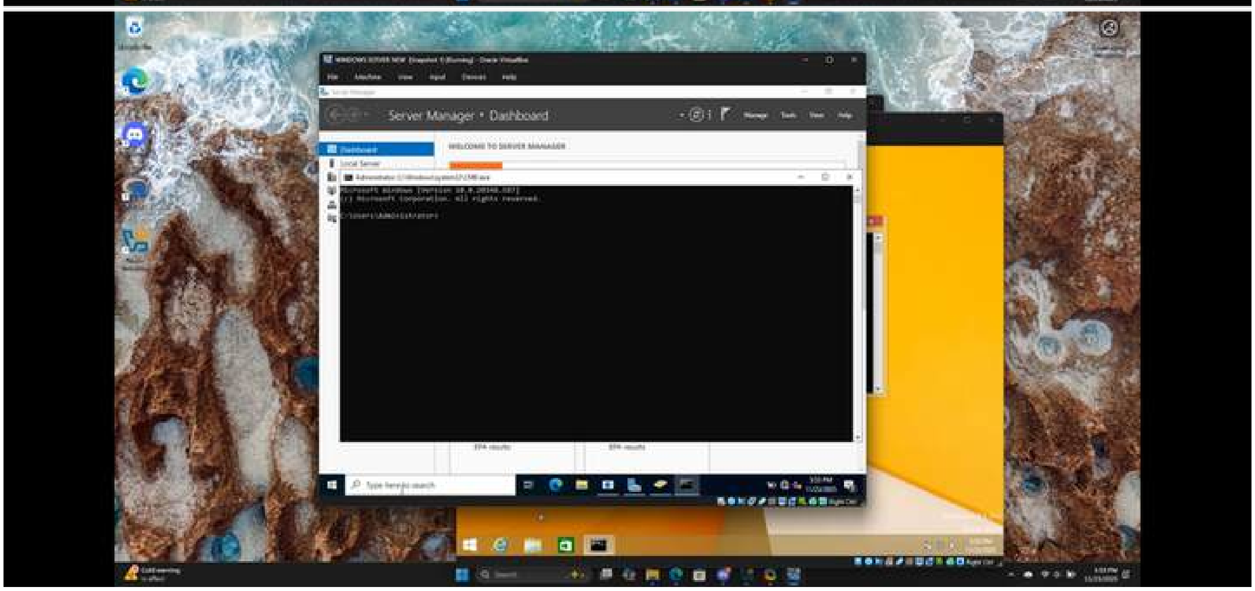


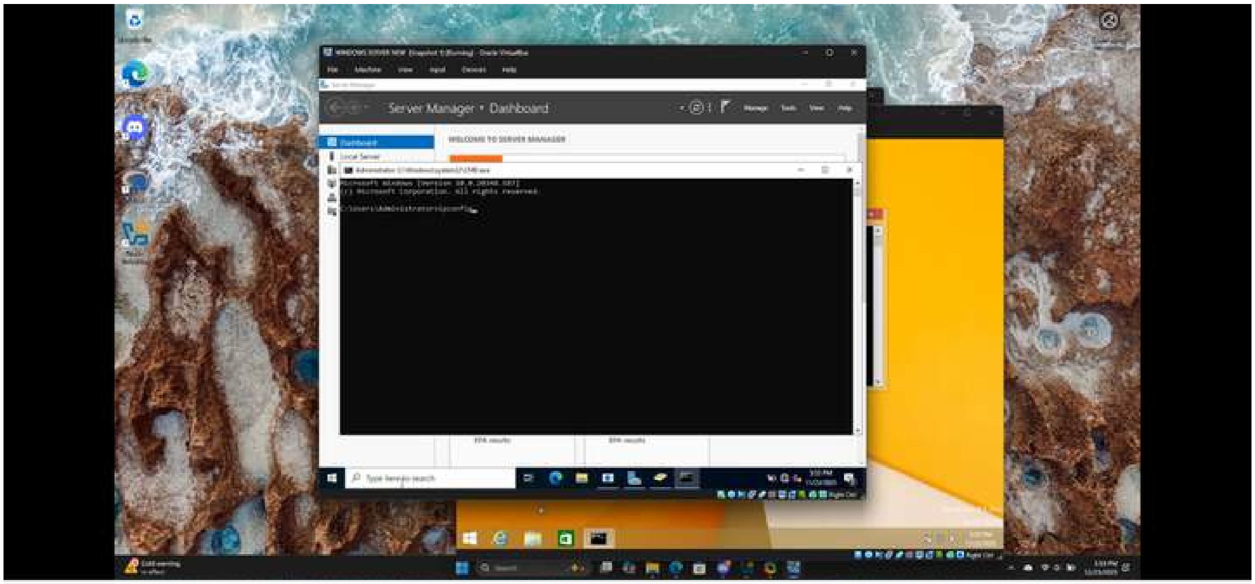
After I type ipconfig and press enter, I got the necessary information.

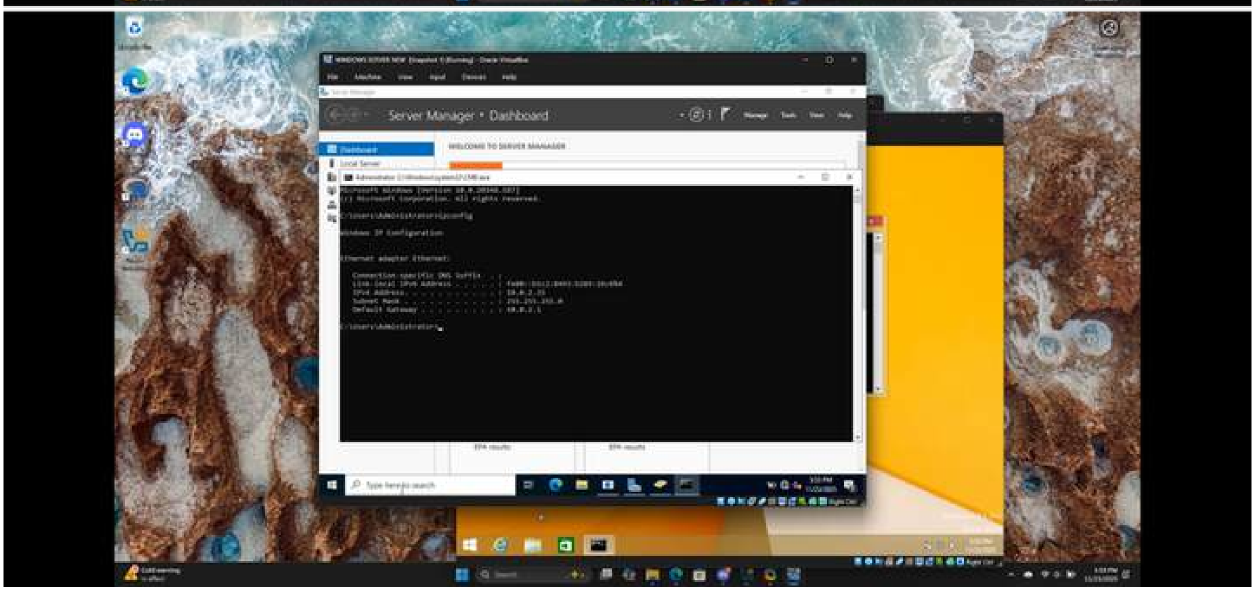


I repeat the same process on the windows server also.

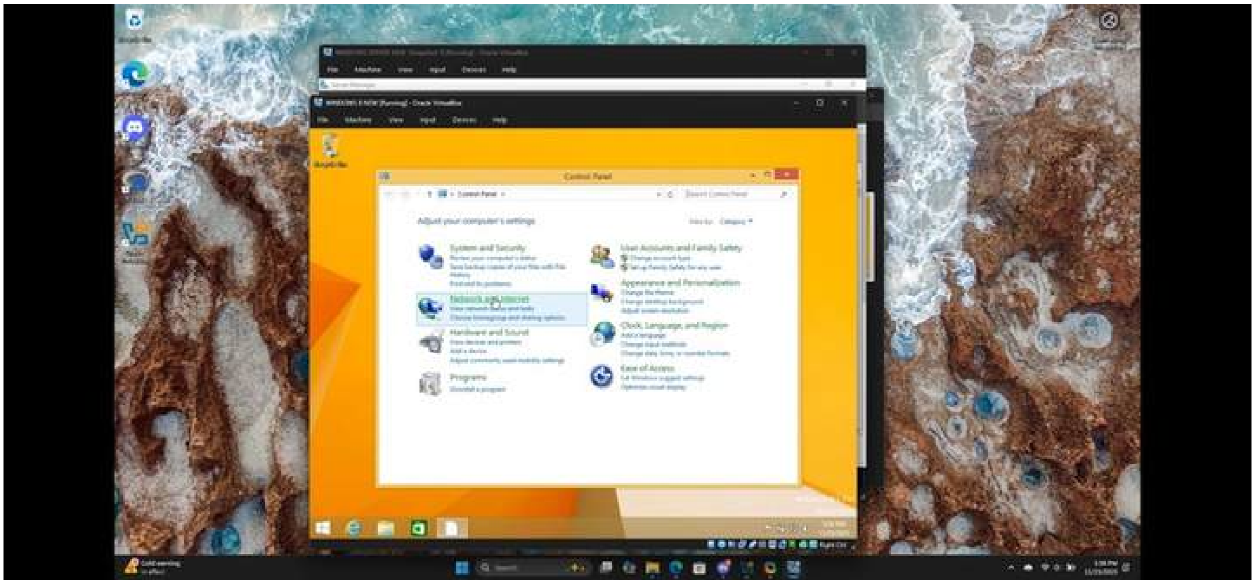




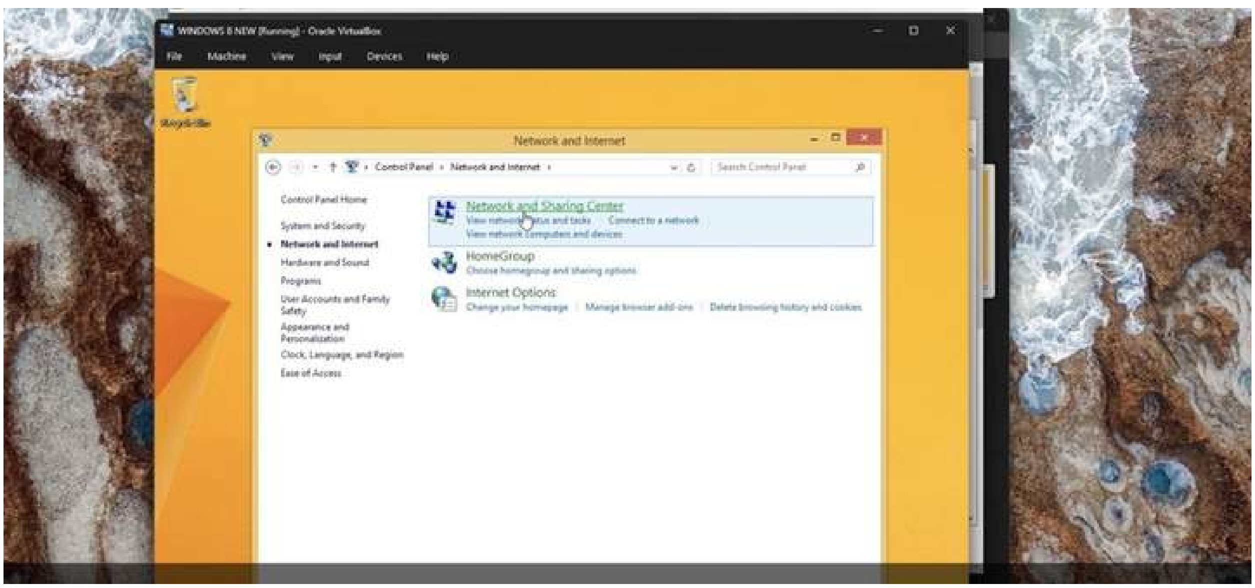




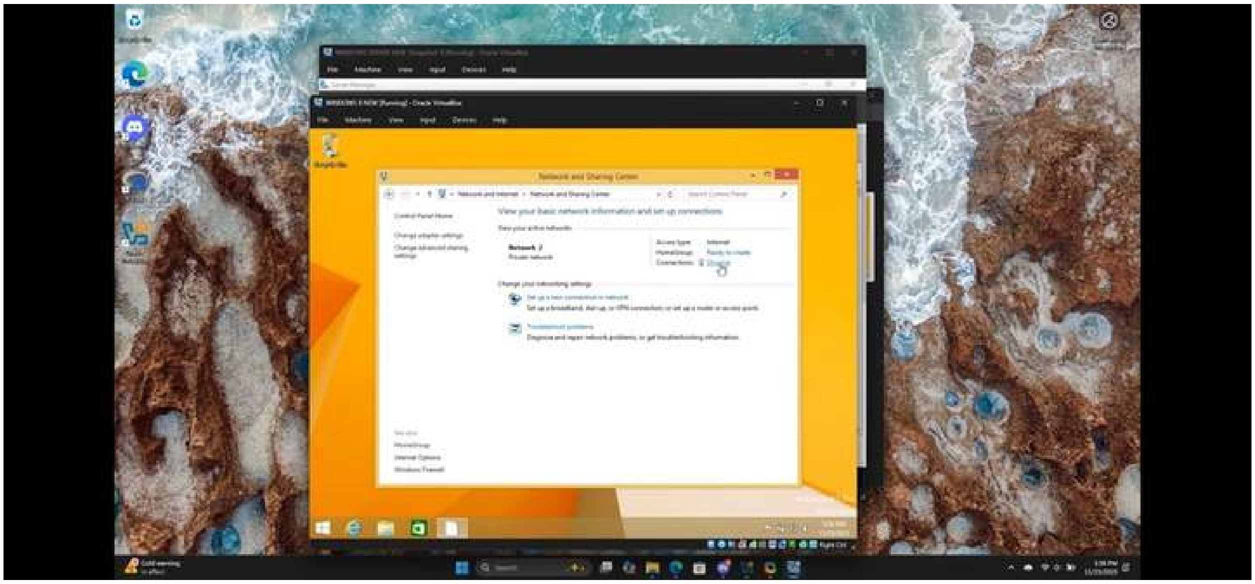
So to ensure the client can locate the Domain Controller, its DNS settings must point to the server.i navigated to the Windows 8 Network and internet



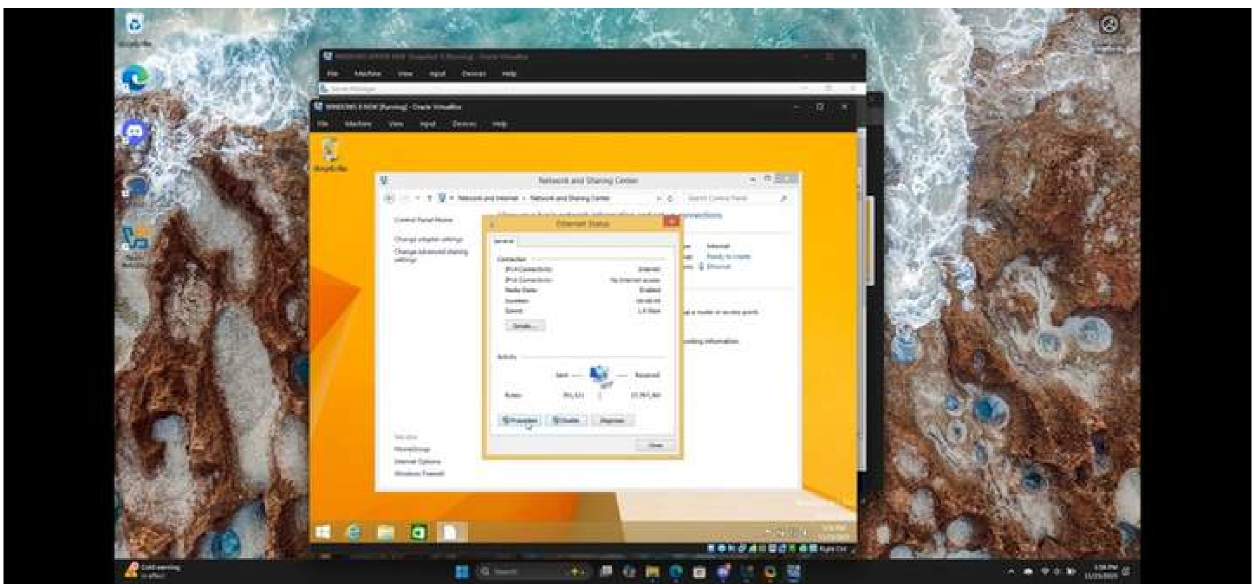
Then to Network and Sharing Center



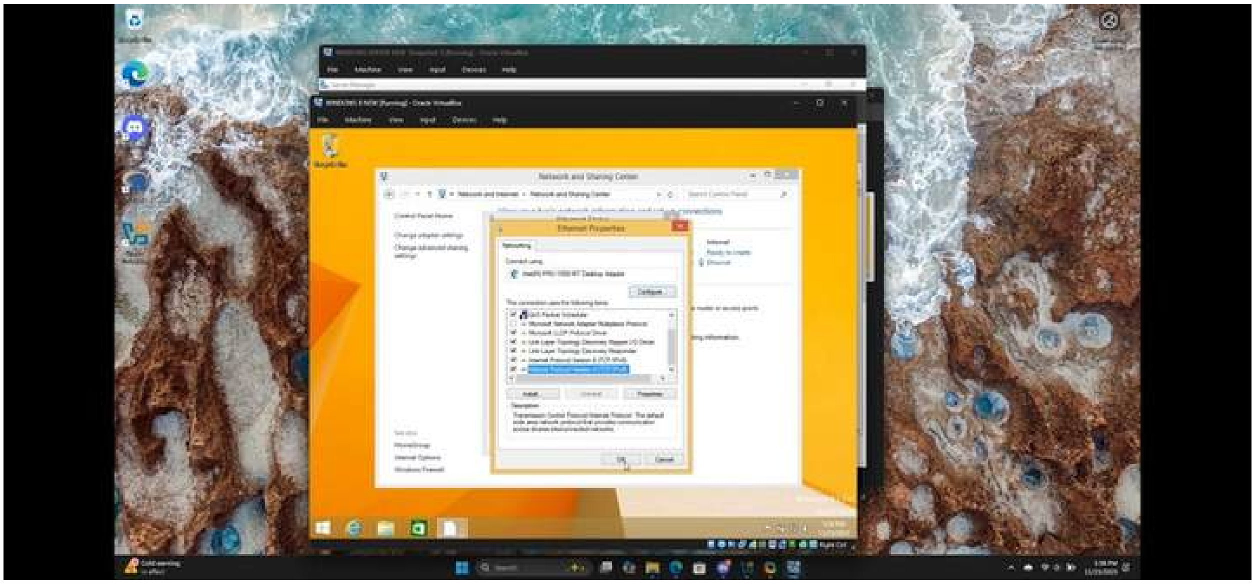
Then to Ethernet >



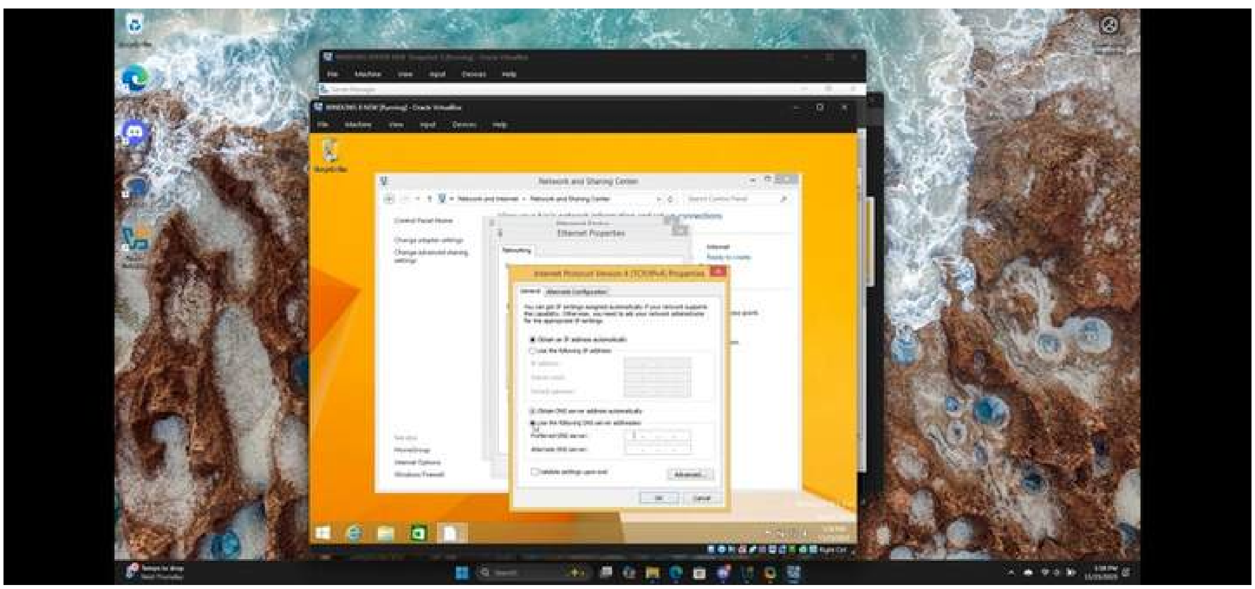
Then I clicked on Properties >

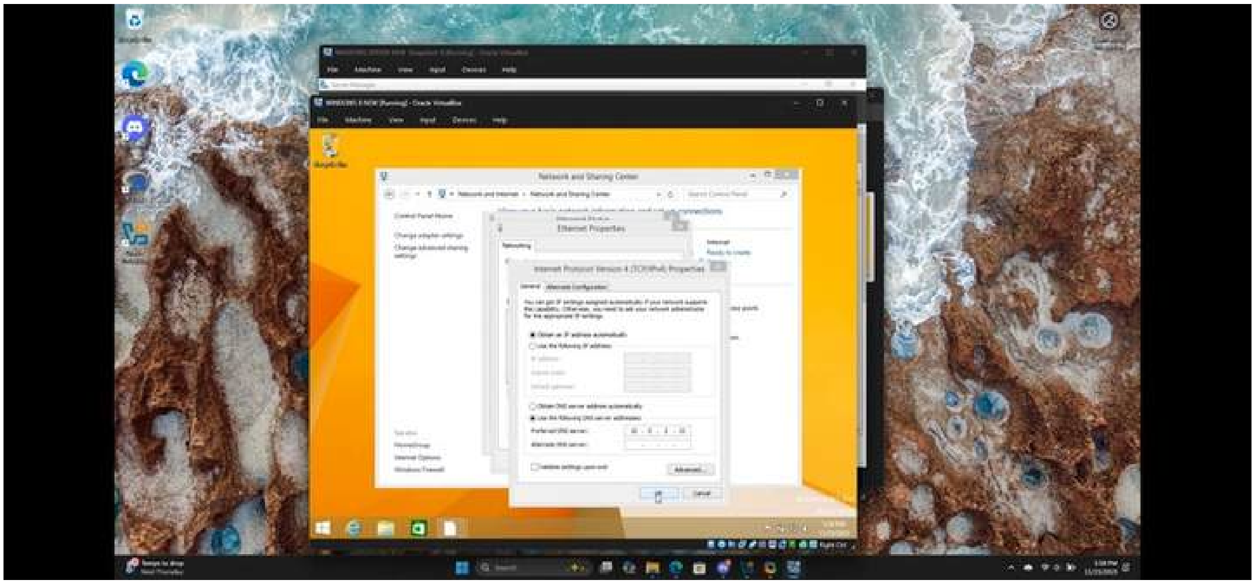


Then I located Internet Protocol Version 4 (TCP/IPv4) Properties.

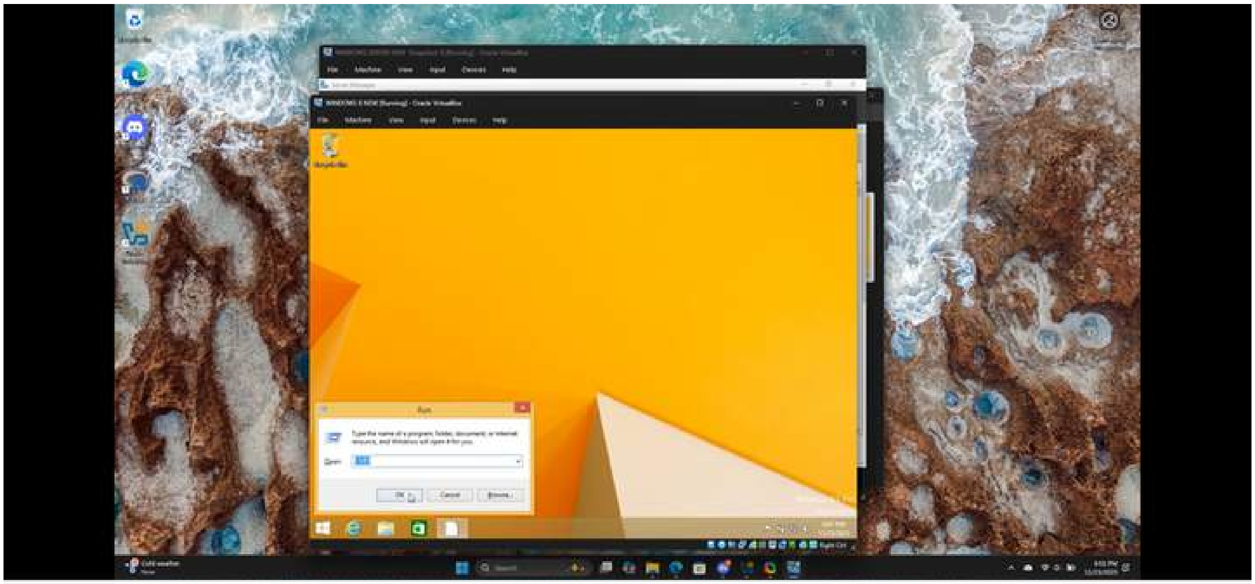


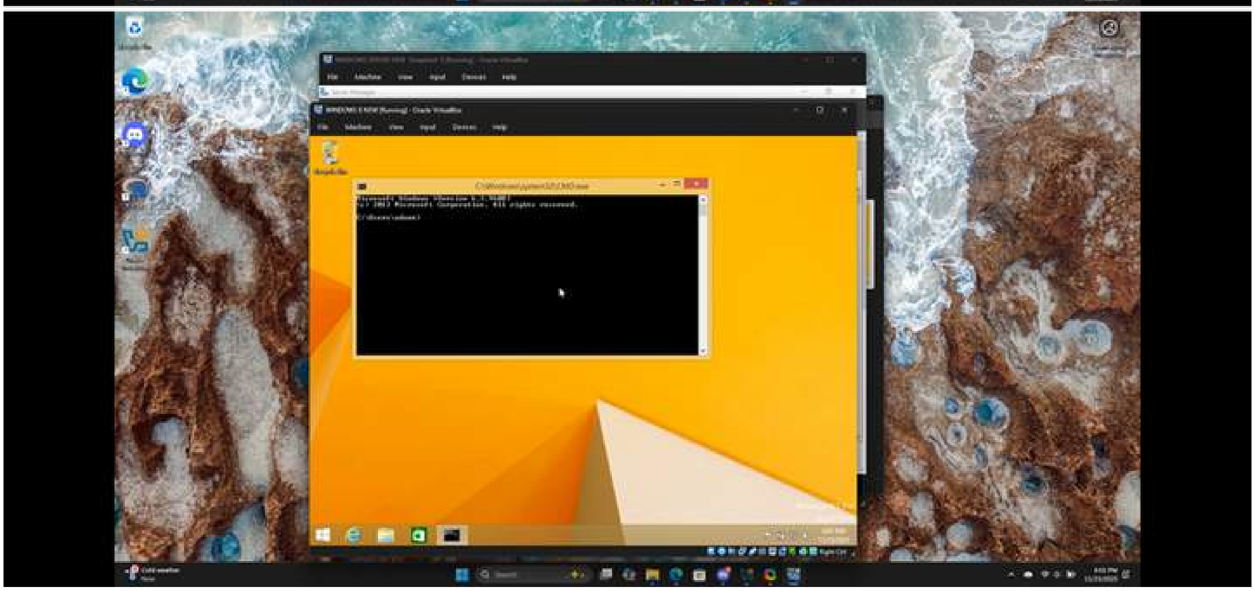
The setting was changed to Use the following DNS server address, and the IP address of the Windows Server (which was identified as 10.0.2.15 via ipconfig) was inserted as the Preferred DNS Server.

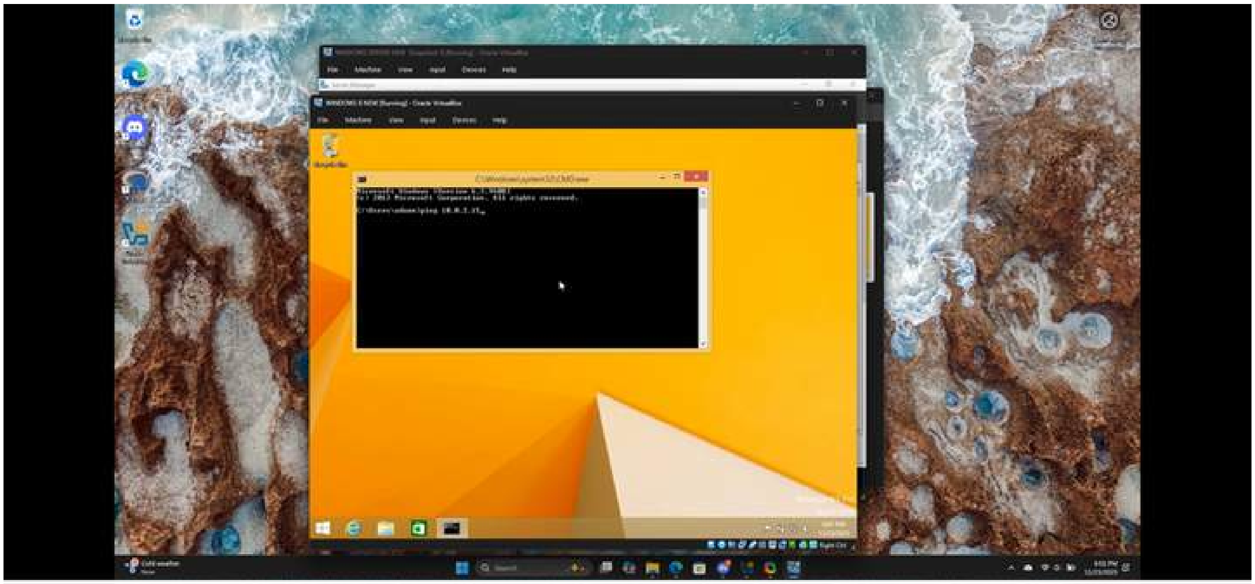
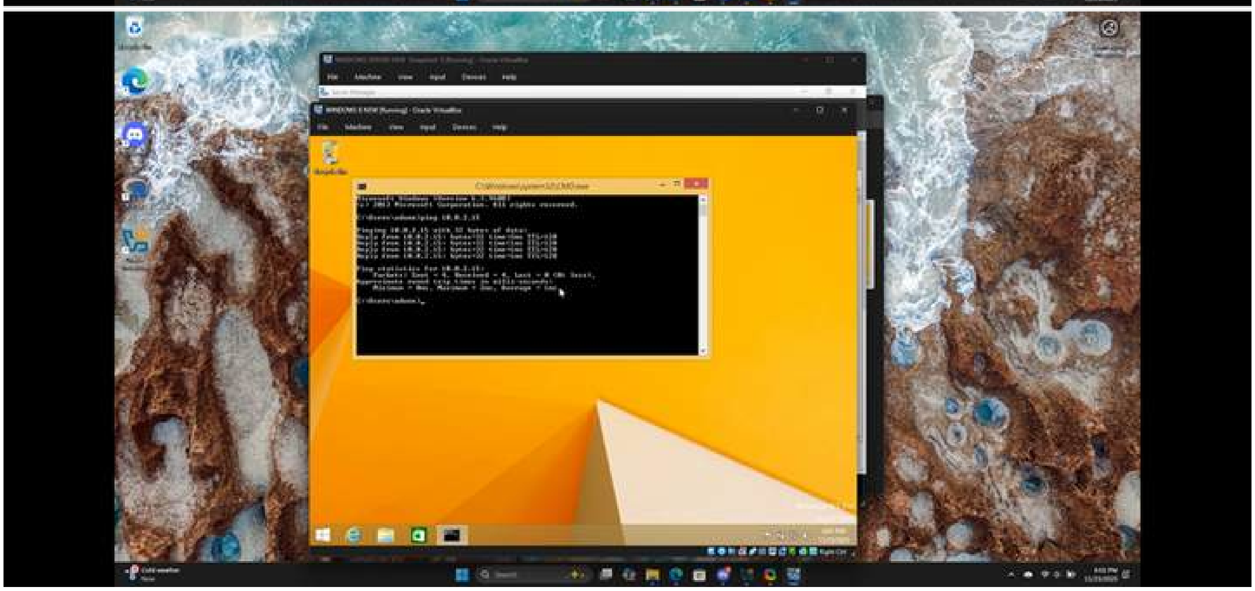




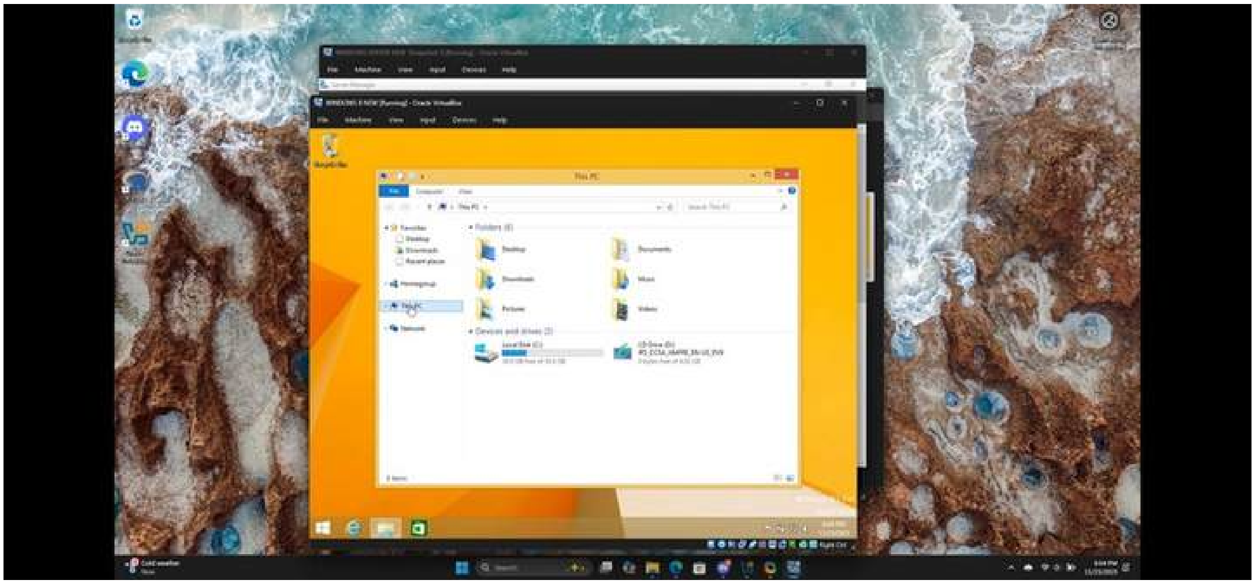
A ping test from the client to the server's IP address (10.0.2.15) confirmed successful communication.



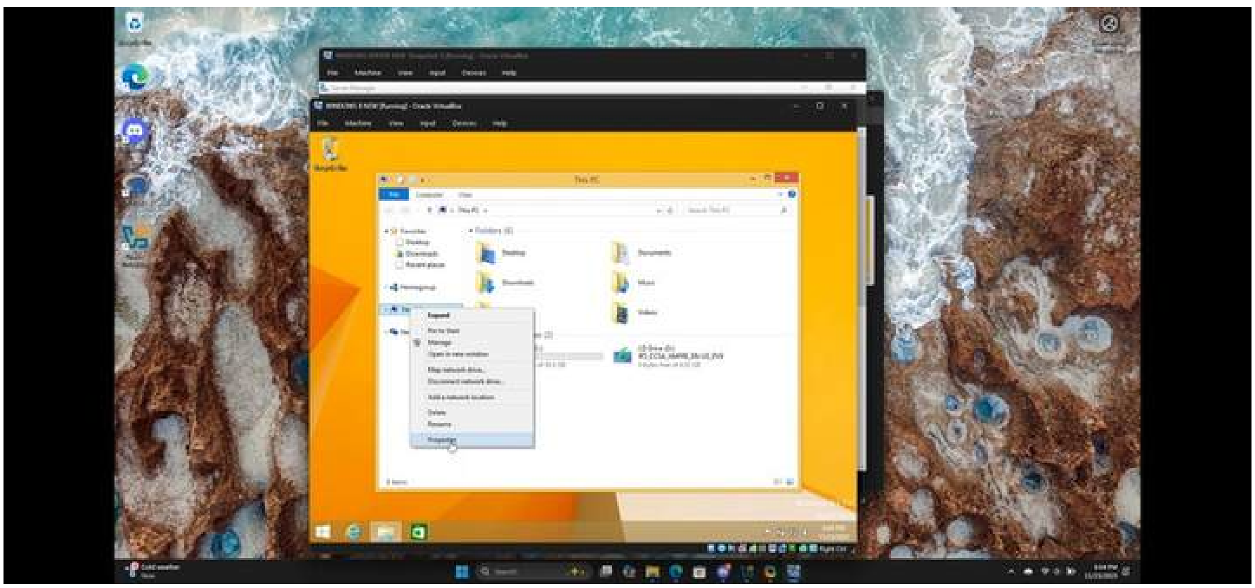


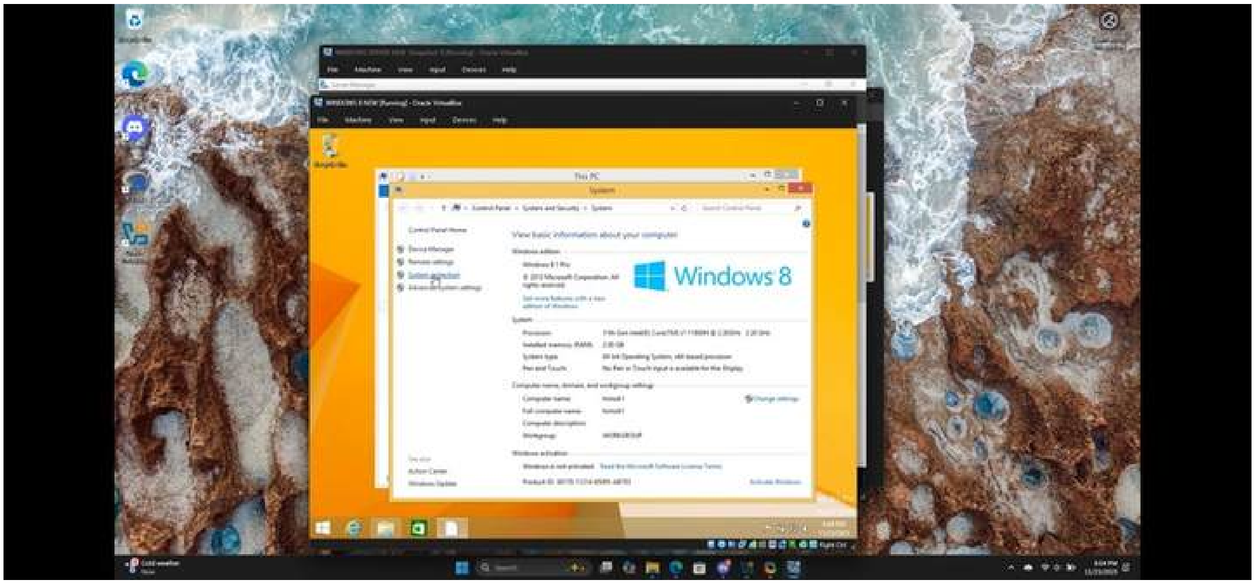
This next stage is to Initiate the Domain Join I mean to link windows 8 pc which i designated for debz.IT . To do that, on the Windows 8 client, i accessed the system properties by pressing the Windows Key and right-click This PC >



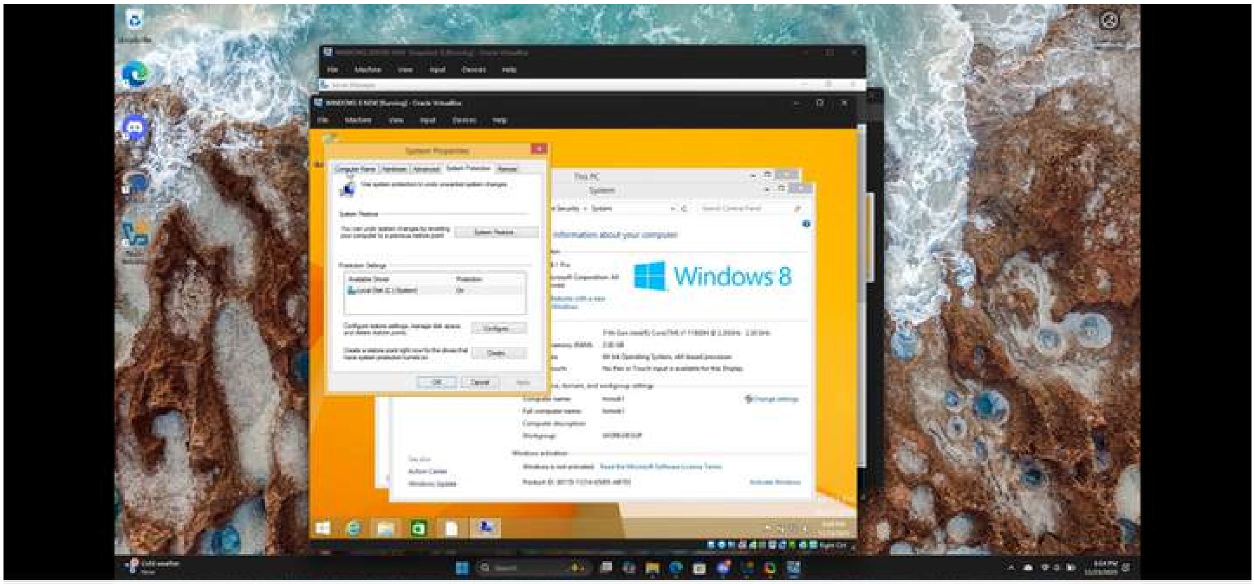
The click on Properties >

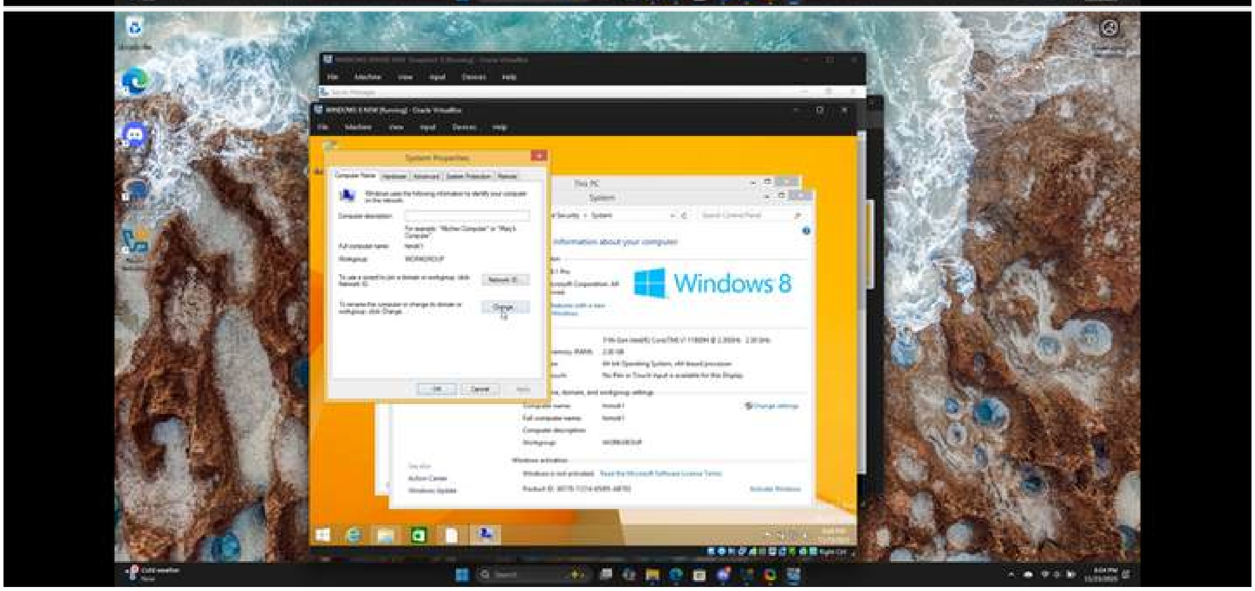


Inavigated to System Protection >

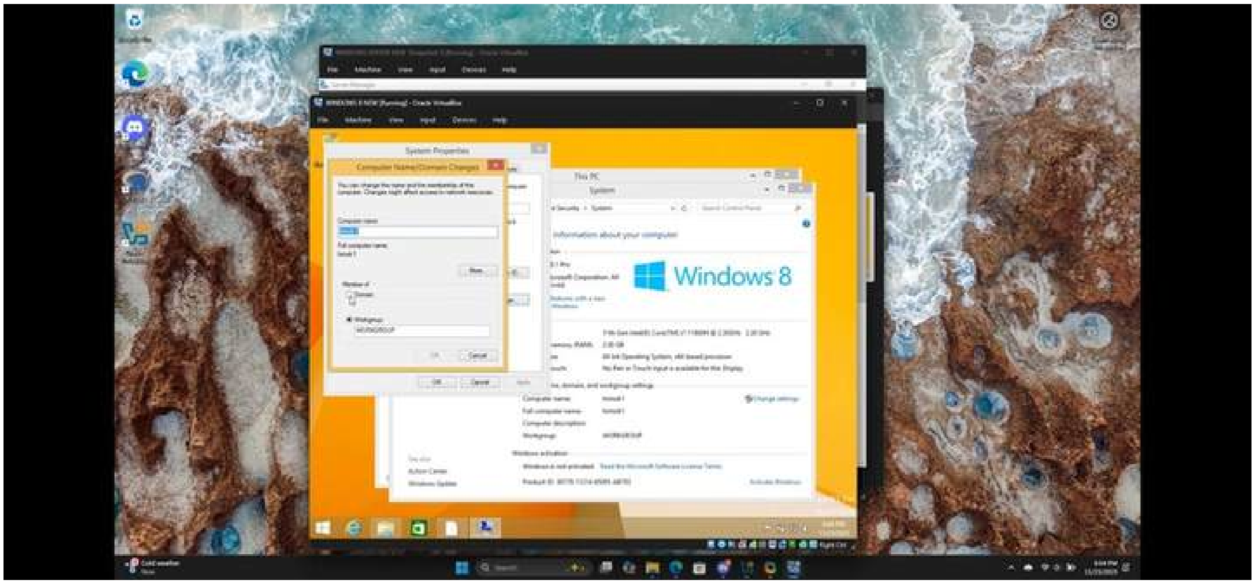


Computer Name and clicked Change.

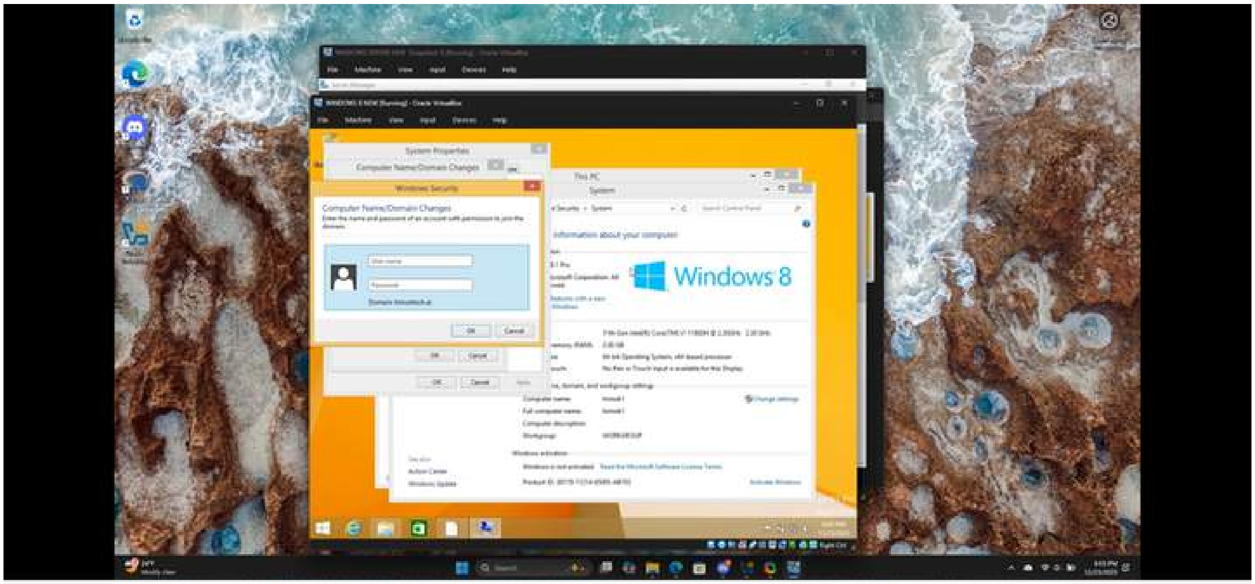
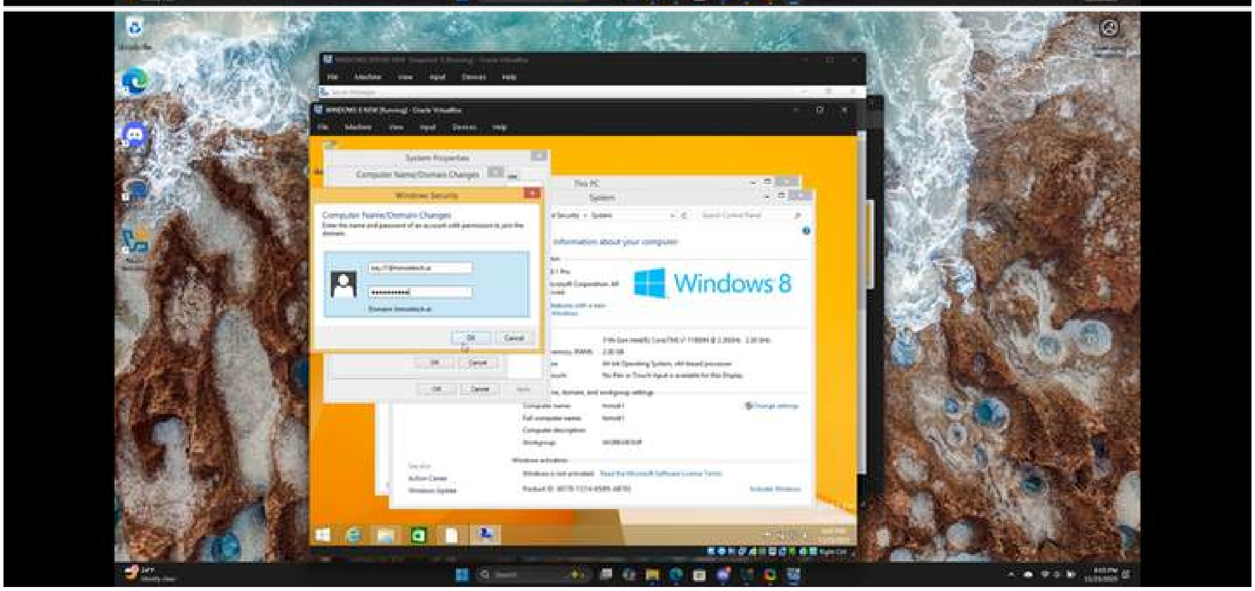


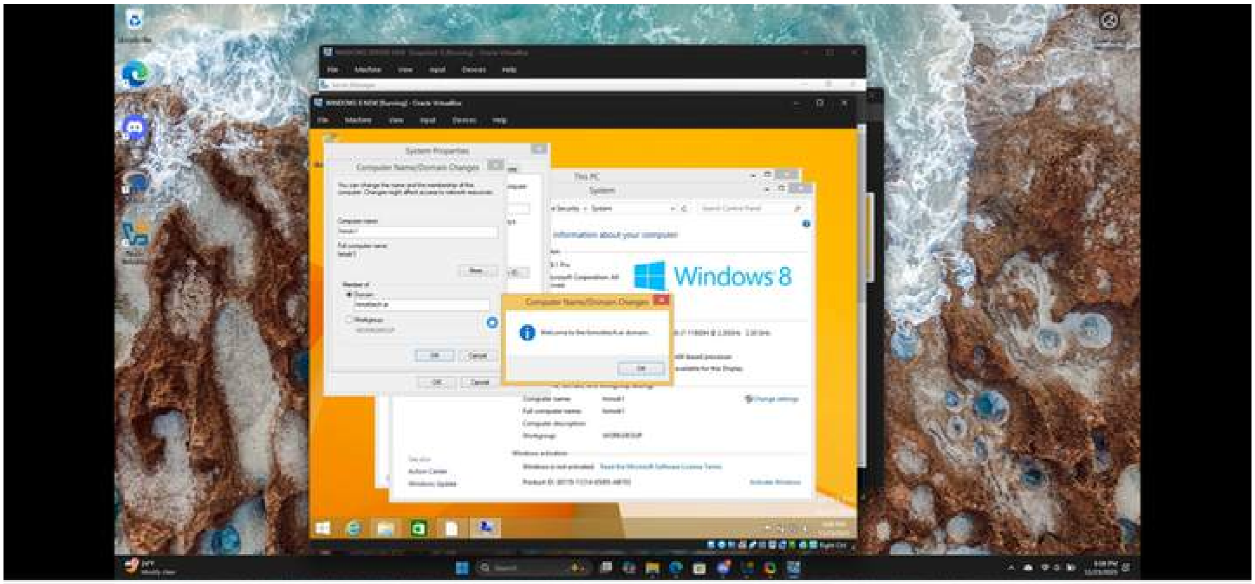


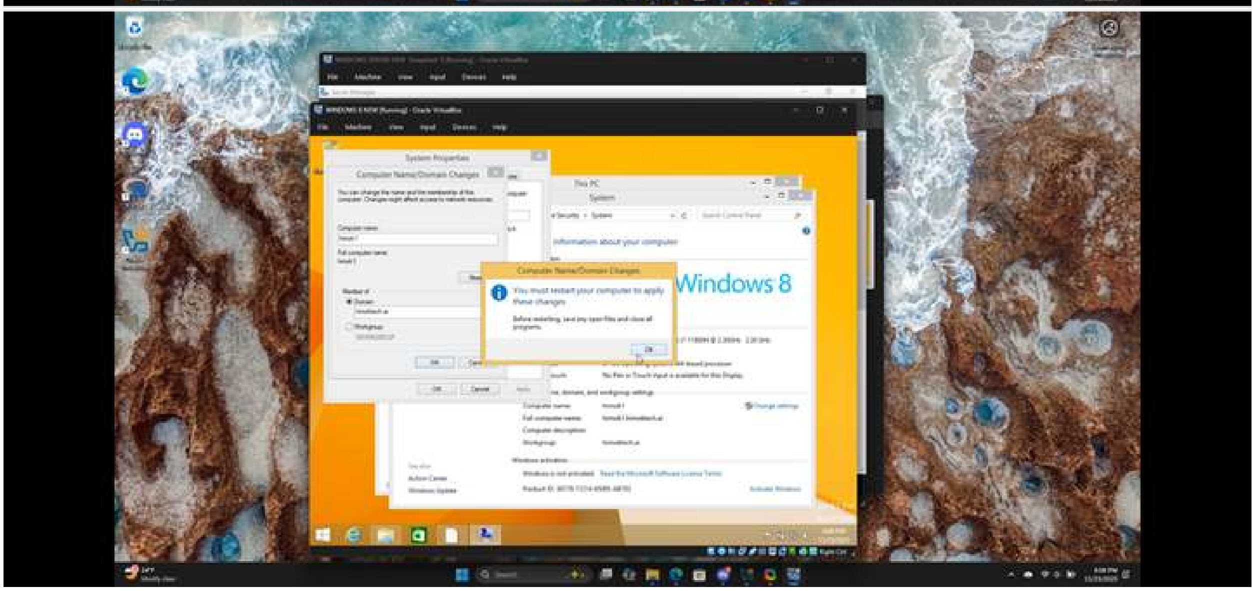
I specify that the client machine to join a Domain, and the fully qualified domain name nxgluxurylimited.local was entered.

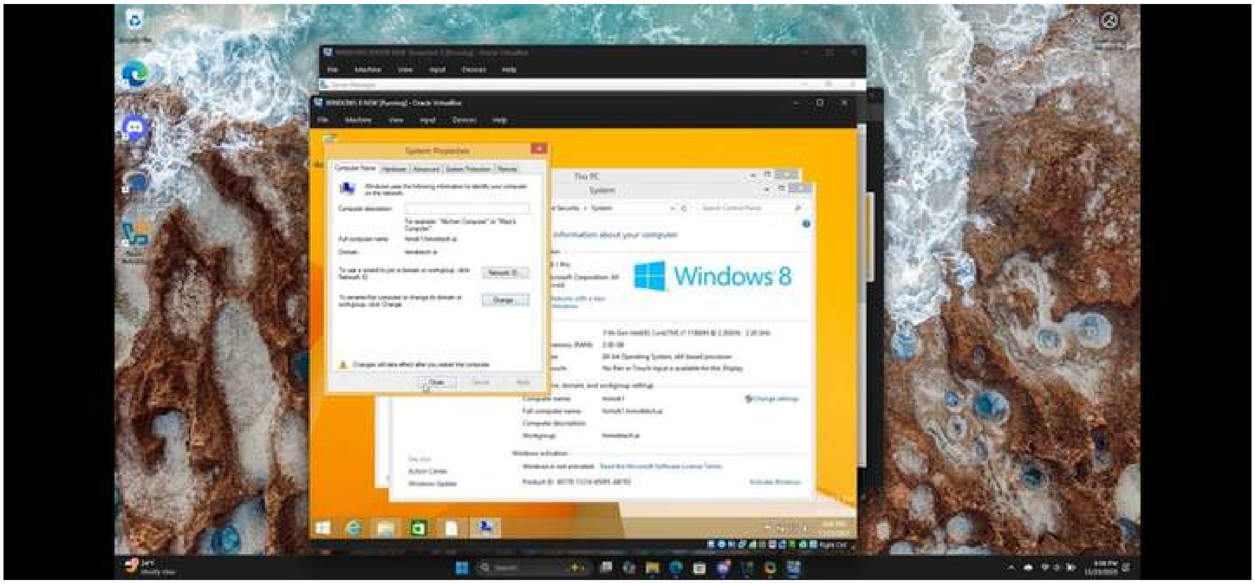


The next prompt was Domain Authentication: A security prompt requested credentials to authorize the machine join. The full User Principal Name of the newly created user (Jay.Sales@nxgluxurylimited.local) and the corresponding password were used for authentication.

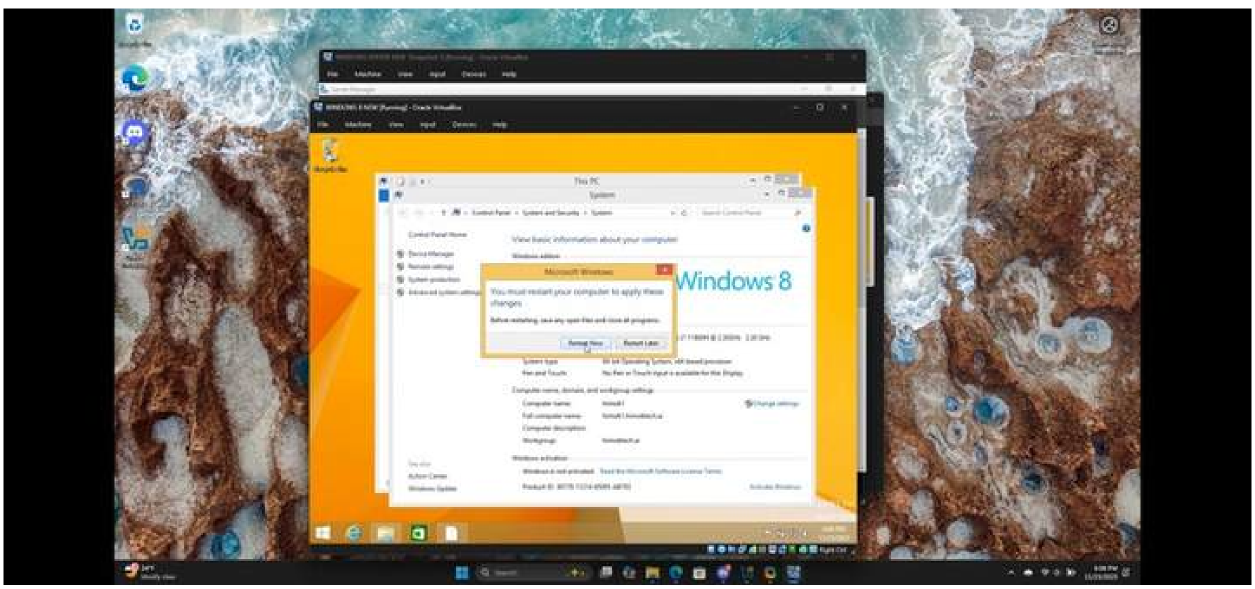

System Properties
Computer Name/Domain Changes
Windows Security
Computer Name/Domain Changes
OK
Cancel
This PC
System
Windows 8

System Properties
Computer Name/Domain Changes
Windows Security
Computer Name/Domain Changes
OK
Cancel
This PC
System
Windows 8

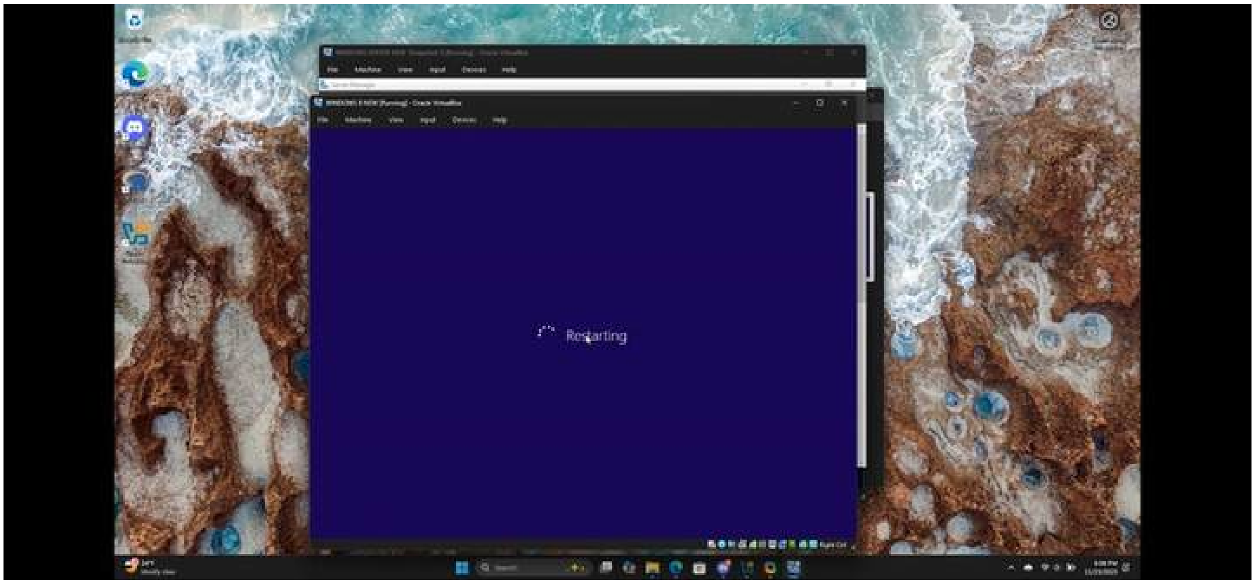

System Properties
Computer Name/Domain Changes
This PC
System
Windows 8
OK
Cancel


System Properties
Computer Name/Domain Changes
This PC
System
You must restart your computer to apply these changes
Windows 8
OK
Cancel

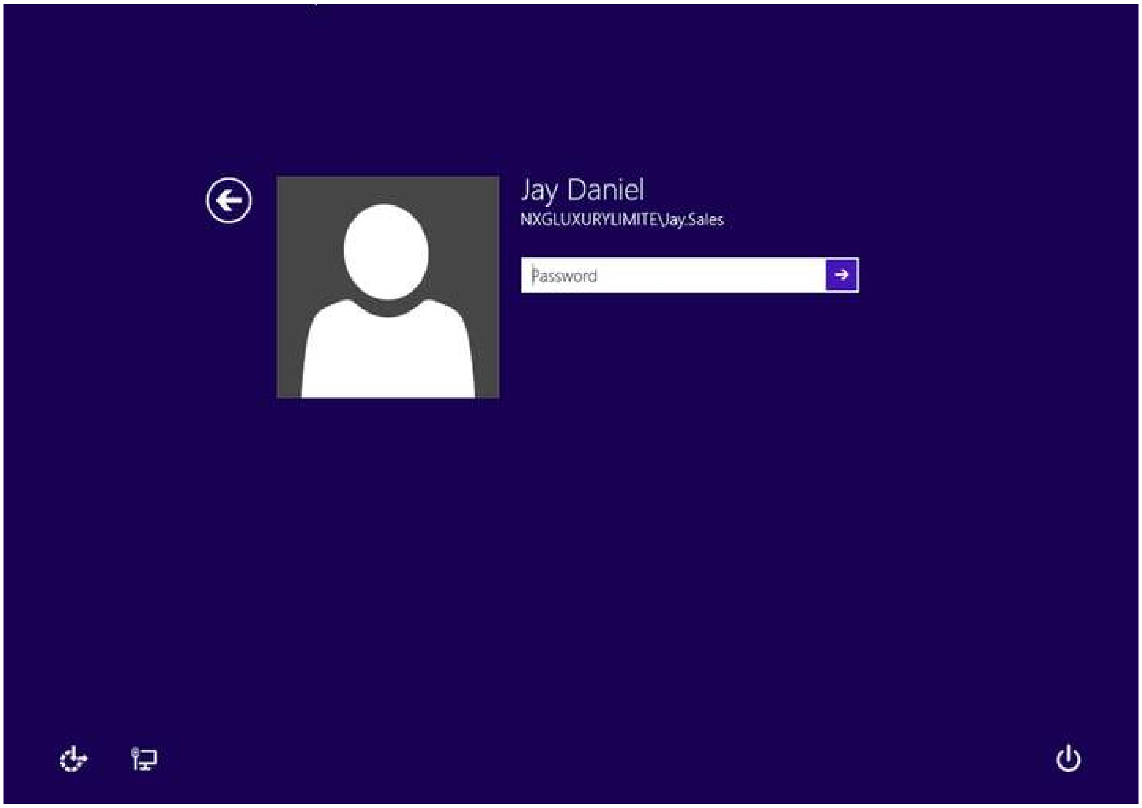


The successful join was confirmed. The client was then restarted as required, completing its integration into the nxgluxurylimited.local domain structure.

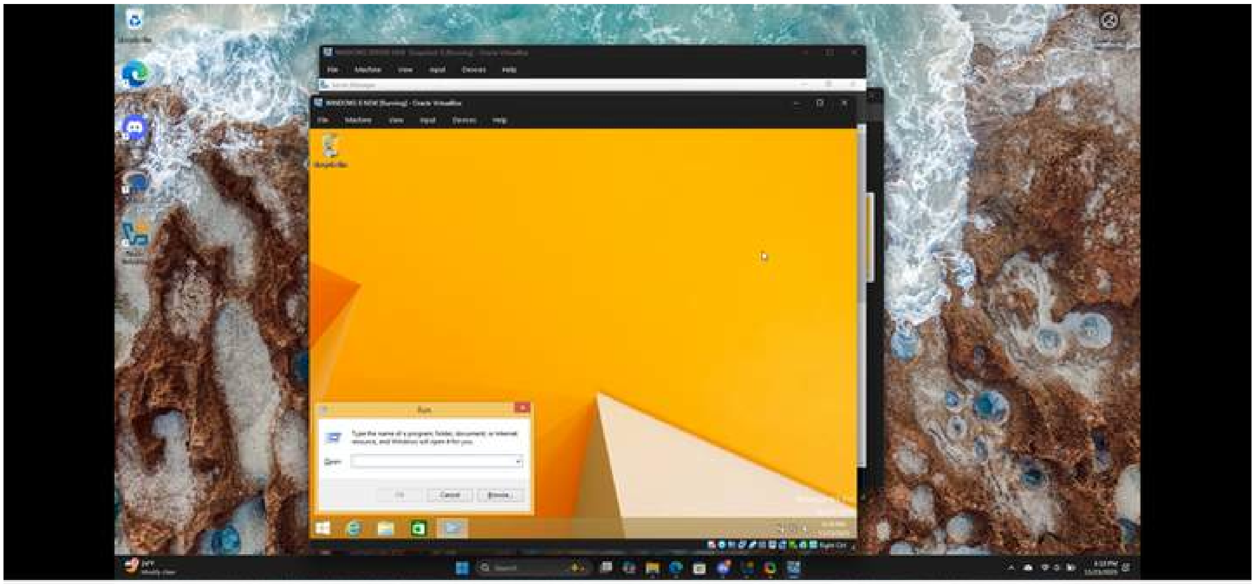


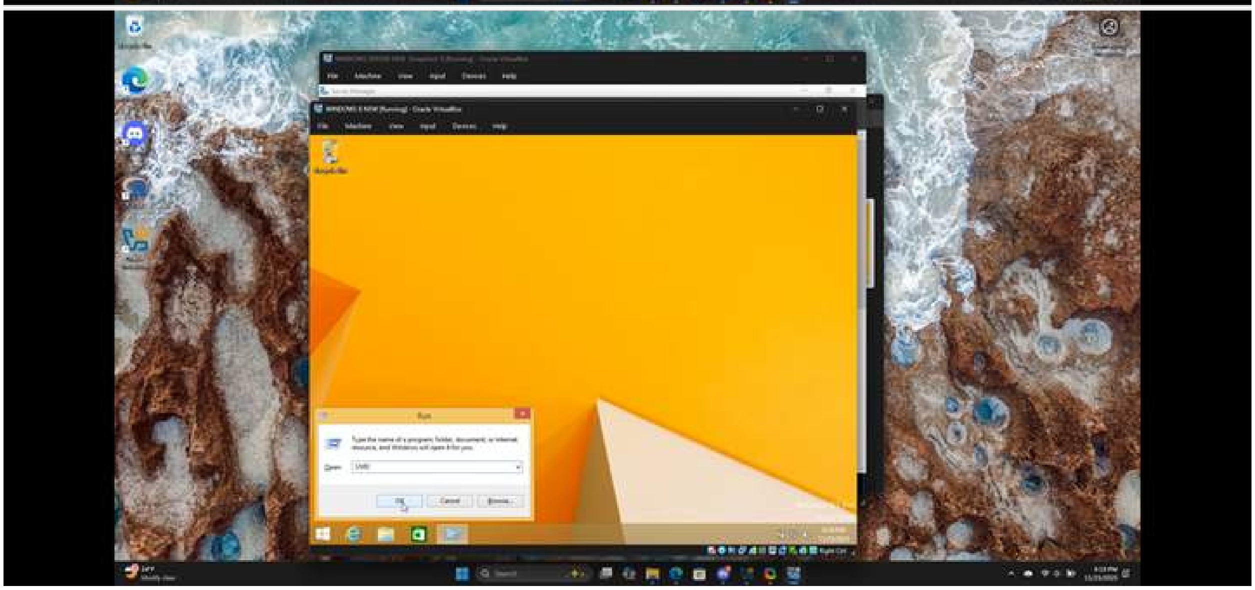


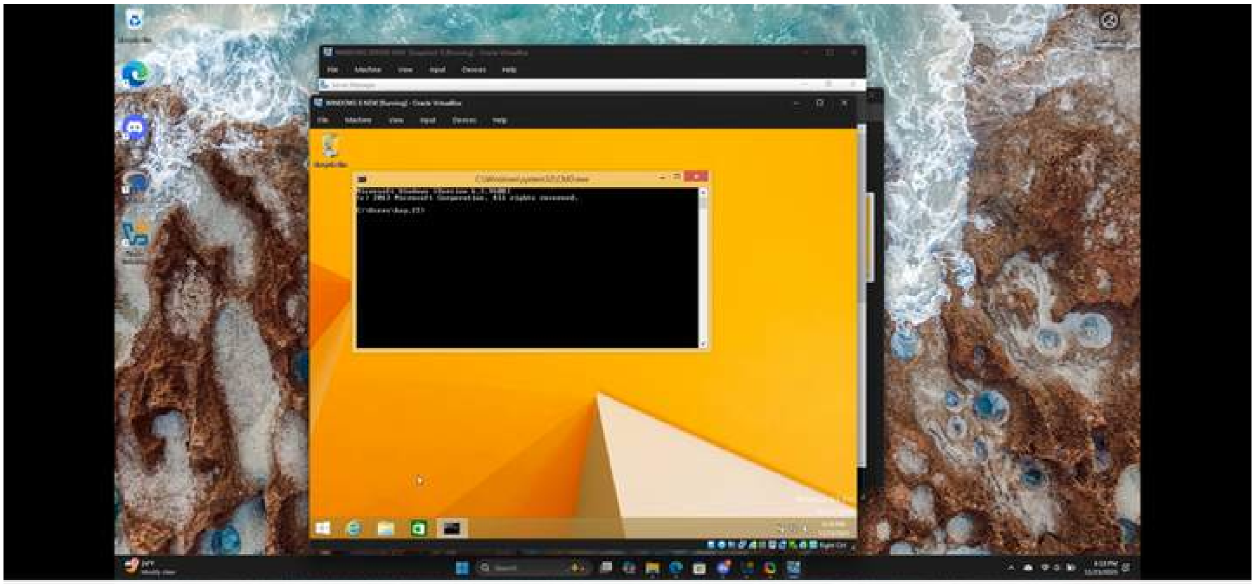
I click on the back arrow , The user Jay.Sales can now log into the Windows 8 machine using domain credentials.

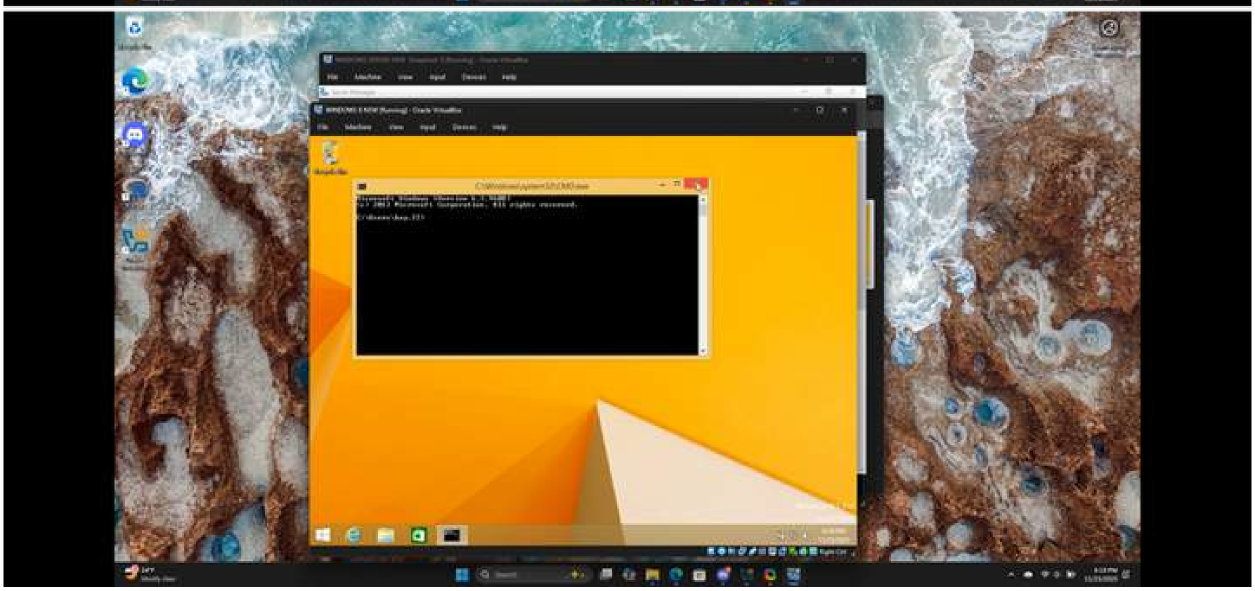

Jay Daniel
NXGLUXURYLIMITE\Jay.Sales
Password


Jay Daniel
NXGLUXURYLIMITE\Jay.Sales
Other user



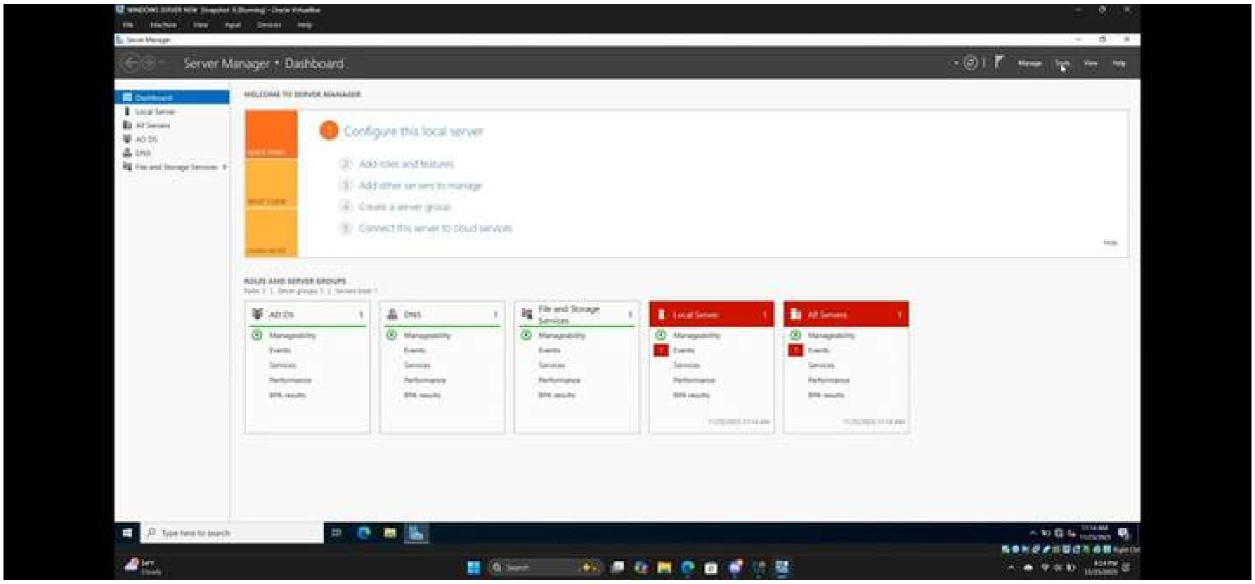




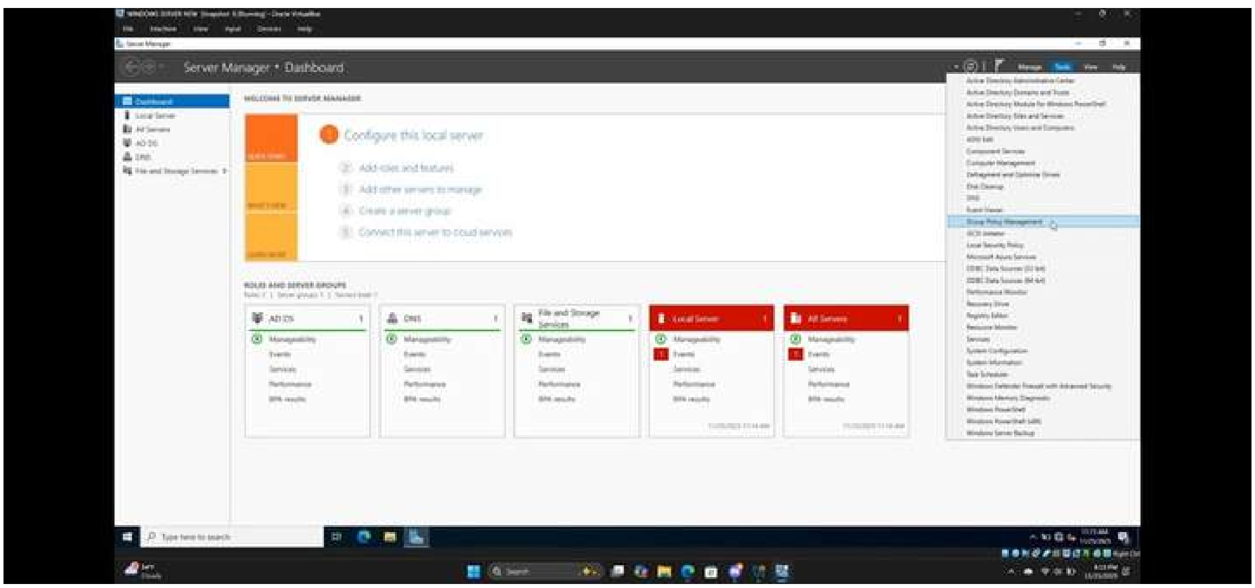


The next phase is to focuses on utilizing Group Policy within the Active Directory to define and enforce standardized settings and security configurations across all domain-joined client endpoint. So I started by creating the Foundational Group Policy Objects (GPOs). The process begins in the Group Policy Management Console (GPMC), the central interface for all policy administration. Let me walk you through the process;

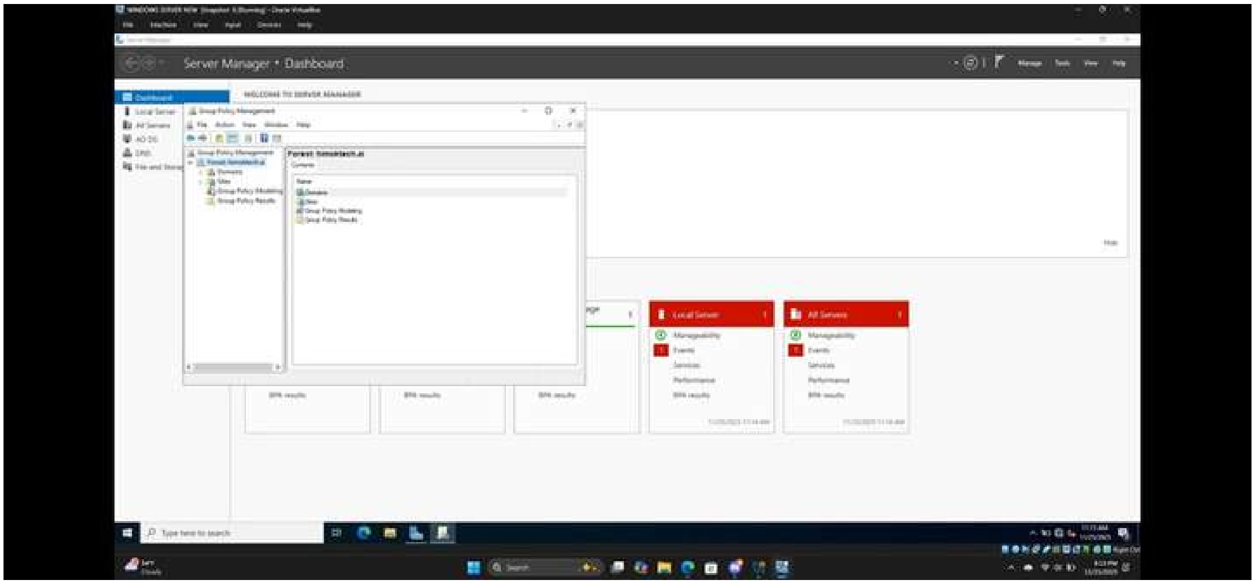
So From the Windows Server dashboard, i navigated to Tools



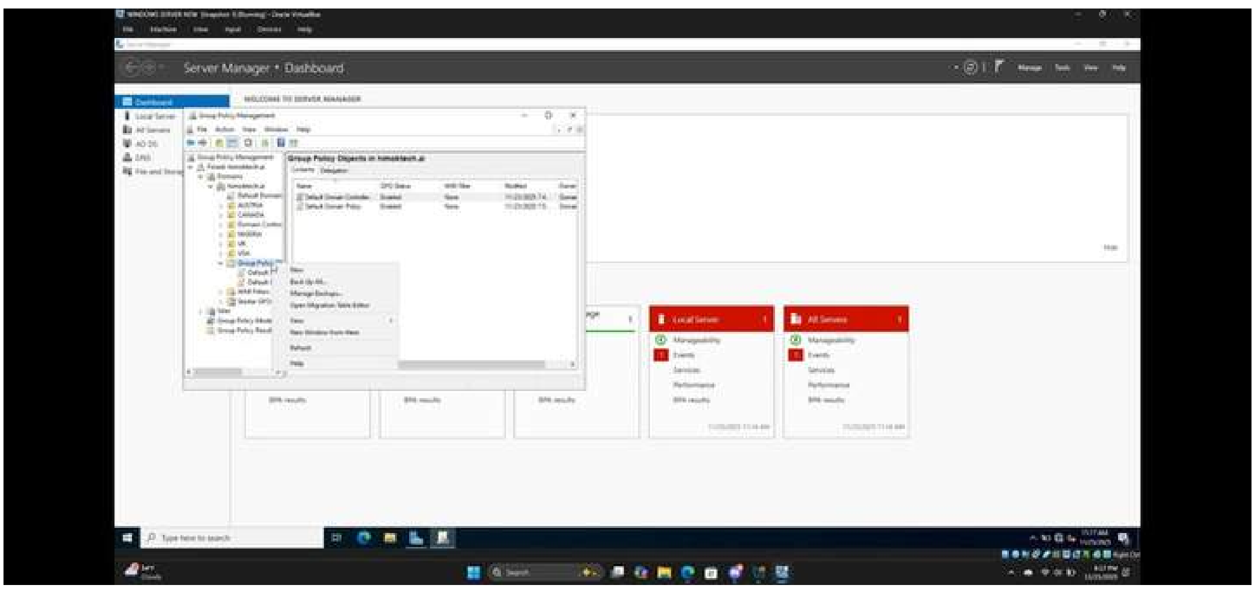
and launched Group Policy Management.



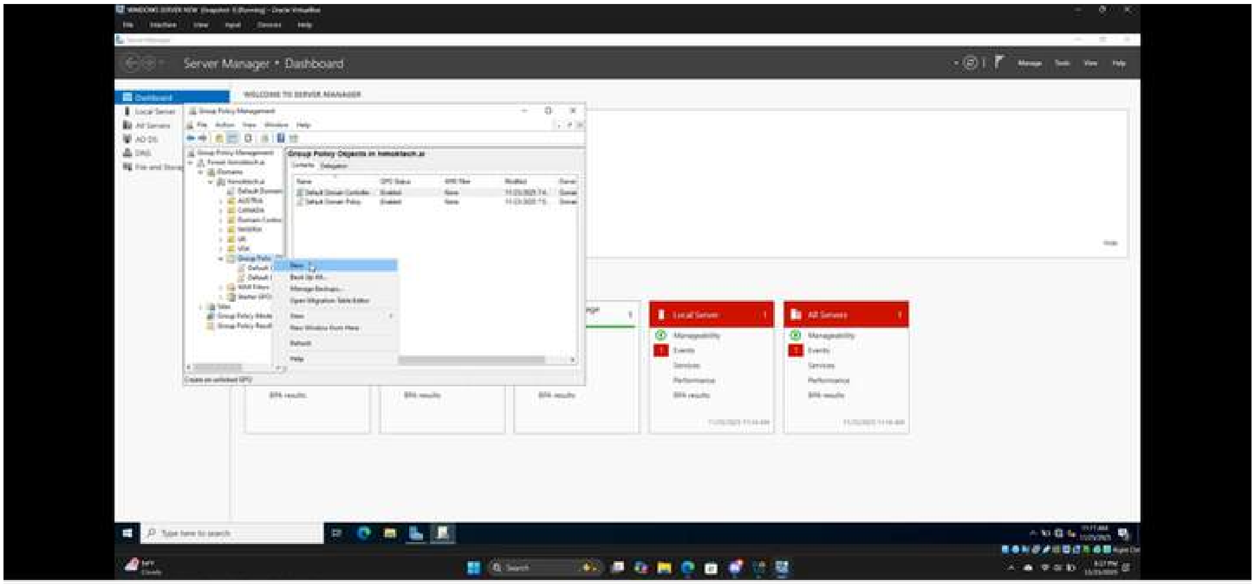
The console loaded, displaying the organizational structure of the himoktech.ai forest.

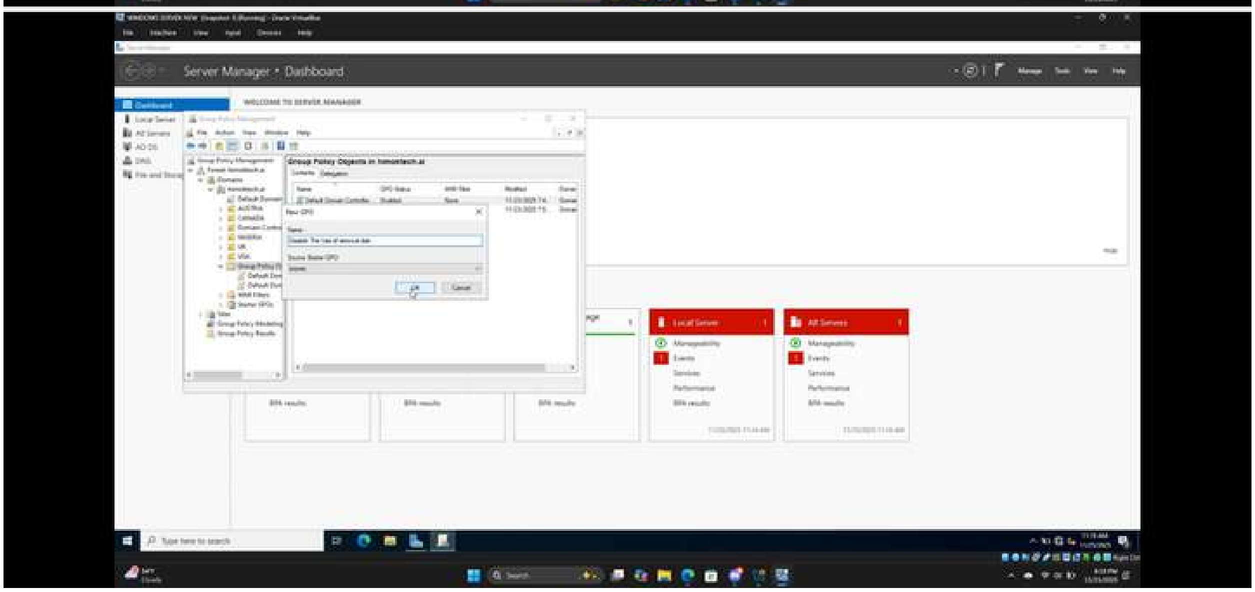


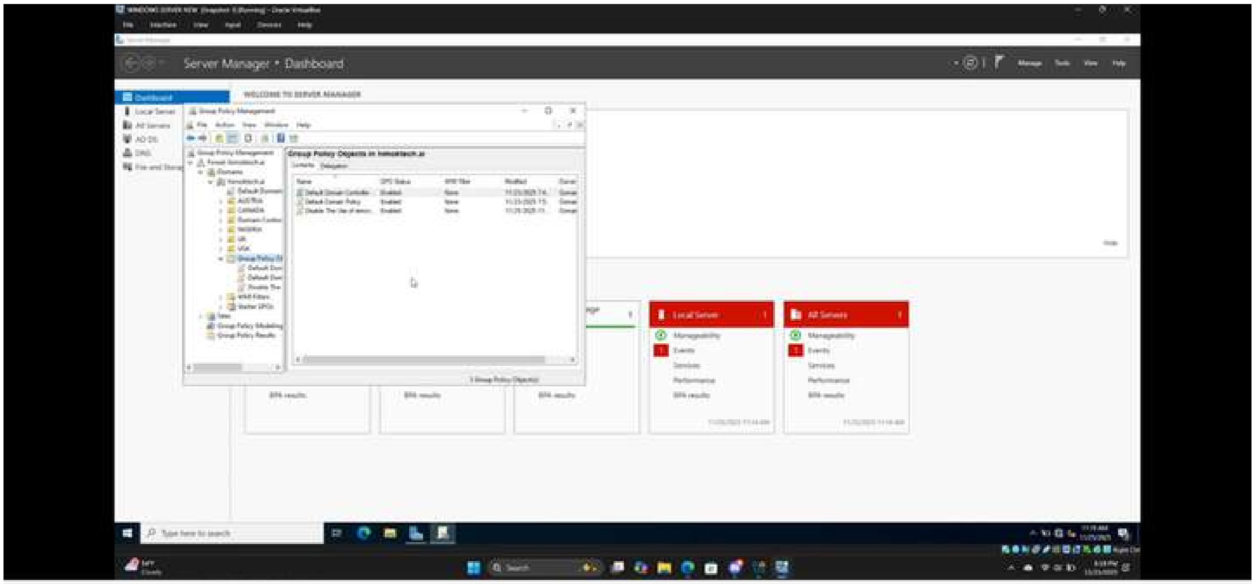
Next,I navigatedandexpandedthedomainstructuretolocatetheGroupPolicyObjects container.

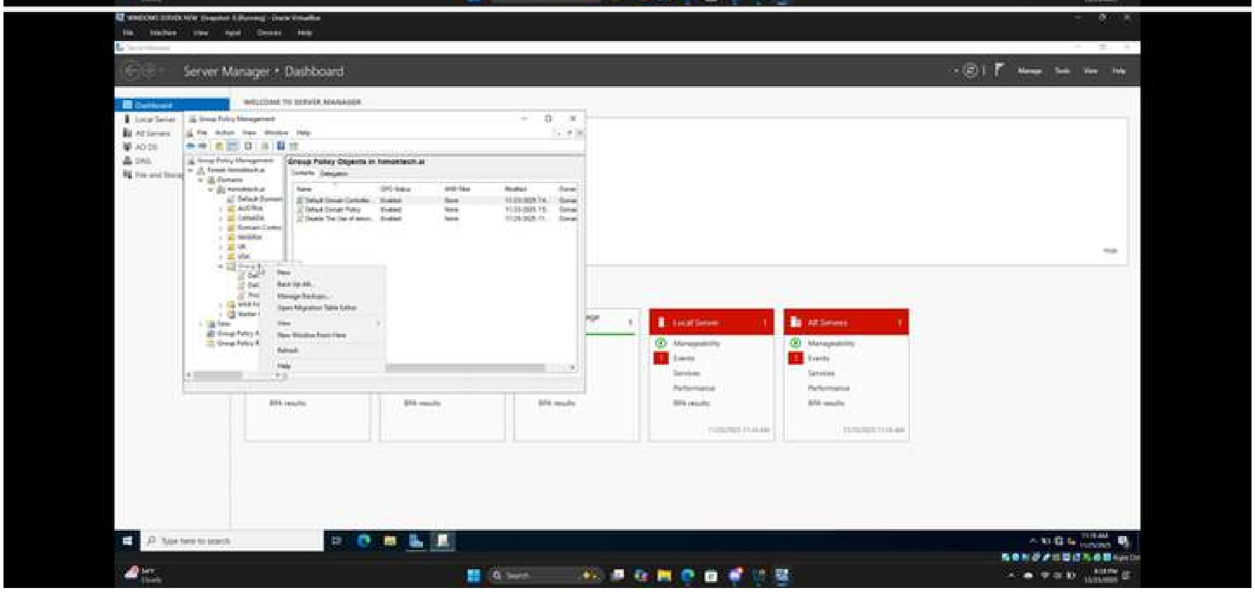


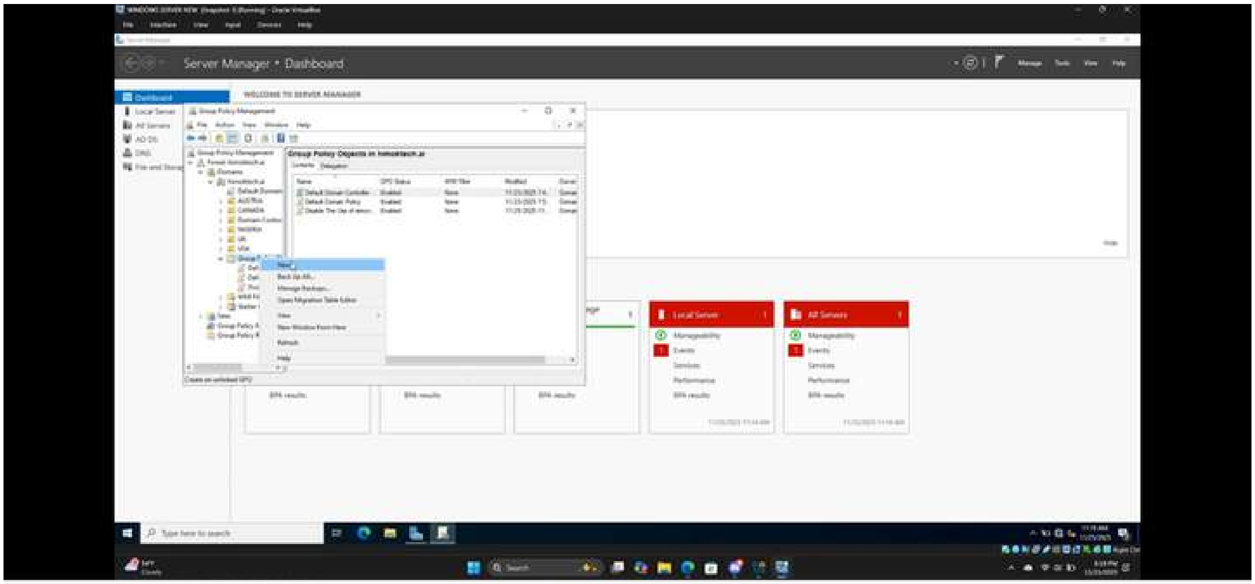
A key administrative principle was established I.e Group Policy Object (GPO) must first be created to house and define specific policy settings before those settings can be applied to an Organizational Unit (OU) or a Group or user. So I systematically created three distinct GPOs by right-clicking the Group Policy Objects container, selecting New, and providing a descriptive name for each security objective:


Server Manager • Dashboard
WELCOME TO SERVER MANAGER
Dashboard
Group Policy Management
Local Server
All Servers
Manageability
Events
Services
Performance
BPA results

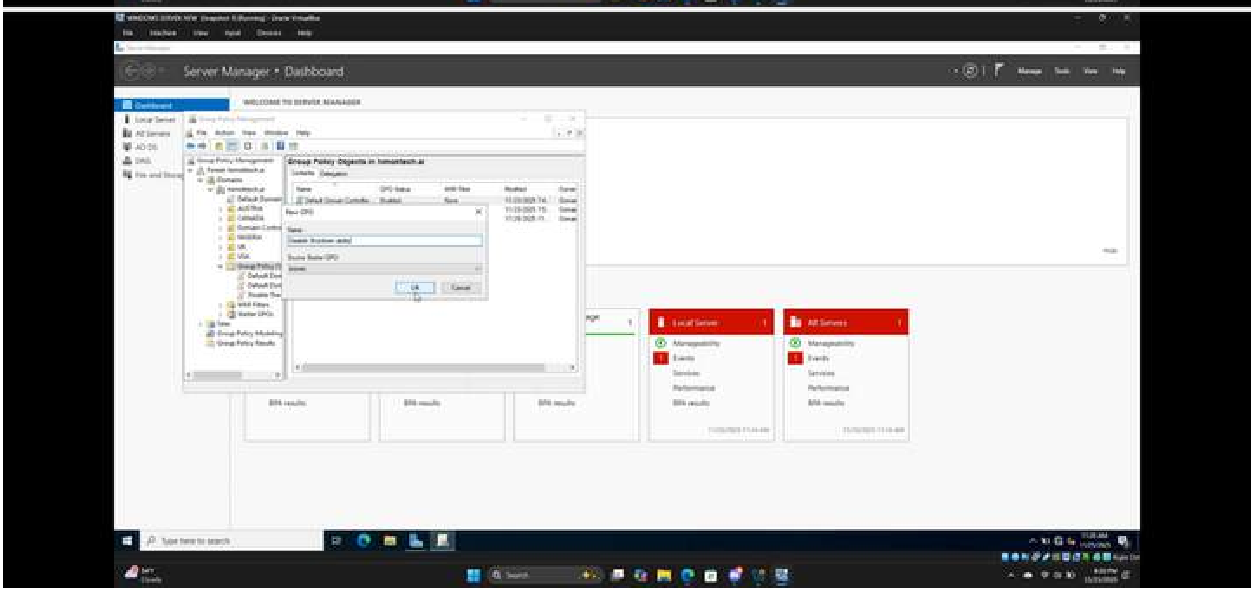

Server Manager • Dashboard
WELCOME TO SERVER MANAGER
Dashboard
New GPO
OK
Cancel
Local Server
All Servers
Manageability
Events
Services
Performance
BPA results


Server Manager • Dashboard
Group Policy Management
Local Server
All Servers
Manageability
Events
Services
Performance
BPA results

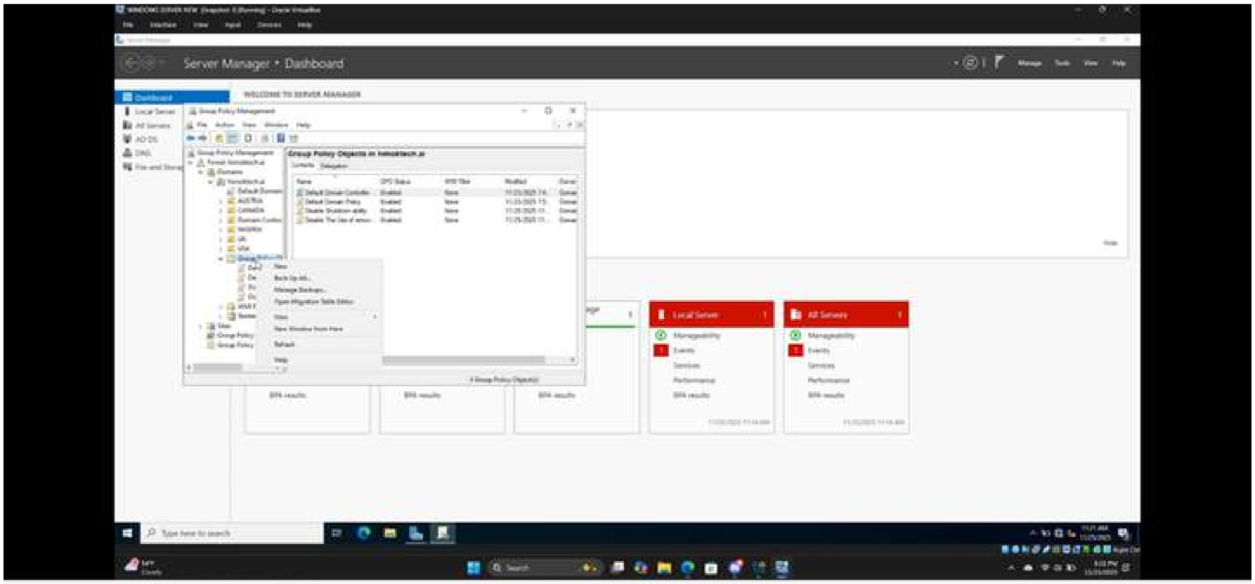

Server Manager • Dashboard
Group Policy Management
Local Server
All Servers
Manageability
Events
Services
Performance
BPA results

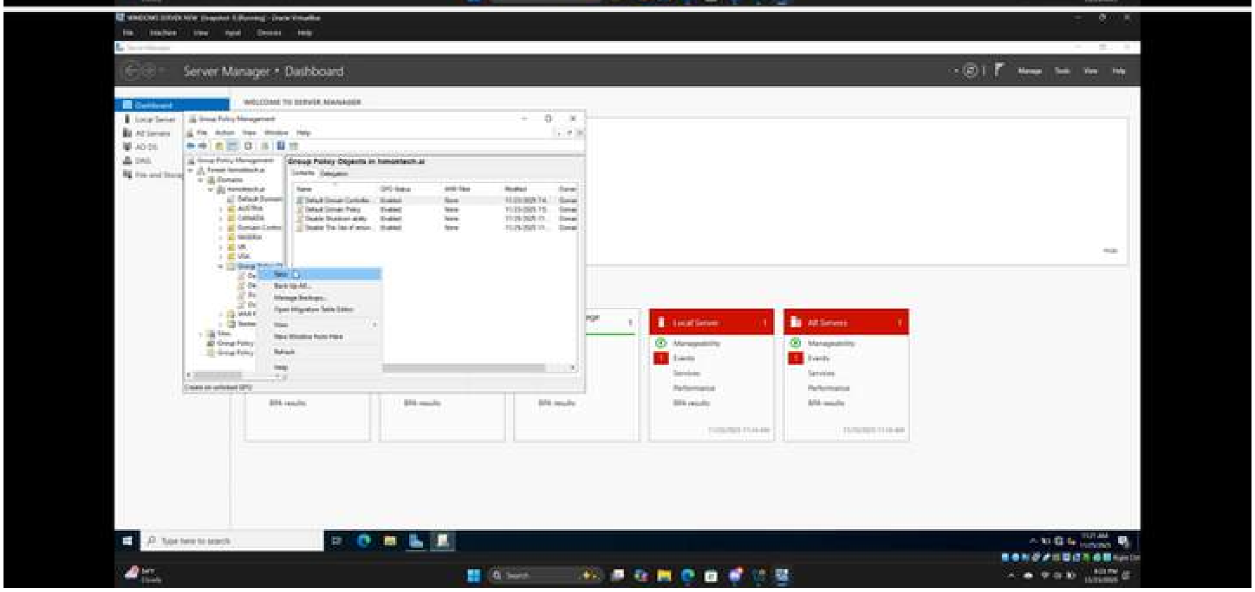


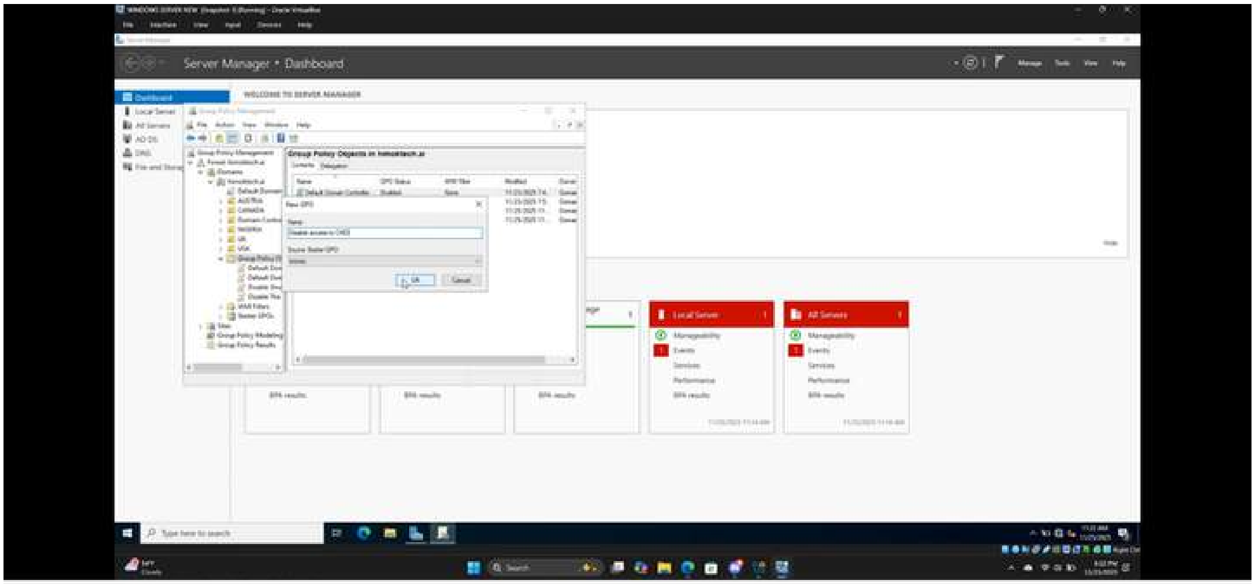
Server Manager • Dashboard
WELCOME TO SERVER MANAGER
Group Policy Management
Local Server
All Servers

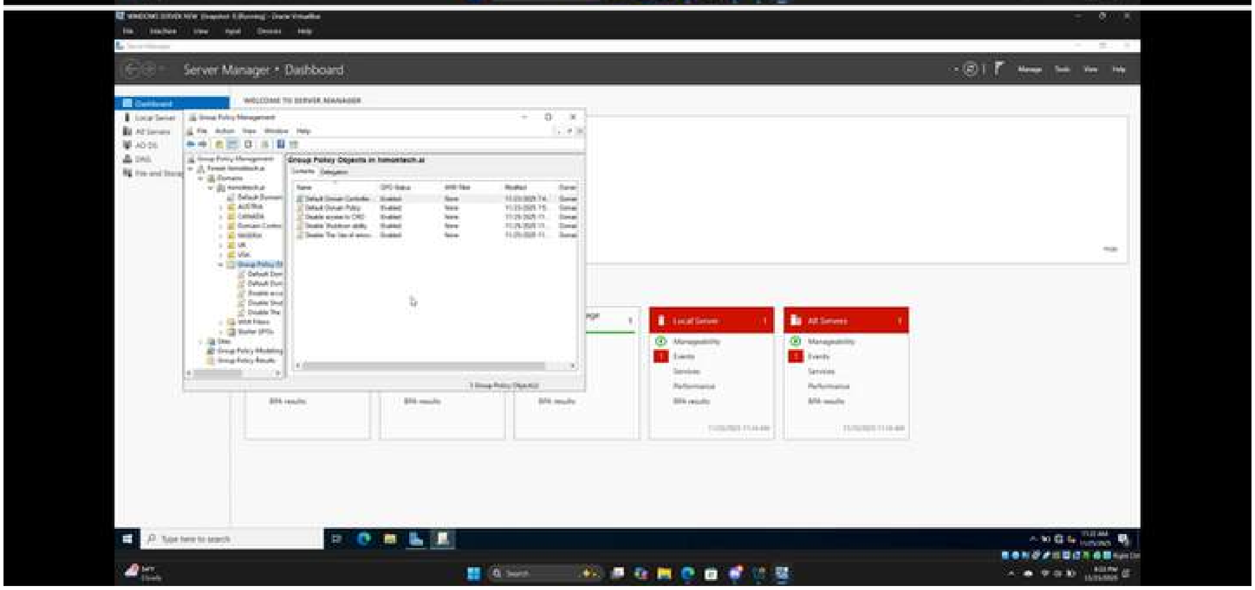


Server Manager • Dashboard
WELCOME TO SERVER MANAGER
Local Server
All Servers


Server Manager • Dashboard
WELCOME TO SERVER MANAGER
Local Server
All Servers


Server Manager • Dashboard
WELCOME TO SERVER MANAGER
Local Server
All Servers



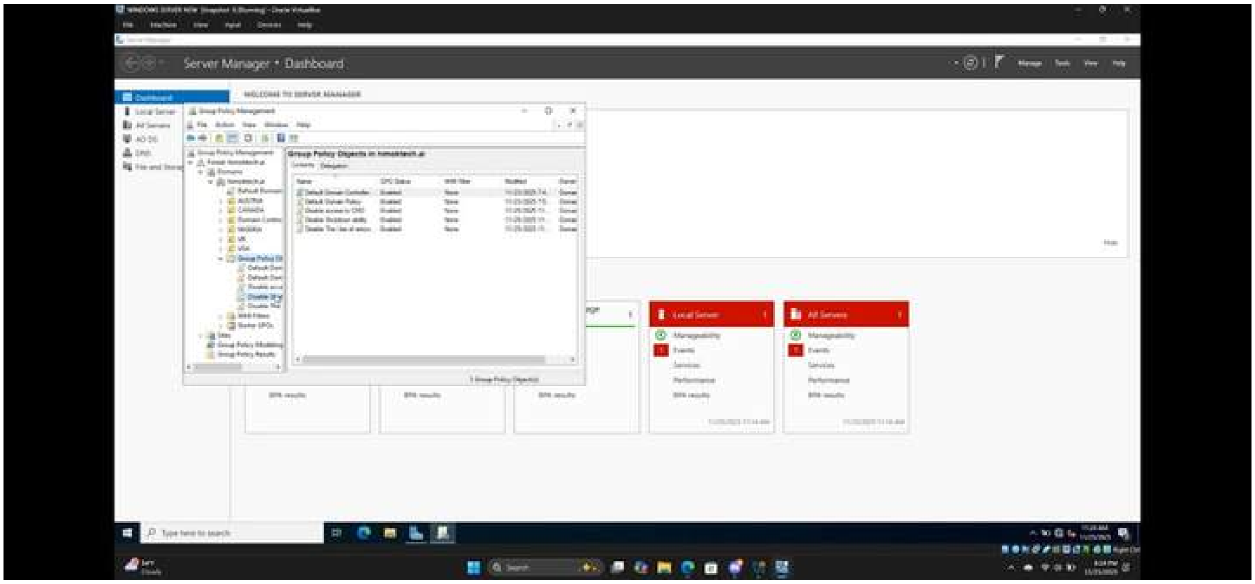


GPO 1 Name: "Disable the use of removable disk" (Policy aimed at preventing data leakage).

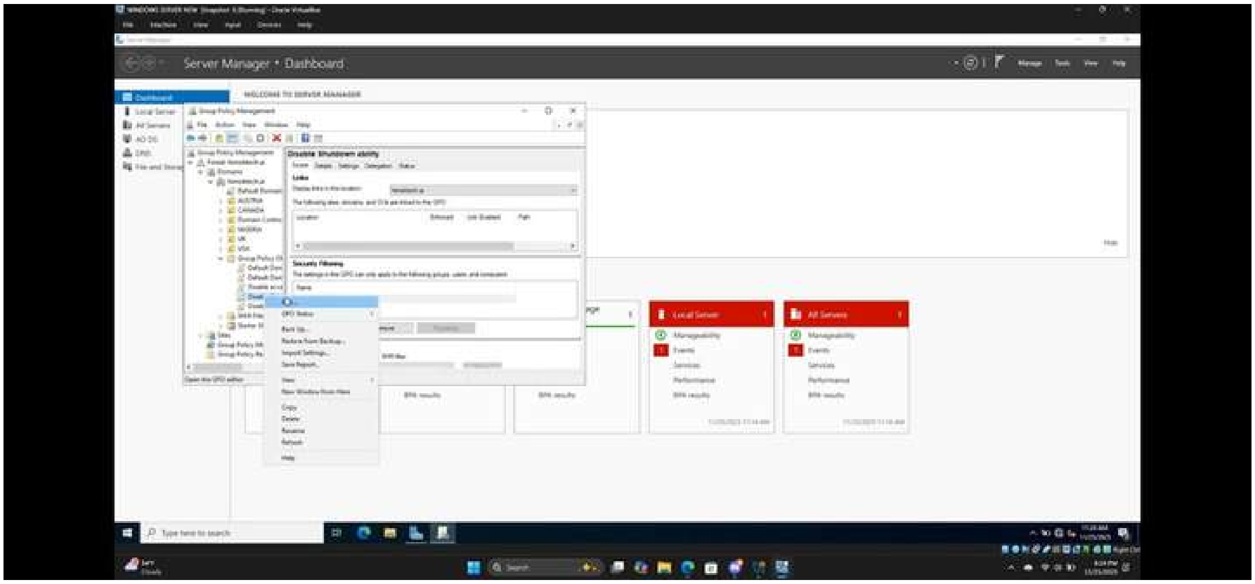
GPO 2 Name: "Disable shutdown ability" (Policy aimed at maintaining system uptime/control).

GPO 3 Name: "Disable access to CMD" (Policy aimed at restricting unauthorized system access/command-line tools).

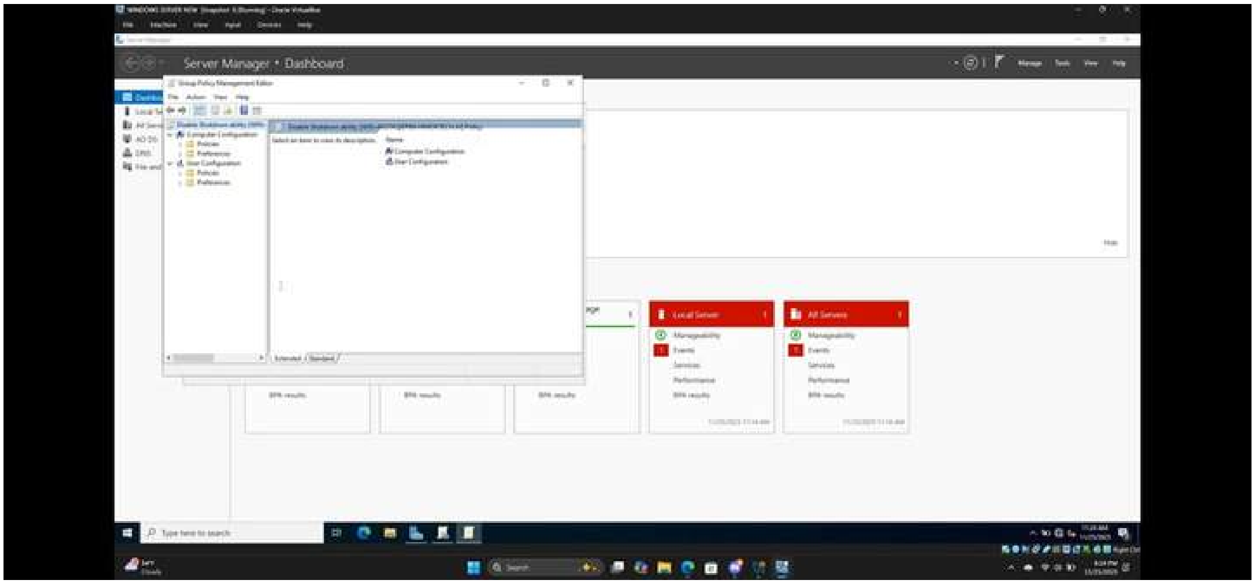
So with the GPOs defined, the next step is to configure the specific technical settings within them. This requires utilizing the Group Policy Editor. I selected one of the newly created objects, "Disable shutdown ability,"



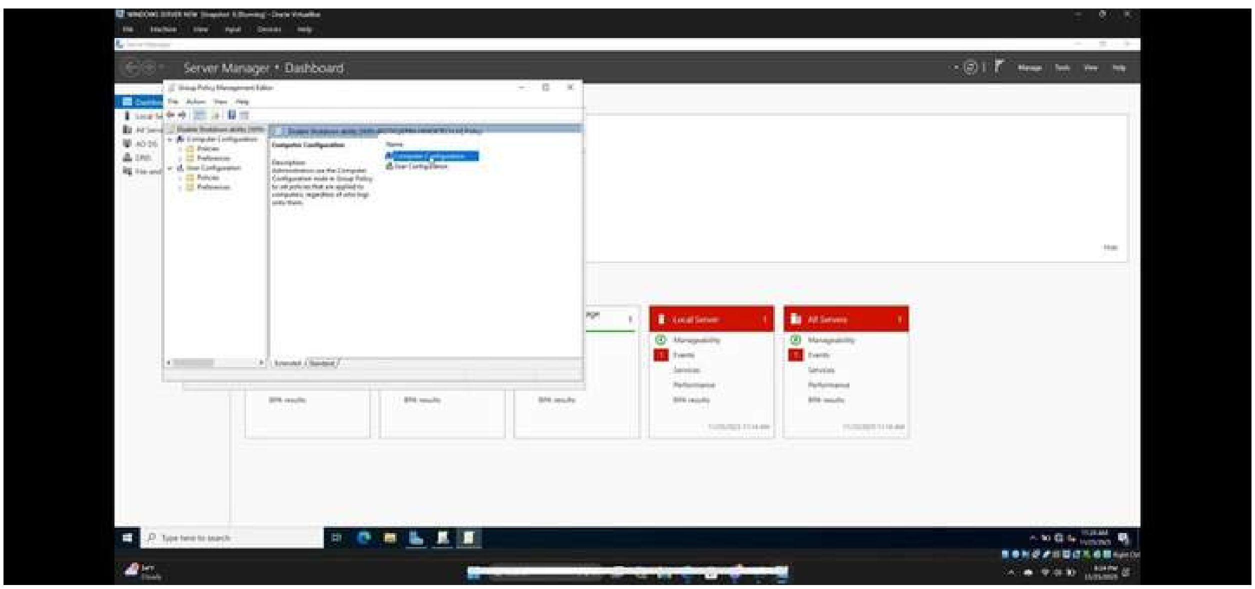
right-clicked, and selected Edit.



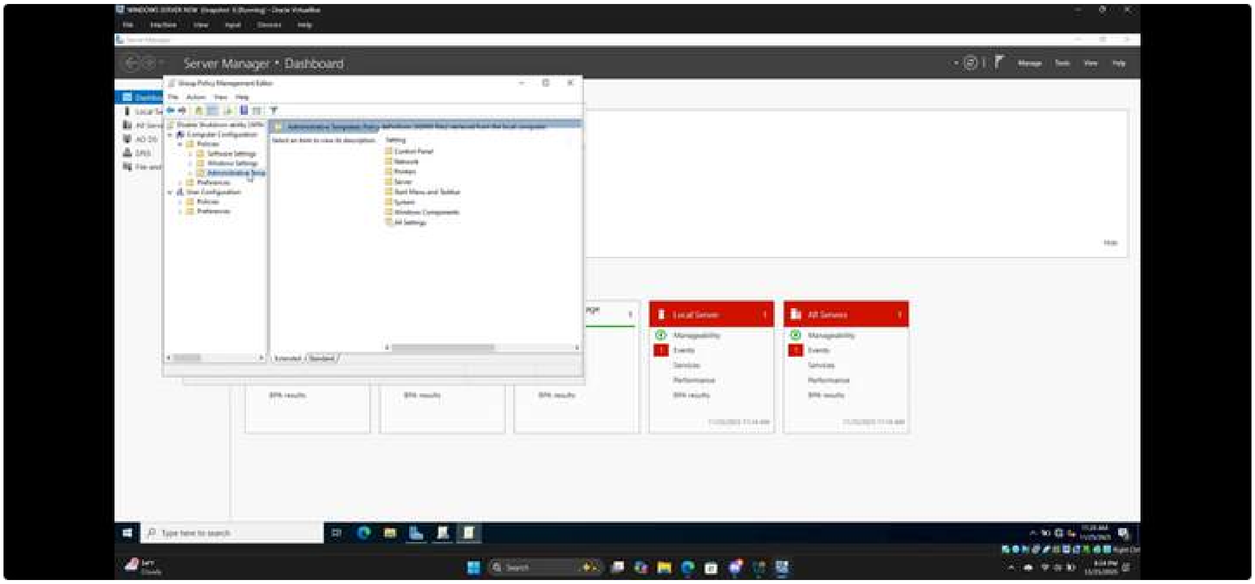
This opened the dedicated Group Policy Management Editor.



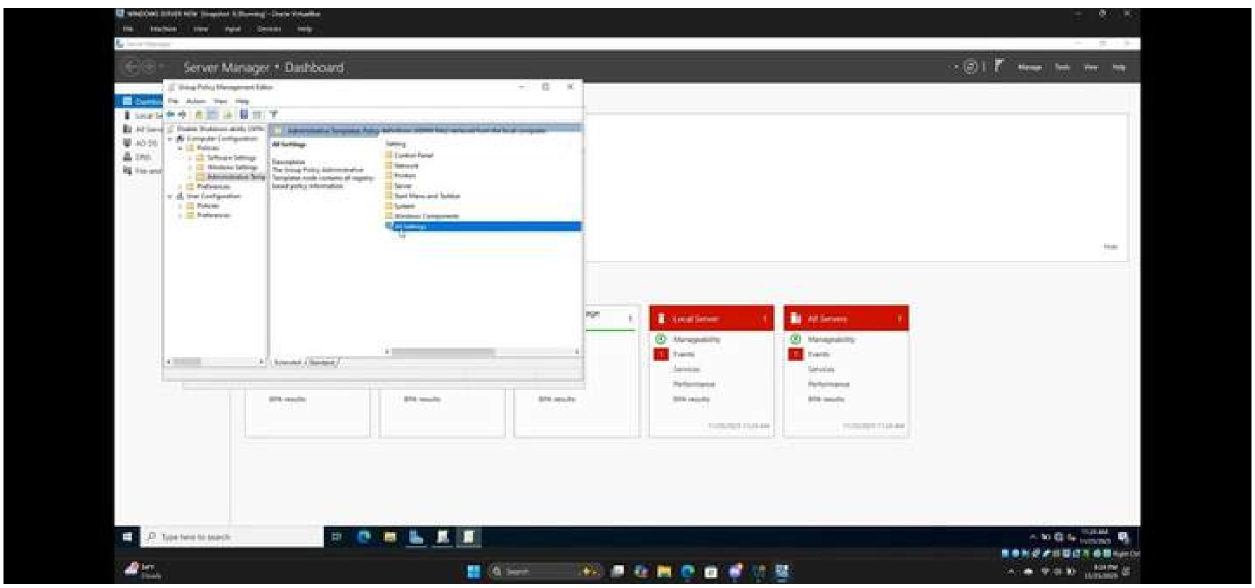
The editor presents two primary configuration areas, acknowledging the dual scope of policy application:User Configuration and Computer Configuration: I navigated the hierarchy within Computer Configuration >



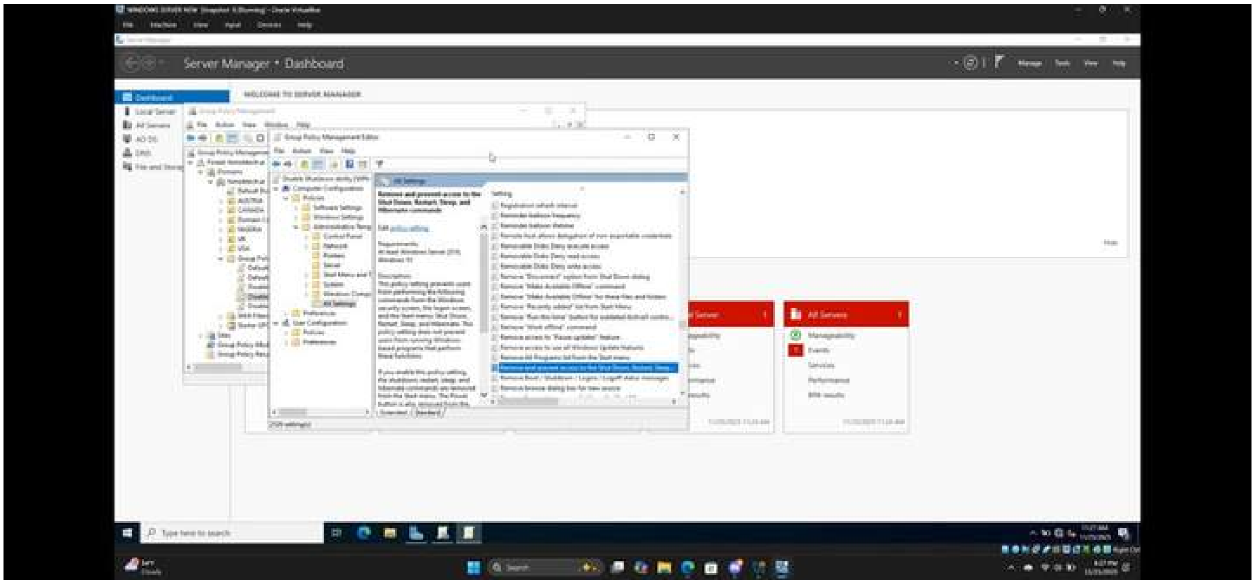
Policies > Administrative Templates



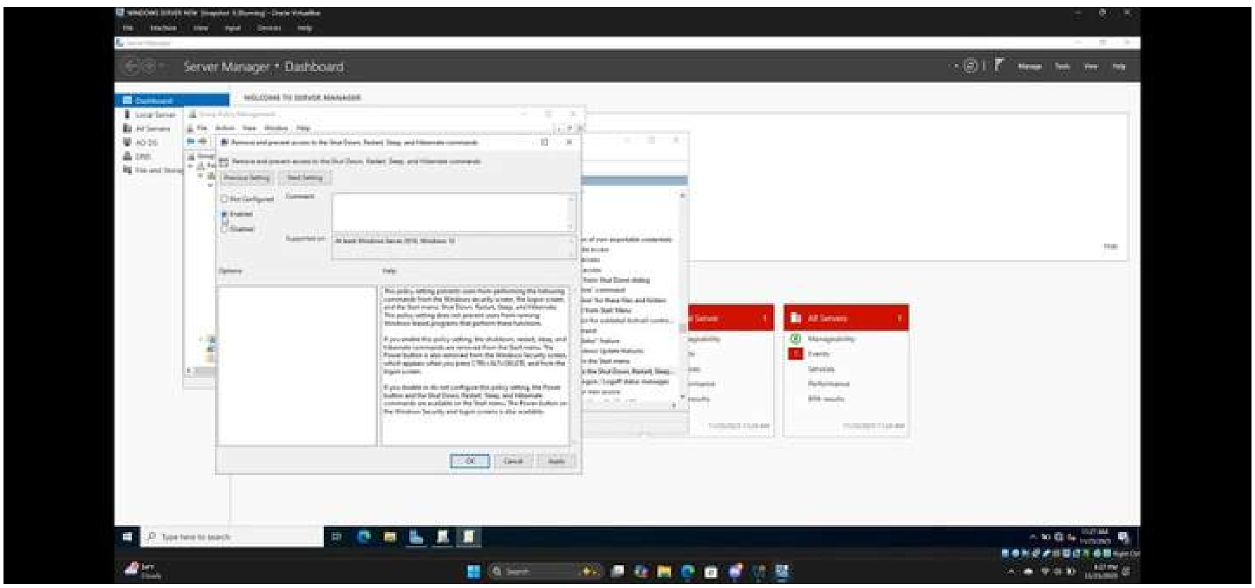
and selected All Settings.



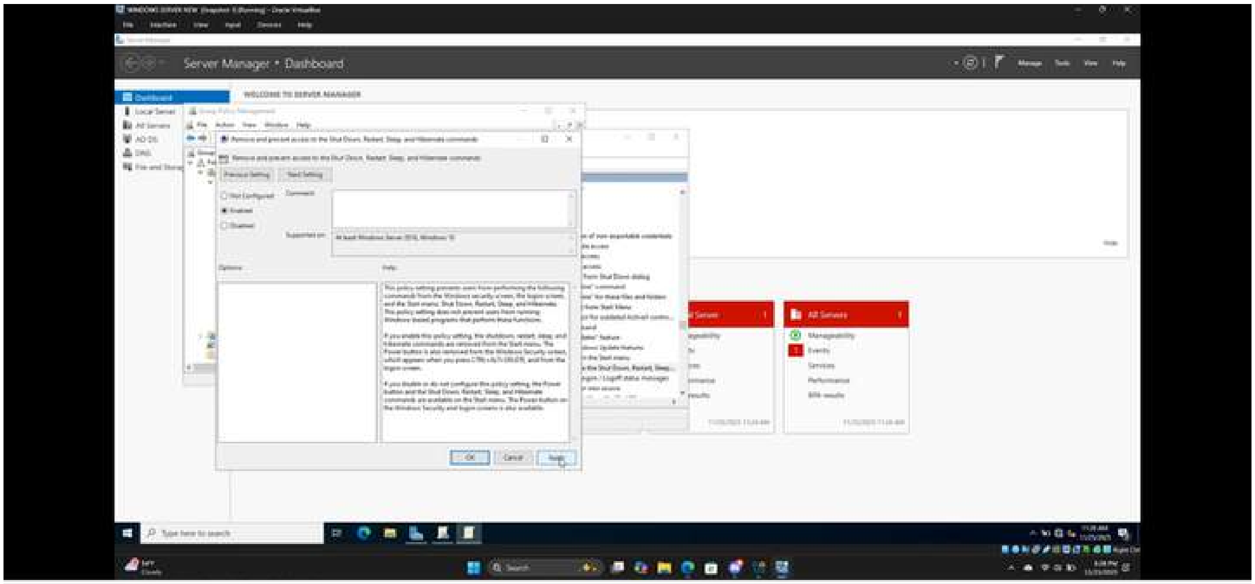
I initiated a search to locate the precise policy responsible for controlling shutdown capabilities, identifying the setting titled "Remove prevent access to the Shut Down, Restart, Sleep."

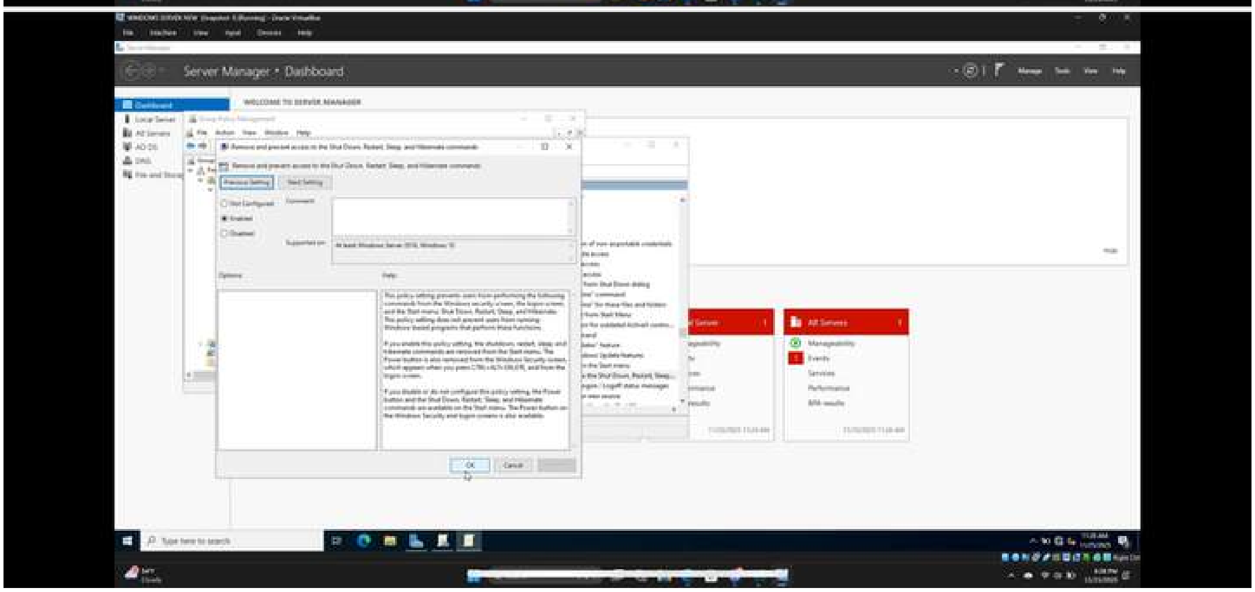


The status was changed to Enabled,

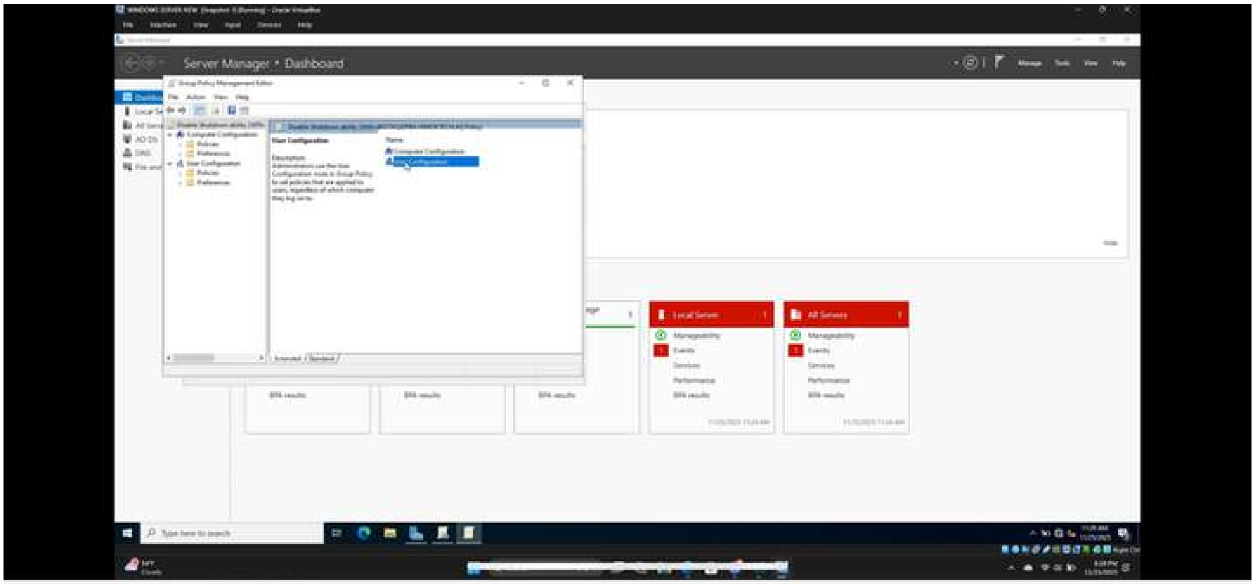


effectively activating the restriction on the client computer itself. This change was saved by clicking Apply and then OK.





For the User Configuration Policy Enforcement: I replicated this configuration within the User Configuration node of the Group Policy Editor. Configuring the policy in both areas ensures the restriction applies whether the rule targets the machine or the authenticated user.


Server Manager • Dashboard
Group Policy Management Editor
Local Server
All Servers
Manageability
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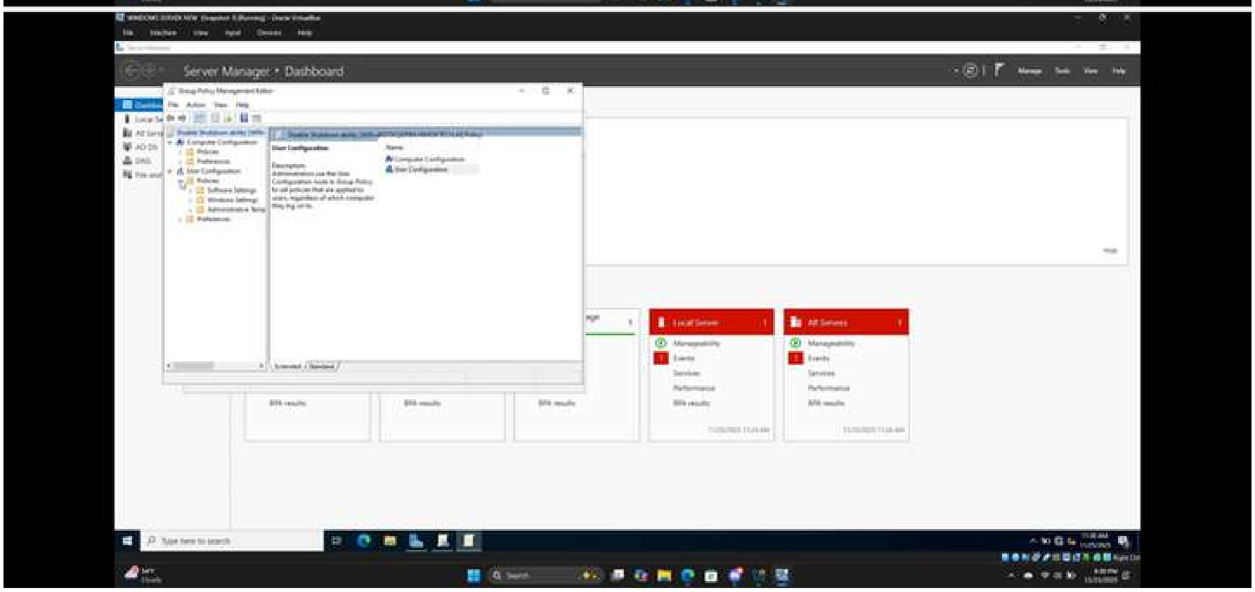
BPA results


Server Manager • Dashboard
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Local Server
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Services
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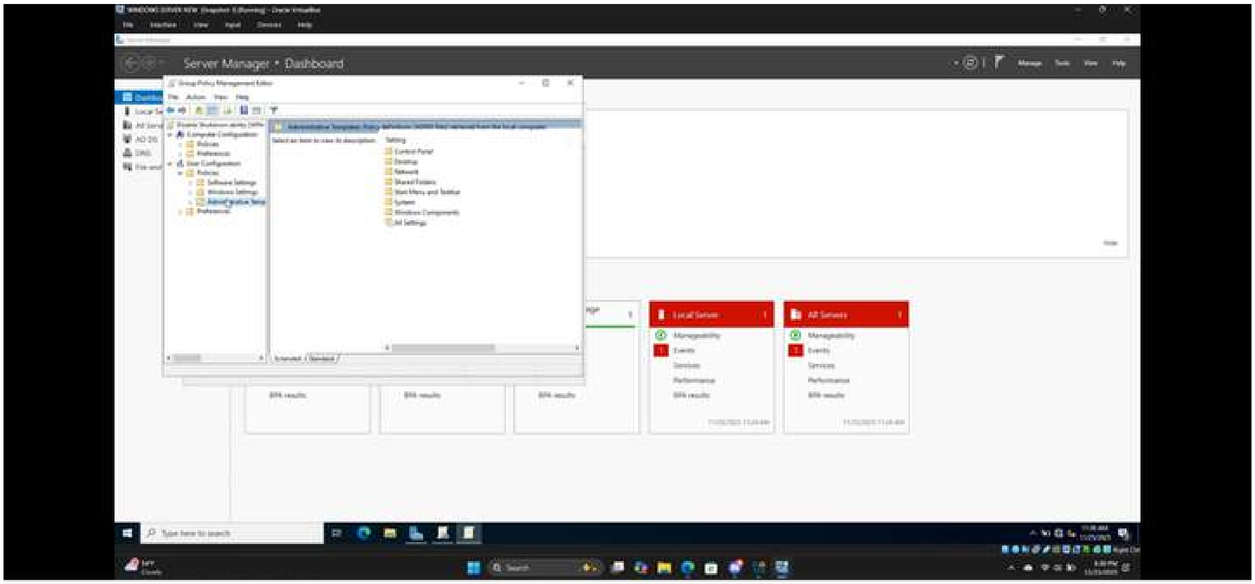
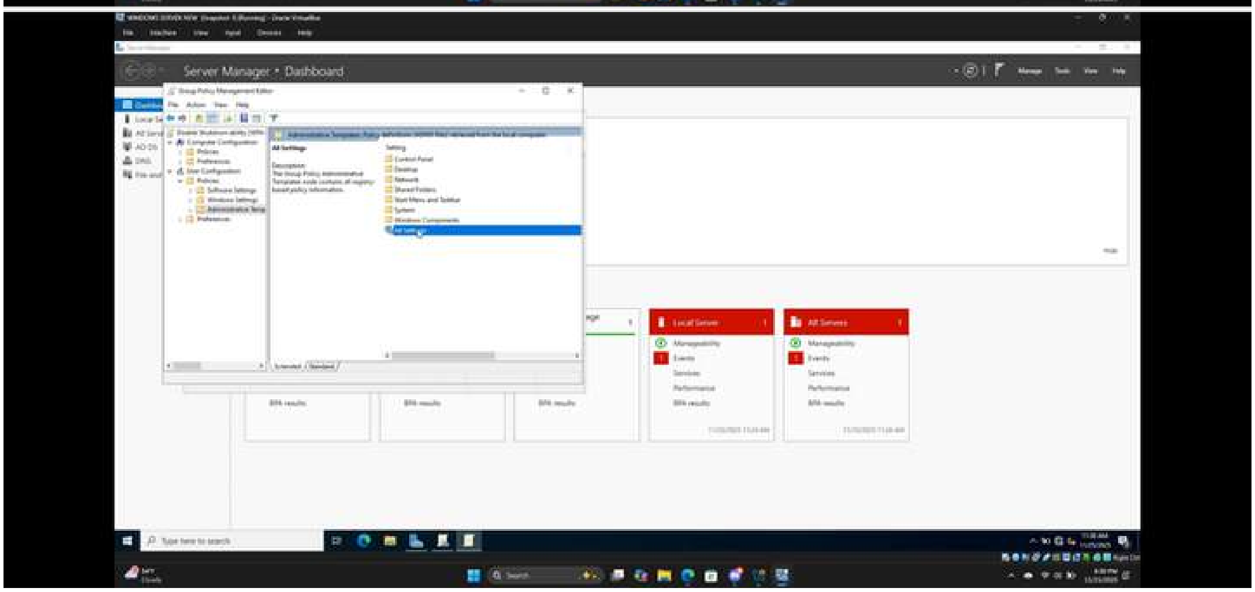
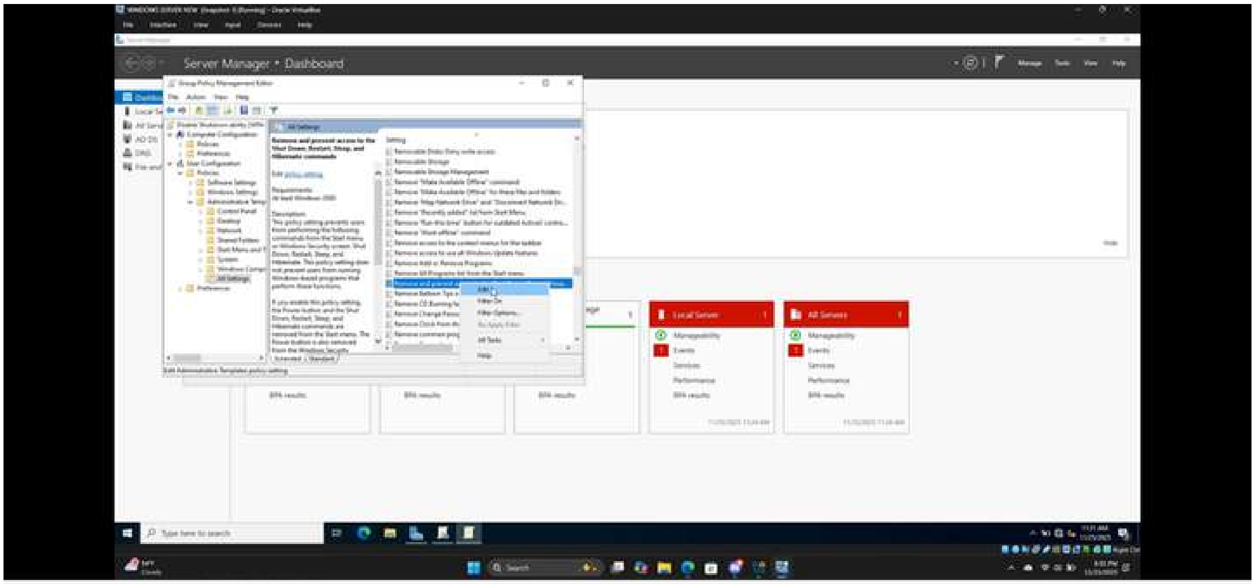
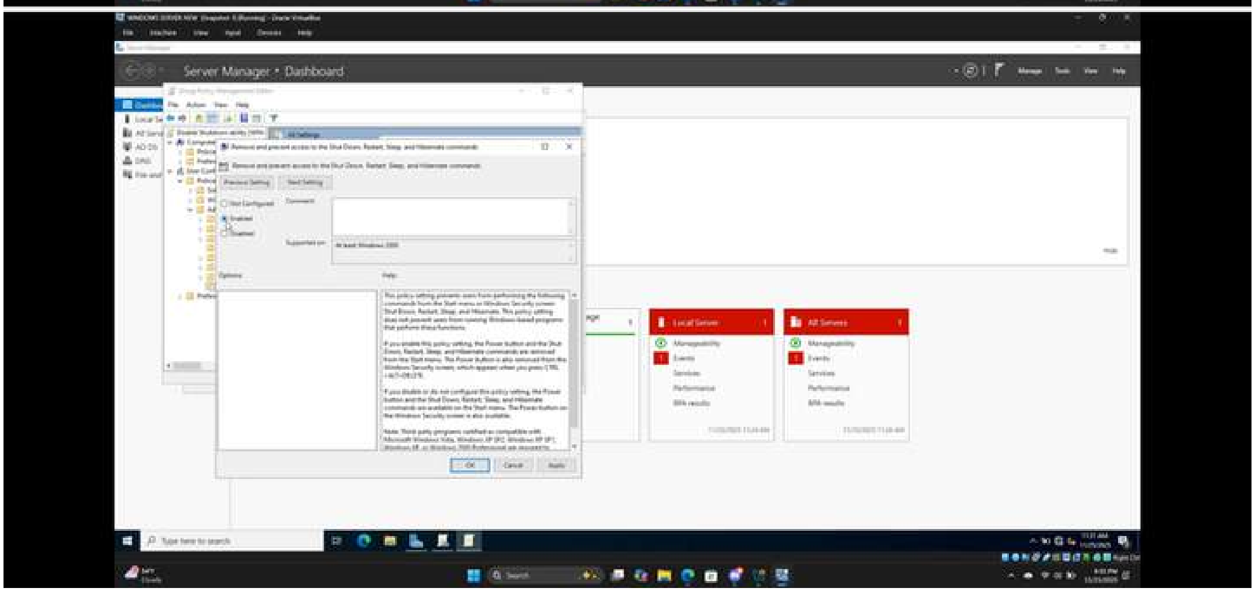
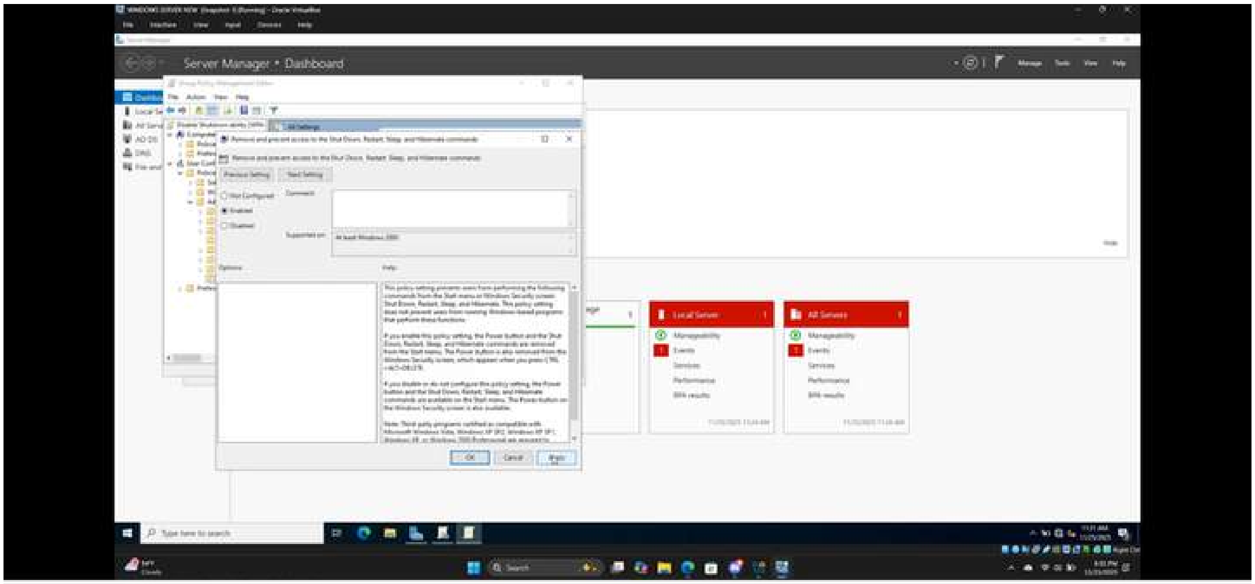
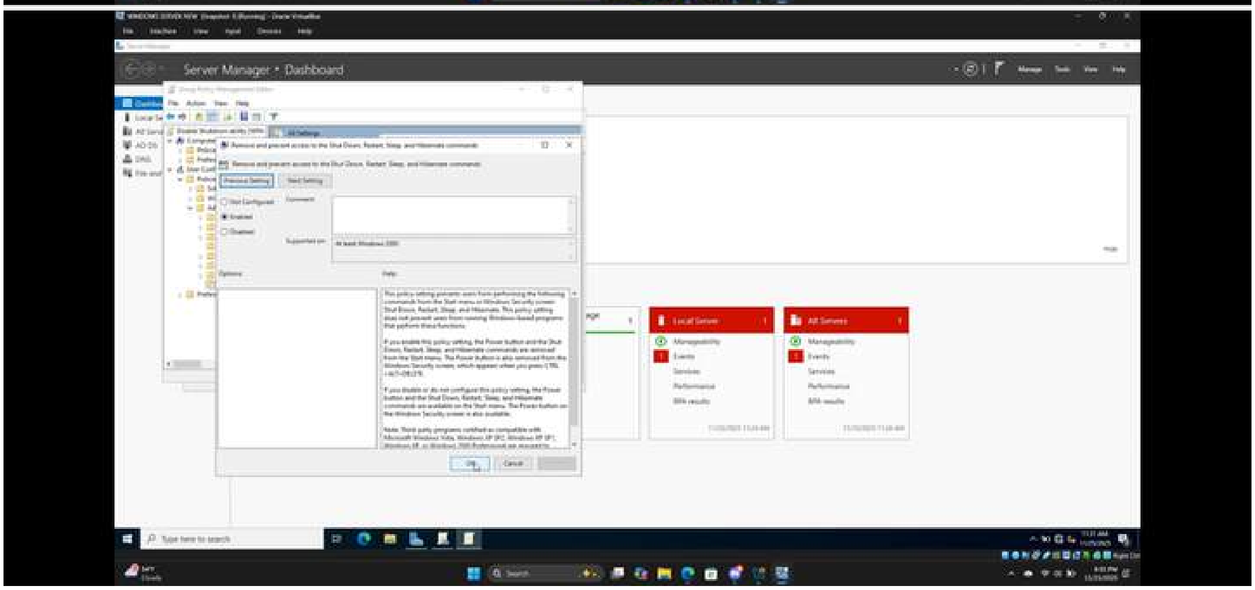

Server Manager • Dashboard
Group Policy Management Editor
Control Panel
Desktop
Network
Shared Folders
Start Menu and Taskbar
System
Windows Components
All Settings
Local Server
All Servers
Manageability
Events
Services
Performance
BPA results


Server Manager • Dashboard
Group Policy Management Editor
All Settings
Control Panel
Desktop
Network
Shared Folders
Start Menu and Taskbar
System
Windows Components
Local Server
All Servers
Manageability
Events
Services
Performance
BPA results

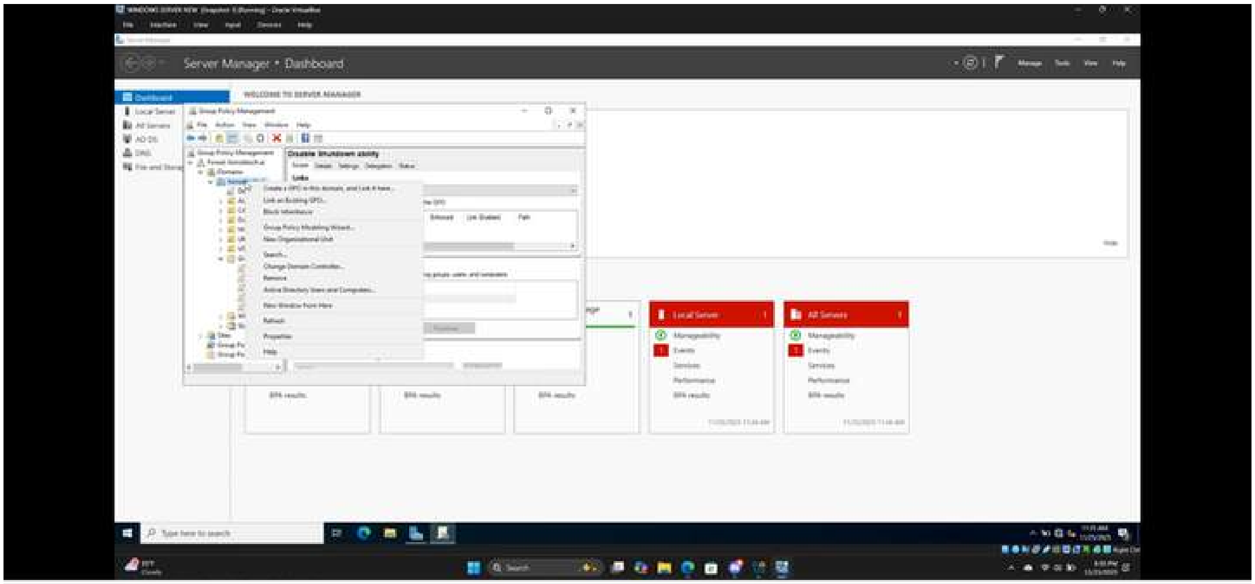

Server Manager • Dashboard

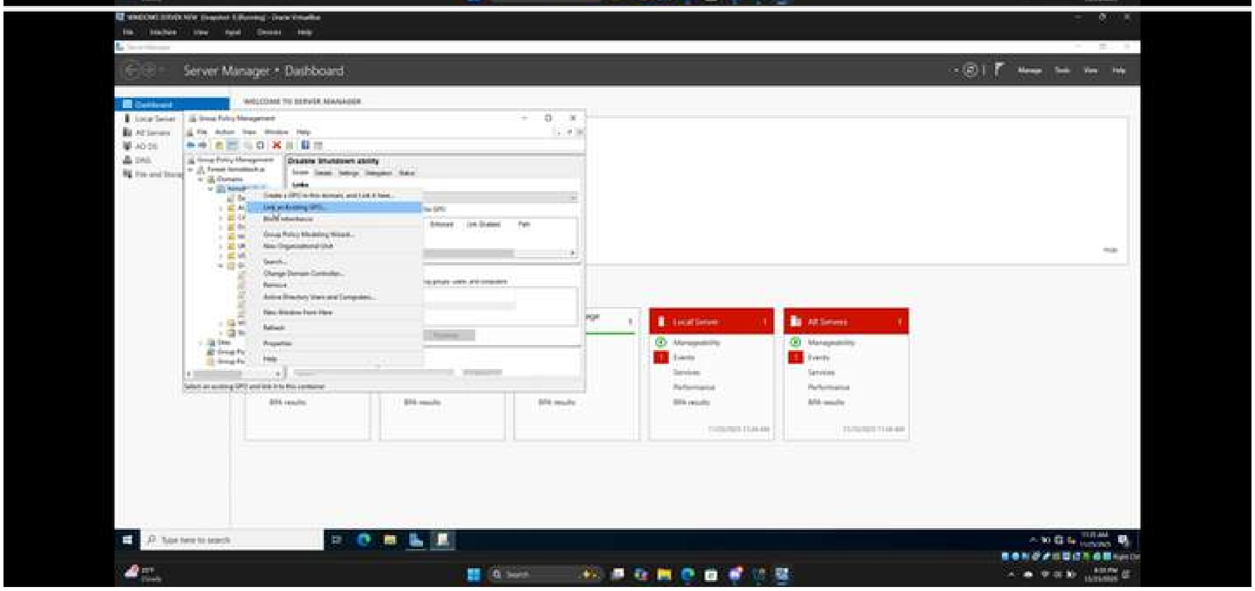

Server Manager • Dashboard

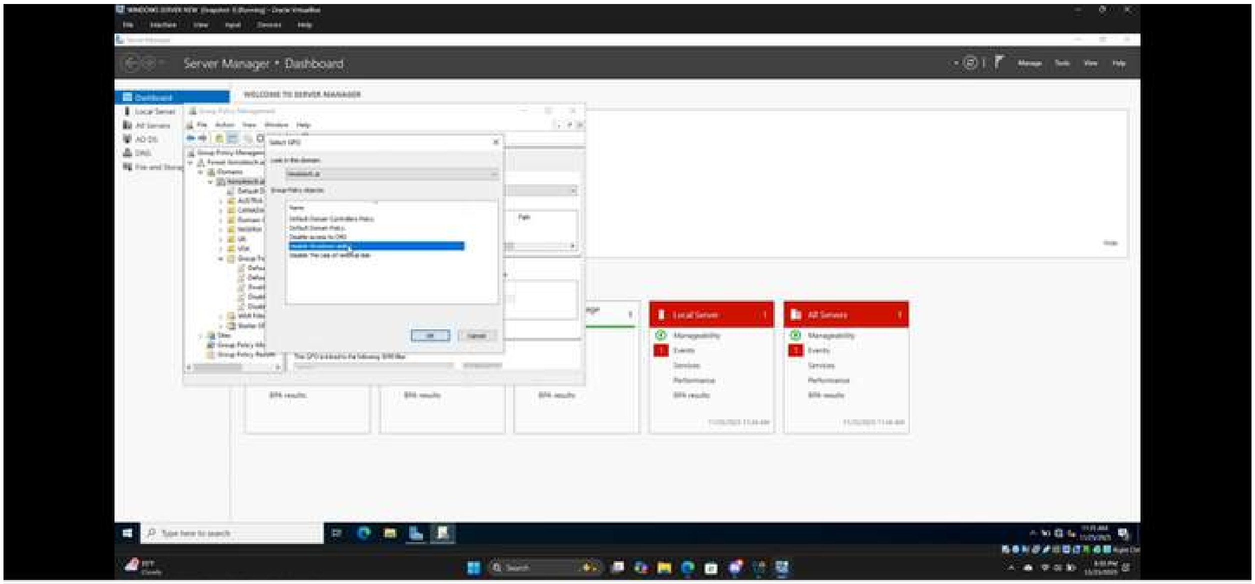


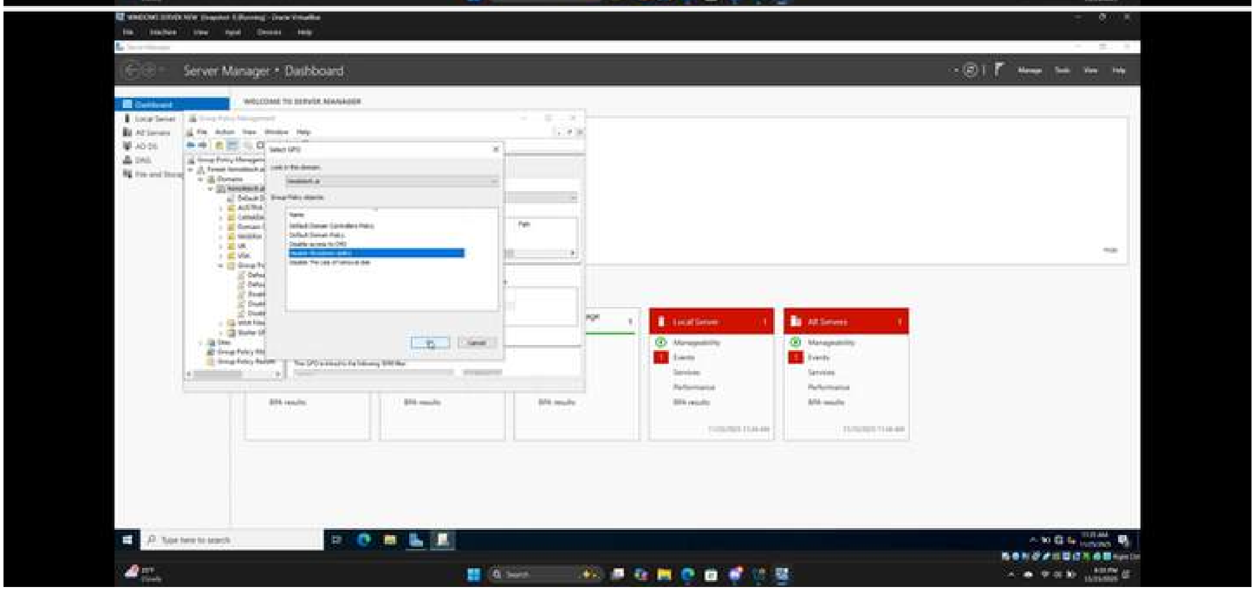


Next is to Link Group Policy Objects (GPOs) to the Domain. To do this , I navigated to the domain root (himoktech.ai) the GPMC, right-clicked, and selected Link an Existing GPO....


Server Manager • Dashboard
Local Server
All Servers


Server Manager • Dashboard
Local Server
All Servers

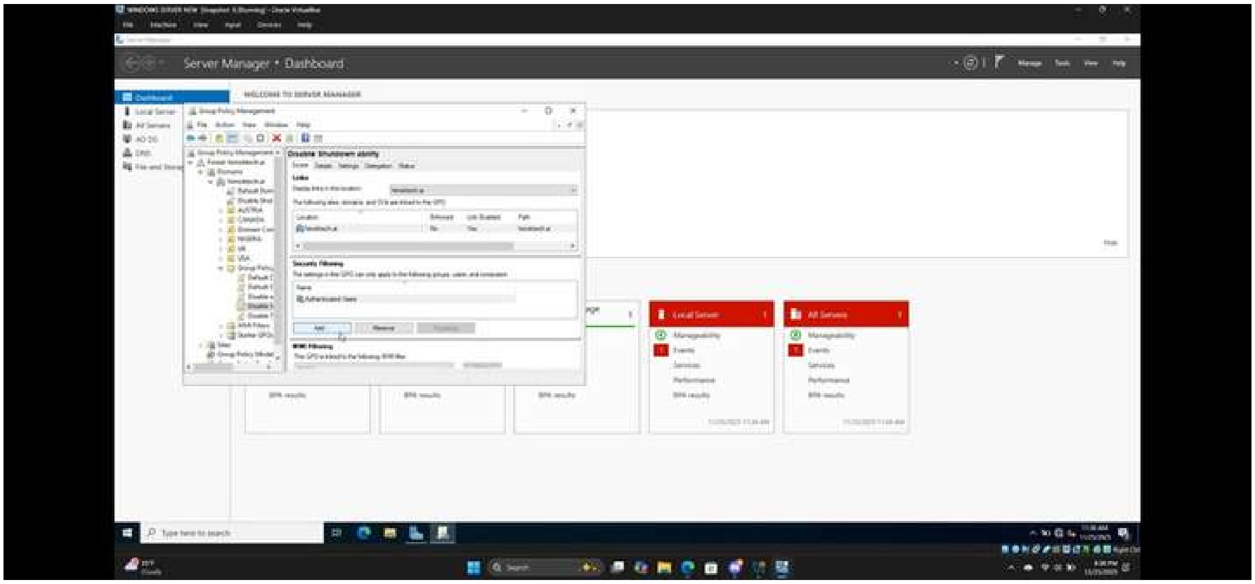




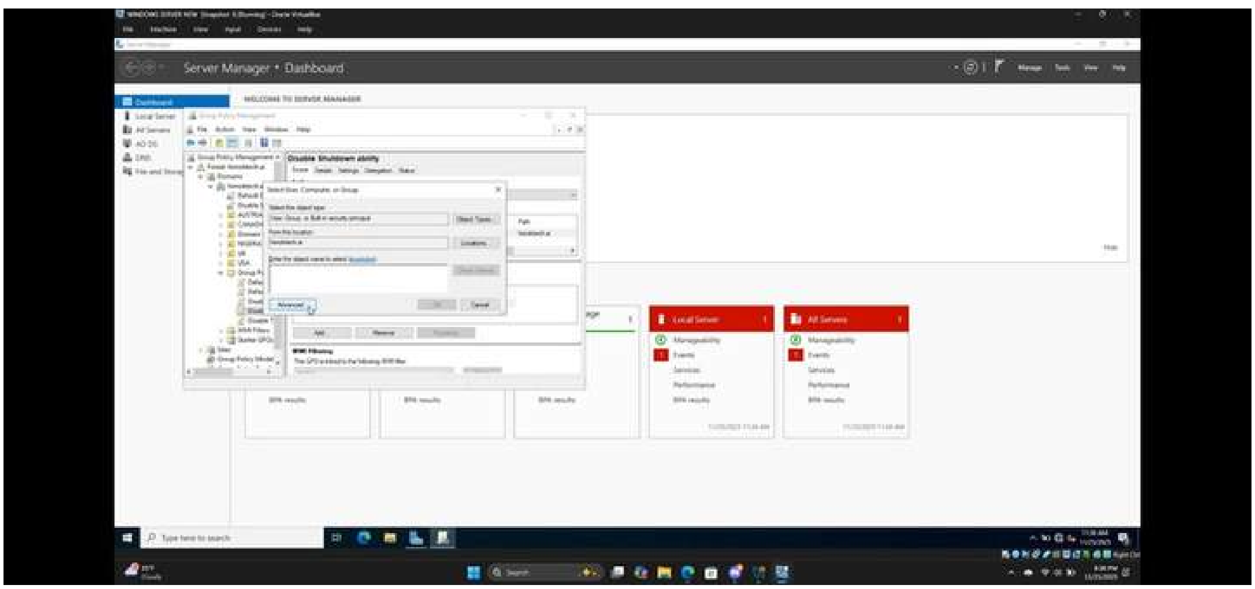
I selected each of the previously created GPOs "Disable the use of removable disk," "Disable shutdown ability," and "Disable access to CMD" one by one and linked them to the domain.

Next is assigned each GPO to every user cos Linking a policy to the entire domain makes it technically accessible to everyone, but Security Filtering is required to control who actually processes and applies the policy.

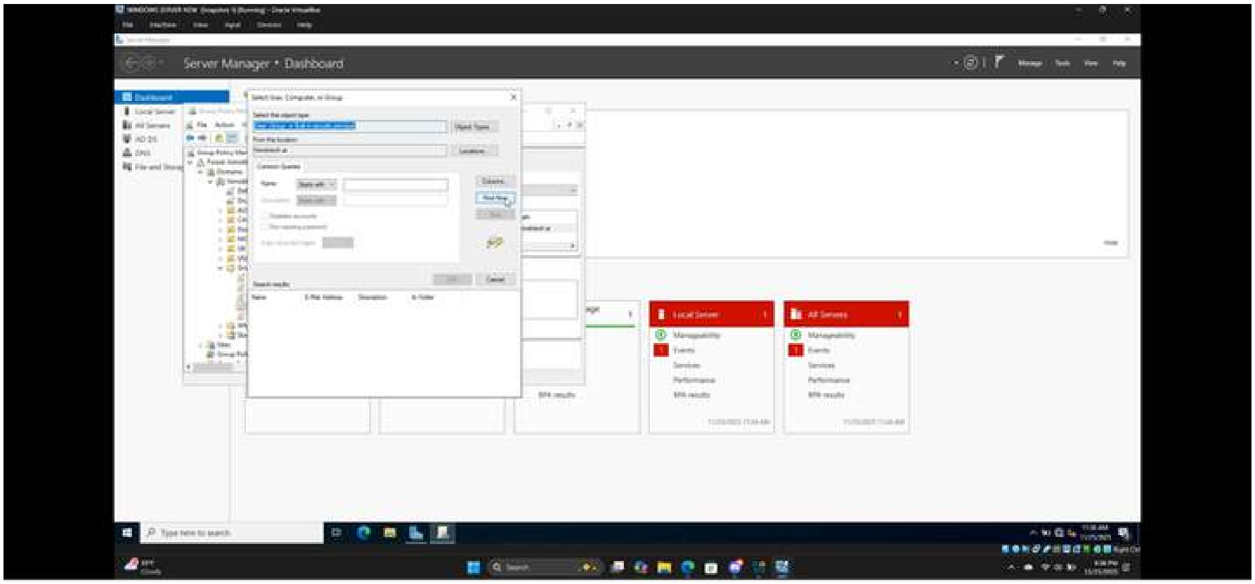
The final stage of policy implementation involves securing the policy by targeting specific users or computers and ensuring immediate propagation across the network.I selected the "Disable shutdown ability" GPO and navigated to the Security Filtering section.used the Add function,

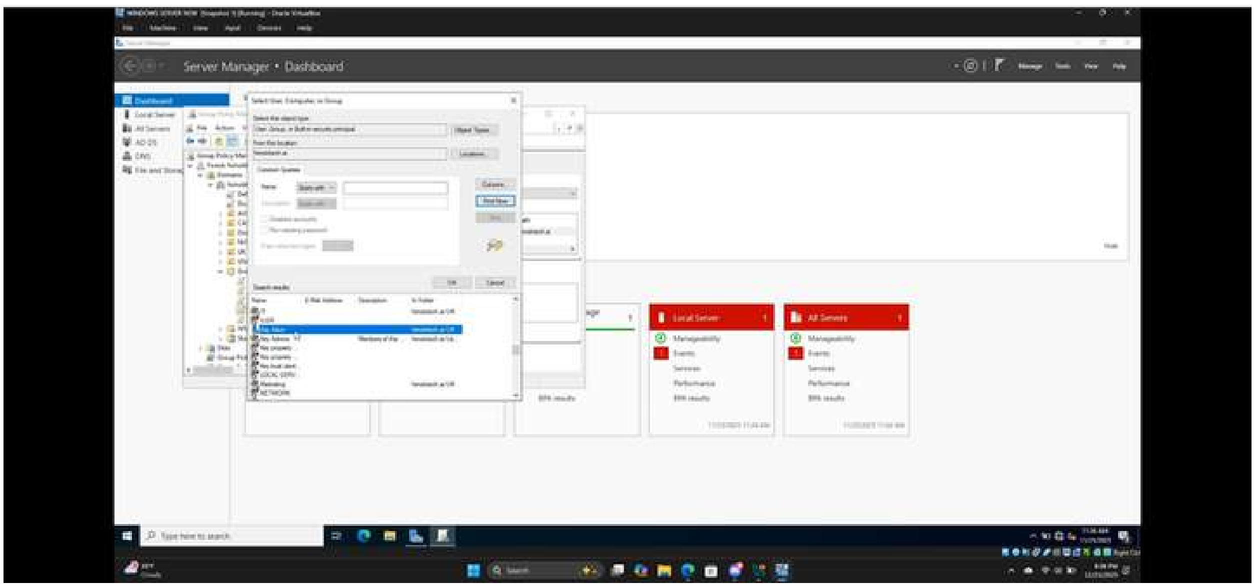


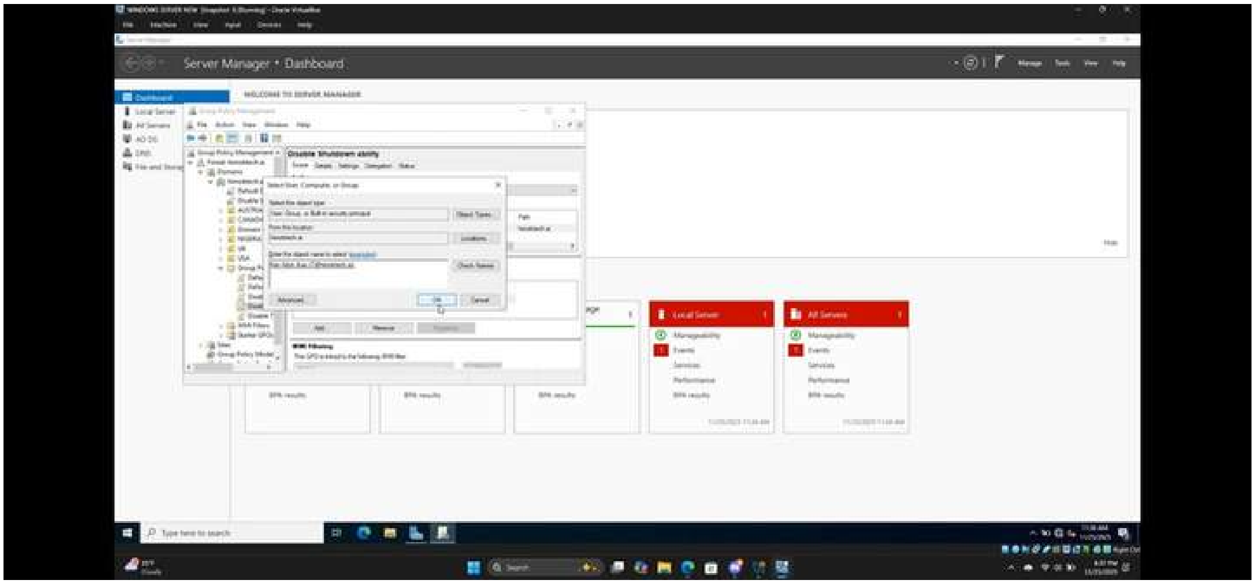
proceeded to Advanced,



and utilized Find Now to explicitly target the security principal (Kay Adun's computer or user account, as implied by the action).


Server Manager • Dashboard
Dashboard
Local Server
All Servers
AD DS
DNS
Select User, Computer, or Group
Object Types...
Locations...
Common Queries
Columns...
Find Now
Stop
OK
Cancel
Search results:
Local Server
Manageability
Events
Services
Performance
BPA results
All Servers
Type here to search


Server Manager • Dashboard
Dashboard
Local Server
All Servers
AD DS
DNS
Select User, Computer, or Group
Object Types...
Locations...
Common Queries
Columns...
Find Now
Stop
OK
Cancel
Search results:
Local Server
Manageability
Events
Services
Performance
BPA results
All Servers
Type here to search



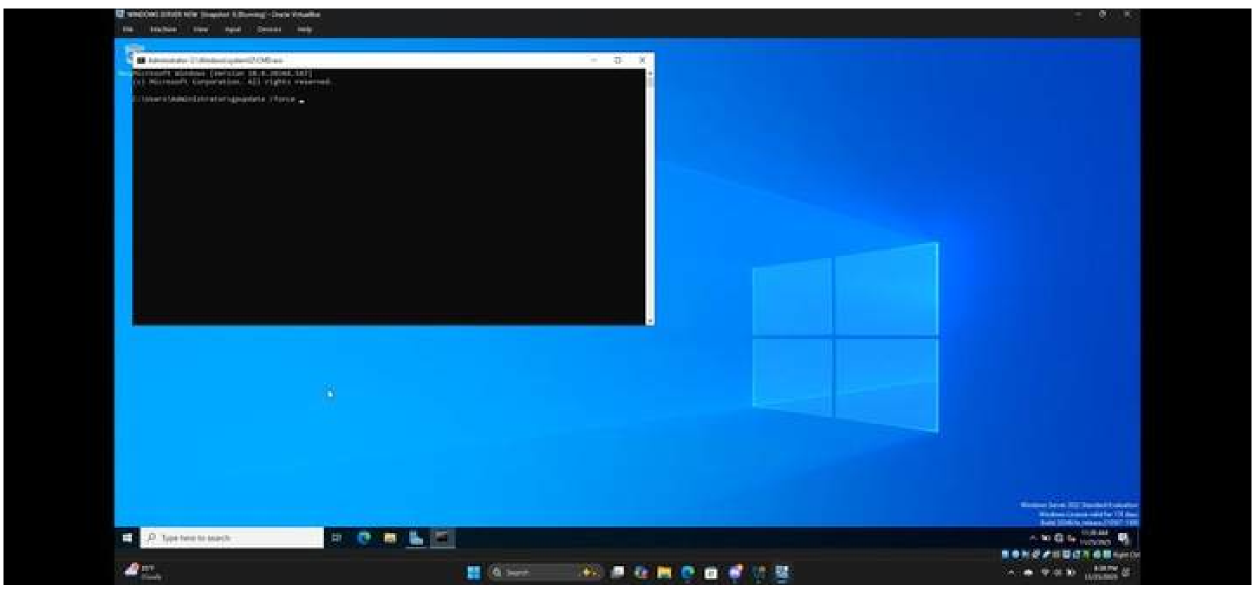
This action modifies the Scope of the GPO, restricting its application only to the specified user or computer, rather than the default of all authenticated users.

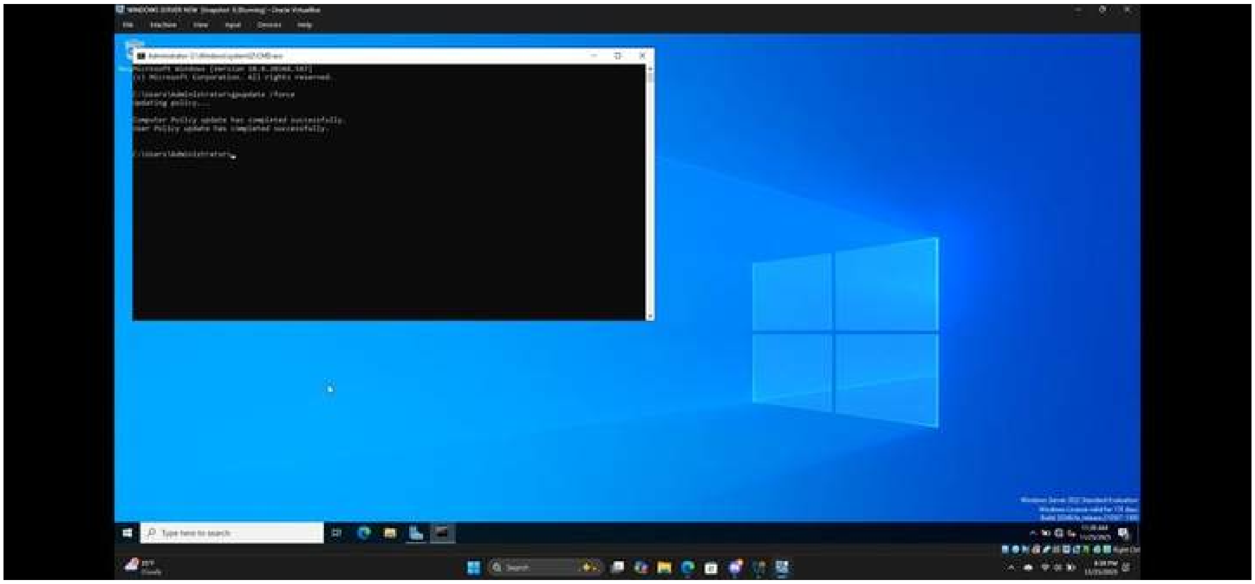
When the policy has been assigned to debz.IT I logged into the client PC assigned to Kay.IT to verify if the shutdown restriction was active.The initial check confirmed that the policy was not yet effective, which is a standard result because clients do not immediately refresh their policy settings.

So to ensure immediate enforcement, i returned to the Server PC

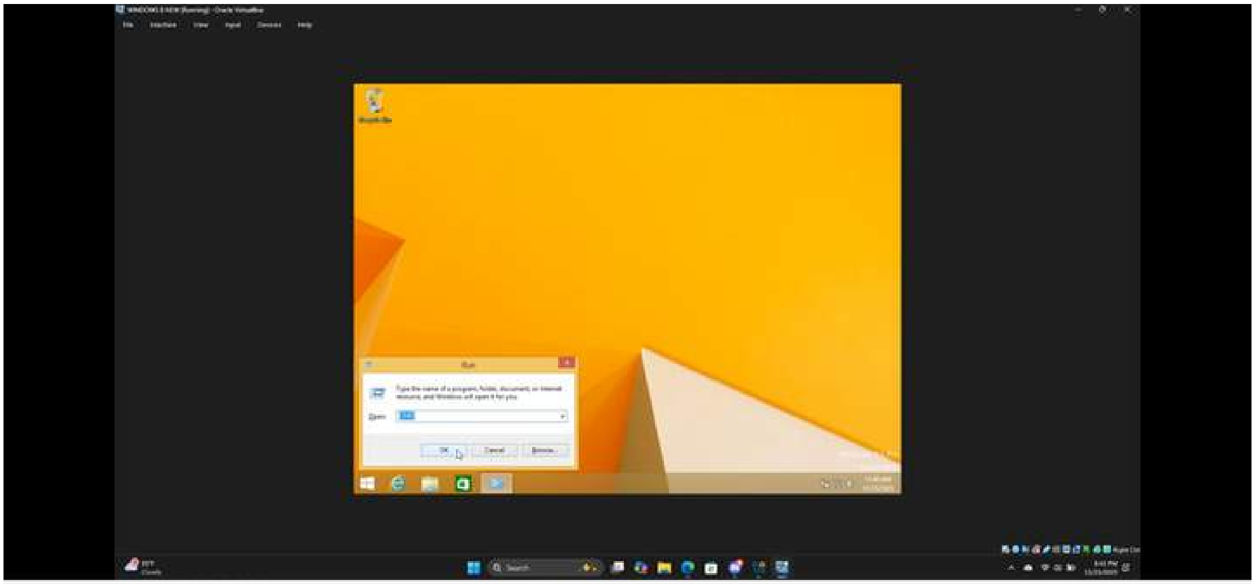
and opened the Command Prompt (Windows Key + R, then CMD).

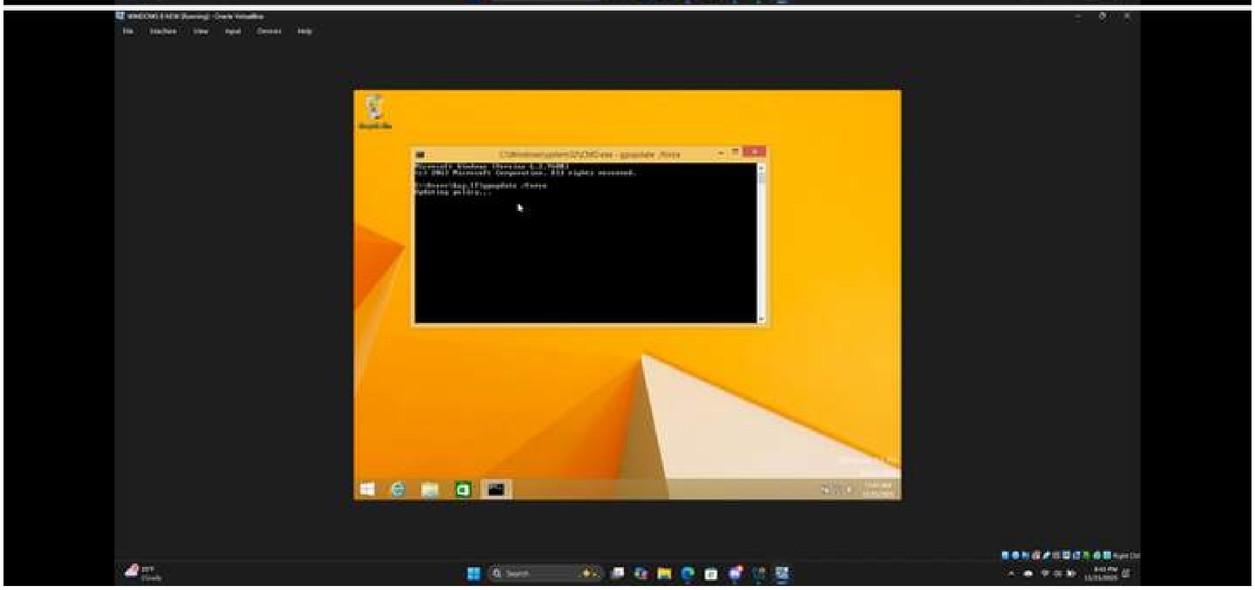
The administrative command “gpupdate /force” was executed. This command bypasses the normal update cycle and forces the server to immediately re-process and push all current Group Policy settings.

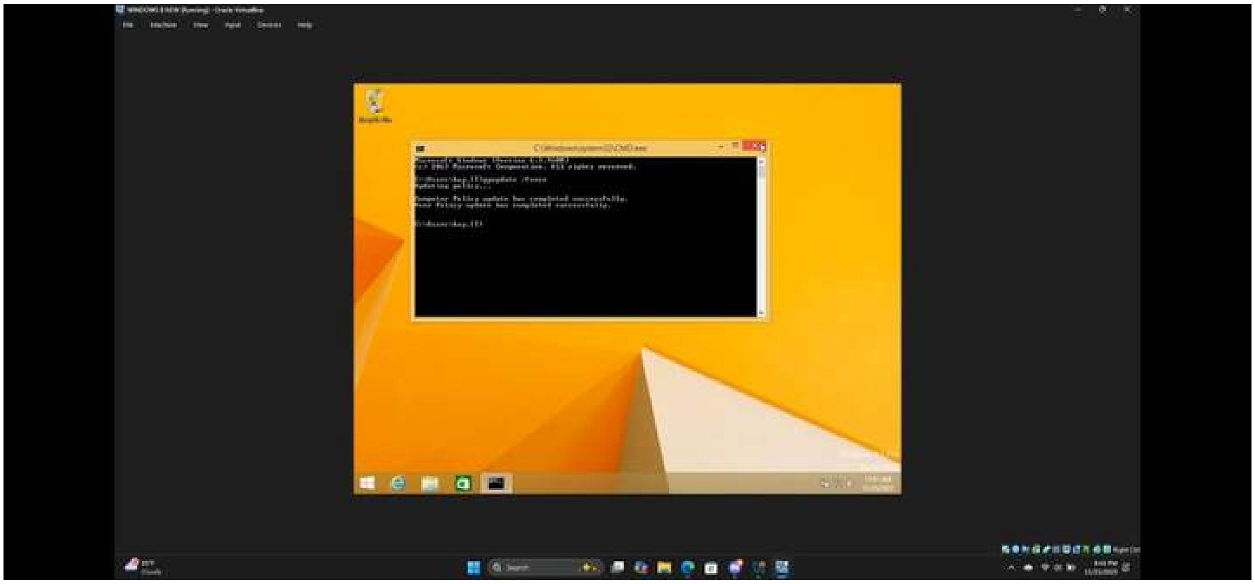




This gpupdate /force command was also executed on the client PC (Deborah Samson machine). This instructs the client to immediately reach out to the server, download, and apply the new restricted settings.







After the forced updates were completed on both the server and the client, a final check on the client machine confirmed that the shutdown ability was successfully disabled, verifying the GPO's effective implementation and security filtering.

End Result (Security and Functional Summary)

The project concluded with a secure, hardened domain environment demonstrating core administrative control.

• Centralized Authentication: The client machine successfully authenticates all users against the Domain Controller, eliminating local user accounts.

• Role-Based Access: All users are assigned to specific security groups, allowing future permissions to be managed efficiently based on their department or role (e.g., IT Group).

• Effective Hardening: The critical phase of hardening was successful. The GPOs restricting fundamental operating system functions, specifically the ability to shut down the machine and execute the command line were successfully enforced on the client endpoint, verified immediately using the gpupdate /force command. This confirms the security policies are active and the domain is being managed centrally.